Packages and Features

Release 10.10

The Sage Development Team

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CHAPTER

ONE

STANDARD PACKAGES

The Sage distribution includes most programs and libraries on which Sage depends. It installs them automatically if it does not find equivalent system packages.

FC

Fortran compiler command passed to `./configure`.

1.1 Mathematics

- *brial: Boolean Ring Algebra implementation using binary decision diagrams*
- *cddlib: Double description method for polyhedral representation conversion*
- *cliquer: Routines for clique searching*
- *conway_polynomials: Python interface to Frank Lübeck's Conway polynomial database*
- *cvxopt: Python software for convex optimization*
- *cypari2: Python interface to the number theory library libpari*
- *eclib: Enumerating and computing with elliptic curves defined over the rational numbers*
- *ecm: Elliptic curve method for integer factorization*
- *elliptic_curves: Databases of elliptic curves*
- *fflas_ffpack: Dense linear algebra over word-size finite fields*
- *flint: Fast Library for Number Theory*
- *fplll: Lattice algorithms, including LLL with floating-point orthogonalization*
- *fpylll: Python interface for FPLLL*
- *gap: Groups, Algorithms, Programming - a system for computational discrete algebra*
- *gf2x: Fast arithmetic in GF(2)[x] and searching for irreducible/primitive trinomials*
- *gfan: Groebner fans and tropical varieties*
- *givaro: C++ library for arithmetic and algebraic computations*
- *glpk: GNU Linear Programming Kit*
- *gmp: Library for arbitrary precision arithmetic*
- *gmpy2: Python interface to GMP/MPIR, MPFR, and MPC*
- *graphs: A database of combinatorial graphs*
- *gsl: The GNU Scientific Library*

- *highs: Linear optimization solver*
- *iml: Integer Matrix Library*
- *lcalc: L-function calculator*
- *libbraiding: Computing with braids*
- *libhomfly: Compute the homfly polynomial of knots and links*
- *linbox: Linear algebra with dense, sparse, structured matrices over the integers and finite fields*
- *lrcalc: Littlewood-Richardson calculator*
- *lrcalc_python: Littlewood-Richardson calculator*
- *m4ri: fast arithmetic with dense matrices over GF(2)*
- *m4rie: Arithmetic with dense matrices over GF(2^e)*
- *maxima: System for manipulating symbolic and numerical expressions*
- *mpc: Arithmetic of complex numbers with arbitrarily high precision and correct rounding*
- *mpfi: Multiple precision interval arithmetic library based on MPFR*
- *mpfr: Multiple-precision floating-point computations with correct rounding*
- *mpmath: Pure Python library for multiprecision floating-point arithmetic*
- *nauty: Find automorphism groups of graphs, generate non-isomorphic graphs*
- *networkx: Python package for complex networks*
- *ntl: A library for doing number theory*
- *numpy: Package for scientific computing with Python*
- *openblas: An optimized implementation of BLAS (Basic Linear Algebra Subprograms)*
- *palp: A package for Analyzing Lattice Polytopes*
- *pari: Computer algebra system for fast computations in number theory*
- *pari_galdata: PARI data package needed to compute Galois groups in degrees 8 through 11*
- *pari_seadata_small: PARI data package needed by ellap for large primes (small version)*
- *planarity: Planarity-related graph algorithms*
- *polytopes_db: Databases of 2- and 3-dimensional reflexive polytopes*
- *ppl: Parma Polyhedra Library*
- *pplpy: Python interface to the Parma Polyhedra Library*
- *primecount: Algorithms for counting primes*
- *primecountpy: Cython interface for C++ primecount library*
- *primesieve: CLI program and C/C++ library for generating primes*
- *qhull: Compute convex hulls, Delaunay triangulations, Voronoi diagrams*
- *rpy2: Python interface to R*
- *rw: Compute rank-width and rank-decompositions*
- *scipy: Scientific tools for Python*
- *singular: Computer algebra system for polynomial computations, algebraic geometry, singularity theory*

- *suitesparse: A suite of sparse matrix software*
- *symmetrica: Library for representation theory*
- *sympow: Computes special values of symmetric power elliptic curve L-functions*
- *sympy: Python library for symbolic mathematics*

1.2 Front-end, graphics, document preparation

- *ipympl: Matplotlib Jupyter Extension*
- *ipython: Interactive computing environment with an enhanced interactive Python shell*
- *ipywidgets: Interactive HTML widgets for Jupyter notebooks and the IPython kernel*
- *jupyterlab: JupyterLab computational environment*
- *jupyterlab_widgets: Jupyter interactive widgets for JupyterLab*
- *matplotlib: Python 2D plotting library*
- *matplotlib_inline: Inline Matplotlib backend for Jupyter*
- *nbconvert: Converting Jupyter Notebooks*
- *notebook: Jupyter notebook, a web-based notebook environment for interactive computing*
- *sagenb_export: Convert legacy SageNB notebooks to Jupyter notebooks and other formats*
- *sagetex: Embed code, results of computations, and plots from Sage into LaTeX documents*
- *tachyon: A ray tracing system*
- *threejs: JavaScript library to display 3D graphics in the browser*

1.3 Other dependencies

- *_prereq: Represents system packages required for installing SageMath from source*
- *accessible_pygments: Accessible Pygments styles*
- *alabaster: Default theme for the Sphinx documentation system*
- *anyio: High level compatibility layer for multiple asynchronous event loop implementations*
- *appdirs: Small Python module for determining appropriate platform-specific dirs, e.g. a "user data dir"*
- *appnope: Disable App Nap on macOS >= 10.9*
- *argon2_cffi: The secure Argon2 password hashing algorithm*
- *argon2_cffi_bindings: Low-level CFFI bindings for Argon2*
- *arrow: Better dates & times for Python*
- *asttokens: Annotate AST trees with source code positions*
- *async_lru: Simple LRU cache for asyncio*
- *attrs: Decorator for Python classes with attributes*
- *babel: Internationalization utilities*
- *beautifulsoup4: Screen-scraping library*
- *beniget: Extract semantic information about static Python code*

- *bleach: An HTML-sanitizing tool*
- *cachetools: Extensible memoizing collections and decorators*
- *calver: Setuptools extension for CalVer package versions*
- *certifi: Python package for providing Mozilla's CA Bundle*
- *cffi: Foreign Function Interface for Python calling C code*
- *chardet: Universal encoding detector for Python 3*
- *charset_normalizer: The Real First Universal Charset Detector. Open, modern and actively maintained alternative to Chardet.*
- *cmake: A cross-platform build system generator*
- *colorama: Cross-platform colored terminal text*
- *comm: Jupyter Python Comm implementation, for usage in ipykernel, xeus-python etc.*
- *contourpy: Python library for calculating contours of 2D quadrilateral grids*
- *cppy: C++ headers for C extension development*
- *curl: Multiprotocol data transfer library and utility*
- *cycler: Composable cycles*
- *cysignals: Interrupt and signal handling for Cython*
- *cython: C-Extensions for Python, an optimizing static compiler*
- *dateutil: Extensions to the standard Python module datetime*
- *debugpy: Implementation of the Debug Adapter Protocol for Python*
- *decorator: Python library providing decorators*
- *defusedxml: Addresses vulnerabilities of XML parsers and XML libraries*
- *distlib: Distribution utilities*
- *docutils: Processing plaintext documentation into useful formats, such as HTML or LaTeX*
- *ecl: An implementation of the Common Lisp language*
- *execnet: Rapid multi-Python deployment*
- *executing: Get the currently executing AST node of a frame, and other information*
- *fastjsonschema: Fastest Python implementation of JSON schema*
- *filelock: Platform independent file lock*
- *flit_core: Distribution-building parts of Flit. See flit package for more information*
- *fonttools: Tools to manipulate font files*
- *fqdn: Validates fully-qualified domain names against RFC 1123, so that they are acceptable to modern bowsers*
- *freetype: A free, high-quality, and portable font engine*
- *furo: A clean customizable Sphinx documentation theme*
- *gast: Python AST that abstracts the underlying Python version*
- *gc: The Boehm-Demers-Weiser conservative garbage collector*
- *gcc: The GNU Compiler Collection or other suitable C and C++ compilers*

- *gengetopt: getopt_long parser generator*
- *gfortran: Fortran compiler from the GNU Compiler Collection*
- *gnulib: Modules imported from Gnulib*
- *gnumake_tokenpool: Jobclient and jobserver for the GNU make tokenpool protocol*
- *h11: A pure-Python, bring-your-own-I/O implementation of HTTP/1.1*
- *hatchling: Modern, extensible Python build backend*
- *hatchling: Modern, extensible Python build backend*
- *hatchling: Modern, extensible Python build backend*
- *httpcore: Minimal low-level HTTP client*
- *httpx: The next generation HTTP client.*
- *hypothesis: A library for property-based testing*
- *iconv: Library for language/country-dependent character encodings*
- *idna: Internationalized Domain Names in Applications (IDNA)*
- *imagesize: Getting image size from png/jpeg/jpeg2000/gif file*
- *info: stand-alone Info documentation reader*
- *iniconfig: Brain-dead simple config-ini parsing*
- *ipykernel: IPython Kernel for Jupyter*
- *ipython_genutils: Vestigial utilities from IPython*
- *ipython_pygments_lexers: Pygments lexers for IPython code*
- *isoduration: Operations with ISO 8601 durations*
- *jedi: Static analysis tool providing IDE support for Python*
- *jinja2: General purpose template engine for Python*
- *json5: Python implementation of the JSON5 data format*
- *jsonpointer: Identify specific nodes in a JSON document (RFC 6901)*
- *jsonschema: Python implementation of JSON Schema*
- *jsonschema_specifications: The JSON Schema meta-schemas and vocabularies, exposed as a Registry*
- *jupyter_client: Jupyter protocol implementation and client libraries*
- *jupyter_core: Jupyter core package*
- *jupyter_events: Jupyter Event System library*
- *jupyter_lsp: Multi-Language Server WebSocket proxy for Jupyter Notebook/Lab server*
- *jupyter_server: The backend (core services, APIs, REST endpoints) to Jupyter web applications*
- *jupyter_server_terminals: A Jupyter Server Extension Providing Terminals*
- *jupyter_sphinx: Jupyter Sphinx Extension*
- *jupyterlab_mathjax2: A MathJax Typesetting provider for JupyterLab 4 and above*
- *jupyterlab_pygments: Pygments theme using JupyterLab CSS variables*
- *jupyterlab_server: Set of server components for JupyterLab and JupyterLab like applications*

- *kiwisolver: Fast implementation of the Cassowary constraint solver*
- *libatomic_ops: Access hardware-provided atomic memory update operations*
- *libffi: A portable foreign-function interface library*
- *libgd: Dynamic graphics generation tool*
- *libpng: Bitmap image support*
- *markupsafe: Safely add untrusted strings to HTML/XML markup*
- *mathjax: A JavaScript library for displaying mathematical formulas*
- *memory_allocator: An extension class to allocate memory easily with Cython*
- *meson: A high performance build system*
- *meson_python: Meson Python build backend (PEP 517)*
- *mistune: Sane and fast Markdown parser with useful plugins and renderers*
- *nbclient: Client library for executing notebooks. Formerly nbconvert's ExecutePreprocessor*
- *nbformat: The Jupyter Notebook format*
- *ncurses: Classic terminal output library*
- *nest_asyncio: Patch asyncio to allow nested event loops*
- *ninja_build: A build system with a focus on speed*
- *notebook_shim: A shim layer for notebook traits and config*
- *openssl: Implementation of the SSL and TLS protocols*
- *overrides: Decorator to automatically detect mismatch when overriding a method*
- *packaging: Core utilities for Python packages*
- *pandocfilters: A Python module for writing pandoc filters*
- *parso: Python Parser*
- *patchelf: A small utility to modify the dynamic linker and RPATH of ELF executables*
- *pathspec: Utility library for gitignore style pattern matching of file paths*
- *pexpect: Python module for controlling and automating other programs*
- *pickleshare: A 'shelve' like datastore with concurrency support*
- *pillow: Python Imaging Library*
- *pip: Tool for installing and managing Python packages*
- *pkgconfig: Python interface to pkg-config*
- *platformdirs: Small Python package for determining appropriate platform-specific dirs, e.g. a "user data dir"*
- *pluggy: Plugin and hook calling mechanisms for python*
- *ply: Python Lex & Yacc*
- *pplpy_doc: Python interface to the Parma Polyhedra Library (documentation)*
- *prometheus_client: Python client for the systems monitoring and alerting toolkit Prometheus*
- *prompt_toolkit: Interactive command lines for Python*
- *psutil: Cross-platform lib for process and system monitoring in Python.*

- *ptyprocess: Python interaction with subprocesses in a pseudoterminal*
- *pure_eval: Safely evaluate AST nodes without side effects*
- *py: Library with cross-python path, ini-parsing, io, code, log facilities*
- *pybind11: Create Python bindings to C++ code*
- *pycparser: Parser of the C language in Python*
- *pygments: Generic syntax highlighter*
- *pyparsing: A Python parsing module*
- *pyproject_api: API to interact with the python pyproject.toml based projects*
- *pyproject_hooks: Wrappers to call pyproject.toml-based build backend hooks.*
- *pyproject_metadata: PEP 621 metadata parsing*
- *pytest: Simple powerful testing with Python*
- *pytest_mock: Thin-wrapper around the mock package for easier use with pytest*
- *pytest_xdist: Pytest xdist plugin for distributed testing, most importantly across multiple CPUs*
- *python_build: A simple, correct PEP517 package builder*
- *python_discovery: Python interpreter discovery*
- *python_json_logger: Python library adding a json log formatter*
- *pythran: Ahead of Time compiler for numeric kernels*
- *pytz: Timezone definitions for Python*
- *pytz_deprecation_shim: Shims to make deprecation of pytz easier*
- *pyyaml: YAML parser and emitter for Python*
- *pyzmq: Python bindings for the zeromq networking library*
- *readline: Command line editing library*
- *referencing: JSON Referencing + Python*
- *requests: An HTTP library for Python*
- *rfc3339_validator: A pure python RFC3339 validator*
- *rfc3986_validator: Pure python rfc3986 validator*
- *roman-numerals: Manipulate well-formed Roman numerals*
- *rpds_py: Python bindings to Rust's persistent data structures*
- *sage_docbuild: Build system of the Sage documentation*
- *sage-setup: Build system of the SageMath library*
- *scikit_build_core: Build backend for CMake based projects*
- *send2trash: Send file to trash natively under Mac OS X, Windows and Linux*
- *setuptools: Build system for Python packages*
- *setuptools_scm: Python build system extension to obtain package version from version control*
- *six: Python 2 and 3 compatibility utilities*
- *sniffio: Sniff out which async library your code is running under*

- *snowballstemmer: Stemmer algorithms for natural language processing in Python*
- *sortedcontainers: sorted collections library*
- *soupsieve: Modern CSS selector implementation for Beautiful Soup*
- *sphinx: Python documentation generator*
- *sphinx_basic_ng: Modern skeleton for Sphinx themes*
- *sphinx_copybutton: Add a copy button to each of your code cells*
- *sphinx_inline_tabs: Add inline tabbed content to your Sphinx documentation*
- *sphinxcontrib_applehelp: Sphinx extension which outputs Apple help book*
- *sphinxcontrib_devhelp: Sphinx extension which outputs Devhelp documents*
- *sphinxcontrib_htmlhelp: Sphinx extension which outputs HTML help book*
- *sphinxcontrib_jsmath: Sphinx extension which renders display math in HTML via JavaScript*
- *sphinxcontrib_qthelp: Sphinx extension which outputs QtHelp documents*
- *sphinxcontrib_serializinghtml: Sphinx extension which outputs serialized HTML files*
- *sphinxcontrib_websupport: Sphinx API for Web apps*
- *stack_data: Extract data from python stack frames and tracebacks for informative displays*
- *terminado: Tornado websocket backend for the term.js Javascript terminal emulator library*
- *tinycss2: A tiny CSS parser*
- *tomli_w: A TOML writer for Python*
- *tornado: Python web framework and asynchronous networking library*
- *tox: tox is a generic virtualenv management and test command line tool*
- *traitlets: Traitlets Python configuration system*
- *trove_classifiers: Canonical source for classifiers on PyPI (pypi.org)*
- *types_python_dateutil: Typing stubs for python-dateutil*
- *typing_extensions: Backported and Experimental Type Hints for Python 3.8+*
- *tzdata: Provider of IANA time zone data*
- *tzlocal: Python timezone information for the local timezone*
- *uri_template: RFC 6570 URI Template Processor*
- *urllib3: HTTP library with thread-safe connection pooling, file post, and more*
- *vcs_versioning: Version management from VCS metadata*
- *virtualenv: Virtual Python Environment builder*
- *wcwidth: Measures the displayed width of unicode strings in a terminal*
- *webcolors: Library for working with the color formats defined by HTML and CSS*
- *webencodings: Character encoding aliases for legacy web content*
- *websocket_client: WebSocket client for Python with low level API options*
- *wheel: A built-package format for Python*
- *widgetsnbextension: Jupyter interactive widgets for Jupyter Notebook*

- *zeromq: A modern networking library*
- *zipp: A pathlib-compatible zipfile object wrapper*

CHAPTER

TWO

OPTIONAL PACKAGES

For additional functionality, you can install some of the following optional packages.

2.1 Mathematics

- *4ti2: Algebraic, geometric and combinatorial problems on linear spaces*
- *admcycles: Computation in the tautological ring of the moduli space of curves*
- *benzene: Generate fusenes and benzenoids with a given number of faces*
- *bliss: Computing automorphism groups and canonical forms of graphs*
- *buckygen: Efficient generation of nonisomorphic fullerenes*
- *cbc: COIN-OR branch and cut solver for mixed-integer programs*
- *clarabel: Clarabel Conic Interior Point Solver for Rust / Python*
- *coxeter3: Library for Coxeter groups, Bruhat ordering, Kazhdan-Lusztig polynomials*
- *csdp: Solver for semidefinite programs*
- *cunningham_tables: List of the prime numbers occuring in the Cunningham table*
- *cvxpy: Domain-specific language for modeling convex optimization problems in Python*
- *cylp: Python interface for CLP, CBC, and CGL*
- *database_cremona_ellcurve: Database of elliptic curves*
- *database_cubic_hecke: Ivan Marin's representations of the cubic Hecke algebra*
- *database_jones_numfield: Table of number fields*
- *database_knotinfo: Content of the KnotInfo and LinkInfo databases as lists of dictionaries*
- *database_kohel: Database of modular and Hilbert polynomials*
- *database_mutation_class: Database of exceptional mutation classes of quivers*
- *database_odlyzko_zeta: Table of zeros of the Riemann zeta function*
- *database_stein_watkins: Database of elliptic curves (full version)*
- *database_stein_watkins_mini: Database of elliptic curves (small version)*
- *database_symbolic_data: Database from the SymbolicData project*
- *dsdp: Semidefinite programming solver*
- *e_antic: Real embedded number fields*

- *ecos_python: Embedded Cone Solver (Python wrapper)*
- *fricas: A general purpose computer algebra system*
- *frobby: Computations on monomial ideals*
- *gap_jupyter: Jupyter kernel for GAP*
- *gap_packages: A collection of GAP packages*
- *giac: A general purpose computer algebra system*
- *glucose: A SAT solver*
- *gp2c: A compiler for translating GP routines to C*
- *igraph: A library for creating and manipulating graphs*
- *isl: Sets and relations of integer points bounded by affine constraints*
- *jupymake: A Python wrapper for the polymake shell*
- *kenzo: Construct topological spaces and compute homology groups*
- *kissat: SAT solver*
- *latte_int: Count lattice points, compute volumes, and integrate over convex polytopes*
- *libnauty: Find automorphism groups of graphs, generate non-isomorphic graphs (callable library)*
- *libsemigroups: Library for semigroups and monoids*
- *lidia: A library for computational number theory*
- *lrslib: Reverse search algorithm for vertex enumeration and convex hull problems*
- *mathics: General-purpose computer algebra system*
- *matroid_database: Python interface to matroid database*
- *mcqd: An exact algorithm for finding a maximum clique in an undirected graph*
- *meataxe: Library for computing with modular representations*
- *modular_resolution: Modular cohomology rings of finite groups*
- *mpfrcx: Arithmetic of univariate polynomials over arbitrary precision real or complex numbers*
- *msolve: Multivariate polynomial system solver*
- *normaliz: Computations in affine monoids, vector configurations, lattice polytopes, and rational cones*
- *ore_algebra: Ore algebra*
- *osqp_python: The Operator Splitting QP Solver (Python wrapper)*
- *p_group_cohomology: Modular cohomology rings of finite groups*
- *papilo: Parallel presolve for integer and linear optimization*
- *pari_elldata: PARI data package for elliptic curves*
- *pari_galpol: PARI data package for polynomials defining Galois extensions of the rationals*
- *pari_jupyter: Jupyter kernel for PARI/GP*
- *pari_nftables: PARI data package for number fields*
- *pari_seadata: PARI data package needed by ellap for large primes (full version)*
- *phitigra: Graph editor for SageMath/Jupyter*

- *plantri: Generate non-isomorphic sphere-embdedded graphs*
- *polymake: Computations with polyhedra, fans, simplicial complexes, matroids, graphs, tropical hypersurfaces*
- *polytopes_db_4d: Database of 4-dimensional reflexive polytopes*
- *pycosat: SAT solver picosat with Python bindings*
- *pycryptosat: Python module of cryptominisat*
- *pynormaliz: Python bindings for the normaliz library*
- *pyscipopt: Python interface and modeling environment for SCIP*
- *pysingular: A basic Python interface to Singular*
- *python_igraph: Python bindings for igraph*
- *qdldl_python: QDLDL, a free LDL factorization routine (Python wrapper)*
- *qepcad: Quantifier elimination by partial cylindrical algebraic decomposition*
- *r: A free software environment for statistical computing and graphics*
- *rubiks: Programs for Rubik's cube*
- *saclib: Computations with real algebraic numbers*
- *sage_flatsurf: Flat surfaces in SageMath*
- *sage_numerical_backends_coin: COIN-OR backend for Sage MixedIntegerLinearProgram*
- *sage_numerical_backends_cplex: Cplex backend for Sage MixedIntegerLinearProgram*
- *sage_numerical_backends_gurobi: Gurobi backend for Sage MixedIntegerLinearProgram*
- *sbcl: a lisp compiler and runtime system*
- *scip: Mixed integer programming solver*
- *scip_sdp: Mixed integer semidefinite programming plugin for SCIP*
- *scs: Splitting conic solver*
- *semigroups: An optional GAP package*
- *singular_jupyter: Jupyter kernel for Singular*
- *sirocco: Compute topologically certified root continuation of bivariate polynomials*
- *slabbe: Sébastien Labbé's Research code*
- *snappy: Topology and geometry of 3-manifolds, with a focus on hyperbolic structures*
- *soplex: Linear optimization solver using the revised simplex method*
- *surface_dynamics: dynamics on surfaces (measured foliations, interval exchange transformation, Teichmüller flow, etc)*
- *symengine: A C++ symbolic manipulation library*
- *symengine_py: Python wrappers for SymEngine*
- *tdlib: Algorithms for computing tree decompositions of graphs*
- *tides: Integration of ODEs*
- *topcom: Compute triangulations of point configurations and oriented matroids*

2.2 Front-end, graphics, document preparation

- *dot2tex: Create PGF/TikZ commands from Graphviz output*
- *graphviz: Graph visualization software*
- *jmol: Java viewer for chemical structures in 3D*
- *jupyter_jsmol: JSmol viewer widget for Jupyter*
- *pandoc: A document converter*
- *pygraphviz: Python interface to Graphviz*
- *rst2ipynb: Convert reStructuredText files to Jupyter notebooks*
- *texlive: A comprehensive TeX system*

2.3 Other dependencies

- *_bootstrap: Represents system packages required for running the top-level bootstrap script*
- *_develop: Represents system packages recommended for development*
- *_recommended: Represents system packages recommended for additional functionality*
- *_sagemath: Downstream package of Sage in distributions*
- *auditwheel_or_delocate: Repair wheels on Linux or macOS*
- *biopython: Tools for computational molecular biology*
- *ccache: A compiler cache*
- *d3js: JavaScript library for manipulating documents based on data*
- *ffmpeg: ffmpeg video converter*
- *free_fonts: a free family of scalable outline fonts*
- *gdb: The GNU Project debugger*
- *git: Version control system*
- *github_cli: Command-line interface for GitHub*
- *gitpython: GitPython is a python library used to interact with Git repositories*
- *graph_genus: Graph genus algorithms*
- *highspy: Python interface to HiGHS optimizer*
- *imagemagick: A collection of tools and libraries for many image file formats*
- *joblib: Lightweight pipelining with Python functions*
- *khoca: Khoca as pip installable package*
- *libgraphviz: Graph visualization software (callable library)*
- *libjpeg: JPEG image support*
- *libogg: Library for the Ogg multimedia container format*
- *libxml2: XML parser and toolkit*
- *llvm: The LLVM Compiler Infrastructure, including the Clang C/C++/Objective-C compiler*

- *nibabel: Access a multitude of neuroimaging data formats*
- *notedown: Create IPython notebooks from markdown*
- *onetbb: oneAPI Threading Building Blocks*
- *pandoc_attributes: A parser and generator for pandoc block attributes*
- *perl_cpan_polymake_prereq: Represents all Perl packages that are prerequisites for polymake*
- *perl_mongodb: A prerequisite for polymake's PolyDB feature*
- *perl_term_readline_gnu: Perl extension for the GNU Readline/History libraries*
- *pybtex: A BibTeX-compatible bibliography processor in Python*
- *pyppeteer: Headless chrome/chromium automation library*
- *python_flint: Bindings for FLINT and Arb*
- *pyx: Generate PostScript, PDF, and SVG files in Python*
- *regina: Low-dimensional topology: triangulations, knots, links, normal surfaces, angle structures*
- *rpy2_rinterface: Low-level interface from Python to R*
- *rpy2_robjects: High-level interface from Python to R*
- *sqlalchemy: A database abstraction library*
- *texlive_luatex: LuaTeX packages*
- *texttable: Python module for creating simple ASCII tables*
- *valgrind: Memory error detector, call graph generator, runtime profiler*
- *xindy: a general-purpose index processor*

CHAPTER

THREE

FEATURES

3.1 Testing for features of the environment at runtime

A computation can require a certain package to be installed in the runtime environment. Abstractly such a package describes a *Feature* which can be tested for at runtime. It can be of various kinds, most prominently an *Executable* in the PATH, a *PythonModule*, or an additional package for some installed system such as a *GapPackage*.

AUTHORS:

- Julian Rüth (2016-04-07): Initial version
- Jeroen Demeyer (2018-02-12): Refactoring and clean up

EXAMPLES:

Some generic features are available for common cases. For example, to test for the existence of a binary, one can use an *Executable* feature:

```
sage: from sage.features import Executable
sage: Executable(name='sh', executable='sh').is_present()
FeatureTestResult('sh', True)
```

```
>>> from sage.all import *
>>> from sage.features import Executable
>>> Executable(name='sh', executable='sh').is_present()
FeatureTestResult('sh', True)
```

Here we test whether the grape GAP package is available:

```
sage: from sage.features.gap import GapPackage
sage: GapPackage("grape", spkg='gap_packages').is_present()  # optional - gap_package_
↪grape
FeatureTestResult('gap_package_grape', True)
```

```
>>> from sage.all import *
>>> from sage.features.gap import GapPackage
>>> GapPackage("grape", spkg='gap_packages').is_present()  # optional - gap_package_
↪grape
FeatureTestResult('gap_package_grape', True)
```

Note that a *FeatureTestResult* acts like a bool in most contexts:

```
sage: if Executable(name='sh', executable='sh').is_present(): "present."
'present.'
```

```
>>> from sage.all import *
>>> if Executable(name='sh', executable='sh').is_present(): "present."
'present.'
```

When one wants to raise an error if the feature is not available, one can use the `require` method:

```
sage: Executable(name='sh', executable='sh').require()

sage: Executable(name='random', executable='randomOochoz6x', spkg='random', url=
↪'http://rand.om').require() # optional - sage_spkg
Traceback (most recent call last):
...
FeatureNotPresentError: random is not available.
Executable 'randomOochoz6x' not found on PATH.
...try to run...sage -i random...
Further installation instructions might be available at http://rand.om.
```

```
>>> from sage.all import *
>>> Executable(name='sh', executable='sh').require()

>>> Executable(name='random', executable='randomOochoz6x', spkg='random', url='http://
↪rand.om').require() # optional - sage_spkg
Traceback (most recent call last):
...
FeatureNotPresentError: random is not available.
Executable 'randomOochoz6x' not found on PATH.
...try to run...sage -i random...
Further installation instructions might be available at http://rand.om.
```

As can be seen above, features try to produce helpful error messages.

class sage.features.**Executable**(**args*, ***kwds*)

Bases: *FileFeature*

A feature describing an executable in the `PATH`.

The executable is searched first in `SAGE_LOCAL/bin`, then in `PATH`.

Note

Overwrite *is_functional()* if you also want to check whether the executable shows proper behaviour.

Calls to *is_present()* are cached. You might want to cache the *Executable* object to prevent unnecessary calls to the executable.

EXAMPLES:

```
sage: from sage.features import Executable
sage: Executable(name='sh', executable='sh').is_present()
FeatureTestResult('sh', True)
sage: Executable(name='does-not-exist', executable='does-not-exist-xxxxyxyyxyy').
↪is_present()
FeatureTestResult('does-not-exist', False)
```

```
>>> from sage.all import *
>>> from sage.features import Executable
>>> Executable(name='sh', executable='sh').is_present()
FeatureTestResult('sh', True)
>>> Executable(name='does-not-exist', executable='does-not-exist-xxxxyxyyxyy').is_
↪present()
FeatureTestResult('does-not-exist', False)
```

absolute_filename()

The absolute path of the executable as a string.

EXAMPLES:

```
sage: from sage.features import Executable
sage: Executable(name='sh', executable='sh').absolute_filename()
'/...bin/sh'
```

```
>>> from sage.all import *
>>> from sage.features import Executable
>>> Executable(name='sh', executable='sh').absolute_filename()
'/...bin/sh'
```

A *FeatureNotPresentError* is raised if the file cannot be found:

```
sage: Executable(name='does-not-exist', executable='does-not-exist-xxxxyxyyxyy
↪').absolute_filename()
Traceback (most recent call last):
...
sage.features.FeatureNotPresentError: does-not-exist is not available.
Executable 'does-not-exist-xxxxyxyyxyy' not found on PATH.
```

```
>>> from sage.all import *
>>> Executable(name='does-not-exist', executable='does-not-exist-xxxxyxyyxyy
↪').absolute_filename()
Traceback (most recent call last):
...
sage.features.FeatureNotPresentError: does-not-exist is not available.
Executable 'does-not-exist-xxxxyxyyxyy' not found on PATH.
```

is_functional()

Return whether an executable in the path is functional.

This method is used internally and can be overridden in subclasses in order to implement a feature test. It should not be called directly. Use *Feature.is_present()* instead.

EXAMPLES:

The function returns `True` unless explicitly overwritten:

```
sage: from sage.features import Executable
sage: Executable(name='sh', executable='sh').is_functional()
FeatureTestResult('sh', True)
```

```
>>> from sage.all import *
>>> from sage.features import Executable
>>> Executable(name='sh', executable='sh').is_functional()
FeatureTestResult('sh', True)
```

class sage.features.**Feature**(**args*, ***kwds*)

Bases: *TrivialUniqueRepresentation*

A feature of the runtime environment.

INPUT:

- `name` – string; name of the feature. This should be suitable as an optional tag for the Sage doctester, i.e., lowercase alphanumeric with underscores (_) allowed; features that correspond to Python modules/packages may use periods (.)
- `spkg` – string; name of the SPKG providing the feature
- `description` – string (optional); plain English description of the feature
- `url` – a URL for the upstream package providing the feature
- `type` – string; one of `'standard'`, `'optional'` (default), `'experimental'`

Overwrite `_is_present` to add feature checks.

EXAMPLES:

```
sage: from sage.features.gap import GapPackage
sage: GapPackage("grape", spkg='gap_packages')  # indirect doctest
Feature('gap_package_grape')
```

```
>>> from sage.all import *
>>> from sage.features.gap import GapPackage
>>> GapPackage("grape", spkg='gap_packages')  # indirect doctest
Feature('gap_package_grape')
```

For efficiency, features are unique:

```
sage: GapPackage("grape") is GapPackage("grape")
True
```

```
>>> from sage.all import *
>>> GapPackage("grape") is GapPackage("grape")
True
```

hide()

Hide this feature. For example this is used when the doctest option `--hide` is set. Setting an installed feature as hidden pretends that it is not available. To revert this use *unhide()*.

EXAMPLES:

Benzene is an optional SPKG. The following test fails if it is hidden or not installed. Thus, in the second invocation the optional tag is needed:

```
sage: from sage.features.graph_generators import Benzene
sage: Benzene().hide()
sage: len(list(graphs.fusenes(2)))                                              #␣
```

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```
↪needs sage.graphs
Traceback (most recent call last):
...
FeatureNotPresentError: benzene is not available.
Feature `benzene` is hidden.
Use method `unhide` to make it available again.

sage: Benzene().unhide()            # optional - benzene, needs sage.graphs
sage: len(list(graphs.fusenes(2)))  # optional - benzene, needs sage.graphs
1
```

```
>>> from sage.all import *
>>> from sage.features.graph_generators import Benzene
>>> Benzene().hide()
>>> len(list(graphs.fusenes(Integer(2))))
↪     # needs sage.graphs
Traceback (most recent call last):
...
FeatureNotPresentError: benzene is not available.
Feature `benzene` is hidden.
Use method `unhide` to make it available again.

>>> Benzene().unhide()            # optional - benzene, needs sage.graphs
>>> len(list(graphs.fusenes(Integer(2))))  # optional - benzene, needs sage.
↪graphs
1
```

is_hidden()

Return whether `self` is present but currently hidden.

EXAMPLES:

sage: from sage.features.sagemath import sage__plot sage: sage__plot().hide() sage: sage__plot().is_hidden() # needs sage.plot True sage: sage__plot().unhide() sage: sage__plot().is_hidden() False

is_optional()

Return whether this feature corresponds to an optional SPKG.

EXAMPLES:

```
sage: from sage.features.databases import DatabaseCremona
sage: DatabaseCremona().is_optional()
True
```

```
>>> from sage.all import *
>>> from sage.features.databases import DatabaseCremona
>>> DatabaseCremona().is_optional()
True
```

is_present()

Return whether the feature is present.

OUTPUT:

A *FeatureTestResult* which can be used as a boolean and contains additional information about the feature test.

EXAMPLES:

```
sage: from sage.features.gap import GapPackage
sage: GapPackage("grape", spkg='gap_packages').is_present()  # optional - gap_
↪package_grape
FeatureTestResult('gap_package_grape', True)
sage: GapPackage("NOT_A_PACKAGE", spkg='gap_packages').is_present()
FeatureTestResult('gap_package_NOT_A_PACKAGE', False)
```

```
>>> from sage.all import *
>>> from sage.features.gap import GapPackage
>>> GapPackage("grape", spkg='gap_packages').is_present()  # optional - gap_
↪package_grape
FeatureTestResult('gap_package_grape', True)
>>> GapPackage("NOT_A_PACKAGE", spkg='gap_packages').is_present()
FeatureTestResult('gap_package_NOT_A_PACKAGE', False)
```

The result is cached:

```
sage: from sage.features import Feature
sage: class TestFeature(Feature):
....:     def _is_present(self):
....:         print("checking presence")
....:         return True
sage: TestFeature("test").is_present()
checking presence
FeatureTestResult('test', True)
sage: TestFeature("test").is_present()
FeatureTestResult('test', True)
sage: TestFeature("other").is_present()
checking presence
FeatureTestResult('other', True)
sage: TestFeature("other").is_present()
FeatureTestResult('other', True)
```

```
>>> from sage.all import *
>>> from sage.features import Feature
>>> class TestFeature(Feature):
...     def _is_present(self):
...         print("checking presence")
...         return True
>>> TestFeature("test").is_present()
checking presence
FeatureTestResult('test', True)
>>> TestFeature("test").is_present()
FeatureTestResult('test', True)
>>> TestFeature("other").is_present()
checking presence
FeatureTestResult('other', True)
>>> TestFeature("other").is_present()
FeatureTestResult('other', True)
```

is_standard()

Return whether this feature corresponds to a standard SPKG.

EXAMPLES:

```
sage: from sage.features.databases import DatabaseCremona
sage: DatabaseCremona().is_standard()
False
```

```
>>> from sage.all import *
>>> from sage.features.databases import DatabaseCremona
>>> DatabaseCremona().is_standard()
False
```

joined_features()

Return a list of features that `self` is the join of.

OUTPUT:

A (possibly empty) list of instances of *Feature*.

EXAMPLES:

```
sage: from sage.features.graphviz import Graphviz
sage: Graphviz().joined_features()
[Feature('dot'), Feature('neato'), Feature('twopi')]
sage: from sage.features.sagemath import sage__rings__function_field
sage: sage__rings__function_field().joined_features()
[Feature('sage.rings.function_field.function_field_polymod'),
Feature('sage.libs.singular'),
Feature('sage.libs.singular.singular'),
Feature('sage.interfaces.singular')]
sage: from sage.features.interfaces import Mathematica
sage: Mathematica().joined_features()
[]
```

```
>>> from sage.all import *
>>> from sage.features.graphviz import Graphviz
>>> Graphviz().joined_features()
[Feature('dot'), Feature('neato'), Feature('twopi')]
>>> from sage.features.sagemath import sage__rings__function_field
>>> sage__rings__function_field().joined_features()
[Feature('sage.rings.function_field.function_field_polymod'),
Feature('sage.libs.singular'),
Feature('sage.libs.singular.singular'),
Feature('sage.interfaces.singular')]
>>> from sage.features.interfaces import Mathematica
>>> Mathematica().joined_features()
[]
```

require()

Raise a *FeatureNotPresentError* if the feature is not present.

EXAMPLES:

```
sage: from sage.features.gap import GapPackage
sage: GapPackage("ve1EeThu").require()                                          #␣
↪needs sage.libs.gap
Traceback (most recent call last):
...
FeatureNotPresentError: gap_package_ve1EeThu is not available.
`LoadPackage("ve1EeThu")` evaluated to `fail` in GAP.
```

```
>>> from sage.all import *
>>> from sage.features.gap import GapPackage
>>> GapPackage("ve1EeThu").require()                                            #␣
↪needs sage.libs.gap
Traceback (most recent call last):
...
FeatureNotPresentError: gap_package_ve1EeThu is not available.
`LoadPackage("ve1EeThu")` evaluated to `fail` in GAP.
```

resolution()

Return a suggestion on how to make *is_present()* pass if it did not pass.

OUTPUT: string

EXAMPLES:

```
sage: from sage.features import Executable
sage: Executable(name='CSDP', spkg='csdp', executable='theta', url='https://
↪github.com/dimpase/csdp').resolution()  # optional - sage_spkg
'...To install CSDP...you can try to run...sage -i csdp...Further␣
↪installation instructions might be available at https://github.com/dimpase/
↪csdp.'
```

```
>>> from sage.all import *
>>> from sage.features import Executable
>>> Executable(name='CSDP', spkg='csdp', executable='theta', url='https://
↪github.com/dimpase/csdp').resolution()  # optional - sage_spkg
'...To install CSDP...you can try to run...sage -i csdp...Further␣
↪installation instructions might be available at https://github.com/dimpase/
↪csdp.'
```

unhide()

Revert what *hide()* did.

EXAMPLES:

sage: from sage.features.sagemath import sage__plot sage: sage__plot().hide() sage: sage__plot().is_present() FeatureTestResult('sage.plot', False) sage: sage__plot().unhide() # needs sage.plot sage: sage__plot().is_present() # needs sage.plot FeatureTestResult('sage.plot', True)

exception sage.features.**FeatureNotPresentError**(*feature*, *reason=None*, *resolution=None*)

Bases: RuntimeError

A missing feature error.

EXAMPLES:

```
sage: from sage.features import Feature, FeatureTestResult
sage: class Missing(Feature):
....:     def _is_present(self):
....:         return False

sage: Missing(name='missing').require()
Traceback (most recent call last):
...
FeatureNotPresentError: missing is not available.
```

```
>>> from sage.all import *
>>> from sage.features import Feature, FeatureTestResult
>>> class Missing(Feature):
...     def _is_present(self):
...         return False

>>> Missing(name='missing').require()
Traceback (most recent call last):
...
FeatureNotPresentError: missing is not available.
```

property resolution

Initialize self. See help(type(self)) for accurate signature.

class sage.features.**FeatureTestResult**(*feature*, *is_present*, *reason=None*, *resolution=None*)

Bases: object

The result of a *Feature.is_present()* call.

Behaves like a boolean with some extra data which may explain why a feature is not present and how this may be resolved.

EXAMPLES:

```
sage: from sage.features.gap import GapPackage
sage: presence = GapPackage("NOT_A_PACKAGE").is_present(); presence  # indirect␣
↪doctest
FeatureTestResult('gap_package_NOT_A_PACKAGE', False)
sage: bool(presence)
False
```

```
>>> from sage.all import *
>>> from sage.features.gap import GapPackage
>>> presence = GapPackage("NOT_A_PACKAGE").is_present(); presence  # indirect␣
↪doctest
FeatureTestResult('gap_package_NOT_A_PACKAGE', False)
>>> bool(presence)
False
```

Explanatory messages might be available as `reason` and `resolution`:

```
sage: presence.reason                                                          #␣
↪needs sage.libs.gap
'`LoadPackage("NOT_A_PACKAGE")` evaluated to `fail` in GAP.'
```

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```
sage: bool(presence.resolution)
False
```

```
>>> from sage.all import *
>>> presence.reason                                                    #␣
↪needs sage.libs.gap
'`LoadPackage("NOT_A_PACKAGE")` evaluated to `fail` in GAP.'
>>> bool(presence.resolution)
False
```

If a feature is not present, `resolution` defaults to `feature.resolution()` if this is defined. If you do not want to use this default you need explicitly set `resolution` to a string:

```
sage: from sage.features import FeatureTestResult
sage: package = GapPackage("NOT_A_PACKAGE", spkg='no_package')
sage: str(FeatureTestResult(package, True).resolution)  # optional - sage_spkg
'...To install gap_package_NOT_A_PACKAGE...you can try to run...sage -i no_
↪package...'
sage: str(FeatureTestResult(package, False).resolution) # optional - sage_spkg
'...To install gap_package_NOT_A_PACKAGE...you can try to run...sage -i no_
↪package...'
sage: FeatureTestResult(package, False, resolution='rtm').resolution
'rtm'
```

```
>>> from sage.all import *
>>> from sage.features import FeatureTestResult
>>> package = GapPackage("NOT_A_PACKAGE", spkg='no_package')
>>> str(FeatureTestResult(package, True).resolution)  # optional - sage_spkg
'...To install gap_package_NOT_A_PACKAGE...you can try to run...sage -i no_
↪package...'
>>> str(FeatureTestResult(package, False).resolution) # optional - sage_spkg
'...To install gap_package_NOT_A_PACKAGE...you can try to run...sage -i no_
↪package...'
>>> FeatureTestResult(package, False, resolution='rtm').resolution
'rtm'
```

property resolution

Initialize self. See help(type(self)) for accurate signature.

class sage.features.**FileFeature**(**args*, ***kwds*)

Bases: *Feature*

Base class for features that describe a file or directory in the file system.

A subclass should implement a method *absolute_filename()*.

EXAMPLES:

Two direct concrete subclasses of *FileFeature* are defined:

```
sage: from sage.features import StaticFile, Executable, FileFeature
sage: issubclass(StaticFile, FileFeature)
True
```

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```
sage: issubclass(Executable, FileFeature)
True
```

```
>>> from sage.all import *
>>> from sage.features import StaticFile, Executable, FileFeature
>>> issubclass(StaticFile, FileFeature)
True
>>> issubclass(Executable, FileFeature)
True
```

To work with the file described by the feature, use the method *absolute_filename()*. A *FeatureNotPresentError* is raised if the file cannot be found:

```
sage: Executable(name='does-not-exist', executable='does-not-exist-xxxxyxyyxyy').
↪absolute_filename()
Traceback (most recent call last):
...
sage.features.FeatureNotPresentError: does-not-exist is not available.
Executable 'does-not-exist-xxxxyxyyxyy' not found on PATH.
```

```
>>> from sage.all import *
>>> Executable(name='does-not-exist', executable='does-not-exist-xxxxyxyyxyy').
↪absolute_filename()
Traceback (most recent call last):
...
sage.features.FeatureNotPresentError: does-not-exist is not available.
Executable 'does-not-exist-xxxxyxyyxyy' not found on PATH.
```

A *FileFeature* also provides the *is_present()* method to test for the presence of the file at run time. This is inherited from the base class *Feature*:

```
sage: Executable(name='sh', executable='sh').is_present()
FeatureTestResult('sh', True)
```

```
>>> from sage.all import *
>>> Executable(name='sh', executable='sh').is_present()
FeatureTestResult('sh', True)
```

absolute_filename()

The absolute path of the file as a string.

Concrete subclasses must override this abstract method.

class sage.features.**PythonModule**(**args*, ***kwds*)

Bases: *Feature*

A *Feature* which describes whether a python module can be imported.

EXAMPLES:

Not all builds of python include the `ssl` module, so you could check whether it is available:

```
sage: from sage.features import PythonModule
sage: PythonModule("ssl").require()  # not tested - output depends on the python␣
↪build
```

```
>>> from sage.all import *
>>> from sage.features import PythonModule
>>> PythonModule("ssl").require()  # not tested - output depends on the python␣
↪build
```

class sage.features.**StaticFile**(**args*, ***kwds*)

Bases: *FileFeature*

A *Feature* which describes the presence of a certain file such as a database.

EXAMPLES:

```
sage: from sage.features import StaticFile
sage: StaticFile(name='no_such_file', filename='KaT1aihu',                #␣
↪optional - sage_spkg
....:            search_path='/', spkg='some_spkg',
....:            url='http://rand.om').require()
Traceback (most recent call last):
...
FeatureNotPresentError: no_such_file is not available.
'KaT1aihu' not found in any of ['/']...
To install no_such_file...you can try to run...sage -i some_spkg...
Further installation instructions might be available at http://rand.om.
```

```
>>> from sage.all import *
>>> from sage.features import StaticFile
>>> StaticFile(name='no_such_file', filename='KaT1aihu',                # optional -
↪ sage_spkg
...            search_path='/', spkg='some_spkg',
...            url='http://rand.om').require()
Traceback (most recent call last):
...
FeatureNotPresentError: no_such_file is not available.
'KaT1aihu' not found in any of ['/']...
To install no_such_file...you can try to run...sage -i some_spkg...
Further installation instructions might be available at http://rand.om.
```

absolute_filename()

The absolute path of the file as a string.

EXAMPLES:

```
sage: from sage.features import StaticFile
sage: from sage.misc.temporary_file import tmp_dir
sage: dir_with_file = tmp_dir()
sage: file_path = os.path.join(dir_with_file, "file.txt")
sage: open(file_path, 'a').close() # make sure the file exists
sage: search_path = ( '/foo/bar', dir_with_file ) # file is somewhere in the␣
↪search path
sage: feature = StaticFile(name='file', filename='file.txt', search_
```

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```
↪path=search_path)
sage: feature.absolute_filename() == file_path
True
```

```
>>> from sage.all import *
>>> from sage.features import StaticFile
>>> from sage.misc.temporary_file import tmp_dir
>>> dir_with_file = tmp_dir()
>>> file_path = os.path.join(dir_with_file, "file.txt")
>>> open(file_path, 'a').close() # make sure the file exists
>>> search_path = ( '/foo/bar', dir_with_file ) # file is somewhere in the␣
↪search path
>>> feature = StaticFile(name='file', filename='file.txt', search_path=search_
↪path)
>>> feature.absolute_filename() == file_path
True
```

A *FeatureNotPresentError* is raised if the file cannot be found:

```
sage: from sage.features import StaticFile
sage: StaticFile(name='no_such_file', filename='KaT1aihu',  # optional - sage_
↪spkg
....:            search_path=(), spkg='some_spkg',
....:            url='http://rand.om').absolute_filename()
Traceback (most recent call last):
...
FeatureNotPresentError: no_such_file is not available.
'KaT1aihu' not found in any of []...
To install no_such_file...you can try to run...sage -i some_spkg...
Further installation instructions might be available at http://rand.om.
```

```
>>> from sage.all import *
>>> from sage.features import StaticFile
>>> StaticFile(name='no_such_file', filename='KaT1aihu',  # optional - sage_
↪spkg
...            search_path=(), spkg='some_spkg',
...            url='http://rand.om').absolute_filename()
Traceback (most recent call last):
...
FeatureNotPresentError: no_such_file is not available.
'KaT1aihu' not found in any of []...
To install no_such_file...you can try to run...sage -i some_spkg...
Further installation instructions might be available at http://rand.om.
```

class sage.features.**TrivialClasscallMetaClass**

Bases: `type`

A trivial version of `sage.misc.classcall_metaclass.ClasscallMetaclass` without Cython dependencies.

class sage.features.**TrivialUniqueRepresentation**(**args*, ***kwds*)

Bases: `object`

A trivial version of `UniqueRepresentation` without Cython dependencies.

sage.features.**package_systems**()

Return a list of *PackageSystem* objects representing the available package systems.

The list is ordered by decreasing preference.

EXAMPLES:

```
sage: from sage.features import package_systems
sage: package_systems()    # random
[Feature('homebrew'), Feature('sage_spkg'), Feature('pip')]
```

```
>>> from sage.all import *
>>> from sage.features import package_systems
>>> package_systems()    # random
[Feature('homebrew'), Feature('sage_spkg'), Feature('pip')]
```

3.2 Join features

class sage.features.join_feature.**JoinFeature**(**args*, ***kwds*)

Bases: *Feature*

Join of several *Feature* instances.

This creates a new feature as the union of the given features. Typically these are executables of an SPKG. For an example, see *Rubiks*.

Furthermore, this can be the union of a single feature. This is used to map the given feature to a more convenient name to be used in `optional` tags of doctests. Thus you can equip a feature such as a *PythonModule* with a tag name that differs from the systematic tag name. As an example for this use case, see *Meataxe*.

EXAMPLES:

```
sage: from sage.features import Executable
sage: from sage.features.join_feature import JoinFeature
sage: F = JoinFeature("shell-boolean",
....:                 (Executable('shell-true', 'true'),
....:                  Executable('shell-false', 'false')))
sage: F.is_present()
FeatureTestResult('shell-boolean', True)
sage: F = JoinFeature("asdfghjkl",
....:                 (Executable('shell-true', 'true'),
....:                  Executable('xxyyyy', 'xxyyyy-does-not-exist')))
sage: F.is_present()
FeatureTestResult('xxyyyy', False)
```

```
>>> from sage.all import *
>>> from sage.features import Executable
>>> from sage.features.join_feature import JoinFeature
>>> F = JoinFeature("shell-boolean",
...                 (Executable('shell-true', 'true'),
...                  Executable('shell-false', 'false')))
>>> F.is_present()
FeatureTestResult('shell-boolean', True)
>>> F = JoinFeature("asdfghjkl",
```

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```
...                 (Executable('shell-true', 'true'),
...                  Executable('xxyyyy', 'xxyyyy-does-not-exist')))
>>> F.is_present()
FeatureTestResult('xxyyyy', False)
```

hide()

Hide this feature and all its joined features.

EXAMPLES:

```
sage: from sage.features.sagemath import sage__groups
sage: f = sage__groups()
sage: f.hide()
sage: f._features[0].is_present()
FeatureTestResult('sage.groups.perm_gps.permgroup', False)

sage: f.require()
Traceback (most recent call last):
...
FeatureNotPresentError: sage.groups is not available.
Feature `sage.groups` is hidden.
Use method `unhide` to make it available again.
```

```
>>> from sage.all import *
>>> from sage.features.sagemath import sage__groups
>>> f = sage__groups()
>>> f.hide()
>>> f._features[Integer(0)].is_present()
FeatureTestResult('sage.groups.perm_gps.permgroup', False)

>>> f.require()
Traceback (most recent call last):
...
FeatureNotPresentError: sage.groups is not available.
Feature `sage.groups` is hidden.
Use method `unhide` to make it available again.
```

unhide()

Revert what *hide()* did.

EXAMPLES:

```
sage: from sage.features.sagemath import sage__groups
sage: f = sage__groups()
sage: f.hide()
sage: f.is_present()
FeatureTestResult('sage.groups', False)
sage: f._features[0].is_present()
FeatureTestResult('sage.groups.perm_gps.permgroup', False)

sage: f.unhide()
sage: f.is_present()    # optional sage.groups
FeatureTestResult('sage.groups', True)
```

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```
sage: f._features[0].is_present() # optional sage.groups
FeatureTestResult('sage.groups.perm_gps.permgroup', True)
```

```
>>> from sage.all import *
>>> from sage.features.sagemath import sage__groups
>>> f = sage__groups()
>>> f.hide()
>>> f.is_present()
FeatureTestResult('sage.groups', False)
>>> f._features[Integer(0)].is_present()
FeatureTestResult('sage.groups.perm_gps.permgroup', False)

>>> f.unhide()
>>> f.is_present()     # optional sage.groups
FeatureTestResult('sage.groups', True)
>>> f._features[Integer(0)].is_present() # optional sage.groups
FeatureTestResult('sage.groups.perm_gps.permgroup', True)
```

3.3 Enumeration of all defined features

sage.features.all.**all_features**()

Return an iterable of all features.

EXAMPLES:

```
sage: from sage.features.all import all_features
sage: sorted(all_features(), key=lambda f: f.name)  # random
[...Feature('sage.combinat')...]
```

```
>>> from sage.all import *
>>> from sage.features.all import all_features
>>> sorted(all_features(), key=lambda f: f.name)  # random
[...Feature('sage.combinat')...]
```

sage.features.all.**module_feature**(*module_name*)

Find a top-level *Feature* that provides the Python module of the given `module_name`.

Only features known to *all_features()* are considered.

INPUT:

- `module_name` – string

OUTPUT: a *Feature* or `None`

EXAMPLES:

```
sage: from sage.features.all import module_feature
sage: module_feature('sage.combinat.tableau')                                  #␣
↪needs sage.combinat
Feature('sage.combinat')
sage: module_feature('sage.combinat.posets.poset')                             #␣
↪needs sage.graphs
```

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```
Feature('sage.graphs')
sage: module_feature('sage.schemes.toric.variety')                          #␣
↪needs sage.geometry.polyhedron
Feature('sage.geometry.polyhedron')
sage: module_feature('scipy')                                               #␣
↪needs scipy
Feature('scipy')
sage: print(module_feature('sage.structure.element'))
None
sage: print(module_feature('sage.does_not_exist'))
None
```

```
>>> from sage.all import *
>>> from sage.features.all import module_feature
>>> module_feature('sage.combinat.tableau')                                 #␣
↪needs sage.combinat
Feature('sage.combinat')
>>> module_feature('sage.combinat.posets.poset')                            #␣
↪needs sage.graphs
Feature('sage.graphs')
>>> module_feature('sage.schemes.toric.variety')                            #␣
↪needs sage.geometry.polyhedron
Feature('sage.geometry.polyhedron')
>>> module_feature('scipy')                                                 #␣
↪needs scipy
Feature('scipy')
>>> print(module_feature('sage.structure.element'))
None
>>> print(module_feature('sage.does_not_exist'))
None
```

sage.features.all.**name_feature**(*name*, *toplevel=None*)

Find a top-level *Feature* that provides the top-level `name`.

Only features known to *all_features()* are considered.

INPUT:

- `name` – string
- `toplevel` – a module or other namespace

OUTPUT: a *Feature* or `None`

EXAMPLES:

```
sage: from sage.features.all import name_feature
sage: name_feature('QuadraticField')                                        #␣
↪needs sage.rings.number_field
Feature('sage.rings.number_field')
sage: name_feature('line')                                                  #␣
↪needs sage.plot
Feature('sage.plot')
sage: print(name_feature('ZZ'))
None
```

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```
sage: print(name_feature('does_not_exist'))
None
```

```
>>> from sage.all import *
>>> from sage.features.all import name_feature
>>> name_feature('QuadraticField')                                    #␣
↪needs sage.rings.number_field
Feature('sage.rings.number_field')
>>> name_feature('line')                                              #␣
↪needs sage.plot
Feature('sage.plot')
>>> print(name_feature('ZZ'))
None
>>> print(name_feature('does_not_exist'))
None
```

3.4 Features that can be explicitly enabled or disabled at build time

These features are unique in that, if they are enabled or disabled at build-time, then we usually do not want to detect them on-the-fly. Instead we trust the value supplied (or detected) at build-time. There is however an overrride to defer these checks to run-time (the classic behavior) for use on binary distros or anywhere it is desirable to enable/disable features without rebuilding sage.

This is an implementation of Option 3 in Github discussion 41067.

class sage.features.build_feature.**BuildFeature**(**args*, ***kwds*)

Bases: *Feature*

A class for features that can be enabled or disabled at build-time.

The current implementation refers to build features that are configurable in meson. For example:

```
option(
  'foo',
  type: 'feature',
  value: 'auto',
  description: 'support for foo'
)
```

At build time, support for this "foo" will be automatically detected, and either enabled or disabled depending on whether or not its requirements are met. Alternatively, users may pass either `-Dfoo=enabled` or `-Dfoo=disabled` to explicitly enable or disable the feature. Features may be disabled regardless of whether or not they are installed, but usually features may only be enabled if their dependencies are present and usable.

In any event, after `meson setup`, support for "foo" is either enabled or disabled, and a boolean variable called something like `foo_enabled` is written to `sage.config`. In your subclass, you should set the member variable `_enabled_in_build` to the value of that config variable.

The `_is_present` method for this class will return the value of that config variable unless `defer_feature_checks` is set to `True` in `sage.config`. If checks are deferred, the `_is_present` method will try to return the value of *is_present_at_runtime* instead. If your feature can be detected at run-time, you should implement that check in *is_present_at_runtime*. Otherwise, leave it unimplemented; and `_is_present` will return `False`.

EXAMPLES:

```
sage: from sage.features.build_feature import BuildFeature
sage: BuildFeature("foo")
Feature('foo')
```

```
>>> from sage.all import *
>>> from sage.features.build_feature import BuildFeature
>>> BuildFeature("foo")
Feature('foo')
```

is_runtime_detectable()

Return whether or not this feature can (and should) be detected at runtime.

A feature is runtime detectable if both of the following hold:

- Deferred feature checks have been enabled globally by passing `-Ddefer_feature_checks=true` to `meson setup`.
- An *is_present_at_runtime* method has been implemented for the feature.

EXAMPLES:

The method returns `False` if you have not implemented *is_present_at_runtime*:

```
sage: from sage.features.build_feature import BuildFeature
sage: bf = BuildFeature("example")
sage: bf.is_runtime_detectable()
False
```

```
>>> from sage.all import *
>>> from sage.features.build_feature import BuildFeature
>>> bf = BuildFeature("example")
>>> bf.is_runtime_detectable()
False
```

class sage.features.build_feature.**BuildModule**(**args*, ***kwds*)

Bases: *BuildFeature*

A *Feature* indicating the presence of a python (or cython) module, with explicit support from the build system.

This subclass encapsulates the runtime import check that is standard for most python and cython modules. Its constructor takes a `module_name` parameter that (optionally) differs from the name of the feature. It is this `module_name` that we actually try to import when feature checks are deferred to runtime.

is_present_at_runtime()

Check for the module at runtime.

This uses `importlib.util.find_spec()` rather than (say) `importlib.import_module()` because we are only checking for the _presence_ of the feature; if there's an error, then so be it, but that doesn't mean that the feature is nonexistent! Perhaps more importantly, this allows us to check for the presence of a feature cheaply, without triggering a module import. If the feature check performed the import, doing a lazy import inside of `if foo.is_present(): ...` would be pointless.

3.5 Features for testing the presence of Python modules in the Sage library

All of these features are present in a monolithic installation of the Sage library, such as the one made by the SageMath distribution.

The features are defined for the purpose of separately testing modularized distributions such as *sagemath-categories* and *sagemath-repl*.

Often, doctests in a module of the Sage library illustrate the interplay with a range of different objects; this is a form of integration testing. These objects may come from modules shipped in other distributions. For example, `sage.structure.element` (shipped by *sagemath-objects*, one of the most fundamental distributions) contains the doctest:

```
sage: G = SymmetricGroup(4)                                                 #␣
↪needs sage.groups
sage: g = G([2, 3, 4, 1])                                                   #␣
↪needs sage.groups
sage: g.powers(4)                                                           #␣
↪needs sage.groups
[(), (1,2,3,4), (1,3)(2,4), (1,4,3,2)]
```

```
>>> from sage.all import *
>>> G = SymmetricGroup(Integer(4))                                          ␣
↪     # needs sage.groups
>>> g = G([Integer(2), Integer(3), Integer(4), Integer(1)])                 ␣
↪                                 # needs sage.groups
>>> g.powers(Integer(4))                                                    ␣
↪     # needs sage.groups
[(), (1,2,3,4), (1,3)(2,4), (1,4,3,2)]
```

This test cannot pass when the distribution *sagemath-objects* is tested separately (in a virtual environment): In this situation, `SymmetricGroup` is not defined anywhere (and thus not present in the top-level namespace). Hence, we conditionalize this doctest on the presence of the feature *sage.groups*.

`sage.features.sagemath.`**`all_features`**`()`

Return features corresponding to parts of the Sage library.

These features are named after Python packages/modules (e.g., `sage.symbolic`), not distribution packages (**sagemath-symbolics**).

This design is motivated by a separation of concerns: The author of a module that depends on some functionality provided by a Python module usually already knows the name of the Python module, so we do not want to force the author to also know about the distribution package that provides the Python module.

Instead, we associate distribution packages to Python modules in *sage.features.sagemath* via the `spkg` parameter of *Feature*.

EXAMPLES:

```
sage: from sage.features.sagemath import all_features
sage: list(all_features())
[...Feature('sage.combinat'), ...]
```

```
>>> from sage.all import *
>>> from sage.features.sagemath import all_features
```

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```
>>> list(all_features())
[...Feature('sage.combinat'), ...]
```

class sage.features.sagemath.**sage__combinat**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing the presence of sage.combinat.

EXAMPLES:

Python modules that provide elementary combinatorial objects such as `sage.combinat.subset`, `sage.combinat.composition`, `sage.combinat.permutation` are always available; there is no need for an `# optional/needs` tag:

```
sage: Permutation([1,2,3]).is_even()
True
sage: Permutation([6,1,4,5,2,3]).bruhat_inversions()
[[0, 1], [0, 2], [0, 3], [2, 4], [2, 5], [3, 4], [3, 5]]
```

```
>>> from sage.all import *
>>> Permutation([Integer(1),Integer(2),Integer(3)]).is_even()
True
>>> Permutation([Integer(6),Integer(1),Integer(4),Integer(5),Integer(2),
↪Integer(3)]).bruhat_inversions()
[[0, 1], [0, 2], [0, 3], [2, 4], [2, 5], [3, 4], [3, 5]]
```

Use `# needs sage.combinat` for doctests that use any other Python modules from sage.combinat, for example `sage.combinat.tableau_tuple`:

```
sage: TableauTuple([[[7,8,9]],[],[[1,2,3],[4,5],[6]]]).shape()                #␣
↪needs sage.combinat
([3], [], [3, 2, 1])
```

```
>>> from sage.all import *
>>> TableauTuple([[[Integer(7),Integer(8),Integer(9)]],[],[[Integer(1),Integer(2),
↪Integer(3)],[Integer(4),Integer(5)],[Integer(6)]]]).shape()                   #␣
↪needs sage.combinat
([3], [], [3, 2, 1])
```

Doctests that use Python modules from sage.combinat that involve trees, graphs, hypergraphs, posets, quivers, combinatorial designs, finite state machines etc. should be marked `# needs sage.combinat sage.graphs`:

```
sage: L = Poset({0: [1], 1: [2], 2:[3], 3:[4]})                               #␣
↪needs sage.combinat sage.graphs
sage: L.is_chain()                                                             #␣
↪needs sage.combinat sage.graphs
True
```

```
>>> from sage.all import *
>>> L = Poset({Integer(0): [Integer(1)], Integer(1): [Integer(2)],␣
↪Integer(2):[Integer(3)], Integer(3):[Integer(4)]})                            ␣
↪    # needs sage.combinat sage.graphs
>>> L.is_chain()                                                              #␣
```

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```
↪needs sage.combinat sage.graphs
True
```

Doctests that use combinatorial modules/algebras, or root systems should use the tag `# needs sage.combinat sage.modules`:

```
sage: # needs sage.combinat sage.modules
sage: A = SchurAlgebra(QQ, 2, 3)
sage: a = A.an_element(); a
2*S((1, 1, 1), (1, 1, 1)) + 2*S((1, 1, 1), (1, 1, 2))
 + 3*S((1, 1, 1), (1, 2, 2))
sage: L = RootSystem(['A',3,1]).root_lattice()
sage: PIR = L.positive_imaginary_roots(); PIR
Positive imaginary roots of type ['A', 3, 1]
```

```
>>> from sage.all import *
>>> # needs sage.combinat sage.modules
>>> A = SchurAlgebra(QQ, Integer(2), Integer(3))
>>> a = A.an_element(); a
2*S((1, 1, 1), (1, 1, 1)) + 2*S((1, 1, 1), (1, 1, 2))
 + 3*S((1, 1, 1), (1, 2, 2))
>>> L = RootSystem(['A',Integer(3),Integer(1)]).root_lattice()
>>> PIR = L.positive_imaginary_roots(); PIR
Positive imaginary roots of type ['A', 3, 1]
```

Doctests that use lattices, semilattices, or Dynkin diagrams should use the tag `# needs sage.combinat sage.graphs sage.modules`:

```
sage: L = LatticePoset({0: [1,2], 1: [3], 2: [3,4], 3: [5], 4: [5]})            #␣
↪needs sage.combinat sage.graphs sage.modules
sage: L.meet_irreducibles()                                                      #␣
↪needs sage.combinat sage.graphs sage.modules
[1, 3, 4]
```

```
>>> from sage.all import *
>>> L = LatticePoset({Integer(0): [Integer(1),Integer(2)], Integer(1):␣
↪[Integer(3)], Integer(2): [Integer(3),Integer(4)], Integer(3): [Integer(5)],␣
↪Integer(4): [Integer(5)]})            # needs sage.combinat sage.graphs sage.
↪modules
>>> L.meet_irreducibles()                                                        #␣
↪needs sage.combinat sage.graphs sage.modules
[1, 3, 4]
```

class sage.features.sagemath.**sage__geometry__polyhedron**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing the presence of `sage.geometry.polyhedron`.

EXAMPLES:

Doctests that use polyhedra, cones, geometric complexes, triangulations, etc. should use the tag `# needs sage.geometry.polyhedron`:

```
sage: co = polytopes.truncated_tetrahedron()                                #␣
↪needs sage.geometry.polyhedron
sage: co.volume()                                                           #␣
↪needs sage.geometry.polyhedron
184/3
```

```
>>> from sage.all import *
>>> co = polytopes.truncated_tetrahedron()                                  #␣
↪needs sage.geometry.polyhedron
>>> co.volume()                                                             #␣
↪needs sage.geometry.polyhedron
184/3
```

Some constructions of polyhedra require additional tags:

```
sage: # needs sage.combinat sage.geometry.polyhedron sage.rings.number_field
sage: perm_a3_reg_nf = polytopes.generalized_permutahedron(
....:     ['A',3], regular=True, backend='number_field'); perm_a3_reg_nf
A 3-dimensional polyhedron in AA^3 defined as the convex hull of 24 vertices
```

```
>>> from sage.all import *
>>> # needs sage.combinat sage.geometry.polyhedron sage.rings.number_field
>>> perm_a3_reg_nf = polytopes.generalized_permutahedron(
...     ['A',Integer(3)], regular=True, backend='number_field'); perm_a3_reg_nf
A 3-dimensional polyhedron in AA^3 defined as the convex hull of 24 vertices
```

class sage.features.sagemath.**sage__graphs**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing the presence of sage.graphs.

EXAMPLES:

Doctests that use anything from sage.graphs (`Graph`, `DiGraph`, …) should be marked `# needs sage.graphs`. The same applies to any doctest that uses a `Poset()`, cluster algebra quiver, finite state machines, abelian sandpiles, or Dynkin diagrams:

```
sage: g = graphs.PetersenGraph()                                            #␣
↪needs sage.graphs
sage: r, s = g.is_weakly_chordal(certificate=True); r                       #␣
↪needs sage.graphs
False
```

```
>>> from sage.all import *
>>> g = graphs.PetersenGraph()                                              #␣
↪needs sage.graphs
>>> r, s = g.is_weakly_chordal(certificate=True); r                         #␣
↪needs sage.graphs
False
```

Also any use of tree classes defined in sage.combinat (`BinaryTree`, `RootedTree`, …) in doctests should be marked the same.

By way of generalization, any use of `SimplicialComplex` or other abstract complexes from `sage.topology`, hypergraphs, and combinatorial designs, should be marked `# needs sage.graphs` as well:

```
sage: X = SimplicialComplex([[0,1,2], [1,2,3]])                                 #␣
↪needs sage.graphs
sage: X.link(Simplex([0]))                                                      #␣
↪needs sage.graphs
Simplicial complex with vertex set (1, 2) and facets {(1, 2)}

sage: IncidenceStructure([[1,2,3],[1,4]]).degrees(2)                            #␣
↪needs sage.graphs
{(1, 2): 1, (1, 3): 1, (1, 4): 1, (2, 3): 1, (2, 4): 0, (3, 4): 0}
```

```
>>> from sage.all import *
>>> X = SimplicialComplex([[Integer(0),Integer(1),Integer(2)], [Integer(1),
↪Integer(2),Integer(3)]])                                  # needs sage.graphs
>>> X.link(Simplex([Integer(0)]))                                                ␣
↪     # needs sage.graphs
Simplicial complex with vertex set (1, 2) and facets {(1, 2)}

>>> IncidenceStructure([[Integer(1),Integer(2),Integer(3)],[Integer(1),
↪Integer(4)]]).degrees(Integer(2))                          # needs sage.graphs
{(1, 2): 1, (1, 3): 1, (1, 4): 1, (2, 3): 1, (2, 4): 0, (3, 4): 0}
```

On the other hand, matroids are not implemented as posets in Sage but are instead closely tied to linear algebra over fields; hence use `# needs sage.modules` instead:

```
sage: # needs sage.modules
sage: M = Matroid(Matrix(QQ, [[1, 0, 0, 0, 1, 1, 1],
....:                         [0, 1, 0, 1, 0, 1, 1],
....:                         [0, 0, 1, 1, 1, 0, 1]]))
sage: N = (M / [2]).delete([3, 4])
sage: sorted(N.groundset())
[0, 1, 5, 6]
```

```
>>> from sage.all import *
>>> # needs sage.modules
>>> M = Matroid(Matrix(QQ, [[Integer(1), Integer(0), Integer(0), Integer(0),␣
↪Integer(1), Integer(1), Integer(1)],
...                         [Integer(0), Integer(1), Integer(0), Integer(1),␣
↪Integer(0), Integer(1), Integer(1)],
...                         [Integer(0), Integer(0), Integer(1), Integer(1),␣
↪Integer(1), Integer(0), Integer(1)]]))
>>> N = (M / [Integer(2)]).delete([Integer(3), Integer(4)])
>>> sorted(N.groundset())
[0, 1, 5, 6]
```

However, many constructions (and some methods) of matroids do involve graphs:

```
sage: # needs sage.modules
sage: W = matroids.Wheel(3)     # despite the name, not created via graphs
sage: W.is_isomorphic(N)          # goes through a graph isomorphism test      #␣
↪needs sage.graphs
False
sage: K4 = matroids.CompleteGraphic(4)    # this one is created via graphs      #␣
↪needs sage.graphs
```

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```
sage: K4.is_isomorphic(W)                                                    #␣
↪needs sage.graphs
True
```

```
>>> from sage.all import *
>>> # needs sage.modules
>>> W = matroids.Wheel(Integer(3))      # despite the name, not created via graphs
>>> W.is_isomorphic(N)            # goes through a graph isomorphism test      #␣
↪needs sage.graphs
False
>>> K4 = matroids.CompleteGraphic(Integer(4))     # this one is created via graphs␣
↪     # needs sage.graphs
>>> K4.is_isomorphic(W)                                                        #␣
↪needs sage.graphs
True
```

class sage.features.sagemath.**sage__groups**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing the presence of `sage.groups`.

EXAMPLES:

Permutations and sets of permutations are always available, but permutation groups are implemented in Sage using the *GAP* system and require the tag `# needs sage.groups`:

```
sage: p = Permutation([2,1,4,3])
sage: p.to_permutation_group_element()                                       #␣
↪needs sage.groups
(1,2)(3,4)
```

```
>>> from sage.all import *
>>> p = Permutation([Integer(2),Integer(1),Integer(4),Integer(3)])
>>> p.to_permutation_group_element()                                         #␣
↪needs sage.groups
(1,2)(3,4)
```

class sage.features.sagemath.**sage__libs__flint**(**args*, ***kwds*)

Bases: *JoinFeature*

A *sage.features.Feature* describing the presence of `sage.libs.flint` and other modules depending on FLINT.

In addition to the modularization purposes that this tag serves, it also provides attribution to the upstream project.

class sage.features.sagemath.**sage__libs__gap**(**args*, ***kwds*)

Bases: *JoinFeature*

A *sage.features.Feature* describing the presence of `sage.libs.gap` (the library interface to *GAP*) and `sage.interfaces.gap` (the pexpect interface to GAP). By design, we do not distinguish between these two, in order to facilitate the conversion of code from the pexpect interface to the library interface.

See also

Features for GAP packages

class sage.features.sagemath.**sage__libs__giac**(**args*, ***kwds*)

Bases: *JoinFeature*

A *sage.features.Feature* describing the presence of `sage.libs.giac`.

In addition to the modularization purposes that this tag serves, it also provides attribution to the upstream project.

class sage.features.sagemath.**sage__libs__linbox**(**args*, ***kwds*)

Bases: *JoinFeature*

A *sage.features.Feature* describing the presence of `sage.libs.linbox` and other modules depending on Givaro, FFLAS-FFPACK, LinBox.

In addition to the modularization purposes that this tag serves, it also provides attribution to the upstream project.

class sage.features.sagemath.**sage__libs__m4ri**(**args*, ***kwds*)

Bases: *JoinFeature*

A *sage.features.Feature* describing the presence of Cython modules depending on the M4RI and/or M4RIe libraries.

In addition to the modularization purposes that this tag serves, it also provides attribution to the upstream project.

class sage.features.sagemath.**sage__libs__ntl**(**args*, ***kwds*)

Bases: *JoinFeature*

A *sage.features.Feature* describing the presence of `sage.libs.ntl` and other modules depending on NTL.

In addition to the modularization purposes that this tag serves, it also provides attribution to the upstream project.

class sage.features.sagemath.**sage__libs__pari**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing the presence of `sage.libs.pari`.

SageMath uses the *PARI* library (via *cypari2*) for numerous purposes. Doctests that involves such features should be marked `# needs sage.libs.pari`.

In addition to the modularization purposes that this tag serves, it also provides attribution to the upstream project.

EXAMPLES:

```
sage: R.<a> = QQ[]
sage: S.<x> = R[]
sage: f = x^2 + a; g = x^3 + a
sage: r = f.resultant(g); r                                                 #␣
↪needs sage.libs.pari
a^3 + a^2
```

```
>>> from sage.all import *
>>> R = QQ['a']; (a,) = R._first_ngens(1)
>>> S = R['x']; (x,) = S._first_ngens(1)
>>> f = x**Integer(2) + a; g = x**Integer(3) + a
>>> r = f.resultant(g); r                                                   #␣
```

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```
↪needs sage.libs.pari
a^3 + a^2
```

class sage.features.sagemath.**sage__libs__singular**(**args*, ***kwds*)

Bases: *JoinFeature*

A *sage.features.Feature* describing the presence of `sage.libs.singular` (the library interface to Singular) and `sage.interfaces.singular` (the pexpect interface to Singular). By design, we do not distinguish between these two, in order to facilitate the conversion of code from the pexpect interface to the library interface.

See also

Feature singular

class sage.features.sagemath.**sage__modular**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing the presence of `sage.modular`.

class sage.features.sagemath.**sage__modules**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing the presence of `sage.modules`.

EXAMPLES:

All uses of implementations of vector spaces / free modules in SageMath, whether `FreeModule()`, `sage.combinat.free_module.CombinatorialFreeModule`, `sage.tensor.modules.finite_rank_free_module.FiniteRankFreeModule`, or additive abelian groups, should be marked `# needs sage.modules`.

The same holds for matrices, tensors, algebras, quadratic forms, point lattices, root systems, matrix/affine/Weyl/Coxeter groups, matroids, and ring derivations.

Likewise, all uses of sage.coding, `sage.crypto`, and `sage.homology` in doctests should be marked `# needs sage.modules`.

class sage.features.sagemath.**sage__numerical__mip**(**args*, ***kwds*)

Bases: *PythonModule*

A *Feature* describing the presence of `sage.numerical.mip`.

class sage.features.sagemath.**sage__plot**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing the presence of `sage.plot`.

class sage.features.sagemath.**sage__rings__complex_double**(**args*, ***kwds*)

Bases: *PythonModule*

A *Feature* describing the presence of `sage.rings.complex_double`.

class sage.features.sagemath.**sage__rings__finite_rings**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing the presence of `sage.rings.finite_rings`; specifically, the element implementations using the *PARI* library.

class sage.features.sagemath.**sage__rings__function_field**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing the presence of `sage.rings.function_field`.

EXAMPLES:

Rational function fields are always available:

```
sage: K.<x> = FunctionField(QQ)
sage: K.maximal_order()
Maximal order of Rational function field in x over Rational Field
```

```
>>> from sage.all import *
>>> K = FunctionField(QQ, names=('x',)); (x,) = K._first_ngens(1)
>>> K.maximal_order()
Maximal order of Rational function field in x over Rational Field
```

Use the tag `# needs sage.rings.function_field` whenever extensions of function fields (by adjoining a root of a univariate polynomial) come into play:

```
sage: R.<y> = K[]
sage: L.<y> = K.extension(y^5 - (x^3 + 2*x*y + 1/x)); L                    #␣
↪needs sage.rings.function_field
Function field in y defined by y^5 - 2*x*y + (-x^4 - 1)/x
```

```
>>> from sage.all import *
>>> R = K['y']; (y,) = R._first_ngens(1)
>>> L = K.extension(y**Integer(5) - (x**Integer(3) + Integer(2)*x*y + Integer(1)/
↪x), names=('y',)); (y,) = L._first_ngens(1); L                    # needs␣
↪sage.rings.function_field
Function field in y defined by y^5 - 2*x*y + (-x^4 - 1)/x
```

Such extensions of function fields are implemented using Gröbner bases of polynomial rings; Sage makes essential use of the *Singular* system for this. (It is not necessary to use the tag `# needs sage.libs.singular`; it is implied by `# needs sage.rings.function_field`.)

class sage.features.sagemath.**sage__rings__number_field**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing the presence of `sage.rings.number_field`.

Number fields are implemented in Sage using a complicated mixture of various libraries, including *FLINT*, *GAP*, *MPFI*, *NTL*, and *PARI*.

EXAMPLES:

Rational numbers are, of course, always available:

```
sage: QQ in NumberFields()
True
```

```
>>> from sage.all import *
>>> QQ in NumberFields()
True
```

Doctests that construct algebraic number fields should be marked `# needs sage.rings.number_field`:

```
sage: # needs sage.rings.number_field
sage: K.<cuberoot2> = NumberField(x^3 - 2)
sage: L.<cuberoot3> = K.extension(x^3 - 3)
sage: S.<sqrt2> = L.extension(x^2 - 2); S
Number Field in sqrt2 with defining polynomial x^2 - 2 over its base field

sage: # needs sage.rings.number_field
sage: K.<zeta> = CyclotomicField(15)
sage: CC(zeta)
0.913545457642601 + 0.406736643075800*I
```

```
>>> from sage.all import *
>>> # needs sage.rings.number_field
>>> K = NumberField(x**Integer(3) - Integer(2), names=('cuberoot2',)); (cuberoot2,
↪) = K._first_ngens(1)
>>> L = K.extension(x**Integer(3) - Integer(3), names=('cuberoot3',)); (cuberoot3,
↪) = L._first_ngens(1)
>>> S = L.extension(x**Integer(2) - Integer(2), names=('sqrt2',)); (sqrt2,) = S._
↪first_ngens(1); S
Number Field in sqrt2 with defining polynomial x^2 - 2 over its base field

>>> # needs sage.rings.number_field
>>> K = CyclotomicField(Integer(15), names=('zeta',)); (zeta,) = K._first_ngens(1)
>>> CC(zeta)
0.913545457642601 + 0.406736643075800*I
```

Doctests that make use of the algebraic field `QQbar` or the algebraic real field `AA` should be marked likewise:

```
sage: # needs sage.rings.number_field
sage: AA(-1)^(1/3)
-1
sage: QQbar(-1)^(1/3)
0.500000000000000? + 0.866025403784439?*I
```

```
>>> from sage.all import *
>>> # needs sage.rings.number_field
>>> AA(-Integer(1))**(Integer(1)/Integer(3))
-1
>>> QQbar(-Integer(1))**(Integer(1)/Integer(3))
0.500000000000000? + 0.866025403784439?*I
```

Use of the universal cyclotomic field should be marked `# needs sage.libs.gap sage.rings.number_field`.

sage: # needs sage.libs.gap sage.rings.number_field sage: UCF = UniversalCyclotomicField(); UCF Universal Cyclotomic Field sage: E = UCF.gen sage: f = E(2) + E(3); f 2*E(3) + E(3)^2 sage: f.galois_conjugates() [2*E(3) + E(3)^2, E(3) + 2*E(3)^2]

class sage.features.sagemath.**sage__rings__padics**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing the presence of `sage.rings.padics`.

class sage.features.sagemath.**sage__rings__real_mpfr**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing the presence of `sage.rings.real_mpfr`.

class sage.features.sagemath.**sage__schemes**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing the presence of `sage.schemes`.

class sage.features.sagemath.**sage__symbolic**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing the presence of `sage.symbolic`.

EXAMPLES:

The symbolics subsystem of Sage will be provided by the distribution sagemath-symbolics, in preparation at Issue #35095. If it is not installed, Sage will be able to provide installation advice:

```
sage: from sage.features.sagemath import sage__symbolic
sage: print(sage__symbolic().resolution())                                  #␣
↪optional - sage_spkg, not tested
...To install sagemath_symbolics...you can try to run...
pip install sagemath-symbolics
...
```

```
>>> from sage.all import *
>>> from sage.features.sagemath import sage__symbolic
>>> print(sage__symbolic().resolution())                                    #␣
↪optional - sage_spkg, not tested
...To install sagemath_symbolics...you can try to run...
pip install sagemath-symbolics
...
```

class sage.features.sagemath.**sagemath_doc_html**(**args*, ***kwds*)

Bases: *StaticFile*

A *Feature* which describes the presence of the documentation of the Sage library in HTML format.

Developers often use `make build` instead of `make` to avoid the long time it takes to compile the documentation. Although commands such as `make ptest` build the documentation before testing, other test commands such as `make ptestlong-nodoc` or `./sage -t --all` do not.

All doctests that refer to the built documentation need to be marked `# needs sagemath_doc_html`.

3.6 Features for testing the presence of package systems `sage_spkg`, `conda`, `pip`, `debian`, `fedora` etc.

class sage.features.pkg_systems.**PackageSystem**(**args*, ***kwds*)

Bases: *Feature*

A *Feature* describing a system package manager.

EXAMPLES:

```
sage: from sage.features.pkg_systems import PackageSystem
sage: PackageSystem('conda')
Feature('conda')
```

```
>>> from sage.all import *
>>> from sage.features.pkg_systems import PackageSystem
>>> PackageSystem('conda')
Feature('conda')
```

spkg_installation_hint(*spkgs*, *prompt*, *feature*)

Return a string that explains how to install `feature`.

EXAMPLES:

```
sage: from sage.features.pkg_systems import PackageSystem
sage: homebrew = PackageSystem('homebrew')
sage: homebrew.spkg_installation_hint('openblas')  # optional - SAGE_ROOT
'To install openblas using the homebrew package manager, you can try to run:\
↪n!brew install openblas'
```

```
>>> from sage.all import *
>>> from sage.features.pkg_systems import PackageSystem
>>> homebrew = PackageSystem('homebrew')
>>> homebrew.spkg_installation_hint('openblas')  # optional - SAGE_ROOT
'To install openblas using the homebrew package manager, you can try to run:\
↪n!brew install openblas'
```

class sage.features.pkg_systems.**PipPackageSystem**(**args*, ***kwds*)

Bases: *PackageSystem*

A *Feature* describing the Pip package manager.

EXAMPLES:

```
sage: from sage.features.pkg_systems import PipPackageSystem
sage: PipPackageSystem()
Feature('pip')
```

```
>>> from sage.all import *
>>> from sage.features.pkg_systems import PipPackageSystem
>>> PipPackageSystem()
Feature('pip')
```

class sage.features.pkg_systems.**SagePackageSystem**(**args*, ***kwds*)

Bases: *PackageSystem*

A *Feature* describing the package manager of the SageMath distribution.

EXAMPLES:

```
sage: from sage.features.pkg_systems import SagePackageSystem
sage: SagePackageSystem()
Feature('sage_spkg')
```

```
>>> from sage.all import *
>>> from sage.features.pkg_systems import SagePackageSystem
>>> SagePackageSystem()
Feature('sage_spkg')
```

3.7 Features for testing the presence of bliss

class sage.features.bliss.**Bliss**(**args*, ***kwds*)

Bases: *BuildModule*

A *Feature* which describes whether the sage.graphs.bliss module is available in this installation of Sage.

EXAMPLES:

```
sage: from sage.features.bliss import Bliss
sage: Bliss().is_present()  # needs bliss
FeatureTestResult('bliss', True)
sage: Bliss().is_present()  # needs !bliss
FeatureTestResult('bliss', False)
```

```
>>> from sage.all import *
>>> from sage.features.bliss import Bliss
>>> Bliss().is_present()  # needs bliss
FeatureTestResult('bliss', True)
>>> Bliss().is_present()  # needs !bliss
FeatureTestResult('bliss', False)
```

A runtime check. We only check the "present" case because, if feature checks are _not_ deferred, the needs !bliss can be satisfied (disabled at build time) at the same time we are able to import a module that was installed for a previous build of sage:

```
sage: from sage.features.bliss import Bliss
sage: Bliss().is_present_at_runtime()  # needs bliss
FeatureTestResult('bliss', True)
```

```
>>> from sage.all import *
>>> from sage.features.bliss import Bliss
>>> Bliss().is_present_at_runtime()  # needs bliss
FeatureTestResult('bliss', True)
```

sage.features.bliss.**all_features**()

3.8 Feature for testing the presence of (lib)brial

class sage.features.brial.**Brial**(**args*, ***kwds*)

Bases: *BuildModule*

A *sage.features.Feature* describing the presence of pbori.

The pbori module in turn depends on the presence and usability of libbrial – a slightly more modern fork of PolyBoRi, which hopefully explains the name.

EXAMPLES:

```
sage: from sage.features.brial import Brial
sage: Brial().is_present()  # needs brial
FeatureTestResult('brial', True)
sage: Brial().is_present()  # needs !brial
FeatureTestResult('brial', False)
```

```
>>> from sage.all import *
>>> from sage.features.brial import Brial
>>> Brial().is_present()  # needs brial
FeatureTestResult('brial', True)
>>> Brial().is_present()  # needs !brial
FeatureTestResult('brial', False)
```

A runtime check. We only check the "present" case because, if feature checks are _not_ deferred, the `needs !brial` can be satisfied (disabled at build time) at the same time we are able to import a module that was installed for a previous build of sage:

```
sage: from sage.features.brial import Brial
sage: Brial().is_present_at_runtime()  # needs brial
FeatureTestResult('brial', True)
```

```
>>> from sage.all import *
>>> from sage.features.brial import Brial
>>> Brial().is_present_at_runtime()  # needs brial
FeatureTestResult('brial', True)
```

sage.features.brial.**all_features**()

3.9 Features for testing the presence of `coxeter3`

class sage.features.coxeter3.**Coxeter3**(**args*, ***kwds*)

Bases: *BuildModule*

A *Feature* which describes whether the `sage.libs.coxeter3` module is available in this installation of Sage.

EXAMPLES:

```
sage: from sage.features.coxeter3 import Coxeter3
sage: Coxeter3().is_present()  # needs coxeter3
FeatureTestResult('coxeter3', True)
sage: Coxeter3().is_present()  # needs !coxeter3
FeatureTestResult('coxeter3', False)
```

```
>>> from sage.all import *
>>> from sage.features.coxeter3 import Coxeter3
>>> Coxeter3().is_present()  # needs coxeter3
FeatureTestResult('coxeter3', True)
>>> Coxeter3().is_present()  # needs !coxeter3
FeatureTestResult('coxeter3', False)
```

A runtime check. We only check the "present" case because, if feature checks are _not_ deferred, the `needs !coxeter3` can be satisfied (disabled at build time) at the same time we are able to import a module that was installed for a previous build of sage:

```
sage: from sage.features.coxeter3 import Coxeter3
sage: Coxeter3().is_present_at_runtime()  # needs coxeter3
FeatureTestResult('coxeter3', True)
```

```
>>> from sage.all import *
>>> from sage.features.coxeter3 import Coxeter3
>>> Coxeter3().is_present_at_runtime()  # needs coxeter3
FeatureTestResult('coxeter3', True)
```

sage.features.coxeter3.**all_features**()

3.10 Feature for testing the presence of csdp

class sage.features.csdp.**CSDP**(**args*, ***kwds*)

Bases: *Executable*

A *Feature* which checks for the `theta` binary of *CSDP*.

EXAMPLES:

```
sage: from sage.features.csdp import CSDP
sage: CSDP().is_present()  # optional - csdp
FeatureTestResult('csdp', True)
```

```
>>> from sage.all import *
>>> from sage.features.csdp import CSDP
>>> CSDP().is_present()  # optional - csdp
FeatureTestResult('csdp', True)
```

is_functional()

Check whether `theta` works on a trivial example.

EXAMPLES:

```
sage: from sage.features.csdp import CSDP
sage: CSDP().is_functional()  # optional - csdp
FeatureTestResult('csdp', True)
```

```
>>> from sage.all import *
>>> from sage.features.csdp import CSDP
>>> CSDP().is_functional()  # optional - csdp
FeatureTestResult('csdp', True)
```

sage.features.csdp.**all_features**()

3.11 Features for testing the presence of various databases

class sage.features.databases.**DatabaseCremona**(**args*, ***kwds*)

Bases: *StaticFile*

A *Feature* which describes the presence of *John Cremona's database of elliptic curves*.

INPUT:

- `name` – either `'cremona'` (the default) for the full large database or `'cremona_mini'` for the small database

EXAMPLES:

```
sage: from sage.features.databases import DatabaseCremona
sage: DatabaseCremona('cremona_mini', type='standard').is_present()
FeatureTestResult('database_cremona_mini_ellcurve', True)
sage: DatabaseCremona().is_present()                                    #␣
↪optional - database_cremona_ellcurve
FeatureTestResult('database_cremona_ellcurve', True)
```

```
>>> from sage.all import *
>>> from sage.features.databases import DatabaseCremona
>>> DatabaseCremona('cremona_mini', type='standard').is_present()
FeatureTestResult('database_cremona_mini_ellcurve', True)
>>> DatabaseCremona().is_present()                                    # optional -
↪ database_cremona_ellcurve
FeatureTestResult('database_cremona_ellcurve', True)
```

class sage.features.databases.**DatabaseCubicHecke**(**args*, ***kwds*)

Bases: *PythonModule*

A *Feature* which describes the presence of the *Cubic Hecke algebra database package*.

The home of this database is the web-page Cubic Hecke algebra on 4 strands of Ivan Marin.

EXAMPLES:

```
sage: from sage.features.databases import DatabaseCubicHecke
sage: DatabaseCubicHecke().is_present()  # optional - database_cubic_hecke
FeatureTestResult('database_cubic_hecke', True)
```

```
>>> from sage.all import *
>>> from sage.features.databases import DatabaseCubicHecke
>>> DatabaseCubicHecke().is_present()  # optional - database_cubic_hecke
FeatureTestResult('database_cubic_hecke', True)
```

class sage.features.databases.**DatabaseEllcurves**(**args*, ***kwds*)

Bases: *StaticFile*

A *Feature* which describes the presence of William Stein's database of interesting curves.

EXAMPLES:

```
sage: from sage.features.databases import DatabaseEllcurves
sage: bool(DatabaseEllcurves().is_present())  # optional - database_ellcurves
True
```

```
>>> from sage.all import *
>>> from sage.features.databases import DatabaseEllcurves
>>> bool(DatabaseEllcurves().is_present())  # optional - database_ellcurves
True
```

class sage.features.databases.**DatabaseGraphs**(**args*, ***kwds*)

Bases: *StaticFile*

A *Feature* which describes the presence of the graphs database.

EXAMPLES:

```
sage: from sage.features.databases import DatabaseGraphs
sage: bool(DatabaseGraphs().is_present())  # optional - database_graphs
True
```

```
>>> from sage.all import *
>>> from sage.features.databases import DatabaseGraphs
>>> bool(DatabaseGraphs().is_present())  # optional - database_graphs
True
```

class sage.features.databases.**DatabaseJones**(**args*, ***kwds*)

Bases: *StaticFile*

A *Feature* which describes the presence of *John Jones's tables of number fields*.

EXAMPLES:

```
sage: from sage.features.databases import DatabaseJones
sage: bool(DatabaseJones().is_present())  # optional - database_jones_numfield
True
```

```
>>> from sage.all import *
>>> from sage.features.databases import DatabaseJones
>>> bool(DatabaseJones().is_present())  # optional - database_jones_numfield
True
```

class sage.features.databases.**DatabaseKnotInfo**(**args*, ***kwds*)

Bases: *PythonModule*

A *Feature* which describes the presence of the *package providing the KnotInfo and LinkInfo databases*.

The homes of these databases are the web-pages KnotInfo and LinkInfo.

EXAMPLES:

```
sage: from sage.features.databases import DatabaseKnotInfo
sage: DatabaseKnotInfo().is_present()  # optional - database_knotinfo
FeatureTestResult('database_knotinfo', True)
```

```
>>> from sage.all import *
>>> from sage.features.databases import DatabaseKnotInfo
>>> DatabaseKnotInfo().is_present()  # optional - database_knotinfo
FeatureTestResult('database_knotinfo', True)
```

class sage.features.databases.**DatabaseMatroids**(**args*, ***kwds*)

Bases: *PythonModule*

A *Feature* which describes the presence of *Yoshitake Matsumoto's Database of Matroids*.

EXAMPLES:

```
sage: from sage.features.databases import DatabaseMatroids
sage: DatabaseMatroids().is_present()                                          #␣
↪optional - matroid_database
FeatureTestResult('matroid_database', True)
```

```
>>> from sage.all import *
>>> from sage.features.databases import DatabaseMatroids
>>> DatabaseMatroids().is_present()                                          #␣
↪optional - matroid_database
FeatureTestResult('matroid_database', True)
```

REFERENCES:

[Mat2012]

class sage.features.databases.**DatabaseReflexivePolytopes**(**args*, ***kwds*)

Bases: *StaticFile*

A *Feature* which describes the presence of the *PALP databases of reflexive three-dimensional and four-dimensional lattice polytopes*.

EXAMPLES:

```
sage: from sage.features.databases import DatabaseReflexivePolytopes
sage: bool(DatabaseReflexivePolytopes().is_present())                     #␣
↪optional - polytopes_db
True
sage: bool(DatabaseReflexivePolytopes('polytopes_db_4d').is_present())   #␣
↪optional - polytopes_db_4d
True
```

```
>>> from sage.all import *
>>> from sage.features.databases import DatabaseReflexivePolytopes
>>> bool(DatabaseReflexivePolytopes().is_present())                    # optional -
↪ polytopes_db
True
>>> bool(DatabaseReflexivePolytopes('polytopes_db_4d').is_present())  # optional -
↪ polytopes_db_4d
True
```

sage.features.databases.**all_features**()

3.12 Feature for testing the presence of `dvipng`

sage.features.dvipng.**all_features**()

class sage.features.dvipng.**dvipng**(**args*, ***kwds*)

Bases: *Executable*

A *Feature* describing the presence of `dvipng`.

EXAMPLES:

```
sage: from sage.features.dvipng import dvipng
sage: dvipng().is_present()              # optional - dvipng
FeatureTestResult('dvipng', True)
```

```
>>> from sage.all import *
>>> from sage.features.dvipng import dvipng
```

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```
>>> dvipng().is_present()              # optional - dvipng
FeatureTestResult('dvipng', True)
```

3.13 Feature for testing the presence of ffmpeg

class sage.features.ffmpeg.**FFmpeg**(**args*, ***kwds*)

Bases: *Executable*

A *Feature* describing the presence of *ffmpeg*.

EXAMPLES:

```
sage: from sage.features.ffmpeg import FFmpeg
sage: FFmpeg().is_present()  # optional - ffmpeg
FeatureTestResult('ffmpeg', True)
```

```
>>> from sage.all import *
>>> from sage.features.ffmpeg import FFmpeg
>>> FFmpeg().is_present()  # optional - ffmpeg
FeatureTestResult('ffmpeg', True)
```

is_functional()

Return whether command `ffmpeg` in the path is functional.

EXAMPLES:

```
sage: from sage.features.ffmpeg import FFmpeg
sage: FFmpeg().is_functional()   # optional - ffmpeg
FeatureTestResult('ffmpeg', True)
```

```
>>> from sage.all import *
>>> from sage.features.ffmpeg import FFmpeg
>>> FFmpeg().is_functional()   # optional - ffmpeg
FeatureTestResult('ffmpeg', True)
```

sage.features.ffmpeg.**all_features**()

3.14 Features for testing the presence of 4ti2

class sage.features.four_ti_2.**FourTi2**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing the presence of all *4ti2* executables.

EXAMPLES:

```
sage: from sage.features.four_ti_2 import FourTi2
sage: FourTi2().is_present()  # optional - 4ti2
FeatureTestResult('4ti2', True)
```

```
>>> from sage.all import *
>>> from sage.features.four_ti_2 import FourTi2
>>> FourTi2().is_present()  # optional - 4ti2
FeatureTestResult('4ti2', True)
```

class sage.features.four_ti_2.**FourTi2Executable**(**args*, ***kwds*)

Bases: *Executable*

A *Feature* for the *4ti2* executables.

sage.features.four_ti_2.**all_features**()

3.15 Features for testing the presence of the SageMath interfaces to `gap` and of GAP packages

class sage.features.gap.**GapPackage**(**args*, ***kwds*)

Bases: *Feature*

A *Feature* describing the presence of a GAP package.

A GAP package is "present" if it *can be* loaded, not if it *has been* loaded.

See also

Feature sage.libs.gap

EXAMPLES:

```
sage: from sage.features.gap import GapPackage
sage: GapPackage("grape", spkg='gap_packages')
Feature('gap_package_grape')
```

```
>>> from sage.all import *
>>> from sage.features.gap import GapPackage
>>> GapPackage("grape", spkg='gap_packages')
Feature('gap_package_grape')
```

sage.features.gap.**all_features**()

3.16 Features for testing the presence of graph generator programs `benzene`, `buckygen`, `plantri`

class sage.features.graph_generators.**Benzene**(**args*, ***kwds*)

Bases: *Executable*

A *Feature* which checks for the *benzene* binary.

EXAMPLES:

```
sage: from sage.features.graph_generators import Benzene
sage: Benzene().is_present()  # optional - benzene
FeatureTestResult('benzene', True)
```

```
>>> from sage.all import *
>>> from sage.features.graph_generators import Benzene
>>> Benzene().is_present()  # optional - benzene
FeatureTestResult('benzene', True)
```

is_functional()

Check whether `benzene` works on trivial input.

EXAMPLES:

```
sage: from sage.features.graph_generators import Benzene
sage: Benzene().is_functional()  # optional - benzene
FeatureTestResult('benzene', True)
```

```
>>> from sage.all import *
>>> from sage.features.graph_generators import Benzene
>>> Benzene().is_functional()  # optional - benzene
FeatureTestResult('benzene', True)
```

class sage.features.graph_generators.**Buckygen**(**args*, ***kwds*)

Bases: *Executable*

A *Feature* which checks for the *buckygen* binary.

EXAMPLES:

```
sage: from sage.features.graph_generators import Buckygen
sage: Buckygen().is_present()  # optional - buckygen
FeatureTestResult('buckygen', True)
```

```
>>> from sage.all import *
>>> from sage.features.graph_generators import Buckygen
>>> Buckygen().is_present()  # optional - buckygen
FeatureTestResult('buckygen', True)
```

is_functional()

Check whether `buckygen` works on trivial input.

EXAMPLES:

```
sage: from sage.features.graph_generators import Buckygen
sage: Buckygen().is_functional()  # optional - buckygen
FeatureTestResult('buckygen', True)
```

```
>>> from sage.all import *
>>> from sage.features.graph_generators import Buckygen
>>> Buckygen().is_functional()  # optional - buckygen
FeatureTestResult('buckygen', True)
```

class sage.features.graph_generators.**Plantri**(**args*, ***kwds*)

Bases: *Executable*

A *Feature* which checks for the *plantri* binary.

EXAMPLES:

```
sage: from sage.features.graph_generators import Plantri
sage: Plantri().is_present()  # optional - plantri
FeatureTestResult('plantri', True)
```

```
>>> from sage.all import *
>>> from sage.features.graph_generators import Plantri
>>> Plantri().is_present()  # optional - plantri
FeatureTestResult('plantri', True)
```

is_functional()

Check whether `plantri` works on trivial input.

EXAMPLES:

```
sage: from sage.features.graph_generators import Plantri
sage: Plantri().is_functional()  # optional - plantri
FeatureTestResult('plantri', True)
```

```
>>> from sage.all import *
>>> from sage.features.graph_generators import Plantri
>>> Plantri().is_functional()  # optional - plantri
FeatureTestResult('plantri', True)
```

sage.features.graph_generators.**all_features**()

3.17 Features for testing the presence of `graphviz`

class sage.features.graphviz.**Graphviz**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing the presence of the *dot*, *neato*, and *twopi* executables from the *graphviz* package.

EXAMPLES:

```
sage: from sage.features.graphviz import Graphviz
sage: Graphviz().is_present()  # optional - graphviz
FeatureTestResult('graphviz', True)
```

```
>>> from sage.all import *
>>> from sage.features.graphviz import Graphviz
>>> Graphviz().is_present()  # optional - graphviz
FeatureTestResult('graphviz', True)
```

sage.features.graphviz.**all_features**()

class sage.features.graphviz.**dot**(**args*, ***kwds*)

Bases: *Executable*

A *Feature* describing the presence of `dot`.

class sage.features.graphviz.**neato**(**args*, ***kwds*)

Bases: *Executable*

A *Feature* describing the presence of `neato`.

class sage.features.graphviz.**twopi**(**args*, ***kwds*)

Bases: *Executable*

A *Feature* describing the presence of `twopi`.

3.18 Feature for testing the presence of `imagemagick`

Currently we only check for the presence of `convert` or `magick`. When needed, other commands like `magick-script`, `mogrify`, `identify`, `composite`, `montage`, `compare`, etc. could be also checked in this module.

class sage.features.imagemagick.**ImageMagick**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing the presence of *ImageMagick*

Currently, only the availability of the `magick` (or `convert`) program is checked.

EXAMPLES:

```
sage: from sage.features.imagemagick import ImageMagick
sage: ImageMagick().is_present()  # optional - imagemagick
FeatureTestResult('imagemagick', True)
```

```
>>> from sage.all import *
>>> from sage.features.imagemagick import ImageMagick
>>> ImageMagick().is_present()  # optional - imagemagick
FeatureTestResult('imagemagick', True)
```

class sage.features.imagemagick.**Magick**(**args*, ***kwds*)

Bases: *Executable*

A *Feature* describing the presence of `magick` or the deprecated `convert`.

EXAMPLES:

```
sage: from sage.features.imagemagick import Magick
sage: Magick().is_present()  # optional - imagemagick
FeatureTestResult('magick', True)
```

```
>>> from sage.all import *
>>> from sage.features.imagemagick import Magick
>>> Magick().is_present()  # optional - imagemagick
FeatureTestResult('magick', True)
```

is_functional()

Return whether command `magick` or `convert` in the path is functional.

EXAMPLES:

```
sage: from sage.features.imagemagick import Magick
sage: Magick().is_functional()  # optional - imagemagick
FeatureTestResult('magick', True)
```

```
>>> from sage.all import *
>>> from sage.features.imagemagick import Magick
```

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```
>>> Magick().is_functional()    # optional - imagemagick
FeatureTestResult('magick', True)
```

sage.features.imagemagick.**all_features**()

3.19 Features for testing whether interpreter interfaces to `magma`, `maple`, `mathematica` etc. are functional

class sage.features.interfaces.**InterfaceFeature**(**args*, ***kwds*)

Bases: *Feature*

A *Feature* describing whether an `Interface` is present and functional.

class sage.features.interfaces.**Macaulay2**(**args*, ***kwds*)

Bases: *InterfaceFeature*

A *Feature* describing whether `sage.interfaces.macaulay2.Macaulay2` is present and functional.

EXAMPLES:

```
sage: from sage.features.interfaces import Macaulay2
sage: Macaulay2().is_present()  # random
FeatureTestResult('macaulay2', False)
```

```
>>> from sage.all import *
>>> from sage.features.interfaces import Macaulay2
>>> Macaulay2().is_present()  # random
FeatureTestResult('macaulay2', False)
```

class sage.features.interfaces.**Magma**(**args*, ***kwds*)

Bases: *InterfaceFeature*

A *Feature* describing whether `sage.interfaces.magma.Magma` is present and functional.

EXAMPLES:

```
sage: from sage.features.interfaces import Magma
sage: Magma().is_present()  # random
FeatureTestResult('magma', False)
```

```
>>> from sage.all import *
>>> from sage.features.interfaces import Magma
>>> Magma().is_present()  # random
FeatureTestResult('magma', False)
```

class sage.features.interfaces.**Maple**(**args*, ***kwds*)

Bases: *InterfaceFeature*

A *Feature* describing whether `sage.interfaces.maple.Maple` is present and functional.

EXAMPLES:

```
sage: from sage.features.interfaces import Maple
sage: Maple().is_present()  # random
FeatureTestResult('maple', False)
```

```
>>> from sage.all import *
>>> from sage.features.interfaces import Maple
>>> Maple().is_present()  # random
FeatureTestResult('maple', False)
```

class sage.features.interfaces.**Mathematica**(**args*, ***kwds*)

Bases: *InterfaceFeature*

A *Feature* describing whether sage.interfaces.mathematica.Mathematica is present and functional.

EXAMPLES:

```
sage: from sage.features.interfaces import Mathematica
sage: Mathematica().is_present()  # not tested
FeatureTestResult('mathematica', False)
```

```
>>> from sage.all import *
>>> from sage.features.interfaces import Mathematica
>>> Mathematica().is_present()  # not tested
FeatureTestResult('mathematica', False)
```

class sage.features.interfaces.**Matlab**(**args*, ***kwds*)

Bases: *InterfaceFeature*

A *Feature* describing whether sage.interfaces.matlab.Matlab is present and functional.

EXAMPLES:

```
sage: from sage.features.interfaces import Matlab
sage: Matlab().is_present()  # random
FeatureTestResult('matlab', False)
```

```
>>> from sage.all import *
>>> from sage.features.interfaces import Matlab
>>> Matlab().is_present()  # random
FeatureTestResult('matlab', False)
```

class sage.features.interfaces.**Octave**(**args*, ***kwds*)

Bases: *InterfaceFeature*

A *Feature* describing whether sage.interfaces.octave.Octave is present and functional.

EXAMPLES:

```
sage: from sage.features.interfaces import Octave
sage: Octave().is_present()  # random
FeatureTestResult('octave', False)
```

```
>>> from sage.all import *
>>> from sage.features.interfaces import Octave
```

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```
>>> Octave().is_present()  # random
FeatureTestResult('octave', False)
```

class sage.features.interfaces.**Scilab**(**args*, ***kwds*)

Bases: *InterfaceFeature*

A *Feature* describing whether sage.interfaces.scilab.Scilab is present and functional.

EXAMPLES:

```
sage: from sage.features.interfaces import Scilab
sage: Scilab().is_present()  # random
FeatureTestResult('scilab', False)
```

```
>>> from sage.all import *
>>> from sage.features.interfaces import Scilab
>>> Scilab().is_present()  # random
FeatureTestResult('scilab', False)
```

sage.features.interfaces.**all_features**()

Return features corresponding to interpreter interfaces.

EXAMPLES:

```
sage: from sage.features.interfaces import all_features
sage: list(all_features())
[Feature('magma'),
 Feature('matlab'),
 Feature('mathematica'),
 Feature('maple'),
 Feature('macaulay2'),
 Feature('octave'),
 Feature('scilab')]
```

```
>>> from sage.all import *
>>> from sage.features.interfaces import all_features
>>> list(all_features())
[Feature('magma'),
 Feature('matlab'),
 Feature('mathematica'),
 Feature('maple'),
 Feature('macaulay2'),
 Feature('octave'),
 Feature('scilab')]
```

3.20 Feature for testing if the Internet is available

class sage.features.internet.**Internet**(**args*, ***kwds*)

Bases: *Feature*

A *Feature* describing if Internet is available.

Failure of connecting to the site "https://www.sagemath.org" within a second is regarded as internet being not available.

EXAMPLES:

```
sage: from sage.features.internet import Internet
sage: Internet()
Feature('internet')
```

```
>>> from sage.all import *
>>> from sage.features.internet import Internet
>>> Internet()
Feature('internet')
```

sage.features.internet.**all_features**()

3.21 Feature for testing the presence of `kenzo`

class sage.features.kenzo.**Kenzo**(**args*, ***kwds*)

Bases: *Feature*

A *Feature* describing the presence of *Kenzo*.

EXAMPLES:

```
sage: from sage.features.kenzo import Kenzo
sage: Kenzo().is_present()  # optional - kenzo
FeatureTestResult('kenzo', True)
```

```
>>> from sage.all import *
>>> from sage.features.kenzo import Kenzo
>>> Kenzo().is_present()  # optional - kenzo
FeatureTestResult('kenzo', True)
```

sage.features.kenzo.**all_features**()

3.22 Features for testing the presence of `latex` and equivalent programs

class sage.features.latex.**LaTeX**(**args*, ***kwds*)

Bases: *Executable*

A *Feature* describing the presence of `latex`.

EXAMPLES:

```
sage: from sage.features.latex import latex
sage: latex().is_present()              # optional - latex
FeatureTestResult('latex', True)
```

```
>>> from sage.all import *
>>> from sage.features.latex import latex
```

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```
>>> latex().is_present()             # optional - latex
FeatureTestResult('latex', True)
```

is_functional()

Return whether `latex` in the path is functional.

EXAMPLES:

```
sage: from sage.features.latex import latex
sage: latex().is_functional()             # optional - latex
FeatureTestResult('latex', True)
```

```
>>> from sage.all import *
>>> from sage.features.latex import latex
>>> latex().is_functional()             # optional - latex
FeatureTestResult('latex', True)
```

When the feature is not functional, more information on the reason can be obtained as follows:

```
sage: result = latex().is_functional()    # not tested
sage: print(result.reason)                # not tested
Running latex on a sample file
(with command='latex -interaction=nonstopmode tmp_wmpos8ak.tex')
returned nonzero exit status='1' with stderr=''
and stdout='This is pdfTeX,
...
Runaway argument?
{document
! File ended while scanning use of \end.
...
No pages of output.
Transcript written on tmp_wmpos8ak.log.'
```

```
>>> from sage.all import *
>>> result = latex().is_functional()    # not tested
>>> print(result.reason)                # not tested
Running latex on a sample file
(with command='latex -interaction=nonstopmode tmp_wmpos8ak.tex')
returned nonzero exit status='1' with stderr=''
and stdout='This is pdfTeX,
...
Runaway argument?
{document
! File ended while scanning use of \end.
...
No pages of output.
Transcript written on tmp_wmpos8ak.log.'
```

class sage.features.latex.**LaTeXPackage**(**args*, ***kwds*)

Bases: *TeXFile*

A *sage.features.Feature* describing the presence of a LaTeX package (`.sty` file).

EXAMPLES:

```
sage: from sage.features.latex import LaTeXPackage
sage: LaTeXPackage('graphics').is_present()  # optional - latex
FeatureTestResult('latex_package_graphics', True)
```

```
>>> from sage.all import *
>>> from sage.features.latex import LaTeXPackage
>>> LaTeXPackage('graphics').is_present()  # optional - latex
FeatureTestResult('latex_package_graphics', True)
```

class sage.features.latex.**TeXFile**(**args*, ***kwds*)

Bases: *StaticFile*

A *sage.features.Feature* describing the presence of a TeX file.

EXAMPLES:

```
sage: from sage.features.latex import TeXFile
sage: TeXFile('x', 'x.tex').is_present()  # optional - latex
FeatureTestResult('x', True)
```

```
>>> from sage.all import *
>>> from sage.features.latex import TeXFile
>>> TeXFile('x', 'x.tex').is_present()  # optional - latex
FeatureTestResult('x', True)
```

absolute_filename()

The absolute path of the file.

EXAMPLES:

```
sage: from sage.features.latex import TeXFile
sage: feature = TeXFile('latex_class_article', 'article.cls')
sage: feature.absolute_filename()  # optional - latex
'.../latex/base/article.cls'
```

```
>>> from sage.all import *
>>> from sage.features.latex import TeXFile
>>> feature = TeXFile('latex_class_article', 'article.cls')
>>> feature.absolute_filename()  # optional - latex
'.../latex/base/article.cls'
```

sage.features.latex.**all_features**()

class sage.features.latex.**dvips**(**args*, ***kwds*)

Bases: *Executable*

A *Feature* describing the presence of `dvips`.

EXAMPLES:

```
sage: from sage.features.latex import dvips
sage: dvips().is_present()              # optional - dvips
FeatureTestResult('dvips', True)
```

```
>>> from sage.all import *
>>> from sage.features.latex import dvips
>>> dvips().is_present()              # optional - dvips
FeatureTestResult('dvips', True)
```

class sage.features.latex.**latex**(**args*, ***kwds*)

Bases: *LaTeX*

A *Feature* describing the presence of `latex`.

EXAMPLES:

```
sage: from sage.features.latex import latex
sage: latex().is_present()              # optional - latex
FeatureTestResult('latex', True)
```

```
>>> from sage.all import *
>>> from sage.features.latex import latex
>>> latex().is_present()              # optional - latex
FeatureTestResult('latex', True)
```

class sage.features.latex.**lualatex**(**args*, ***kwds*)

Bases: *LaTeX*

A *Feature* describing the presence of `lualatex`.

EXAMPLES:

```
sage: from sage.features.latex import lualatex
sage: lualatex().is_present()              # optional - lualatex
FeatureTestResult('lualatex', True)
```

```
>>> from sage.all import *
>>> from sage.features.latex import lualatex
>>> lualatex().is_present()              # optional - lualatex
FeatureTestResult('lualatex', True)
```

class sage.features.latex.**pdflatex**(**args*, ***kwds*)

Bases: *LaTeX*

A *Feature* describing the presence of `pdflatex`.

EXAMPLES:

```
sage: from sage.features.latex import pdflatex
sage: pdflatex().is_present()              # optional - pdflatex
FeatureTestResult('pdflatex', True)
```

```
>>> from sage.all import *
>>> from sage.features.latex import pdflatex
>>> pdflatex().is_present()              # optional - pdflatex
FeatureTestResult('pdflatex', True)
```

class sage.features.latex.**xelatex**(**args*, ***kwds*)

Bases: *LaTeX*

A *Feature* describing the presence of `xelatex`.

EXAMPLES:

```
sage: from sage.features.latex import xelatex
sage: xelatex().is_present()              # optional - xelatex
FeatureTestResult('xelatex', True)
```

```
>>> from sage.all import *
>>> from sage.features.latex import xelatex
>>> xelatex().is_present()              # optional - xelatex
FeatureTestResult('xelatex', True)
```

3.23 Features for testing the presence of `latte_int`

class sage.features.latte.**Latte**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing the presence of executables from *LattE integrale*.

EXAMPLES:

```
sage: from sage.features.latte import Latte
sage: Latte().is_present()  # optional - latte_int
FeatureTestResult('latte_int', True)
```

```
>>> from sage.all import *
>>> from sage.features.latte import Latte
>>> Latte().is_present()  # optional - latte_int
FeatureTestResult('latte_int', True)
```

class sage.features.latte.**Latte_count**(**args*, ***kwds*)

Bases: *Executable*

Feature for the executable `count` from *LattE integrale*.

class sage.features.latte.**Latte_integrate**(**args*, ***kwds*)

Bases: *Executable*

Feature for the executable `integrate` from *LattE integrale*.

sage.features.latte.**all_features**()

3.24 Feature for testing the presence of libbraiding

class sage.features.libbraiding.**Libbraiding**(**args*, ***kwds*)

Bases: *BuildModule*

A *sage.features.Feature* describing the presence of `sage.libs.braiding`, the interface to libbraiding.

EXAMPLES:

```
sage: from sage.features.libbraiding import Libbraiding
sage: Libbraiding().is_present()  # needs libbraiding
FeatureTestResult('libbraiding', True)
sage: Libbraiding().is_present()  # needs !libbraiding
FeatureTestResult('libbraiding', False)
```

```
>>> from sage.all import *
>>> from sage.features.libbraiding import Libbraiding
>>> Libbraiding().is_present()  # needs libbraiding
FeatureTestResult('libbraiding', True)
>>> Libbraiding().is_present()  # needs !libbraiding
FeatureTestResult('libbraiding', False)
```

A runtime check. We only check the "present" case because, if feature checks are _not_ deferred, the `needs !libbraiding` can be satisfied (disabled at build time) at the same time we are able to import a module that was installed for a previous build of sage:

```
sage: from sage.features.libbraiding import Libbraiding
sage: Libbraiding().is_present_at_runtime()  # needs libbraiding
FeatureTestResult('libbraiding', True)
```

```
>>> from sage.all import *
>>> from sage.features.libbraiding import Libbraiding
>>> Libbraiding().is_present_at_runtime()  # needs libbraiding
FeatureTestResult('libbraiding', True)
```

`sage.features.libbraiding.`**`all_features`**`()`

3.25 Feature for testing the presence of libhomfly

class `sage.features.libhomfly.`**`Libhomfly`**`(`*args*, ***kwds*`)`

Bases: *BuildModule*

A *sage.features.Feature* describing the presence of `sage.libs.homfly`, the interface to libhomfly.

EXAMPLES:

```
sage: from sage.features.libhomfly import Libhomfly
sage: Libhomfly().is_present()  # needs libhomfly
FeatureTestResult('libhomfly', True)
sage: Libhomfly().is_present()  # needs !libhomfly
FeatureTestResult('libhomfly', False)
```

```
>>> from sage.all import *
>>> from sage.features.libhomfly import Libhomfly
>>> Libhomfly().is_present()  # needs libhomfly
FeatureTestResult('libhomfly', True)
>>> Libhomfly().is_present()  # needs !libhomfly
FeatureTestResult('libhomfly', False)
```

A runtime check. We only check the "present" case because, if feature checks are _not_ deferred, the `needs !libhomfly` can be satisfied (disabled at build time) at the same time we are able to import a module that was installed for a previous build of sage:

```
sage: from sage.features.libhomfly import Libhomfly
sage: Libhomfly().is_present_at_runtime()  # needs libhomfly
FeatureTestResult('libhomfly', True)
```

```
>>> from sage.all import *
>>> from sage.features.libhomfly import Libhomfly
>>> Libhomfly().is_present_at_runtime()  # needs libhomfly
FeatureTestResult('libhomfly', True)
```

sage.features.libhomfly.**all_features**()

3.26 Feature for testing the presence of `lrslib`

class sage.features.lrs.**Lrs**(**args*, ***kwds*)

Bases: *Executable*

A *Feature* describing the presence of the `lrs` binary which comes as a part of `lrslib`.

EXAMPLES:

```
sage: from sage.features.lrs import Lrs
sage: Lrs().is_present()  # optional - lrslib
FeatureTestResult('lrs', True)
```

```
>>> from sage.all import *
>>> from sage.features.lrs import Lrs
>>> Lrs().is_present()  # optional - lrslib
FeatureTestResult('lrs', True)
```

is_functional()

Test whether `lrs` works on a trivial input.

EXAMPLES:

```
sage: from sage.features.lrs import Lrs
sage: Lrs().is_functional()  # optional - lrslib
FeatureTestResult('lrs', True)
```

```
>>> from sage.all import *
>>> from sage.features.lrs import Lrs
>>> Lrs().is_functional()  # optional - lrslib
FeatureTestResult('lrs', True)
```

class sage.features.lrs.**LrsNash**(**args*, ***kwds*)

Bases: *Executable*

A *Feature* describing the presence of the `lrsnash` binary which comes as a part of `lrslib`.

EXAMPLES:

```
sage: from sage.features.lrs import LrsNash
sage: LrsNash().is_present()  # optional - lrslib
FeatureTestResult('lrsnash', True)
```

```
>>> from sage.all import *
>>> from sage.features.lrs import LrsNash
>>> LrsNash().is_present()  # optional - lrslib
FeatureTestResult('lrsnash', True)
```

is_functional()

Test whether `lrsnash` works on a trivial input.

EXAMPLES:

```
sage: from sage.features.lrs import LrsNash
sage: LrsNash().is_functional()  # optional - lrslib
FeatureTestResult('lrsnash', True)
```

```
>>> from sage.all import *
>>> from sage.features.lrs import LrsNash
>>> LrsNash().is_functional()  # optional - lrslib
FeatureTestResult('lrsnash', True)
```

class sage.features.lrs.**Lrslib**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing the presence of the executables *lrs* and *lrsnash* provided by the *lrslib* package.

EXAMPLES:

```
sage: from sage.features.lrs import Lrslib
sage: Lrslib().is_present()  # optional - lrslib
FeatureTestResult('lrslib', True)
```

```
>>> from sage.all import *
>>> from sage.features.lrs import Lrslib
>>> Lrslib().is_present()  # optional - lrslib
FeatureTestResult('lrslib', True)
```

sage.features.lrs.**all_features**()

3.27 Features for testing the presence of mcqd

class sage.features.mcqd.**Mcqd**(**args*, ***kwds*)

Bases: *BuildModule*

A *Feature* describing the presence of the `sage.graphs.mcqd` module, which is the SageMath interface to the *mcqd* library

EXAMPLES:

```
sage: from sage.features.mcqd import Mcqd
sage: Mcqd().is_present()  # needs mcqd
FeatureTestResult('mcqd', True)
sage: Mcqd().is_present()  # needs !mcqd
FeatureTestResult('mcqd', False)
```

```
>>> from sage.all import *
>>> from sage.features.mcqd import Mcqd
>>> Mcqd().is_present()  # needs mcqd
FeatureTestResult('mcqd', True)
>>> Mcqd().is_present()  # needs !mcqd
FeatureTestResult('mcqd', False)
```

A runtime check. We only check the "present" case because, if feature checks are _not_ deferred, the `needs !mcqd` can be satisfied (disabled at build time) at the same time we are able to import a module that was installed for a previous build of sage:

```
sage: from sage.features.mcqd import Mcqd
sage: Mcqd().is_present_at_runtime()  # needs mcqd
FeatureTestResult('mcqd', True)
```

```
>>> from sage.all import *
>>> from sage.features.mcqd import Mcqd
>>> Mcqd().is_present_at_runtime()  # needs mcqd
FeatureTestResult('mcqd', True)
```

sage.features.mcqd.**all_features**()

3.28 Feature for testing the presence of `meataxe`

class sage.features.meataxe.**Meataxe**(**args*, ***kwds*)

Bases: *BuildModule*

A *Feature* describing the presence of the Sage modules that depend on the *meataxe* library.

EXAMPLES:

```
sage: from sage.features.meataxe import Meataxe
sage: Meataxe().is_present()  # needs meataxe
FeatureTestResult('meataxe', True)
sage: Meataxe().is_present()  # needs !meataxe
FeatureTestResult('meataxe', False)
```

```
>>> from sage.all import *
>>> from sage.features.meataxe import Meataxe
>>> Meataxe().is_present()  # needs meataxe
FeatureTestResult('meataxe', True)
>>> Meataxe().is_present()  # needs !meataxe
FeatureTestResult('meataxe', False)
```

A runtime check. We only check the "present" case because, if feature checks are _not_ deferred, the `needs !meataxe` can be satisfied (disabled at build time) at the same time we are able to import a module that was installed for a previous build of sage:

```
sage: from sage.features.meataxe import Meataxe
sage: Meataxe().is_present_at_runtime()  # needs meataxe
FeatureTestResult('meataxe', True)
```

```
>>> from sage.all import *
>>> from sage.features.meataxe import Meataxe
>>> Meataxe().is_present_at_runtime()  # needs meataxe
FeatureTestResult('meataxe', True)
```

sage.features.meataxe.**all_features**()

3.29 Features for testing the presence of MixedIntegerLinearProgram backends

class sage.features.mip_backends.**COIN**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing whether the MixedIntegerLinearProgram backend COIN is available.

class sage.features.mip_backends.**CPLEX**(**args*, ***kwds*)

Bases: *MIPBackend*

A *Feature* describing whether the MixedIntegerLinearProgram backend CPLEX is available.

class sage.features.mip_backends.**CVXOPT**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing whether the MixedIntegerLinearProgram backend CVXOPT is available.

class sage.features.mip_backends.**Gurobi**(**args*, ***kwds*)

Bases: *MIPBackend*

A *Feature* describing whether the MixedIntegerLinearProgram backend Gurobi is available.

class sage.features.mip_backends.**MIPBackend**(**args*, ***kwds*)

Bases: *Feature*

A *Feature* describing whether a MixedIntegerLinearProgram backend is available.

sage.features.mip_backends.**all_features**()

3.30 Feature for testing the presence of pynormaliz

class sage.features.normaliz.**PyNormaliz**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing the presence of the Python package *PyNormaliz*.

EXAMPLES:

```
sage: from sage.features.normaliz import PyNormaliz
sage: PyNormaliz().is_present()                    # optional - pynormaliz
FeatureTestResult('pynormaliz', True)
```

```
>>> from sage.all import *
>>> from sage.features.normaliz import PyNormaliz
>>> PyNormaliz().is_present()                    # optional - pynormaliz
FeatureTestResult('pynormaliz', True)
```

sage.features.normaliz.**all_features**()

3.31 Feature for testing the presence of pandoc

class sage.features.pandoc.**Pandoc**(**args*, ***kwds*)

Bases: *Executable*

A *Feature* describing the presence of *pandoc*.

EXAMPLES:

```
sage: from sage.features.pandoc import Pandoc
sage: Pandoc().is_present()  # optional - pandoc
FeatureTestResult('pandoc', True)
```

```
>>> from sage.all import *
>>> from sage.features.pandoc import Pandoc
>>> Pandoc().is_present()  # optional - pandoc
FeatureTestResult('pandoc', True)
```

sage.features.pandoc.**all_features**()

3.32 Feature for testing the presence of jupymake, the Python interface to polymake

class sage.features.polymake.**JuPyMake**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing the presence of the *JuPyMake* module, a Python interface to the *polymake* library.

EXAMPLES:

```
sage: from sage.features.polymake import JuPyMake
sage: JuPyMake().is_present()  # optional - jupymake
FeatureTestResult('jupymake', True)
```

```
>>> from sage.all import *
>>> from sage.features.polymake import JuPyMake
>>> JuPyMake().is_present()  # optional - jupymake
FeatureTestResult('jupymake', True)
```

sage.features.polymake.**all_features**()

3.33 Feature for the rank-width graph decomposition

class sage.features.rankwidth.**RankWidth**(**args*, ***kwds*)

Bases: *BuildModule*

A *Feature* indicating whether or not the rank-width graph decomposition is available.

This is a proxy for the build-time availability of the rankwidth (librw) library from the "rw" package; the associated module sage.graphs.graph_decompositions.rankwidth is built if and only if the library is detected, usable, and enabled.

EXAMPLES:

```
sage: from sage.features.rankwidth import RankWidth
sage: RankWidth().is_present()  # needs rankwidth
FeatureTestResult('rankwidth', True)
sage: RankWidth().is_present()  # needs !rankwidth
FeatureTestResult('rankwidth', False)
```

```
>>> from sage.all import *
>>> from sage.features.rankwidth import RankWidth
>>> RankWidth().is_present()  # needs rankwidth
FeatureTestResult('rankwidth', True)
>>> RankWidth().is_present()  # needs !rankwidth
FeatureTestResult('rankwidth', False)
```

A runtime check. We only check the "present" case because, if feature checks are _not_ deferred, the `needs !rankwidth` can be satisfied (disabled at build time) at the same time we are able to import a module that was installed for a previous build of sage:

```
sage: from sage.features.rankwidth import RankWidth
sage: RankWidth().is_present_at_runtime()  # needs rankwidth
FeatureTestResult('rankwidth', True)
```

```
>>> from sage.all import *
>>> from sage.features.rankwidth import RankWidth
>>> RankWidth().is_present_at_runtime()  # needs rankwidth
FeatureTestResult('rankwidth', True)
```

sage.features.rankwidth.**all_features**()

3.34 Features for testing the presence of `rubiks`

class sage.features.rubiks.**Rubiks**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing the presence of the *cu2*, *cubex*, *dikcube*, *mcube*, *optimal*, and *size222* programs from the *rubiks* package.

EXAMPLES:

```
sage: from sage.features.rubiks import Rubiks
sage: Rubiks().is_present()  # optional - rubiks
FeatureTestResult('rubiks', True)
```

```
>>> from sage.all import *
>>> from sage.features.rubiks import Rubiks
>>> Rubiks().is_present()  # optional - rubiks
FeatureTestResult('rubiks', True)
```

sage.features.rubiks.**all_features**()

class sage.features.rubiks.**cu2**(**args*, ***kwds*)

Bases: *Executable*

A *Feature* describing the presence of `cu2`.

EXAMPLES:

```
sage: from sage.features.rubiks import cu2
sage: cu2().is_present()  # optional - rubiks
FeatureTestResult('cu2', True)
```

```
>>> from sage.all import *
>>> from sage.features.rubiks import cu2
>>> cu2().is_present()  # optional - rubiks
FeatureTestResult('cu2', True)
```

class sage.features.rubiks.**cubex**(**args*, ***kwds*)

Bases: *Executable*

A *Feature* describing the presence of `cubex`.

EXAMPLES:

```
sage: from sage.features.rubiks import cubex
sage: cubex().is_present()  # optional - rubiks
FeatureTestResult('cubex', True)
```

```
>>> from sage.all import *
>>> from sage.features.rubiks import cubex
>>> cubex().is_present()  # optional - rubiks
FeatureTestResult('cubex', True)
```

class sage.features.rubiks.**dikcube**(**args*, ***kwds*)

Bases: *Executable*

A *Feature* describing the presence of `dikcube`.

EXAMPLES:

```
sage: from sage.features.rubiks import dikcube
sage: dikcube().is_present()  # optional - rubiks
FeatureTestResult('dikcube', True)
```

```
>>> from sage.all import *
>>> from sage.features.rubiks import dikcube
>>> dikcube().is_present()  # optional - rubiks
FeatureTestResult('dikcube', True)
```

class sage.features.rubiks.**mcube**(**args*, ***kwds*)

Bases: *Executable*

A *Feature* describing the presence of `mcube`.

EXAMPLES:

```
sage: from sage.features.rubiks import mcube
sage: mcube().is_present()  # optional - rubiks
FeatureTestResult('mcube', True)
```

```
>>> from sage.all import *
>>> from sage.features.rubiks import mcube
>>> mcube().is_present()  # optional - rubiks
FeatureTestResult('mcube', True)
```

class sage.features.rubiks.**optimal**(**args*, ***kwds*)

Bases: *Executable*

A *Feature* describing the presence of `optimal`.

EXAMPLES:

```
sage: from sage.features.rubiks import optimal
sage: optimal().is_present()  # optional - rubiks
FeatureTestResult('optimal', True)
```

```
>>> from sage.all import *
>>> from sage.features.rubiks import optimal
>>> optimal().is_present()  # optional - rubiks
FeatureTestResult('optimal', True)
```

class sage.features.rubiks.**size222**(**args*, ***kwds*)

Bases: *Executable*

A *Feature* describing the presence of `size222`.

EXAMPLES:

```
sage: from sage.features.rubiks import size222
sage: size222().is_present()  # optional - rubiks
FeatureTestResult('size222', True)
```

```
>>> from sage.all import *
>>> from sage.features.rubiks import size222
>>> size222().is_present()  # optional - rubiks
FeatureTestResult('size222', True)
```

3.35 Features for testing the presence of `sirocco`

class sage.features.sirocco.**Sirocco**(**args*, ***kwds*)

Bases: *BuildModule*

A *Feature* which describes whether the `sage.libs.sirocco` module is available in this installation of Sage.

EXAMPLES:

```
sage: from sage.features.sirocco import Sirocco
sage: Sirocco().is_present()  # needs sirocco
FeatureTestResult('sirocco', True)
sage: Sirocco().is_present()  # needs !sirocco
FeatureTestResult('sirocco', False)
```

```
>>> from sage.all import *
>>> from sage.features.sirocco import Sirocco
>>> Sirocco().is_present()  # needs sirocco
FeatureTestResult('sirocco', True)
>>> Sirocco().is_present()  # needs !sirocco
FeatureTestResult('sirocco', False)
```

A runtime check. We only check the "present" case because, if feature checks are _not_ deferred, the `needs !sirocco` can be satisfied (disabled at build time) at the same time we are able to import a module that was installed for a previous build of sage:

```
sage: from sage.features.sirocco import Sirocco
sage: Sirocco().is_present_at_runtime()  # needs sirocco
FeatureTestResult('sirocco', True)
```

```
>>> from sage.all import *
>>> from sage.features.sirocco import Sirocco
>>> Sirocco().is_present_at_runtime()  # needs sirocco
FeatureTestResult('sirocco', True)
```

sage.features.sirocco.**all_features**()

3.36 Features for testing the presence of `singular` and the SageMath interfaces to it

class sage.features.singular.**Singular**(**args*, ***kwds*)

Bases: *Executable*

A *Feature* describing the presence of the *singular* executable.

See also

Feature sage.libs.singular

EXAMPLES:

```
sage: from sage.features.singular import Singular
sage: Singular().is_present()                                                  #␣
↪needs singular
FeatureTestResult('singular', True)
```

```
>>> from sage.all import *
>>> from sage.features.singular import Singular
>>> Singular().is_present()                                                  #␣
↪needs singular
FeatureTestResult('singular', True)
```

sage.features.singular.**all_features**()

3.37 Features for testing the presence of tdlib

class sage.features.tdlib.**Tdlib**(**args*, ***kwds*)

Bases: *BuildModule*

A *Feature* describing the presence of the SageMath interface to the treedec (formerly tdlib) library.

EXAMPLES:

```
sage: from sage.features.tdlib import Tdlib
sage: Tdlib().is_present()  # needs tdlib
FeatureTestResult('tdlib', True)
sage: Tdlib().is_present()  # needs !tdlib
FeatureTestResult('tdlib', False)
```

```
>>> from sage.all import *
>>> from sage.features.tdlib import Tdlib
>>> Tdlib().is_present()  # needs tdlib
FeatureTestResult('tdlib', True)
>>> Tdlib().is_present()  # needs !tdlib
FeatureTestResult('tdlib', False)
```

A runtime check. We only check the "present" case because, if feature checks are _not_ deferred, the `needs !tdlib` can be satisfied (disabled at build time) at the same time we are able to import a module that was installed for a previous build of sage:

```
sage: from sage.features.tdlib import Tdlib
sage: Tdlib().is_present_at_runtime()  # needs tdlib
FeatureTestResult('tdlib', True)
```

```
>>> from sage.all import *
>>> from sage.features.tdlib import Tdlib
>>> Tdlib().is_present_at_runtime()  # needs tdlib
FeatureTestResult('tdlib', True)
```

sage.features.tdlib.**all_features**()

3.38 Features for testing the presence of topcom executables

class sage.features.topcom.**TOPCOM**(**args*, ***kwds*)

Bases: *JoinFeature*

A *Feature* describing the presence of the executables which comes as a part of *TOPCOM*.

EXAMPLES:

```
sage: from sage.features.topcom import TOPCOM
sage: TOPCOM().is_present()                              # optional - topcom
FeatureTestResult('topcom', True)
```

```
>>> from sage.all import *
>>> from sage.features.topcom import TOPCOM
>>> TOPCOM().is_present()                              # optional - topcom
FeatureTestResult('topcom', True)
```

class sage.features.topcom.**TOPCOMExecutable**(**args*, ***kwds*)

Bases: *Executable*

A *Feature* which checks for executables from the *TOPCOM* package.

EXAMPLES:

```
sage: from sage.features.topcom import TOPCOMExecutable
sage: TOPCOMExecutable('points2allfinetriangs').is_present()     # optional -␣
↪topcom
FeatureTestResult('topcom_points2allfinetriangs', True)
```

```
>>> from sage.all import *
>>> from sage.features.topcom import TOPCOMExecutable
>>> TOPCOMExecutable('points2allfinetriangs').is_present()     # optional - topcom
FeatureTestResult('topcom_points2allfinetriangs', True)
```

sage.features.topcom.**all_features**()

CHAPTER

FOUR

DISTRIBUTION PACKAGES OF THE SAGE LIBRARY

- *sagemath_categories: Sage categories and basic rings*
- *sagemath_doc_html: SageMath documentation in HTML format*
- *sagemath_doc_pdf: SageMath documentation in PDF format*
- *sagemath_environment: System and software environment*
- *sagemath_giac: Giac integration*
- *sagemath_objects: Sage objects, elements, parents, categories, coercion, metaclasses*
- *sagemath_repl: IPython kernel, Sage preparser, doctester*

CHAPTER

FIVE

EXPERIMENTAL PACKAGES

Some packages that provide additional functionality are marked as "experimental". Developers are needed in order to improve the integration of these packages into the Sage distribution.

5.1 Mathematics

- *awali: Computation of/with finite state machines*
- *barvinok: Projections of integer point sets of parametric polytopes*
- *cocoalib: Computations in commutative algebra*
- *deformation: Count points on hypersurfaces using the deformation method*
- *gap3: A minimal distribution of GAP 3 containing packages that have no equivalent in GAP 4*
- *lie: Library for the representation theory of complex semisimple Lie groups and algebras*
- *polylib: Operations on unions of polyhedra*
- *r_jupyter: Jupyter kernel for R*

5.2 Other dependencies

CHAPTER

SIX

ALL EXTERNAL PACKAGES

6.1 Details of external packages

Packages are in alphabetical order.

6.1.1 4ti2: Algebraic, geometric and combinatorial problems on linear spaces

Description

A software package for algebraic, geometric and combinatorial problems on linear spaces. Available at https://4ti2.github.io/.

License

4ti2 is released under a GPL v2 license.

Upstream Contact

- https://4ti2.github.io/
- Raymond Hemmecke, TU Munich, Germany
- Matthias Köppe, UC Davis, CA, USA

Type

optional

Dependencies

- $(MP_LIBRARY)
- *glpk: GNU Linear Programming Kit*

Version Information

package-version.txt:

```
1.6.10
```

Equivalent System Packages

```
$ sudo pacman -S 4ti2
```

```
$ conda install 4ti2
```

```
$ sudo apt-get install 4ti2
```

```
$ sudo dnf install 4ti2 4ti2-devel
```

```
$ sudo pkg install math/4ti2
```

```
$ sudo emerge sci-mathematics/4ti2
```

```
$ brew install Macaulay2/tap/fourtitwo
```

```
$ sudo zypper install 4ti2 4ti2-devel
```

See https://repology.org/project/4ti2/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.2 _bootstrap: Represents system packages required for running the top-level bootstrap script

Description

This optional script package represents the requirements (system packages) that are needed in addition to those represented by the `_prereq` package in order to run the top-level `bootstrap` script.

Namely, the following standard tools must be installed on your computer:

- **liblzma/xz**: XZ Utils, the free general-purpose data compression software with a high compression ratio.
- **python**: Python 3.12 or later from your system package manager or another external installation. Sage no longer builds its own Python interpreter; it uses this interpreter to create `SAGE_VENV`. It needs to have the development headers and the following standard modules available: sqlite3, ctypes, math, hashlib, socket, ssl, ensurepip, zlib, setuptools

XZ Utils (liblzma) is available with most supported distributions (package names such as `xz-devel` on Fedora, `liblzma-dev` on Debian/Ubuntu). It can also be built from source from https://github.com/tukaani-project/xz. After downloading and untarring the release archive, and changing to the directory with the sources:

```
$ ./configure --prefix=/usr/local && make && make install
```

will install XZ Utils in `/usr/local`. Instead of `/usr/local` one may choose another location, say `/opt/foo`. A modern xz/liblzma installs a `liblzma.pc` file, and Sage's `./configure` checks for it using a pkg-config macro. To detect xz/liblzma at a non-standard location, add it to `PKG_CONFIG_PATH`:

```
$ export PKG_CONFIG_PATH=$PKG_CONFIG_PATH:/opt/foo/lib/pkgconfig
```

Python 3.12 (or later) with development headers is available with most supported distributions (package names such as `python3-devel` on Fedora, `python3-dev` on Debian/Ubuntu). It can also be built from source from https://www.python.org/downloads/. After downloading and untarring the release archive, and changing to the directory with the sources:

```
$ ./configure --prefix=/usr/local --enable-optimizations --with-ensurepip=install &&␣
↪make && make install
```

will install Python in `/usr/local`; this takes a few minutes on a moderately fast machine. Instead of `/usr/local` one may choose another location, say `/opt/foo`, which then should be passed to Sage's `./configure` with `--with-python=/opt/foo/bin/python3`. The `--enable-optimizations` flag is optional but recommended for performance. Make sure that the prerequisites for the required standard modules (`sqlite3`, `ctypes`, `zlib`, `ssl`, etc.) are installed before building Python; otherwise these modules will be missing.

It is also possible to use a Python installed by uv and pass it to `./configure` with `--with-python=...`.

Type

optional

Dependencies

Version Information

Equivalent System Packages

```
$ apk add bash gettext-dev autoconf automake libtool python3  \
      python3-dev py3-setuptools xz xz-dev
```

```
$ sudo pacman -S autoconf automake libtool python python-setuptools xz
```

```
$ conda install autoconf automake libtool python setuptools xz xz
```

```
$ sudo apt-get install autoconf automake libtool python3 libpython3-dev  \
      python3-setuptools python3-venv xz-utils liblzma-dev
```

```
$ sudo dnf install autoconf automake libtool python3 python3-devel  \
      python3-setuptools xz xz-devel xz
```

```
$ sudo pkg install autoconf automake libtool lang/python  \
      devel/py-setuptools archivers/lzma
```

```
$ sudo emerge dev-build/autoconf dev-build/automake dev-build/libtool  \
      dev-lang/python dev-python/setuptools app-arch/xz-utils
```

```
$ brew install autoconf automake libtool python3 python-setuptools xz
```

```
$ sudo port install gettext autoconf automake libtool python313  \
      py-setuptools xz
```

```
$ nix-env --install autoconf automake libtool python3  \
      python3Packages.setuptools xz
```

```
$ sudo zypper install autoconf automake libtool python3 python3-devel  \
      python3-setuptools 'pkgconfig(liblzma)' xz
```

```
$ sudo slackpkg install autoconf automake libtool python3 xz
```

```
$ sudo xbps-install autoconf automake libtool xtools mk-configure  \
     python3 python3-devel python3-setuptools liblzma-devel xz
```

6.1.3 _develop: Represents system packages recommended for development

Description

Script package representing a list of system packages recommended for developers.

Type

optional

Dependencies

- *_bootstrap: Represents system packages required for running the top-level bootstrap script*
- *git: Version control system*
- *github_cli: Command-line interface for GitHub*
- *pytest: Simple powerful testing with Python*
- *pytest_xdist: Pytest xdist plugin for distributed testing, most importantly across multiple CPUs*

Version Information

Equivalent System Packages

```
$ apk add gnupg-gpgconf openssh-client
```

```
$ sudo pacman -S gnupg openssh
```

```
$ conda install openssh pycodestyle esbonio
```

```
$ sudo apt-get install gpgconf openssh-client
```

```
$ sudo dnf install gnupg2 openssh
```

```
$ sudo pkg install security/gnupg security/openssh-portable
```

```
$ sudo emerge app-crypt/gnupg net-misc/openssh
```

```
$ brew install gnupg
```

```
$ sudo port install gnupg2
```

```
$ nix-env --install gnupg openssh
```

```
$ sudo zypper install gpg2 openssh
```

```
$ sudo slackpkg install gnupg2 openssh
```

```
$ sudo xbps-install gnupg2 openssh
```

See https://repology.org/project/gnupg/versions, https://repology.org/project/openssh/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.4 _prereq: Represents system packages required for installing SageMath from source

Description

This dummy package represents the minimal requirements (system packages) for installing SageMath from source.

In addition to standard POSIX utilities and the bash shell, the following standard command-line development tools must be installed on your computer:

- **C compiler** (**C/C++** - compiler required on macOS): a sufficently modern compiler. Ideally these can be directly used to build Sage. The options are essentially GNU gcc/g++ on Linux, and clang/clang++ on macOS (which conventionally misnames them gcc/g++), on BSDs, and also on Linux.
- **make**: GNU make, version 3.80 or later. Version 3.82 or later is recommended.
- **m4**: GNU m4 1.4.2 or later (non-GNU or older versions might also work).
- **perl**: version 5.8.0 or later.
- **ar** and **ranlib**: can be obtained as part of GNU binutils.
- **tar**: GNU tar version 1.17 or later, or BSD tar (as provided on macOS).
- **patch**.
- **boost**: the library `boost` with its headers.
- **bzip2**: the executable `bzip2` and the library `libbz2` with its headers. (some Linux distros package these separately, e.g. Debian/Ubuntu needs packages `bzip2` and `libbz2-dev`; Fedora needs `bzip2` and `bzip2-devel`.)
- **pkgconf**, also known as `pkg-config`.
- **zlib**: the library `libz` with its headers, and its pkg-config zlib.pc file. (some Linux distros package these separately.) On macOS we provide a basic zlib.pc file in build/platform/macos/pkgconfig, which, if needed, gets prepended to PKG_CONFIG_PATH by ./configure.

Other versions of these may work, but they are untested.

On macOS, suitable versions of most of these tools are provided by the Xcode Command Line Tools. To install them, after installing XCode itself, open a terminal window and run `xcode-select --install`; then click "Install" in the pop-up window. If the Xcode Command Line Tools are already installed, you may want to check if they need to be updated by typing `softwareupdate -l`. The remaining are provided by either one of macOS's "missing package managers", such as Homebrew, or as standalone tools. In particular `pkgconf` is available as pkg-config_pkg.

On Linux, `ar` and `ranlib` are in the binutils package. The other programs are usually located in packages with their respective names.

Boost is available with most supported distribitions. It can also be installed using the Boost's project installer B2. After downloading and untarring the archive, and changing to the directory with the sources, `./bootstrap.sh && ./b2 && ./b2 install --prefix=/usr/local` will install Boost in `/usr/local`; this takes around 5 minutes of wall clock time on a moderately fast M1 Apple Mac. Instead of `/usr/local` one may choose another location, say `/opt/foo`,which then might have to be passed (in case the location is not known to the toolchain) to Sage via its `./configure`, with `--with-boost=/opt/foo` option.

On Redhat-derived systems not all perl components are installed by default and you might have to install the `perl-ExtUtils-MakeMaker` package.

To check if you have the above prerequisites installed, for example `perl`, type:

```
$ command -v perl
```

or:

```
$ which perl
```

on the command line. If it gives an error (or returns nothing), then either `perl` is not installed, or it is installed but not in your PATH.

Type

standard

Dependencies

Version Information

Equivalent System Packages

```
$ apk add binutils make m4 perl python3 tar bc gcc g++ ca-certificates  \
      coreutils patch bzip2 pkgconf zlib-dev boost boost-dev
```

```
$ sudo pacman -S binutils make m4 perl python tar bc gcc bzip2 which  \
      patch pkgconf zlib boost
```

```
$ conda install compilers make m4 perl python tar bc patch bzip2  \
      pkg-config zlib zlib-devel libboost-devel
```

```
$ sudo apt-get install binutils make m4 perl flex python3 python3-dev  \
      python3-setuptools python3-venv tar bc gcc libbz2-dev bzip2 g++  \
      ca-certificates patch pkg-config libz-dev libboost-dev
```

```
$ sudo dnf install binutils make m4 python3 python3-devel  \
      python3-setuptools perl perl-ExtUtils-MakeMaker tar gcc bzip2  \
      bzip2-devel gcc-c++ findutils which diffutils perl-IPC-Cmd flex  \
      patch pkg-config zlib-devel boost-devel
```

```
$ sudo pkg install gmake automake bash dash python patch bzip2  \
      devel/pkgconf zlib devel/boost-libs
```

```
$ sudo emerge sys-devel/binutils sys-libs/binutils-libs dev-build/make  \
      dev-scheme/guile dev-libs/libffi app-arch/tar sys-devel/gcc  \
      dev-libs/mpc sys-libs/glibc sys-kernel/linux-headers  \
      dev-lang/perl sys-devel/m4 sys-devel/bc dev-lang/python  \
      sys-devel/flex app-misc/ca-certificates dev-libs/libxml2  \
      sys-apps/findutils sys-apps/which sys-apps/diffutils  \
      sys-devel/patch app-arch/bzip2 dev-util/pkgconf sys-libs/zlib  \
      dev-libs/boost
```

```
$ brew install pkgconf zlib boost
```

```
$ sudo port install pkgconfig zlib boost
```

```
$ nix-env --install binutils gnumake gnum4 perl python3 gnutar bc gcc  \
      bash gnupatch bzip2 pkg-config zlib boost
```

```
$ sudo zypper install binutils make m4 gawk perl python3 python3-devel  \
      python3-setuptools tar bc which glibc-locale-base gcc gcc-c++  \
      ca-certificates gzip findutils diffutils patch bzip2  \
      'pkgconfig(bzip2)' pkgconf 'pkgconfig(zlib)' boost-base
```

install the following packages: boost-cpp

```
$ sudo slackpkg install binutils make guile gc libffi gcc-13 gcc-g++-13  \
      libmpc glibc kernel-headers perl m4 bc python3 flex  \
      ca-certificates libxml2 cyrus-sasl patch bzip2 pkg-config zlib  \
      boost
```

```
$ sudo xbps-install bc binutils gcc libgomp-devel m4 make perl python3  \
      tar bash which diffutils gzip python3-devel bzip2-devel  \
      libffi-devel zlib-devel libxcrypt-devel patch bzip2-devel pkgconf  \
      boost-devel
```

See https://repology.org/project/boost/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.5 _recommended: Represents system packages recommended for additional functionality

Description

Script package representing a list of system packages recommended to be installed for additional functionality.

Type

optional

Dependencies

- *ffmpeg: ffmpeg video converter*
- *free_fonts: a free family of scalable outline fonts*
- *git: Version control system*
- *imagemagick: A collection of tools and libraries for many image file formats*
- *libjpeg: JPEG image support*
- *pandoc: A document converter*
- *texlive: A comprehensive TeX system*
- *texlive_luatex: LuaTeX packages*
- *xindy: a general-purpose index processor*

Version Information

Equivalent System Packages

```
$ sudo apt-get install default-jdk libavdevice-dev
```

```
$ brew install texinfo
```

```
$ sudo port install texinfo
```

If the system package is installed, `./configure` will check if it can be used.

6.1.6 _sagemath: Downstream package of Sage in distributions

SageMath is available from various distributions and can be installed by package managers.

This dummy package records the names of the system packages that should be installed to provide a standard installation of SageMath, including documentation and Jupyter.

Downstream Contact

See wiki page Distribution

Type

optional

Dependencies

Version Information

Equivalent System Packages

```
$ sudo pacman -S sagemath sagemath-doc
```

```
$ conda install sage
```

```
$ sudo apt-get install sagemath sagemath-doc sagemath-jupyter
```

```
$ sudo dnf install sagemath
```

```
$ sudo pkg install math/sage
```

```
$ brew install sage
```

```
$ nix-env --install sage
```

```
$ sudo xbps-install sagemath
```

See https://repology.org/project/sagemath/versions

6.1.7 accessible_pygments: Accessible Pygments styles

Description

A collection of accessible Pygments styles that meet WCAG AA and AAA contrast ratio guidelines for syntax highlighting, ensuring code is readable by users with visual impairments.

License

BSD-3-Clause

Upstream Contact

https://pypi.org/project/accessible-pygments/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *pygments: Generic syntax highlighter*

Version Information

package-version.txt:

```
0.0.5
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.8 admcycles: Computation in the tautological ring of the moduli space of curves

Description

The SageMath package admcycles offers the possibility to compute in the tautological ring of the Deligne-Mumford compactification of the moduli space of curves. Construction for standard generators are provided (psi, kappa and lambda classes) as well as more advanced algebraic construction (double ramification cycle, strata of differentials).

License

GPLv2+

Upstream Contact

https://pypi.org/project/admcycles/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

requirements.txt:

```
admcycles
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.9 alabaster: Default theme for the Sphinx documentation system

Description

Alabaster is a visually (c)lean, responsive, configurable theme for the Sphinx documentation system. It is Python 2+3 compatible.

It began as a third-party theme, and is still maintained separately, but as of Sphinx 1.3, Alabaster is an install-time dependency of Sphinx and is selected as the default theme.

Live examples of this theme can be seen on paramiko.org, fabfile.org and pyinvoke.org.

Upstream Contact

https://alabaster.readthedocs.io/en/latest/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
1.0.0
```

version_requirements.txt:

```
alabaster >=0.7.12
```

Equivalent System Packages

```
$ conda install alabaster
```

```
$ sudo emerge dev-python/alabaster
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-alabaster'
```

```
$ sudo xbps-install python3-alabaster
```

See https://repology.org/project/alabaster/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.10 anyio: High level compatibility layer for multiple asynchronous event loop implementations

Description

High level compatibility layer for multiple asynchronous event loop implementations

License

MIT

Upstream Contact

https://pypi.org/project/anyio/

Type

standard

Dependencies

- $(PYTHON)
- *idna: Internationalized Domain Names in Applications (IDNA)*
- *pip: Tool for installing and managing Python packages*
- *sniffio: Sniff out which async library your code is running under*
- *typing_extensions: Backported and Experimental Type Hints for Python 3.8+*

Version Information

package-version.txt:

```
4.13.0
```

version_requirements.txt:

```
anyio
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.11 appdirs: Small Python module for determining appropriate platform-specific dirs, e.g. a "user data dir"

Description

Small Python module for determining appropriate platform-specific dirs, e.g. a "user data dir"

License

MIT

Upstream Contact

https://pypi.org/project/appdirs/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
1.4.4
```

version_requirements.txt:

```
appdirs
```

Equivalent System Packages

```
$ conda install appdirs
```

```
$ sudo xbps-install python3-appdirs
```

If the system package is installed, `./configure` will check if it can be used.

6.1.12 appnope: Disable App Nap on macOS >= 10.9

Description

Disable App Nap on macOS >= 10.9

License

BSD

Upstream Contact

https://pypi.org/project/appnope/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
0.1.4
```

version_requirements.txt:

```
appnope >=0.1.0
```

Equivalent System Packages

```
$ conda install appnope
```

```
$ sudo port install py-appnope
```

See https://repology.org/project/python:appnope/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.13 argon2_cffi: The secure Argon2 password hashing algorithm

Description

The secure Argon2 password hashing algorithm.

License

MIT

Upstream Contact

https://pypi.org/project/argon2-cffi/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *argon2_cffi_bindings: Low-level CFFI bindings for Argon2*
- *flit_core: Distribution-building parts of Flit. See flit package for more information*
- *hatchling: Modern, extensible Python build backend*
- *hatchling: Modern, extensible Python build backend*
- *hatchling: Modern, extensible Python build backend*

Version Information

package-version.txt:

```
25.1.0
```

version_requirements.txt:

```
argon2-cffi
```

Equivalent System Packages

```
$ conda install argon2-cffi
```

```
$ sudo emerge dev-python/argon2-cffi
```

```
$ sudo port install py-argon2-cffi
```

```
$ sudo xbps-install python3-argon2
```

See https://repology.org/project/argon2-cffi/versions, https://repology.org/project/python:argon2-cffi/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.14 argon2_cffi_bindings: Low-level CFFI bindings for Argon2

Description

Low-level CFFI bindings for Argon2

License

MIT

Upstream Contact

https://pypi.org/project/argon2-cffi-bindings/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cffi: Foreign Function Interface for Python calling C code*
- *setuptools: Build system for Python packages*
- *setuptools_scm: Python build system extension to obtain package version from version control*

Version Information

package-version.txt:

```
25.1.0
```

version_requirements.txt:

```
argon2-cffi-bindings
```

Equivalent System Packages

```
$ apk add py3-argon2-cffi-bindings
```

```
$ sudo pacman -S python-argon2-cffi-bindings
```

```
$ sudo dnf install python-argon2-cffi-bindings
```

```
$ sudo pkg install security/py-argon2-cffi-bindings
```

```
$ sudo emerge dev-python/argon2-cffi-bindings
```

```
$ sudo port install py-argon2-cffi-bindings
```

```
$ sudo zypper install python-argon2-cffi-bindings
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.15 arrow: Better dates & times for Python

Description

Better dates & times for Python

License

Apache 2.0

Upstream Contact

https://pypi.org/project/arrow/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *dateutil: Extensions to the standard Python module datetime*
- *types_python_dateutil: Typing stubs for python-dateutil*
- *tzdata: Provider of IANA time zone data*

Version Information

package-version.txt:

```
1.4.0
```

version_requirements.txt:

```
arrow
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.16 asttokens: Annotate AST trees with source code positions

Description

Annotate AST trees with source code positions

License

Apache 2.0

Upstream Contact

https://pypi.org/project/asttokens/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
3.0.2
```

version_requirements.txt:

```
asttokens
```

Equivalent System Packages

```
$ conda install asttokens
```

```
$ sudo emerge dev-python/asttokens
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.17 async_lru: Simple LRU cache for asyncio

Description

Simple LRU cache for asyncio

License

MIT License

Upstream Contact

https://pypi.org/project/async-lru/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
2.3.0
```

version_requirements.txt:

```
async-lru
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.18 attrs: Decorator for Python classes with attributes

Description

attrs is the Python package that will bring back the joy of writing classes by relieving you from the drudgery of implementing object protocols (aka dunder methods).

License

MIT License

Upstream Contact

Home page: https://www.attrs.org

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
26.1.0
```

version_requirements.txt:

```
attrs >=19.3.0
```

Equivalent System Packages

```
$ conda install attrs
```

```
$ sudo emerge dev-python/attrs
```

```
$ sudo port install py-attrs
```

```
$ sudo xbps-install python3-attrs
```

See https://repology.org/project/python:attrs/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.19 auditwheel_or_delocate: Repair wheels on Linux or macOS

Description

This package represents `auditwheel` on Linux and `delocate` on macOS.

(Actually, we install `delocate` also on Linux because our script `make -j list-broken-packages` uses a small subroutine of `delocate` even on Linux.)

License

MIT

BSD 2-clause

Upstream Contact

https://pypi.org/project/auditwheel/

https://pypi.org/project/delocate/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

requirements.txt:

```
delocate
auditwheel; sys_platform != 'darwin'
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.20 awali: Computation of/with finite state machines

Description

Awali is a software platform dedicated to the computation of, and with, finite state machines. Here finite state machines is to be understood in the broadest possible sense: finite automata with output — often called transducers then — or even more generally finite automata with multiplicity, that is, automata that not only accept, or recognize, sequences of symbols but compute for every such sequence a 'value' that is associated with it and which can be taken in any semiring. Hence the variety of situations that can thus be modellized.

License

- GPL 3.0

Upstream Contact

- Website: http://vaucanson-project.org/Awali/index.html
- Releases: http://files.vaucanson-project.org/tarballs/

Dependencies

- graphviz must be installed from your distro, and available in the path.

Type

experimental

Dependencies

- $(PYTHON)
- *cmake: A cross-platform build system generator*
- *cython: C-Extensions for Python, an optimizing static compiler*
- *nbconvert: Converting Jupyter Notebooks*
- *ncurses: Classic terminal output library*

Version Information

package-version.txt:

```
1.0.2-190218
```

Equivalent System Packages

See https://repology.org/project/awali/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.21 babel: Internationalization utilities

Description

Internationalization utilities

License

BSD-3-Clause

Upstream Contact

https://pypi.org/project/Babel/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *pytz: Timezone definitions for Python*

Version Information

package-version.txt:

```
2.18.0
```

version_requirements.txt:

```
babel >=2.11.0
```

Equivalent System Packages

```
$ sudo pacman -S python-babel
```

```
$ conda install babel
```

```
$ sudo apt-get install python3-babel
```

```
$ sudo dnf install babel
```

```
$ sudo emerge dev-python/Babel
```

```
$ sudo port install py-babel
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-Babel'
```

```
$ sudo xbps-install python3-Babel
```

See https://repology.org/project/python:babel/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.22 barvinok: Projections of integer point sets of parametric polytopes

Description

barvinok is a library for counting the number of integer points in parametric and non-parametric polytopes as well as projections of such sets.

License

GPL v2

Upstream Contact

- https://sourceforge.net/projects/barvinok/
- https://groups.google.com/group/isl-development

Type

experimental

Dependencies

- *isl: Sets and relations of integer points bounded by affine constraints*
- *ntl: A library for doing number theory*
- *polylib: Operations on unions of polyhedra*

Version Information

package-version.txt:

```
0.41.7
```

Equivalent System Packages

```
$ sudo pkg install math/barvinok
```

```
$ sudo zypper install barvinok 'pkgconfig(barvinok)'
```

See https://repology.org/project/barvinok/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.23 beautifulsoup4: Screen-scraping library

Description

Screen-scraping library

License

MIT

Upstream Contact

https://pypi.org/project/beautifulsoup4/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *soupsieve: Modern CSS selector implementation for Beautiful Soup*
- *typing_extensions: Backported and Experimental Type Hints for Python 3.8+*

Version Information

package-version.txt:

```
4.14.3
```

version_requirements.txt:

```
beautifulsoup4
```

Equivalent System Packages

```
$ apk add py3-beautifulsoup4
```

```
$ sudo pacman -S python-beautifulsoup4
```

```
$ conda install beautifulsoup4
```

```
$ sudo apt-get install python3-bs4
```

```
$ sudo dnf install python3-beautifulsoup4
```

```
$ sudo emerge dev-python/beautifulsoup4
```

```
$ sudo port install py-beautifulsoup4
```

```
$ sudo zypper install python-beautifulsoup4
```

```
$ sudo xbps-install python3-BeautifulSoup4
```

See https://repology.org/project/python:beautifulsoup4/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.24 beniget: Extract semantic information about static Python code

Description

Extract semantic information about static Python code

License

BSD 3-Clause

Upstream Contact

https://pypi.org/project/beniget/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *gast: Python AST that abstracts the underlying Python version*

Version Information

package-version.txt:

```
0.5.0
```

version_requirements.txt:

```
beniget
```

Equivalent System Packages

```
$ conda install beniget
```

```
$ sudo dnf install python3-beniget
```

```
$ sudo emerge dev-python/beniget
```

```
$ sudo xbps-install python3-beniget
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.25 benzene: Generate fusenes and benzenoids with a given number of faces

Description

Benzene is a program for the efficient generation of all nonisomorphic fusenes and benzenoids with a given number of faces. Fusenes are planar polycyclic hydrocarbons with all bounded faces hexagons. Benzenoids are fusenes that are subgraphs of the hexagonal lattice.

License

Benzene is licensed under the GNU General Public License v2 or later (June 2007)

Upstream Contact

Benzene was written by Gunnar Brinkmann and Gilles Caporossi. This version was adapted by Gunnar Brinkmann and Nico Van Cleemput for Grinvin.

http://www.grinvin.org/

Type

optional

Dependencies

Version Information

package-version.txt:

```
20130630
```

Equivalent System Packages

```
$ sudo pacman -S benzene
```

```
$ sudo zypper install benzene
```

See https://repology.org/project/benzene/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.26 biopython: Tools for computational molecular biology

Description

Freely available tools for computational molecular biology.

License

Upstream Contact

https://pypi.org/project/biopython/

http://biopython.org/

Type

optional

Dependencies

Version Information

requirements.txt:

```
biopython
```

Equivalent System Packages

```
$ conda install biopython
```

```
$ sudo dnf install python3-biopython
```

```
$ sudo port install py-biopython
```

See https://repology.org/project/biopython/versions, https://repology.org/project/python:biopython/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.27 bleach: An HTML-sanitizing tool

Description

An easy safelist-based HTML-sanitizing tool.

License

Apache License v2

Upstream Contact

Home Page: https://github.com/mozilla/bleach

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *webencodings: Character encoding aliases for legacy web content*

Version Information

package-version.txt:

```
6.4.0
```

version_requirements.txt:

```
bleach >= 5
```

Equivalent System Packages

```
$ sudo pacman -S python-bleach
```

```
$ conda install bleach
```

```
$ sudo apt-get install python3-bleach
```

```
$ sudo dnf install python3-bleach
```

```
$ sudo emerge dev-python/bleach
```

```
$ sudo port install py-bleach
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-bleach'
```

```
$ sudo xbps-install python3-bleach
```

See https://repology.org/project/python:bleach/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.28 bliss: Computing automorphism groups and canonical forms of graphs

Description

bliss is an open source tool for computing automorphism groups and canonical forms of graphs.

License

LGPL3

Upstream Contact

Bliss is currently being maintained by Tommi Junttila at

https://users.aalto.fi/~tjunttil/bliss/index.html

Bliss used to be maintained by Tommi Junttila and Petteri Kaski up to version 0.73 at

http://www.tcs.tkk.fi/Software/bliss/index.html

Type

optional

Dependencies

- *cmake: A cross-platform build system generator*

Version Information

package-version.txt:

```
0.77
```

Equivalent System Packages

```
$ apk add bliss
```

```
$ sudo pacman -S bliss
```

```
$ conda install 'bliss>=0.77'
```

```
$ sudo dnf install bliss bliss-devel
```

```
$ sudo emerge sci-libs/bliss
```

```
$ sudo zypper install bliss bliss-devel
```

See https://repology.org/project/bliss-graphs/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.29 brial: Boolean Ring Algebra implementation using binary decision diagrams

Description

BRiAl ("Boolean Ring Algebra") is the successor to PolyBoRi.

The core of PolyBoRi is a C++ library, which provides high-level data types for Boolean polynomials and monomials, exponent vectors, as well as for the underlying polynomial rings and subsets of the powerset of the Boolean variables. As a unique approach, binary decision diagrams are used as internal storage type for polynomial structures. On top of this C++-library we provide a Python interface. This allows parsing of complex polynomial systems, as well as sophisticated and extendable strategies for Gröbner base computation. PolyBoRi features a powerful reference implementation for Gröbner basis computation.

License

GPL version 2 or later

Upstream Contact

https://github.com/BRiAl/BRiAl

Type

standard

Dependencies

- *libpng: Bitmap image support*
- *m4ri: fast arithmetic with dense matrices over GF(2)*

Version Information

package-version.txt:

```
1.2.15
```

Equivalent System Packages

```
$ sudo pacman -S brial
```

```
$ conda install brial
```

```
$ sudo apt-get install libbrial-dev libbrial-groebner-dev
```

```
$ sudo dnf install brial brial-devel
```

```
$ sudo pkg install math/brial
```

```
$ sudo emerge sci-libs/brial
```

```
$ nix-env --install brial
```

```
$ sudo zypper install brial-devel
```

```
$ sudo xbps-install brial-devel
```

See https://repology.org/project/brial/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.30 buckygen: Efficient generation of nonisomorphic fullerenes

Description

Buckygen is a program for the efficient generation of all nonisomorphic fullerenes. These are triangulations where all vertices have degree 5 or 6. Or if the dual representation is used: cubic plane graphs where all faces are pentagons or hexagons.

License

Buckygen is licensed under the GNU General Public License v3 (June 2007)

Upstream Contact

Buckygen was mainly written by Jan Goedgebeur, jan.goedgebeur[at]ugent.be.

http://caagt.ugent.be/buckygen/

Type

optional

Dependencies

Version Information

package-version.txt:

```
1.1
```

Equivalent System Packages

```
$ sudo pacman -S buckygen
```

```
$ sudo zypper install buckygen
```

See https://repology.org/project/buckygen/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.31 cachetools: Extensible memoizing collections and decorators

Description

Extensible memoizing collections and decorators

License

MIT

Upstream Contact

https://pypi.org/project/cachetools/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
7.0.5
```

version_requirements.txt:

```
cachetools
```

Equivalent System Packages

(none known)

If the system package is installed, `./configure` will check if it can be used.

6.1.32 calver: Setuptools extension for CalVer package versions

Description

Setuptools extension for CalVer package versions

License

Upstream Contact

https://pypi.org/project/calver/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
2025.10.20
```

version_requirements.txt:

```
calver
```

Equivalent System Packages

```
$ apk add py3-calver
```

```
$ sudo pacman -S python-calver
```

```
$ conda install calver
```

```
$ sudo pkg install devel/py-calver
```

```
$ sudo emerge calver
```

```
$ sudo port install py-calver
```

```
$ sudo zypper install python-calver
```

```
$ sudo xbps-install python3-calver
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.33 cbc: COIN-OR branch and cut solver for mixed-integer programs

Description

The Computational Infrastructure for Operations Research (COIN-OR**, or simply COIN) project is an initiative to spur the development of open-source software for the operations research community.

The COIN Branch and Cut solver (CBC) is an open-source mixed-integer program (MIP) solver written in C++. CBC is intended to be used primarily as a callable library to create customized branch-and-cut solvers. A basic, stand-alone executable version is also available.

License

Eclipse Public License, Version 1.0 (EPL-1.0) (http://opensource.org/licenses/eclipse-1.0)

Upstream Contact

- https://github.com/coin-or/Cbc

Type

optional

Dependencies

- $(BLAS)
- *readline: Command line editing library*

Version Information

package-version.txt:

```
2.9.4.p0
```

Equivalent System Packages

```
$ sudo pacman -S coin-or-cbc
```

```
$ conda install coincbc
```

```
$ sudo apt-get install coinor-cbc coinor-libcbc-dev
```

```
$ sudo dnf install coin-or-Cbc coin-or-Cbc-devel
```

```
$ sudo pkg install math/cbc
```

```
$ sudo emerge sci-libs/coinor-cbc
```

```
$ brew install cbc
```

```
$ nix-env --install cbc
```

```
$ sudo xbps-install CoinMP-devel
```

See https://repology.org/project/coin-or-cbc/versions, https://repology.org/project/cbc/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.34 ccache: A compiler cache

Description

ccache is a compiler cache. It speeds up recompilation by caching previous compilations and detecting when the same compilation is being done again. Supported languages are C, C++, Objective-C and Objective-C++.

License

GNU General Public License version 3 or later

Upstream Contact

- Author: Andrew Tridgell
- Website: http://ccache.samba.org/

Type

optional

Dependencies

- *cmake: A cross-platform build system generator*

Version Information

package-version.txt:

```
4.10.2
```

Equivalent System Packages

```
$ sudo pacman -S ccache
```

```
$ conda install ccache
```

```
$ sudo dnf install ccache
```

```
$ brew install ccache
```

```
$ sudo port install ccache
```

```
$ sudo zypper install ccache
```

```
$ sudo xbps-install ccache
```

See https://repology.org/project/ccache/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.35 cddlib: Double description method for polyhedral representation conversion

Description

The C-library cddlib is a C implementation of the Double Description Method of Motzkin et al. for generating all vertices (i.e. extreme points) and extreme rays of a general convex polyhedron in R^d given by a system of linear inequalities:

P = { x=(x1, …, xd)^T : b - A x >= 0 }

where A is a given m x d real matrix, b is a given m-vector and 0 is the m-vector of all zeros.

The program can be used for the reverse operation (i.e. convex hull computation). This means that one can move back and forth between an inequality representation and a generator (i.e. vertex and ray) representation of a polyhedron with cdd. Also, cdd can solve a linear programming problem, i.e. a problem of maximizing and minimizing a linear function over P.

License

GPL v2

Upstream Contact

https://github.com/cddlib/cddlib

Type

standard

Dependencies

- $(MP_LIBRARY)

Version Information

package-version.txt:

```
0.94n
```

Equivalent System Packages

```
$ sudo pacman -S cddlib
```

```
$ conda install cddlib
```

```
$ sudo apt-get install libcdd-dev libcdd-tools
```

```
$ sudo dnf install cddlib cddlib-devel
```

```
$ sudo pkg install math/cddlib
```

```
$ sudo emerge 'sci-libs/cddlib[tools]'
```

```
$ brew install cddlib
```

```
$ sudo port install cddlib
```

```
$ nix-env --install cddlib
```

```
$ sudo zypper install cddlib-tools 'pkgconfig(cddlib)'
```

```
$ sudo xbps-install cddlib-devel
```

See https://repology.org/project/cddlib/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.36 certifi: Python package for providing Mozilla's CA Bundle

Description

Python package for providing Mozilla's CA Bundle.

License

MPL-2.0

Upstream Contact

https://pypi.org/project/certifi/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
2026.2.25
```

version_requirements.txt:

```
certifi >=2020.6.20
```

Equivalent System Packages

```
$ sudo pacman -S python-certifi
```

```
$ conda install certifi
```

```
$ sudo apt-get install python3-certifi
```

```
$ sudo dnf install python3-certifi
```

```
$ sudo emerge dev-python/certifi
```

```
$ sudo port install py-certifi
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-certifi'
```

```
$ sudo xbps-install python3-certifi
```

See https://repology.org/project/python:certifi/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.37 cffi: Foreign Function Interface for Python calling C code

Description

development website: https://foss.heptapod.net/pypy/cffi

documentation website: https://cffi.readthedocs.io/en/latest/

PyPI page: https://pypi.org/project/cffi/

License

MIT

Upstream Contact

https://foss.heptapod.net/pypy/cffi

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *pycparser: Parser of the C language in Python*

Version Information

package-version.txt:

```
2.0.0
```

version_requirements.txt:

```
cffi >=1.14.0
```

Equivalent System Packages

```
$ conda install cffi
```

```
$ sudo dnf install python3-cffi
```

```
$ sudo emerge virtual/python-cffi
```

```
$ brew install cffi
```

```
$ sudo port install py-cffi
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-cffi'
```

```
$ sudo xbps-install python3-cffi
```

See https://repology.org/project/python:cffi/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.38 chardet: Universal encoding detector for Python 3

Description

Universal encoding detector for Python 3

License

LGPL

Upstream Contact

https://pypi.org/project/chardet/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
7.4.0.post2
```

version_requirements.txt:

```
chardet
```

Equivalent System Packages

(none known)

If the system package is installed, `./configure` will check if it can be used.

6.1.39 charset_normalizer: The Real First Universal Charset Detector. Open, modern and actively maintained alternative to Chardet.

Description

The Real First Universal Charset Detector. Open, modern and actively maintained alternative to Chardet.

License

MIT

Upstream Contact

https://pypi.org/project/charset-normalizer/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
3.4.6
```

version_requirements.txt:

```
charset-normalizer
```

Equivalent System Packages

```
$ conda install charset-normalizer
```

```
$ sudo dnf install python3-charset-normalizer
```

```
$ sudo emerge dev-python/charset_normalizer
```

```
$ sudo xbps-install python3-charset-normalizer
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.40 clarabel: Clarabel Conic Interior Point Solver for Rust / Python

Description

Clarabel Conic Interior Point Solver for Rust / Python

License

Apache-2.0

Upstream Contact

https://pypi.org/project/clarabel/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cffi: Foreign Function Interface for Python calling C code*
- *numpy: Package for scientific computing with Python*
- *scipy: Scientific tools for Python*

Version Information

package-version.txt:

```
0.11.1
```

requirements.txt:

```
clarabel
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.41 cliquer: Routines for clique searching

Description

Cliquer is a set of C routines for finding cliques in an arbitrary weighted graph. It uses an exact branch-and-bound algorithm developed by Patric Östergård.

License

GNU General Public License v2

Upstream Contact

Cliquer was mainly written by Sampo Niskanen, sampo.niskanenQiki.fi (Q=@).

https://users.aalto.fi/~pat/cliquer.html

Patches

- minor config updates (v1.22)
- autotoolized - see https://github.com/dimpase/autocliquer (v1.21)

Type

standard

Dependencies

Version Information

package-version.txt:

```
1.23
```

Equivalent System Packages

```
$ sudo pacman -S cliquer
```

```
$ conda install cliquer
```

```
$ sudo apt-get install cliquer libcliquer-dev
```

```
$ sudo dnf install cliquer cliquer-devel
```

```
$ sudo pkg install math/cliquer
```

```
$ sudo emerge sci-mathematics/cliquer
```

```
$ brew install dimpase/tap/cliquer
```

```
$ nix-env --install cliquer
```

```
$ sudo zypper install cliquer cliquer-devel
```

```
$ sudo xbps-install cliquer-devel
```

See https://repology.org/project/cliquer/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.42 cmake: A cross-platform build system generator

Description

The "cmake" executable is the CMake command-line interface. It may be used to configure projects in scripts. Project configuration settings may be specified on the command line with the -D option. The -i option will cause cmake to interactively prompt for such settings.

CMake is a cross-platform build system generator. Projects specify their build process with platform-independent CMake listfiles included in each directory of a source tree with the name CMakeLists.txt. Users build a project by using CMake to generate a build system for a native tool on their platform.

License

CMake is distributed under the OSI-approved BSD 3-clause License.

Upstream Contact

- https://cmake.org/
- cmake-developers@cmake.org

Type

standard

Dependencies

- *curl: Multiprotocol data transfer library and utility*

Version Information

package-version.txt:

```
3.31.9
```

Equivalent System Packages

```
$ apk add cmake
```

```
$ sudo pacman -S cmake
```

```
$ conda install 'cmake<4'
```

```
$ sudo apt-get install cmake
```

```
$ sudo dnf install cmake
```

```
$ sudo pkg install devel/cmake
```

```
$ sudo emerge dev-build/cmake
```

```
$ brew install cmake
```

No package needed.

```
$ nix-env --install cmake
```

```
$ sudo zypper install cmake
```

```
$ sudo slackpkg install cmake
```

```
$ sudo xbps-install cmake
```

See https://repology.org/project/cmake/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.43 cocoalib: Computations in commutative algebra

Description

CoCoA is a program to compute with numbers and polynomials.

License

- GPL v3

Upstream Contact

- Authors: http://cocoa.dima.unige.it/research/
- Email: cocoa@dima.unige.it
- Website: http://cocoa.dima.unige.it/
- Releases: http://cocoa.dima.unige.it/cocoalib/

Type

experimental

Dependencies

- $(MP_LIBRARY)

Version Information

package-version.txt:

```
0.99564
```

Equivalent System Packages

```
$ sudo dnf install cocoalib cocoalib-devel
```

```
$ sudo pkg install math/cocoalib
```

See https://repology.org/project/cocoalib/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.44 colorama: Cross-platform colored terminal text

Description

Cross-platform colored terminal text

License

Upstream Contact

https://pypi.org/project/colorama/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
0.4.6
```

version_requirements.txt:

```
colorama
```

Equivalent System Packages

(none known)

If the system package is installed, `./configure` will check if it can be used.

6.1.45 comm: Jupyter Python Comm implementation, for usage in ipykernel, xeus-python etc.

Description

Jupyter Python Comm implementation, for usage in ipykernel, xeus-python etc.

License

BSD 3-Clause License

Upstream Contact

https://pypi.org/project/comm/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *traitlets: Traitlets Python configuration system*

Version Information

package-version.txt:

```
0.2.3
```

version_requirements.txt:

```
comm
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.46 configure: Files of the Sage distribution that are autogenerated in the bootstrapping phase

Description

This package contains a tar archive of auto-generated files. They are shipped with Sage in case you do not have a sufficiently recent autotools version installed.

License

GPLv3+

Upstream Contact

Automatically generated by Sage, use trac and/or sage-devel for questions.

Special Update/Build Instructions

This tarball is automatically generated by Sage whenever you run the $SAGE_ROOT/bootstrap -s or the $SAGE_ROOT/tools/update-version script.

Type

base

Dependencies

Version Information

package-version.txt:

```
3801b3c56c2b7de7b3e41ec990b80d40a52b46cd
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.47 contourpy: Python library for calculating contours of 2D quadrilateral grids

Description

Python library for calculating contours of 2D quadrilateral grids

License

BSD-3-Clause

Upstream Contact

https://pypi.org/project/contourpy/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *numpy: Package for scientific computing with Python*
- *pybind11: Create Python bindings to C++ code*

Version Information

package-version.txt:

```
1.3.3
```

version_requirements.txt:

```
contourpy
```

Equivalent System Packages

```
$ sudo dnf install python3-contourpy
```

```
$ sudo emerge dev-python/contourpy
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.48 conway_polynomials: Python interface to Frank Lübeck's Conway polynomial database

Description

This python module evolved from the old SageMath *conway_polynomials* package once hosted at,

http://files.sagemath.org/spkg/upstream/conway_polynomials/

It's still maintained by Sage developers, but having a pip-installable interface to the data will make it easier to install SageMath via pip or another package manager.

License

GPL version 3 or later

Upstream Contact

https://github.com/sagemath/conway-polynomials

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
0.10
```

version_requirements.txt:

```
conway-polynomials >=0.8
```

Equivalent System Packages

```
$ sudo pacman -S sage-data-conway_polynomials
```

```
$ conda install conway-polynomials
```

```
$ sudo emerge dev-python/conway-polynomials
```

See https://repology.org/project/sagemath-conway-polynomials/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.49 coxeter3: Library for Coxeter groups, Bruhat ordering, Kazhdan-Lusztig polynomials

Description

This package wraps Fokko Ducloux's Coxeter 3 C++ library

Features:

- General Coxeter groups, implemented through the combinatorics of reduced words;
- Reduced expression and normal form computations;
- Bruhat ordering;
- Ordinary Kazhdan-Lusztig polynomials;
- Kazhdan-Lusztig polynomials with unequal parameters;
- Inverse Kazhdan-Lusztig polynomials;
- Cells and W-graphs;

http://math.univ-lyon1.fr/~ducloux/coxeter/coxeter3/english/coxeter3_e.html

This is a patched version done by Mike Hansen 2009-2013 and some fixes by Nicolas M. Thiéry and Jean-Pierre Flori.

License

GPL

Upstream Contact

github: https://github.com/tscrim/coxeter

Alas, Fokko Ducloux passed away in 2006.

http://math.univ-lyon1.fr/~ducloux/du_Cloux.html

Special Update/Build Instructions

The source package was created by running

```
commit=8ac9c71723c8ca57a836d6381aed125261e44e9e
git clone https://github.com/tscrim/coxeter.git
cd coxeter
git archive $commit | bzip2 --best >coxeter-$commit.tar.bz2
```

Type

optional

Dependencies

Version Information

package-version.txt:

```
8ac9c71723c8ca57a836d6381aed125261e44e9e.p0
```

Equivalent System Packages

```
$ sudo pacman -S coxeter
```

```
$ sudo dnf install coxeter coxeter-devel coxeter-tools
```

```
$ brew install dimpase/tap/coxeter3
```

```
$ sudo zypper install coxeter
```

See https://repology.org/project/coxeter/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.50 cppy: C++ headers for C extension development

Description

From: https://pypi.org/project/cppy/

A small C++ header library which makes it easier to write Python extension modules. The primary feature is a PyObject smart pointer which automatically handles reference counting and provides convenience methods for performing common object operations.

License

Modified BSD 3-Clause-License

Upstream Contact

https://github.com/nucleic/cppy

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*
- *setuptools: Build system for Python packages*

Version Information

package-version.txt:

```
1.3.1
```

version_requirements.txt:

```
cppy >=1.2.0
```

Equivalent System Packages

```
$ conda install cppy
```

```
$ sudo dnf install python3-cppy
```

```
$ sudo emerge dev-python/cppy
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.51 csdp: Solver for semidefinite programs

Description

This is a fast SDP solver written in C, with a callable library namely, an autotool'ed version of CSDP, by Brian Borchers, see https://projects.coin-or.org/Csdp

License

Common Public License Version 1.0

Upstream Contact

Dmitrii Pasechnik <dimpase+sage@gmail.com>

Special Update/Build Instructions

csdp is an autotool'ed version of CSDP, see https://projects.coin-or.org/Csdp, developed in its own repository at https://github.com/dimpase/csdp.

To update to a new version, you need to bump the version number in configure.ac and rerun autotools (autoreconf -fiv). Any changes should be merged to the upstream repo.

The build is done with NOSHORTS variable defined; this makes it compatible with packages, where NOSHORTS must be defined, e.g. https://github.com/dimpase/pycsdp; also the Sage Cython interface needs NOSHORTS defined.

Detailed steps to build the spkg are as follows. You need

- git
- autotools and libtool (the full autohell suite, version at least 2.67)

With these ready:

- ./spkg-src
- copy the resulting csdp-<version>.tar.gz to SAGE_ROOT/upstream, or somewhere else appropriate

Type

optional

Dependencies

- $(BLAS)

Version Information

package-version.txt:

```
6.2.p1
```

Equivalent System Packages

```
$ sudo pacman -S coin-or-csdp
```

```
$ sudo dnf install csdp csdp-devel csdp-tools
```

```
$ brew install Macaulay2/tap/csdp
```

See https://repology.org/project/coin-or-csdp/versions, https://repology.org/project/csdp/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.52 cunningham_tables: List of the prime numbers occuring in the Cunningham table

The script read_cunningham_prime_factors.py was used to generate the data set from the file http://cage.ugent.be/~jdemeyer/cunningham/main.gz We include a local copy, main.gz (see comments in the file for details)

Type

optional

Dependencies

Version Information

package-version.txt:

```
1.0
```

Equivalent System Packages

See https://repology.org/project/cunningham-tables/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.53 curl: Multiprotocol data transfer library and utility

Description

Multiprotocols data transfer library (and utility).

License

"MIT style license" : see file "COPYING" at the root of the source tarball, explanations at https://curl.haxx.se/docs/copyright.html.

Upstream Contact

According to the file README at the root of the tarball, contact is done by mailing https://curl.haxx.se/mail/

Type

standard

Dependencies

- *openssl: Implementation of the SSL and TLS protocols*

Version Information

package-version.txt:

```
8.16.0
```

Equivalent System Packages

```
$ apk add curl
```

```
$ conda install curl
```

```
$ sudo apt-get install curl libcurl4-openssl-dev
```

```
$ sudo dnf install libcurl-devel curl
```

```
$ sudo pkg install ftp/curl
```

```
$ brew install curl
```

No package needed.

```
$ sudo zypper install curl 'pkgconfig(libcurl)'
```

```
$ sudo slackpkg install curl cyrus-sasl openldap-client libssh2
```

```
$ sudo xbps-install curl libcurl-devel
```

See https://repology.org/project/curl/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.54 cvxopt: Python software for convex optimization

Description

CVXOPT is a free software package for convex optimization based on the Python programming language. It can be used with the interactive Python interpreter, on the command line by executing Python scripts, or integrated in other software via Python extension modules. Its main purpose is to make the development of software for convex optimization applications straightforward by building on Python's extensive standard library and on the strengths of Python as a high-level programming language.

Upstream Contact

- J. Dahl <dahl.joachim@gmail.com>
- L. Vandenberghe <vandenbe@ee.ucla.edu>

https://cvxopt.org/

License

GPLv3 or later. Includes parts under GPLv2, GNU Lesser General Public License, v2.1. See src/LICENSE for more details. (Sage-compatible)

Type

standard

Dependencies

- $(BLAS)
- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *glpk: GNU Linear Programming Kit*
- *gsl: The GNU Scientific Library*
- *numpy: Package for scientific computing with Python*
- *pkgconfig: Python interface to pkg-config*
- *suitesparse: A suite of sparse matrix software*

Version Information

package-version.txt:

```
1.3.3
```

version_requirements.txt:

```
cvxopt >=1.2.5
```

Equivalent System Packages

```
$ sudo pacman -S python-cvxopt
```

```
$ conda install cvxopt
```

```
$ sudo apt-get install python3-cvxopt
```

```
$ sudo dnf install python3-cvxopt
```

```
$ sudo pkg install math/py-cvxopt
```

```
$ sudo emerge dev-python/cvxopt
```

```
$ sudo port install py-cvxopt
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-cvxopt'
```

See https://repology.org/project/python:cvxopt/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.55 cvxpy: Domain-specific language for modeling convex optimization problems in Python

Description

Domain-specific language for modeling convex optimization problems in Python

License

Apache License, Version 2.0

Upstream Contact

https://pypi.org/project/cvxpy/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *clarabel: Clarabel Conic Interior Point Solver for Rust / Python*
- *cvxopt: Python software for convex optimization*
- *ecos_python: Embedded Cone Solver (Python wrapper)*
- *glpk: GNU Linear Programming Kit*
- *highspy: Python interface to HiGHS optimizer*
- *numpy: Package for scientific computing with Python*
- *osqp_python: The Operator Splitting QP Solver (Python wrapper)*

- *scipy: Scientific tools for Python*
- *scs: Splitting conic solver*

Version Information

package-version.txt:

```
1.8.2
```

version_requirements.txt:

```
cvxpy
```

Equivalent System Packages

```
$ conda install cvxpy
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.56 cycler: Composable cycles

Description

Cycler is a small break-off of matplotlib to deal with "composable cycles". It is a required dependency of matplotlib 1.5.0.

License

BSD

Upstream Contact

cycler is developed on github: https://github.com/matplotlib/cycler

A more informative webpage about cycler, its motivation and usage is at http://tacaswell.github.io/cycler/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
0.12.1
```

version_requirements.txt:

```
cycler >=0.12.1
```

Equivalent System Packages

```
$ sudo pacman -S python-cycler
```

```
$ conda install cycler
```

```
$ sudo apt-get install python3-cycler
```

```
$ sudo dnf install python3-cycler
```

```
$ sudo pkg install devel/py-cycler
```

```
$ sudo emerge dev-python/cycler
```

```
$ sudo port install py-cycler
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-Cycler'
```

```
$ sudo xbps-install python3-cycler
```

See https://repology.org/project/cycler/versions, https://repology.org/project/python:cycler/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.57 cylp: Python interface for CLP, CBC, and CGL

Description

Python interface for CLP, CBC, and CGL

License

Eclipse Public License (EPL) version 2 (without a Secondary Licenses Notice).

Note: This license is incompatible with the GPL according to https://www.gnu.org/licenses/license-list.html#EPL2; see also the discussion in Issue #26511.

Upstream Contact

https://pypi.org/project/cylp/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cbc: COIN-OR branch and cut solver for mixed-integer programs*
- *cython: C-Extensions for Python, an optimizing static compiler*

- *numpy: Package for scientific computing with Python*
- *scipy: Scientific tools for Python*

Version Information

package-version.txt:

```
0.94.0
```

version_requirements.txt:

```
cylp
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.58 cypari2: Python interface to the number theory library libpari

Description

A Python interface to the number theory library libpari.

License

GPL version 2 or later

Upstream Contact

https://github.com/defeo/cypari2

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cysignals: Interrupt and signal handling for Cython*
- *cython: C-Extensions for Python, an optimizing static compiler*
- *pari: Computer algebra system for fast computations in number theory*

Version Information

package-version.txt:

```
2.2.4
```

version_requirements.txt:

```
cypari2 >=2.2.1; sys_platform != 'win32'
```

Equivalent System Packages

```
$ conda install cypari2
```

```
$ sudo emerge dev-python/cypari2
```

See https://repology.org/project/python:cypari2/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.59 cysignals: Interrupt and signal handling for Cython

Description

Interrupt and signal handling for Cython

License

LGPL version 3 or later

Upstream Contact

https://github.com/sagemath/cysignals

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cython: C-Extensions for Python, an optimizing static compiler*

Version Information

package-version.txt:

```
1.12.6
```

version_requirements.txt:

```
cysignals >=1.12.1
```

Equivalent System Packages

```
$ conda install cysignals
```

```
$ sudo emerge dev-python/cysignals
```

See https://repology.org/project/cysignals/versions, https://repology.org/project/python:cysignals/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.60 cython: C-Extensions for Python, an optimizing static compiler

Description

Cython is a language that makes writing C extensions for the Python language as easy as Python itself. Cython is based on the well-known Pyrex, but supports more cutting edge functionality and optimizations.

The Cython language is very close to the Python language, but Cython additio- nally supports calling C functions and declaring C types on variables and class attributes. This allows the compiler to generate very efficient C code from Cython code.

This makes Cython the ideal language for wrapping for external C libraries, and for fast C modules that speed up the execution of Python code.

License

Apache License, Version 2.0

Upstream Contact

- http://www.cython.org/
- cython-devel@python.org

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *pythran: Ahead of Time compiler for numeric kernels*

Version Information

package-version.txt:

```
3.2.4
```

version_requirements.txt:

```
cython >=3.1.0
```

Equivalent System Packages

```
$ sudo pacman -S cython
```

```
$ conda install 'cython>=3.0,!=3.0.3,<4.0'
```

```
$ sudo apt-get install cython3
```

```
$ sudo dnf install python3-cython
```

```
$ sudo pkg install lang/cython
```

```
$ sudo emerge dev-python/cython
```

```
$ brew install cython
```

```
$ sudo port install py-cython
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-Cython'
```

```
$ sudo xbps-install python3-Cython
```

See https://repology.org/project/python:cython/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.61 d3js: JavaScript library for manipulating documents based on data

Description

D3.js is a JavaScript library for manipulating documents based on data. The file d3.min.js will be placed into the ${SAGE_SHARE}/d3js/ directory.

License

BSD 3-Clause License

Upstream Contact

- Author: Mike Bostock (http://bost.ocks.org/mike/)
- Home page: http://d3js.org/

Special Update/Build Instructions

Two kind of archives can be downloaded from d3.js website: one with all source code and tests that weights 2,9M (both in zip and tar.gz formats) and one with the final javascript scripts which weights 121K (zip format only). Since testing requires node.js that is not shipped with Sage, we currenlty ship the final js only. Hence we have to transform it from zip to tar.gz format. Running sage-src should do all the repackaging job.

Type

optional

Dependencies

Version Information

package-version.txt:

```
3.4.8
```

Equivalent System Packages

See https://repology.org/project/node:d3/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.62 database_cremona_ellcurve: Database of elliptic curves

Description

John Cremona's database of elliptic curves

See https://github.com/JohnCremona/ecdata

This is an optional package, not included by default.

License

Public Domain

Upstream Contact

- Author: John Cremona
- Email: john.cremona@gmail.com
- Website: http://homepages.warwick.ac.uk/staff/J.E.Cremona/

Update Instructions

Get an up-to-date copy of the git repository ecdata from https://github.com/JohnCremona/ecdata.

If the cremona database has already been installed, remove `SAGE_DATA/cremona/cremona.db`. Then run

The build script expects to find the files in subfolders allcurves, allgens, degphi and allbsd of the ecdata folder. It extracts them and builds the new cremona.db file from the contents.

Finally, copy `SAGE_DATA/cremona/cremona.db` to the src directory of the spkg.

Type

optional

Dependencies

Version Information

package-version.txt:

```
20190911
```

Equivalent System Packages

See https://repology.org/project/sage-data-cremona-ellcurve/versions, https://repology.org/project/sagemath-database-cremona-elliptic-curves/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.63 database_cubic_hecke: Ivan Marin's representations of the cubic Hecke algebra

Description

Ivan Marin's representations of the cubic Hecke algebra on 4 strands as Python dictionaries

License

GPL

Upstream Contact

https://pypi.org/project/database-cubic-hecke/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
2022.4.4
```

version_requirements.txt:

```
database-cubic-hecke
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.64 database_jones_numfield: Table of number fields

Description

This is a table of number fields with bounded ramification and degree at most 6.

License

GPLv2+

Upstream Contact

sage-devel@googlegroups.com

Special Update/Build Instructions

Created by taking the original old-style spkg and removing crud from it.

Type

optional

Dependencies

Version Information

package-version.txt:

```
4
```

Equivalent System Packages

See https://repology.org/project/sage-data-jones-numfield/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.65 database_knotinfo: Content of the KnotInfo and LinkInfo databases as lists of dictionaries

Description

Content of the KnotInfo and LinkInfo databases as lists of dictionaries

License

GPL

Upstream Contact

https://pypi.org/project/database-knotinfo/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
2026.3.1
```

version_requirements.txt:

```
database-knotinfo
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.66 database_kohel: Database of modular and Hilbert polynomials

Description

Database of modular and Hilbert polynomials.

Upstream Contact

- David Kohel <David.Kohel@univ-amu.fr>

Type

optional

Dependencies

Version Information

package-version.txt:

```
20160724
```

Equivalent System Packages

See https://repology.org/project/sage-data-kohel/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.67 database_mutation_class: Database of exceptional mutation classes of quivers

Description

Contains a database of all exceptional mutation classes of quivers.

Every file in the database is of the form `mutation_classes_n.dig6` for some `n` and

- contains a `cPickle.dump` of a dictionary where
- the keys are tuples representing irreducible exceptional quiver mutation types of rank `n`, and

- the values are all quivers in the given mutation class stored in canonical form as `(dig6,edges)` where
- `dig6` is the dig6 data of the given `DiGraph`, and
- `edges` are the non-simply-laced edges thereof.
- is obtained by running the function

  ```
  sage.combinat.cluster_algebra_quiver.quiver_mutation_type._save_data_dig6(n,
  types='Exceptional', verbose=False)
  ```

SPKG Maintainers

- C. Stump <christian.stump@gmail.com>

Type

optional

Dependencies

Version Information

package-version.txt:

```
1.0
```

Equivalent System Packages

See https://repology.org/project/database-mutation-class/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.68 database_odlyzko_zeta: Table of zeros of the Riemann zeta function

Description

Table of zeros of the Riemann zeta function by Andrew Odlyzko.

This package contains the file 'zeros6' with the first 2,001,052 zeros of the Riemann zeta function, accurate to within 4*10^(-9).

Type

optional

Dependencies

- $(SAGERUNTIME)

Version Information

package-version.txt:

```
20061209
```

Equivalent System Packages

See https://repology.org/project/sage-data-odlyzko-zeta/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.69 database_stein_watkins: Database of elliptic curves (full version)

Description

The Stein-Watkins database of elliptic curves (full version)

See http://modular.math.washington.edu/papers/stein-watkins/

This is an optional (huge) package, not included by default.

License

Public Domain

Type

optional

Dependencies

Version Information

package-version.txt:

```
20110713
```

Equivalent System Packages

See https://repology.org/project/database-stein-watkins/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.70 database_stein_watkins_mini: Database of elliptic curves (small version)

Description

The Stein-Watkins database of elliptic curves (small version)

See http://modular.math.washington.edu/papers/stein-watkins/

This is an optional package, not included by default.

License

Public Domain

Type

optional

Dependencies

Version Information

package-version.txt:

```
20070827
```

Equivalent System Packages

See https://repology.org/project/database-stein-watkins-mini/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.71 database_symbolic_data: Database from the SymbolicData project

Description

The SymbolicData project is set out

- to develop concepts and tools for profiling, testing and benchmarking Computer Algebra Software (CAS) and
- to collect and interlink relevant data and activities from different Computer Algebra Communities.

SymbolicData is an

- inter-community project that has its roots in the activities of different Computer Algebra Communities and
- aims at interlinking these activities using modern Semantic Web concepts.

Tools and data are designed to be used both

- on a local site for special testing and profiling purposes
- and to manage a central repository at www.symbolicdata.org.

License

GNU General Public License

Upstream Contact

- Andreas Nareike <nareike@informatik.uni-leipzig.de>

Type

optional

Dependencies

Version Information

package-version.txt:

```
20070206
```

Equivalent System Packages

See https://repology.org/project/database-symbolic-data/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.72 dateutil: Extensions to the standard Python module datetime

Description

The dateutil module provides powerful extensions to the standard datetime module.

License

Simplified BSD License

Upstream Contact

Author: Gustavo Niemeyer <gustavo@niemeyer.net>

Home page: http://labix.org/python-dateutil

https://pypi.org/project/python-dateutil/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*
- *six: Python 2 and 3 compatibility utilities*

Version Information

package-version.txt:

```
2.9.0.post0
```

version_requirements.txt:

```
python-dateutil >=2.8.1
```

Equivalent System Packages

```
$ sudo pacman -S python-dateutil
```

```
$ conda install python-dateutil
```

```
$ sudo apt-get install python3-dateutil
```

```
$ sudo dnf install python3-dateutil
```

```
$ sudo pkg install devel/py-dateutil
```

```
$ sudo emerge dev-python/python-dateutil
```

```
$ sudo port install py-dateutil
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-python-dateutil'
```

```
$ sudo xbps-install python3-dateutil
```

See https://repology.org/project/python:python-dateutil/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.73 debugpy: Implementation of the Debug Adapter Protocol for Python

Description

Implementation of the Debug Adapter Protocol for Python

License

MIT

Upstream Contact

https://pypi.org/project/debugpy/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
1.8.20
```

version_requirements.txt:

```
debugpy
```

Equivalent System Packages

```
$ conda install debugpy
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.74 decorator: Python library providing decorators

Description

Better living through Python with decorators

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
5.2.1
```

version_requirements.txt:

```
decorator >=4.4.0
```

Equivalent System Packages

```
$ sudo pacman -S python-decorator
```

```
$ conda install decorator
```

```
$ sudo apt-get install python3-decorator
```

```
$ sudo dnf install python3-decorator
```

```
$ sudo emerge dev-python/decorator
```

```
$ sudo port install py-decorator
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-decorator'
```

```
$ sudo xbps-install python3-decorator
```

See https://repology.org/project/python:decorator/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.75 deformation: Count points on hypersurfaces using the deformation method

Description

Deformation is a C library for counting points on hypersurfaces using the deformation method, developed by Sebastian Pancratz.

License

GLPv3

Upstream Contact

- Sebastian Pancratz: sebastian.pancratz@gmail.com, sage-devel@googlegroups.com
- We use the fork at https://github.com/sagemath/deformation the fork uses GMP instead of MPIR, and Flint 2.7+.

Type

experimental

Dependencies

- $(MP_LIBRARY)
- *flint: Fast Library for Number Theory*
- *mpfr: Multiple-precision floating-point computations with correct rounding*

Version Information

package-version.txt:

```
20210503
```

Equivalent System Packages

See https://repology.org/project/deformation/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.76 defusedxml: Addresses vulnerabilities of XML parsers and XML libraries

Description

defusedxml addresses vulnerabilities of XML parsers and XML libraries.

It became a dependency of nbconvert starting with nbconvert 5.4.

License

Python Software Foundation License (PSFL)

Upstream Contact

https://pypi.org/project/defusedxml/

Special Update/Build Instructions

None.

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
0.7.1
```

version_requirements.txt:

```
defusedxml >=0.6.0
```

Equivalent System Packages

```
$ conda install defusedxml
```

```
$ sudo dnf install python3-defusedxml
```

```
$ sudo emerge dev-python/defusedxml
```

```
$ sudo port install py-defusedxml
```

```
$ sudo xbps-install python3-defusedxml
```

See https://repology.org/project/python:defusedxml/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.77 distlib: Distribution utilities

Description

Distribution utilities

License

PSF-2.0

Upstream Contact

https://pypi.org/project/distlib/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
0.4.0
```

version_requirements.txt:

```
distlib
```

Equivalent System Packages

```
$ conda install distlib
```

```
$ sudo xbps-install python3-distlib
```

If the system package is installed, `./configure` will check if it can be used.

6.1.78 docutils: Processing plaintext documentation into useful formats, such as HTML or LaTeX

Description

Docutils is a modular system for processing documentation into useful formats, such as HTML, XML, and LaTeX. For input Docutils supports reStructuredText, an easy-to-read, what-you-see-is-what-you-get plaintext markup syntax.

License

public domain, Python, 2-Clause BSD, GPL 3 (see COPYING.txt)

Upstream Contact

https://pypi.org/project/docutils/

Home Page: http://docutils.sourceforge.net/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
0.22.4
```

version_requirements.txt:

```
docutils >=0.14
```

Equivalent System Packages

```
$ conda install docutils
```

```
$ sudo dnf install python3-docutils
```

```
$ sudo emerge dev-python/docutils
```

```
$ brew install docutils
```

```
$ sudo port install py-docutils
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-docutils'
```

```
$ sudo xbps-install python3-docutils
```

See https://repology.org/project/docutils/versions, https://repology.org/project/python:docutils/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.79 dot2tex: Create PGF/TikZ commands from Graphviz output

Description

dot2tex is a Python module, whose purpose is to give graphs generated by Graphviz a more LaTeX friendly look and feel. This is accomplished by converting xdot output from Graphviz to a series of PSTricks or PGF/TikZ commands.

See https://github.com/sagemath/dot2tex/

License

- MIT

Upstream Contact

- Kjell Magne Fauske, km@fauskes.net
- SageMath developers

Dependencies

graphviz (www.graphviz.org) should be installed and in the path (for example via the graphviz spkg).

preview, a LaTeX package for extracting parts of a document.

Self-tests dependencies:

- graphviz
- texlive-latex-base
- texlive-pictures
- texlive-pstricks

Special Update/Build Instructions

Make sure corresponding optional doctests still pass:

sage -t –long –optional=dot2tex,graphviz,sage –all

Type

optional

Dependencies

Version Information

requirements.txt:

```
dot2tex >=2.12.0
```

Equivalent System Packages

```
$ sudo pacman -S dot2tex
```

```
$ conda install dot2tex
```

```
$ sudo dnf install dot2tex
```

```
$ sudo port install dot2tex
```

See https://repology.org/project/dot2tex/versions, https://repology.org/project/python:dot2tex/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.80 dsdp: Semidefinite programming solver

Description

Implementation of an interior-point method for semidefinite programming. It provides primal and dual solutions, exploits low-rank structure and sparsity in the data, and has relatively low memory requirements for an interior-point method. It allows feasible and infeasible starting points and provides approximate certificates of infeasibility when no feasible solution exists. The dual-scaling algorithm implemented in this package has a convergence proof and worst-case polynomial complexity under mild assumptions on the data.

License

Permissive open source license https://www.mcs.anl.gov/hs/software/DSDP/Copyright.txt

Upstream Contact

https://www.mcs.anl.gov/hs/software/DSDP/

Type

optional

Dependencies

- $(BLAS)
- *cmake: A cross-platform build system generator*

Version Information

package-version.txt:

```
5.8
```

Equivalent System Packages

```
$ sudo pacman -S dsdp
```

```
$ conda install dsdp
```

```
$ sudo apt-get install libdsdp-dev
```

```
$ sudo dnf install DSDP-devel
```

```
$ sudo pkg install math/dsdp
```

```
$ sudo emerge sci-libs/dsdp
```

```
$ sudo port install DSDP
```

See https://repology.org/project/dsdp/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.81 e_antic: Real embedded number fields

Description

e-antic is a C library for exact computations with real embedded number field maintained by Vincent Delecroix.

Website: https://github.com/videlec/e-antic

License

e-antic is licensed GPL v3.

Upstream Contact

- https://github.com/videlec/e-antic

Type

optional

Dependencies

- $(MP_LIBRARY)
- *flint: Fast Library for Number Theory*

Version Information

package-version.txt:

```
2.1.0
```

Equivalent System Packages

```
$ sudo pacman -S e-antic
```

```
$ conda install libeantic
```

```
$ sudo apt-get install libeantic-dev
```

```
$ sudo dnf install e-antic-devel
```

```
$ sudo pkg install math/e-antic
```

```
$ sudo zypper install e-antic-devel
```

See https://repology.org/project/e-antic/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.82 ecl: An implementation of the Common Lisp language

Description

ECL is an implementation of the Common Lisp language as defined by the ANSI X3J13 specification. The most relevant features:

- A bytecodes compiler and interpreter.
- Compiles Lisp also with any C/C++ compiler.
- It can build standalone executables and libraries.
- ASDF, Sockets, Gray streams, MOP, and other useful components.
- Extremely portable.
- A reasonable license.

ECL supports the operating systems Linux, FreeBSD, NetBSD, OpenBSD, Solaris and Windows, running on top of the Intel, Sparc, Alpha and PowerPC processors. Porting to other architectures should be rather easy.

Website: https://common-lisp.net/project/ecl/

License

- LGPL V2+ or compatible - for details see

  https://common-lisp.net/project/ecl/static/manual/Copyrights.html#Copyright-of-ECL

Upstream Contact

- the ECL mailing list - see https://mailman.common-lisp.net/listinfo/ecl-devel

Special Update/Build Instructions

- Note: for the time being, ECL is built single threaded library as it seems to interact badly with the pexpect interface and Sage's signal handling when built multithreaded.
- Do NOT quote SAGE_LOCAL when setting CPPFLAGS and/or LDFLAGS, in spkg-install as this caused the build to break. See Issue #10187#comment:117
- TODO: Add the ECL test suite, and an spkg-check file to run it.
- TODO: Make ECL use Sage's Boehm GC on MacOS X as well (but perhaps put some changes from ECL's into Sage's Boehm GC), then remove the src/src/gc directory, too.

Type

standard

Dependencies

- $(MP_LIBRARY)
- *gc: The Boehm-Demers-Weiser conservative garbage collector*
- *info: stand-alone Info documentation reader*
- *libffi: A portable foreign-function interface library*
- *readline: Command line editing library*

Version Information

package-version.txt:

```
26.5.5
```

Equivalent System Packages

```
$ apk add ecl-dev
```

```
$ sudo pacman -S ecl
```

```
$ conda install ecl
```

```
$ sudo apt-get install ecl
```

```
$ sudo dnf install ecl
```

```
$ sudo pkg install lang/ecl
```

```
$ sudo emerge dev-lisp/ecl
```

```
$ brew install ecl
```

```
$ sudo port install ecl
```

```
$ nix-env --install ecl
```

```
$ sudo xbps-install ecl
```

See https://repology.org/project/ecl/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.83 eclib: Enumerating and computing with elliptic curves defined over the rational numbers

Description

John Cremona's programs for enumerating and computing with elliptic curves defined over the rational numbers.

mwrank is a program written in C++ for computing Mordell-Weil groups of elliptic curves over Q via 2-descent. It is available as source code in the eclib package, which may be distributed under the GNU General Public License, version 2, or any later version.

mwrank is now only distributed as part of eclib. eclib is also included in Sage, and for most potential users the easiest way to run mwrank is to install Sage (which also of course gives you much much more). I no longer provide a source code distribution of mwrank by itself: use eclib instead.

License

eclib is licensed GPL v2+.

Upstream Contact

- Author: John Cremona
- Email: john.cremona@gmail.com
- Website: https://johncremona.github.io/mwrank/index.html
- Repository: https://github.com/JohnCremona/eclib

Type

standard

Dependencies

- *flint: Fast Library for Number Theory*
- *ntl: A library for doing number theory*
- *pari: Computer algebra system for fast computations in number theory*

Version Information

package-version.txt:

```
20250627
```

Equivalent System Packages

```
$ sudo pacman -S eclib
```

```
$ conda install eclib
```

```
$ sudo apt-get install libec-dev eclib-tools
```

```
$ sudo dnf install eclib eclib-devel
```

```
$ sudo pkg install math/eclib
```

```
$ sudo emerge 'sci-mathematics/eclib[flint]'
```

```
$ nix-env --install eclib
```

```
$ sudo xbps-install eclib-devel
```

See https://repology.org/project/eclib/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.84 ecm: Elliptic curve method for integer factorization

Description

GMP-ECM - Elliptic Curve Method for Integer Factorization

Sources can be obtained from https://gitlab.inria.fr/zimmerma/ecm

License

LGPL V3+

Upstream Contact

- ecm-discuss@inria.fr

Special Update/Build Instructions

- GMP-ECM comes with a self-tuning feature; we could support that as an option ($SAGE_TUNE_*=yes) in the future.
- ECM currently does not (by itself) use the CC and CFLAGS settings from 'gmp.h' since we pass (other) options in CFLAGS, and CC is set by Sage and might got set by the user. We now at least partially fix that such that "optimized" code generation options ('-mcpu=…', '-mtune=…') are used by gcc. Of course a user can also manually enable them by setting the "global" CFLAGS to e.g. '-march=native' on x86[_64] systems, or '-mcpu=…' and '-mtune=…' on other architectures where "native" isn't supported. Note that this doesn't affect the packages' selection of processor-specific optimized [assembly] code. 'spkg-install' already reads the settings from Sage's and also a system-wide GMP now, but doesn't (yet) use all of them. If SAGE_FAT_BINARY="yes", we should avoid too specific settings of "-mcpu=…", and perhaps pass a more generic "–host=…" to 'configure'.
- We currently work around a linker bug on MacOS X 10.5 PPC (with GCC 4.2.1) which breaks 'configure' if debug symbols are enabled. This *might* get fixed in later upstream releases.
- We could save some space by removing the `src/build.vc10/` directory which isn't used in Sage. (It gets probably more worth in case also directories / files for later versions of Microsoft Visual C get added.)

Type

standard

Dependencies

- $(MP_LIBRARY)
- *primesieve: CLI program and C/C++ library for generating primes*

Version Information

package-version.txt:

```
7.0.7
```

Equivalent System Packages

```
$ sudo pacman -S gmp-ecm
```

```
$ conda install ecm
```

```
$ sudo apt-get install gmp-ecm libecm-dev
```

```
$ sudo dnf install gmp-ecm gmp-ecm-devel
```

```
$ sudo pkg install math/gmp-ecm
```

```
$ brew install gmp-ecm
```

```
$ sudo port install gmp-ecm
```

```
$ nix-env --install ecm
```

```
$ sudo xbps-install ecm-devel
```

See https://repology.org/project/gmp-ecm/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.85 ecos_python: Embedded Cone Solver (Python wrapper)

Description

This is the Python package for ECOS: Embedded Cone Solver.

It vendors ECOS.

License

GPLv3

Upstream Contact

https://pypi.org/project/ecos/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *numpy: Package for scientific computing with Python*
- *scipy: Scientific tools for Python*

Version Information

package-version.txt:

```
2.0.14
```

version_requirements.txt:

```
ecos
```

Equivalent System Packages

```
$ conda install ecos
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.86 elliptic_curves: Databases of elliptic curves

Description

Includes two databases:

- A small subset of John Cremona's database of elliptic curves up to conductor 10000.
- William Stein's database of interesting curves

Upstream Contact

cremona_mini

- Author: John Cremona
- Email: john.cremona@gmail.com
- Website: http://johncremona.github.io/ecdata/

ellcurves

- Author: William Stein
- Email: wstein@gmail.com

Type

standard

Dependencies

- $(PYTHON)

Version Information

package-version.txt:

```
0.8.1
```

Equivalent System Packages

```
$ sudo pacman -S sage-data-elliptic_curves
```

```
$ conda install sagemath-db-elliptic-curves
```

```
$ sudo xbps-install sage-data-elliptic_curves
```

See https://repology.org/project/sagemath-elliptic-curves/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.87 execnet: Rapid multi-Python deployment

Description

Rapid multi-Python deployment

License

Upstream Contact

https://pypi.org/project/execnet/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
2.1.2
```

version_requirements.txt:

```
execnet
```

Equivalent System Packages

```
$ sudo pacman -S python-execnet
```

```
$ sudo apt-get install python3-execnet
```

```
$ sudo dnf install python3-execnet
```

```
$ sudo emerge dev-python/execnet
```

```
$ sudo xbps-install python3-execnet
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.88 executing: Get the currently executing AST node of a frame, and other information

Description

Get the currently executing AST node of a frame, and other information

License

MIT

Upstream Contact

https://pypi.org/project/executing/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
2.2.1
```

version_requirements.txt:

```
executing
```

Equivalent System Packages

```
$ conda install executing
```

```
$ sudo dnf install python3-executing
```

```
$ sudo emerge dev-python/executing
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.89 fastjsonschema: Fastest Python implementation of JSON schema

Description

Fastest Python implementation of JSON schema

License

BSD

Upstream Contact

https://pypi.org/project/fastjsonschema/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
2.21.2
```

version_requirements.txt:

```
fastjsonschema
```

Equivalent System Packages

```
$ apk add py3-fastjsonschema
```

```
$ sudo pacman -S python-fastjsonschema
```

```
$ conda install python-fastjsonschema
```

```
$ sudo apt-get install python3-fastjsonschema
```

```
$ sudo dnf install python3-fastjsonschema
```

```
$ sudo pkg install devel/py-fastjsonschema
```

```
$ sudo emerge dev-python/fastjsonschema
```

```
$ sudo port install py-fastjsonschema
```

```
$ sudo zypper install python-fastjsonschema
```

```
$ sudo xbps-install python3-fastjsonschema
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.90 fflas_ffpack: Dense linear algebra over word-size finite fields

Description

FFLAS-FFPACK is a LGPL-2.1+ source code library for dense linear algebra over word-size finite fields.

http://linbox-team.github.io/fflas-ffpack/

License

LGPL V2.1 or later

Upstream Contact

- <ffpack-devel@googlegroups.com>

Type

standard

Dependencies

- $(BLAS)
- $(MP_LIBRARY)
- *givaro: C++ library for arithmetic and algebraic computations*
- *gsl: The GNU Scientific Library*

Version Information

package-version.txt:

```
2.5.0+sage-2024-05-18b
```

Equivalent System Packages

```
$ sudo pacman -S fflas-ffpack
```

```
$ conda install fflas-ffpack
```

```
$ sudo apt-get install fflas-ffpack
```

```
$ sudo dnf install fflas-ffpack-devel
```

```
$ sudo pkg install math/fflas-ffpack
```

```
$ sudo emerge sci-libs/fflas-ffpack
```

```
$ brew install Macaulay2/tap/fflas-ffpack
```

```
$ nix-env --install fflas-ffpack
```

```
$ sudo zypper install 'pkgconfig(fflas-ffpack)'
```

```
$ sudo xbps-install fflas-ffpack
```

See https://repology.org/project/fflas-ffpack/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.91 ffmpeg: ffmpeg video converter

Description

ffmpeg is a very fast video and audio converter that can also grab from a live audio/video source. It can also convert between arbitrary sample rates and resize video on the fly with a high quality polyphase filter.

License

"FFmpeg is licensed under the GNU Lesser General Public License (LGPL) version 2.1 or later. However, FFmpeg incorporates several optional parts and optimizations that are covered by the GNU General Public License (GPL) version 2 or later. If those parts get used the GPL applies to all of FFmpeg."

http://ffmpeg.org/legal.html

Upstream Contact

http://ffmpeg.org/

Type

optional

Dependencies

Version Information

Equivalent System Packages

```
$ apk add ffmpeg
```

```
$ sudo pacman -S ffmpeg
```

```
$ conda install imageio-ffmpeg
```

```
$ sudo apt-get install ffmpeg
```

```
$ sudo dnf install ffmpeg-free ffmpeg-free-devel
```

```
$ sudo pkg install multimedia/ffmpeg
```

```
$ brew install ffmpeg
```

```
$ sudo port install ffmpeg
```

```
$ nix-env --install ffmpeg
```

```
$ sudo zypper install ffmpeg
```

install the following packages: ffmpeg

```
$ sudo xbps-install ffmpeg
```

See https://repology.org/project/ffmpeg/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.92 filelock: Platform independent file lock

Description

Platform independent file lock

License

Public Domain <http://unlicense.org>

Upstream Contact

https://pypi.org/project/filelock/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
3.25.2
```

version_requirements.txt:

```
filelock
```

Equivalent System Packages

```
$ conda install filelock
```

```
$ sudo xbps-install python3-filelock
```

If the system package is installed, `./configure` will check if it can be used.

6.1.93 flint: Fast Library for Number Theory

Description

FLINT is a C library for doing number theory, maintained by Fredrik Johansson.

Website: http://www.flintlib.org

License

FLINT is licensed GPL v2+.

Upstream Contact

- flint-devel Gougle Group (http://groups.google.co.uk/group/flint-devel)
- Fredrik Johansson

Type

standard

Dependencies

- $(MP_LIBRARY)
- *mpfr: Multiple-precision floating-point computations with correct rounding*

Version Information

package-version.txt:

```
3.4.0
```

Equivalent System Packages

```
$ apk add flint-dev
```

```
$ conda install libflint
```

```
$ sudo apt-get install libflint-dev
```

```
$ sudo dnf install flint flint-devel
```

```
$ sudo pkg install math/flint2
```

```
$ sudo emerge 'sci-mathematics/flint[ntl]'
```

```
$ brew install flint
```

```
$ sudo port install flint
```

```
$ nix-env --install flint
```

```
$ sudo zypper install flint-devel
```

```
$ sudo xbps-install flintlib-devel
```

See https://repology.org/project/flint/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.94 flit_core: Distribution-building parts of Flit. See flit package for more information

Description

Distribution-building parts of Flit. See flit package for more information

License

Upstream Contact

https://pypi.org/project/flit-core/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
3.12.0
```

version_requirements.txt:

```
flit-core >= 3.12.0
```

Equivalent System Packages

```
$ conda install flit-core
```

```
$ sudo dnf install python3-flit-core
```

```
$ sudo emerge dev-python/flit_core
```

```
$ sudo xbps-install python3-flit_core
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.95 fonttools: Tools to manipulate font files

Description

Tools to manipulate font files

License

MIT

Upstream Contact

https://pypi.org/project/fonttools/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cython: C-Extensions for Python, an optimizing static compiler*

Version Information

package-version.txt:

```
4.62.1
```

version_requirements.txt:

```
fonttools
```

Equivalent System Packages

```
$ conda install fonttools
```

```
$ sudo dnf install python3-fonttools
```

```
$ sudo emerge dev-python/fonttools
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.96 fplll: Lattice algorithms, including LLL with floating-point orthogonalization

Description

fplll contains implementations of several lattice algorithms. The implementation relies on floating-point orthogonalization, and LLL is central to the code, hence the name.

Website: https://github.com/fplll/fplll

License

- LGPL V2.1+

Upstream Contact

- Martin Albrecht <martinralbrecht+fplll@googlemail.com>
- Mailing List https://groups.google.com/forum/#!forum/fplll-devel

Type

standard

Dependencies

- $(MP_LIBRARY)
- *mpfr: Multiple-precision floating-point computations with correct rounding*

Version Information

package-version.txt:

```
5.5.0
```

Equivalent System Packages

```
$ sudo pacman -S fplll
```

```
$ conda install fplll
```

```
$ sudo apt-get install libfplll-dev
```

```
$ sudo dnf install libfplll libfplll-devel
```

```
$ sudo pkg install math/fplll
```

```
$ sudo emerge sci-libs/fplll
```

```
$ brew install fplll
```

```
$ sudo zypper install 'pkgconfig(fplll)' fplll-devel fplll
```

```
$ sudo xbps-install fplll-devel
```

See https://repology.org/project/fplll/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.97 fpylll: Python interface for FPLLL

Description

A Python interface for https://github.com/fplll/fplll (Lattice algorithms using floating-point arithmetic)

License

GPL version 2 or later

Upstream Contact

https://github.com/fplll/fpylll

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cysignals: Interrupt and signal handling for Cython*
- *cython: C-Extensions for Python, an optimizing static compiler*
- *fplll: Lattice algorithms, including LLL with floating-point orthogonalization*
- *numpy: Package for scientific computing with Python*

Version Information

package-version.txt:

```
0.6.4
```

version_requirements.txt:

```
fpylll >=0.5.9
```

Equivalent System Packages

```
$ conda install 'fpylll>=0.5.9'
```

```
$ sudo emerge dev-python/fpylll
```

See https://repology.org/project/fpylll/versions, https://repology.org/project/python:fpylll/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.98 fqdn: Validates fully-qualified domain names against RFC 1123, so that they are acceptable to modern bowsers

Description

Validates fully-qualified domain names against RFC 1123, so that they are acceptable to modern bowsers

License

MPL 2.0

Upstream Contact

https://pypi.org/project/fqdn/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
1.5.1
```

version_requirements.txt:

```
fqdn
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.99 free_fonts: a free family of scalable outline fonts

Description

This dummy package represents the GNU free fonts: a free family of scalable outline fonts, suitable for general use on computers and for desktop publishing. It is Unicode-encoded for compatibility with all modern operating systems.

We do not have an SPKG for it. The purpose of this dummy package is to associate system package lists with it.

License

GNU General Public License GPLv3+

Upstream Contact

https://www.gnu.org/software/freefont/

Type

optional

Dependencies

Version Information

Equivalent System Packages

```
$ apk add ttf-freefont
```

```
$ sudo pacman -S gnu-free-fonts
```

```
$ conda install open-fonts
```

```
$ sudo apt-get install fonts-freefont-otf
```

```
$ sudo dnf install gnu-free-mono-fonts gnu-free-sans-fonts  \
      gnu-free-serif-fonts texlive-gnu-freefont
```

```
$ sudo pkg install x11-fonts/freefont-ttf
```

```
$ sudo emerge media-fonts/freefont
```

```
$ sudo port install freefont-ttf
```

```
$ nix-env --install freefont-ttf
```

```
$ sudo pkg_add fonts/freefont-ttf
```

```
$ sudo zypper install gnu-free-fonts
```

```
$ sudo xbps-install freefont-ttf
```

See https://repology.org/project/font-freefont/versions, https://repology.org/project/fonts:gnu-freefont/versions, https://repology.org/project/texlive:gnu-freefont/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.100 freetype: A free, high-quality, and portable font engine

Description

From the documentation:

FreeType is a software font engine that is designed to be small, efficient, highly customizable, and portable while capable of producing high-quality output (glyph images). It can be used in graphics libraries, display servers, font conversion tools, text image generation tools, and many other products as well.

Note that FreeType is a font service and doesn't provide APIs to perform higher-level features like text layout or graphics processing (e.g., colored text rendering, 'hollowing', etc.). However, it greatly simplifies these tasks by providing a simple, easy to use, and uniform interface to access the content of font files.

Please note that 'FreeType' is also called 'FreeType 2', to distinguish it from the old, deprecated 'FreeType 1' library, a predecessor no longer maintained and supported.

The package in Sage is called freetype (in lowercase).

License

- FreeType (BSD-like)
- GNU Public License v2

From the documentation:

FreeType is released under two open-source licenses: our own BSD-like FreeType License and the GNU Public License, Version 2. It can thus be used by any kind of projects, be they proprietary or not.

Upstream Contact

- home: https://www.freetype.org
- repo:
  - official: http://git.savannah.gnu.org/cgit/freetype
  - mirror: https://github.com/aseprite/freetype2/

Type

standard

Dependencies

- *libpng: Bitmap image support*

Version Information

package-version.txt:

```
2.10.4
```

Equivalent System Packages

```
$ apk add freetype-dev
```

```
$ conda install freetype
```

```
$ sudo apt-get install libfreetype-dev
```

```
$ sudo pkg install print/freetype2
```

```
$ brew install freetype
```

No package needed.

```
$ nix-env --install freetype
```

```
$ sudo zypper install 'pkgconfig(freetype2)'
```

```
$ sudo slackpkg install freetype harfbuzz glib glib2
```

```
$ sudo xbps-install freetype-devel
```

See https://repology.org/project/freetype/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.101 fricas: A general purpose computer algebra system

Description

FriCAS is a general purpose computer algebra system.

License

Modified BSD license.

Upstream Contact

http://fricas.sourceforge.net/

Type

optional

Dependencies

- *ecl: An implementation of the Common Lisp language*

Version Information

package-version.txt:

```
1.3.12
```

Equivalent System Packages

```
$ sudo apt-get install fricas
```

```
$ sudo pkg install math/fricas
```

```
$ sudo emerge sci-mathematics/fricas
```

```
$ brew install fricas
```

```
$ sudo port install fricas
```

```
$ sudo zypper install fricas
```

See https://repology.org/project/fricas/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.102 frobby: Computations on monomial ideals

Description

The software package Frobby provides a number of computations on monomial ideals. The current main feature is the socle of a monomial ideal, which is largely equivalent to computing the maximal standard monomials, the Alexander dual or the irreducible decomposition.

Operations on monomial ideals are much faster than algorithms designed for ideals in general, which is what makes a specialized library for these operations on monomial ideals useful.

License

- GPL version 2.0 or later

Upstream Contact

- http://www.broune.com/frobby/
- https://github.com/Macaulay2/frobby

Special Update/Build instructions

Download Frobby at www.broune.com/ and then type "make spkg VER=blah" which wil create an spkg named frobby-VER.spkg in bin/. The files related to doing this is in the sage/ sub-directory of the Frobby source distribution.

Type

optional

Dependencies

- $(MP_LIBRARY)

Version Information

package-version.txt:

```
0.9.0.p2
```

Equivalent System Packages

```
$ sudo dnf install frobby libfrobby libfrobby-devel
```

```
$ brew install Macaulay2/tap/frobby
```

See https://repology.org/project/frobby/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.103 furo: A clean customizable Sphinx documentation theme

Description

A clean customizable Sphinx documentation theme.

License

Copyright (c) 2020 Pradyun Gedam <mail@pradyunsg.me> Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files (the "Software"), to deal in the Software without restriction, including without limitation the rights to use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software, and to permit persons to whom the Software is furnished to do so, subject to the following conditions: The above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software. THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NON-INFRINGEMENT. IN NO EVENT SHALL THE AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE.

Upstream Contact

https://pypi.org/project/furo/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *accessible_pygments: Accessible Pygments styles*
- *beautifulsoup4: Screen-scraping library*
- *pygments: Generic syntax highlighter*
- *sphinx: Python documentation generator*
- *sphinx_basic_ng: Modern skeleton for Sphinx themes*

Version Information

package-version.txt:

```
2025.12.19
```

version_requirements.txt:

```
furo >= 2024.7.18
```

Equivalent System Packages

```
$ sudo pacman -S python-sphinx-furo
```

```
$ conda install furo
```

```
$ sudo apt-get install furo
```

```
$ sudo dnf install python3-furo
```

```
$ sudo pkg install textproc/py-furo
```

```
$ sudo emerge dev-python/furo
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.104 gap: Groups, Algorithms, Programming - a system for computational discrete algebra

Description

GAP is a system for computational discrete algebra, with particular emphasis on Computational Group Theory. GAP provides a programming language, a library of thousands of functions implementing algebraic algorithms written in the GAP language as well as large data libraries of algebraic objects. See also the overview and the description of the mathematical capabilities. GAP is used in research and teaching for studying groups and their representations, rings, vector spaces, algebras, combinatorial structures, and more. The system, including source, is distributed freely. You can study and easily modify or extend it for your special use.

This is a stripped-down version of GAP. The databases, which are architecture-independent, are in a separate package.

Upstream Contact

https://www.gap-system.org

Mailing list at https://mail.gap-system.org/mailman/listinfo/gap

Type

standard

Dependencies

- $(MP_LIBRARY)
- *ncurses: Classic terminal output library*
- *readline: Command line editing library*

Version Information

package-version.txt:

```
4.15.1
```

Equivalent System Packages

```
$ sudo pacman -S gap
```

```
$ conda install gap-defaults
```

```
$ sudo apt-get install gap libgap-dev
```

```
$ sudo dnf install gap gap-core gap-devel gap-libs libgap xgap  \
      gap-pkg-ace gap-pkg-aclib gap-pkg-alnuth gap-pkg-anupq  \
      gap-pkg-atlasrep gap-pkg-autodoc gap-pkg-automata gap-pkg-autpgrp  \
      gap-pkg-browse gap-pkg-caratinterface gap-pkg-circle  \
      gap-pkg-congruence gap-pkg-crisp gap-pkg-crypting  \
      gap-pkg-crystcat gap-pkg-curlinterface gap-pkg-cvec  \
      gap-pkg-datastructures gap-pkg-digraphs gap-pkg-edim  \
      gap-pkg-ferret gap-pkg-fga gap-pkg-fining gap-pkg-float  \
      gap-pkg-format gap-pkg-forms gap-pkg-fplsa gap-pkg-fr  \
      gap-pkg-francy gap-pkg-genss gap-pkg-groupoids gap-pkg-grpconst  \
      gap-pkg-images gap-pkg-io gap-pkg-irredsol gap-pkg-json  \
      gap-pkg-jupyterviz gap-pkg-lpres gap-pkg-nq gap-pkg-openmath  \
      gap-pkg-orb gap-pkg-permut gap-pkg-polenta gap-pkg-polycyclic  \
      gap-pkg-primgrp gap-pkg-profiling gap-pkg-radiroot gap-pkg-recog  \
      gap-pkg-resclasses gap-pkg-scscp gap-pkg-semigroups  \
      gap-pkg-singular gap-pkg-smallgrp gap-pkg-smallsemi  \
      gap-pkg-sophus gap-pkg-spinsym gap-pkg-standardff gap-pkg-tomlib  \
      gap-pkg-transgrp gap-pkg-transgrp-data gap-pkg-utils gap-pkg-uuid  \
      gap-pkg-xmod gap-pkg-zeromqinterface
```

```
$ sudo pkg install math/gap
```

```
$ sudo emerge sci-mathematics/gap
```

```
$ brew install dimpase/gap/gap
```

```
$ nix-env --install gap
```

```
$ sudo xbps-install gap-devel
```

See https://repology.org/project/gap/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.105 gap3: A minimal distribution of GAP 3 containing packages that have no equivalent in GAP 4

Description

This package installs Jean Michel's pre-packaged GAP3, which is a minimal GAP3 distribution containing packages that have no equivalent in GAP4.

Below is the full description from Jean Michel's webpage (accessed 23 July 2015).

> A pre-packaged GAP3 with everything you need
>
> To help people who are just interested in GAP3 because they need a package which has not been ported to GAP4, I have prepared an easy-to install minimal GAP3 distribution containing an up-to-date versions of the packages:
>
> anusq, arep, autag, chevie, cryst, dce, grim, matrix, meataxe, monoid, nq, pcqa, sisyphos, specht, ve, vkcurve.
>
> These packages have been chosen since most have no equivalent in GAP4. They are autoloaded when starting gap.

This distribution includes only partial lists of small groups, 2-groups, 3-groups, character tables from the Atlas and tables of marks. It does not include either the packages:

anupq, grape, kbmag, xgap, cohomolo, gliss, guava, xmod

which have some equivalent in GAP4. You can get these extra features at

http://www.math.rwth-aachen.de/~Frank.Luebeck/gap/GAP3

In this distribution:

- The on-line help includes the documentation of the included packages.
- The html documentation (htm/index.html) also does.
- The manual (manual.pdf) also does.

License

Most parts of the GAP distribution, including the core part of the GAP system, are distributed under the terms of the GNU General Public License (see http://www.gnu.org/licenses/gpl.html or the file GPL in the etc directory of the GAP installation).

SPKG Maintainers

- Christian Stump <christian.stump@gmail.com>

Upstream Contact

Jean Michel <jmichel@math.jussieu.fr> http://webusers.imj-prg.fr/~jean.michel/

Special Update/Build Instructions

The difference between the distributed tarball and Jean Michel's original tarball also contains the binaries

Patches

None

Type

experimental

Dependencies

Version Information

package-version.txt:

```
04jul17
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.106 gap_jupyter: Jupyter kernel for GAP

Description

Jupyter kernel for GAP

This wrapper-kernel is a Jupyter kernel for the GAP Computer Algebra System based on the same ideas as the bash wrapper kernel.

License

3-Clause BSD License

Upstream Contact

- https://github.com/gap-packages/jupyter-gap

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *gap: Groups, Algorithms, Programming - a system for computational discrete algebra*
- *ipython: Interactive computing environment with an enhanced interactive Python shell*

Version Information

package-version.txt:

```
0.9
```

version_requirements.txt:

```
gap_jupyter >=0.9
```

Equivalent System Packages

```
$ conda install gap
```

```
$ sudo dnf install gap-pkg-jupyterkernel
```

See https://repology.org/project/gap-jupyterkernel/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.107 gap_packages: A collection of GAP packages

Description

Several "official" and "undeposited" GAP packages available from https://www.gap-system.org/Packages/packages.html

Installing this SPKG will install the corresponding GAP packages, but before you can use them in Sage, they still have to be loaded into either the GAP interface or libgap:

```
sage: gap.eval('LoadPackage("Grape")')  # optional - gap_packages
'true'
sage: libgap.LoadPackage("Grape")       # optional - gap_packages
true
```

```
>>> from sage.all import *
>>> gap.eval('LoadPackage("Grape")')  # optional - gap_packages
'true'
>>> libgap.LoadPackage("Grape")       # optional - gap_packages
true
```

Those correspond to:

```
gap> LoadPackage("Grape");
```

within the GAP interface and libgap, respectively.

Upstream Contact

Mailing list at https://mail.gap-system.org/mailman/listinfo/gap

Dependencies

- GAP (a standard spkg)

TODO

The crystallographic group packages are untested/untestable. They rely on polymake and the dependency "cryst" is missing. This needs to be cleaned up.

Notes

A brief description of each package follows:

cohomolo - The cohomolo package is a GAP interface to some C programs for computing Schur multipliers and covering groups of finite groups and first and second cohomology groups of finite groups acting on finite modules. (Author: Max Horn, Markus Pfeiffer)

CoReLG - Contains functionality for working with real semisimple Lie algebras. (Author: Heiko Dietrich, Paolo Faccin, Willem Adriaan de Graaf)

crime - package to compute the cohomology ring of finite p-groups, induced maps, and Massey products. (Author: Marcus Bishop)

crypting - The crypting package provides some cryptographic primitives so that the JupyterKernel package works. (Authors: Markus Pfeiffer and The GAP Team)

cryst - Computing with crystallographic groups (Authors: Bettina Eick, Franz Gähler, Werner Nickel)

CTblLib - The GAP Character Table Library (Author: Thomas Breuer)

datastructures - The datastructures package provides some standard data structures. (Authors: Markus Pfeiffer, Max Horn, Christopher Jefferson and Steve Linton)

DESIGN is a package for classifying, partitioning and studying block designs. (Author: Leonard H. Soicher)

Digraphs - GAP package containing methods for graphs, digraphs, and multidigraphs. (Authors: Jan De Beule, Julius Jonusas, James Mitchell, Wilf A. Wilson, Michael Young, Marina Anagnostopoulou-Merkouri, Finn Buck, Stuart Burrell, Graham Campbell, Raiyan Chowdhury, Reinis Cirpons, Ashley Clayton, Tom Conti-Leslie, Joseph Edwards, Luna Elliott, Isuru Fernando, Ewan Gilligan, Sebastian Gutsche, Samantha Harper, Max Horn, Christopher Jefferson, Olexandr Konovalov, Hyeokjun Kwon, Andrea Lee, Saffron McIver, Michael Orlitzky, Matthew Pancer, Markus Pfeiffer, Daniel Pointon, Lea Racine, Christopher Russell, Artur Schaefer, Isabella Scott, Kamran Sharma, Finn Smith, Ben Spiers, Maria Tsalakou, Meike Weiss, Murray Whyte and Fabian Zickgraf)

FactInt is a package providing routines for factoring integers, in particular:

- Pollard's p-1
- Williams' p+1
- Elliptic Curves Method (ECM)
- Continued Fraction Algorithm (CFRAC)
- Multiple Polynomial Quadratic Sieve (MPQS)

(Author: Stefan Kohl)

GAPDoc is a package containing a definition of a structure for GAP documentation, based on XML. It also contains conversion programs for producing text-, DVI-, PDF- or HTML-versions of such documents, with hyperlinks if possible. (Authors: Frank Luebeck, Max Neunhoeffer)

GBNP - The GBNP package provides algorithms for computing Grobner bases of noncommutative polynomials with coefficients from a field implemented in GAP and with respect to the "total degree first then lexicographical" ordering. Further provided are some variations, such as a weighted and truncated version and a tracing facility. The word "algorithm" is to be interpreted loosely here: in general one cannot expect such an algorithm to terminate, as it would imply solvability of the word problem for finitely presented (semi)groups. (Authors: A.M. Cohen, J.W. Knopper)

GRAPE is a package for computing with graphs and groups, and is primarily designed for constructing and analysing graphs related to groups, finite geometries, and designs. (Author: Leonard H. Soicher)

GUAVA is included here, and with Sage standard.

HAP (Homological Algebra Programming) is a GAP package providing some functions for group cohomology computation. (Author: Graham Ellis)

HAPcryst - an extension package for HAP, which allows for group cohomology computation for a wider class of groups. (Author: Marc Roeder)

hecke - Provides functions for calculating decomposition matrices of Hecke algebras of the symmetric groups and q-Schur algebras. Hecke is a port of the GAP 3 package Specht 2.4 to GAP 4. (Author: Dmitriy Traytel)

IO - as its name suggests, provides bindings for GAP to the lower levels of Input/Output functionality in the C library. (Authors: Max Neunhöffer and Max Horn)

LAGUNA - this package provides functionality for calculation of the normalized unit group of the modular group algebra of the finite p-group and for investigation of Lie algebra associated with group algebras and other associative algebras. (Authors :Victor Bovdi, Alexander Konovalov, Richard Rossmanith, Csaba Schneider)

liealgdb - A database of Lie algebras (Author: Serena Cicalo', Willem Adriaan de Graaf, Csaba Schneider)

LiePRing - Database and algorithms for Lie p-rings (Author: Michael Vaughan-Lee, Bettina Eick)

LieRing - contains functionality for working with finitely presented Lie rings and the Lazard correspondence. (Author: Serena Cicalo', Willem Adriaan de Graaf)

LINS - provides an algorithm for computing the normal subgroups of a finitely presented group up to some given index bound. (Author: Friedrich Rober)

loops - Provides researchers in nonassociative algebra with a computational tool that integrates standard notions of loop theory with libraries of loops and group-theoretical algorithms of GAP. The package also expands GAP toward nonassociative structures. (Authors: Gabor Nagy, Petr Vojtechovsky)

mapclass - The package calculates the mapping class group orbits for a given finite group. (Authors: Adam James, Kay Magaard, Sergey Shpectorov, Helmut Volklein)

nq - This package provides access to the ANU nilpotent quotient program for computing nilpotent factor groups of finitely presented groups. (Authors: Max Horn and Werner Nickel)

orb - This package is about enumerating orbits in various ways. (Authors: Juergen Mueller, Max Neunhöffer, Felix Noeske and Max Horn)

polymake - an interface with the (standalone) polymake program used by HAPcryst. (Author: Marc Roeder)

qpa - Quivers and Path Algebras provides data structures and algorithms for doing computations with finite dimensional quotients of path algebras, and finitely generated modules over such algebras. The current version of the QPA package has data structures for quivers, quotients of path algebras, and modules, homomorphisms and complexes of modules over quotients of path algebras. (Authors: Edward Green, Oeyvind Solberg)

quagroup - Contains functionality for working with quantized enveloping algebras of finite-dimensional semisimple Lie algebras. (Author: Willem Adriaan de Graaf)

repsn - The package provides GAP functions for computing characteristic zero matrix representations of finite groups. (Author: Vahid Dabbaghian)

Semigroups - split out as a separate optional spkg

sla - a package for doing computations with simple Lie algebras (Author: Willem Adriaan de Graaf)

SONATA ("System Of Nearrings And Their Applications") is a package which constructs finite nearrings and related objects. (Authors: Erhard Aichinger, Franz Binder, Jürgen Ecker, Peter Mayr, Christof Noebauer)

TORIC is a GAP package for computing with toric varieties. (Author: David Joyner)

Type

optional

Dependencies

- $(SAGERUNTIME)
- *gap: Groups, Algorithms, Programming - a system for computational discrete algebra*
- *planarity: Planarity-related graph algorithms*

Version Information

package-version.txt:

```
4.15.1
```

Equivalent System Packages

```
$ conda install gap
```

```
$ sudo dnf install gap-pkg-cohomolo gap-pkg-corelg gap-pkg-crime  \
      gap-pkg-cryst gap-pkg-ctbllib gap-pkg-design gap-pkg-factint  \
      GAPDoc gap-pkg-gbnp gap-pkg-grape gap-pkg-guava gap-pkg-hap  \
```

(continues on next page)

(continued from previous page)

```
    gap-pkg-hapcryst gap-pkg-hecke gap-pkg-laguna gap-pkg-liealgdb  \
    gap-pkg-liepring gap-pkg-liering gap-pkg-loops gap-pkg-mapclass  \
    gap-pkg-polymaking gap-pkg-qpa gap-pkg-quagroup gap-pkg-repsn  \
    gap-pkg-sla gap-pkg-sonata gap-pkg-toric
```

See https://repology.org/project/gap/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.108 gast: Python AST that abstracts the underlying Python version

Description

Python AST that abstracts the underlying Python version

License

BSD 3-Clause

Upstream Contact

https://pypi.org/project/gast/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
0.7.0
```

version_requirements.txt:

```
gast
```

Equivalent System Packages

```
$ conda install gast
```

```
$ sudo dnf install python3-gast
```

```
$ sudo emerge dev-python/gast
```

```
$ sudo xbps-install python3-gast
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.109 gc: The Boehm-Demers-Weiser conservative garbage collector

Description

The Boehm-Demers-Weiser conservative garbage collector.

License

- MIT-style (https://github.com/bdwgc/bdwgc/blob/master/LICENSE)

Upstream Contact

- Ivan Maidanski

Webpage: - https://github.com/bdwgc/bdwgc/ - https://www.hboehm.info/gc/

Special Update/Build Instructions

None.

Type

standard

Dependencies

- *libatomic_ops: Access hardware-provided atomic memory update operations*

Version Information

package-version.txt:

```
8.2.12
```

Equivalent System Packages

```
$ apk add gc-dev
```

```
$ sudo pacman -S gc
```

```
$ conda install bdw-gc
```

```
$ sudo apt-get install libgc-dev
```

```
$ sudo dnf install gc gc-devel
```

```
$ sudo pkg install devel/boehm-gc devel/boehm-gc-threaded
```

```
$ sudo emerge dev-libs/boehm-gc
```

```
$ brew install bdw-gc
```

```
$ sudo port install boehmgc
```

```
$ sudo zypper install 'pkgconfig(bdw-gc)'
```

```
$ sudo slackpkg install gc
```

```
$ sudo xbps-install gc-devel
```

See https://repology.org/project/boehm-gc/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.110 gcc: The GNU Compiler Collection or other suitable C and C++ compilers

Description

This package represents the required C and C++ compilers.

- Sage's classical build checks for a C compiler that can compile C99 code.
- It also checks for a C++ compiler with C++11 support.
- GCC (GNU Compiler Collection) and Clang (LLVM) are both supported.

The required Fortran compiler is represented by the package `gfortran`.

You can pass the names of compilers to use to `./configure` using the environment variables `CC`, `CXX`, and *`FC`*, for C, C++, and Fortran compilers, respectively.

For example, if your C compiler is `clang`, your C++ compiler is `clang++`, and your Fortran compiler is `flang`, then you would need to run:

```
$ ./configure CC=clang CXX=clang++ FC=flang
```

Vendor and versions of the C and C++ compilers should match.

Users of older Linux distributions should upgrade their systems before attempting to install Sage from source. In particular, users on `ubuntu` should use `ubuntu-jammy` (22.04) or a newer release.

The minimum supported GCC version is 10.3. The following example uses matching version `15` C, C++, and Fortran compilers. On `ubuntu-jammy`, these packages are available from `ppa:ubuntu-toolchain-r/test`:

```
$ sudo add-apt-repository ppa:ubuntu-toolchain-r/test
$ sudo apt-get update
$ sudo apt-get install gcc-15 g++-15 gfortran-15
```

If these packages are already available from the standard repositories of your release, omit the `add-apt-repository` command. After installation, select the compilers explicitly when configuring Sage:

```
$ ./configure CC=gcc-15 CXX=g++-15 FC=gfortran-15
```

This package uses the non-standard default `configure --with-system-gcc=force`, giving an error at `configure` time when no suitable system compilers are configured.

You can override this using `./configure --without-system-gcc`. In this case, Sage builds and installs the GNU Compiler Collection, including the C, C++ and Fortran compiler. This is not recommended. You will need suitable C and

C++ compilers from which GCC can bootstrap itself. There are some known problems with old assemblers, in particular when building the `ecm` and `fflas_ffpack` packages. You should ensure that your assembler understands all instructions for your processor. On Linux, this means you need a recent version of `binutils` (not provided by an SPKG); on macOS you need a recent version of Xcode.

(Installing the `gfortran` SPKG becomes a no-op in this case.)

Building Sage from source on Apple Silicon (M1, M2, M3, M4; arm64) requires the use of Apple's Command Line Tools, and those tools include a suitable compiler. Sage's `gcc` SPKG is not suitable for Apple Silicon; building it will likely fail.

License

GPL version 2 or version 3

Upstream Contact

https://gcc.gnu.org/

Type

standard

Dependencies

- $(MP_LIBRARY)
- *mpc: Arithmetic of complex numbers with arbitrarily high precision and correct rounding*
- *mpfr: Multiple-precision floating-point computations with correct rounding*

Version Information

package-version.txt:

```
14.2.0
```

Equivalent System Packages

```
$ sudo pacman -S gcc
```

```
$ sudo apt-get install gcc g++
```

```
$ sudo dnf install gcc gcc-c++ gcc-gfortran
```

```
$ sudo pkg install lang/gcc9
```

```
$ brew install gcc
```

```
$ sudo zypper install gcc-c++
```

```
$ sudo xbps-install gcc
```

See https://repology.org/project/gcc/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.111 gdb: The GNU Project debugger

Description

GDB, the GNU Project debugger, allows you to see what is going on “inside” another program while it executes – or what another program was doing at the moment it crashed.

License

GPL v3+

Upstream Contact

http://www.gnu.org/software/gdb/

Type

optional

Dependencies

Version Information

Equivalent System Packages

```
$ apk add gdb
```

No package needed.

```
$ sudo dnf install gdb
```

```
$ brew install gdb
```

```
$ sudo port install gdb
```

```
$ sudo zypper install gdb
```

```
$ sudo xbps-install gdb
```

See https://repology.org/project/gdb/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.112 gengetopt: getopt_long parser generator

Description

GNU Gengetopt converts a textual description of your program’s arguments and options into a getopt_long() parser in C (or C++).

Website: https://www.gnu.org/software/gengetopt/

License

GPL-3+ (https://www.gnu.org/software/gengetopt/LICENSE)

Type

standard

Dependencies

Version Information

package-version.txt:

```
2.23
```

Equivalent System Packages

```
$ apk add gengetopt
```

```
$ conda install gengetopt
```

```
$ sudo apt-get install gengetopt
```

```
$ sudo dnf install gengetopt
```

```
$ sudo emerge dev-util/gengetopt
```

```
$ brew install gengetopt
```

```
$ nix-env --install gengetopt
```

```
$ sudo xbps-install gengetopt
```

See https://repology.org/project/gengetopt/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.113 gf2x: Fast arithmetic in GF(2)[x] and searching for irreducible/primitive trinomials

Description

gf2x is a C/C++ software package containing routines for fast arithmetic in GF(2)[x] (multiplication, squaring, GCD) and searching for irreducible/primitive trinomials.

Website: https://gitlab.inria.fr/gf2x/gf2x

License

- GNU GPLv2+.

Upstream Contact

- Richard Brent
- Pierrick Gaudry
- Emmanuel Thomé
- Paul Zimmermann

Type

standard

Dependencies

Version Information

package-version.txt:

```
1.3.0
```

Equivalent System Packages

```
$ apk add gf2x-dev
```

```
$ sudo pacman -S gf2x
```

```
$ conda install gf2x
```

```
$ sudo apt-get install libgf2x-dev
```

```
$ sudo dnf install gf2x gf2x-devel
```

```
$ sudo pkg install math/gf2x
```

```
$ sudo zypper install 'pkgconfig(gf2x)'
```

```
$ sudo xbps-install gf2x-devel
```

See https://repology.org/project/gf2x/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.114 gfan: Groebner fans and tropical varieties

Description

Gfan is a software package for computing Groebner fans and tropical varieties.

These are polyhedral fans associated to polynomial ideals. The maximal cones of a Groebner fan are in bijection with the marked reduced Groebner bases of its defining ideal. The software computes all marked reduced Groebner bases of an ideal. Their union is a universal Groebner basis. The tropical variety of a polynomial ideal is a certain subcomplex of the Groebner fan. Gfan contains algorithms for computing this complex for general ideals and specialized algorithms for tropical curves, tropical hypersurfaces and tropical varieties of prime ideals. In addition to the above core functions the package contains many tools which are useful in the study of Groebner bases, initial ideals and tropical geometry. The

full list of commands can be found in Appendix B of the manual. For ordinary Groebner basis computations Gfan is not competitive in speed compared to programs such as CoCoA, Singular and Macaulay2.

License

- GPL version 2 or version 3 (according to the gfan website)

Upstream Contact

Anders Nedergaard Jensen

https://users-math.au.dk/jensen/software/gfan/gfan.html

Special Update/Build Instructions

Remove the doc, homepage, and examples subdirectories, which take up most of the space.

Type

standard

Dependencies

- $(MP_LIBRARY)
- *cddlib: Double description method for polyhedral representation conversion*

Version Information

package-version.txt:

```
0.6.2.p1
```

Equivalent System Packages

```
$ sudo pacman -S gfan
```

```
$ conda install gfan
```

```
$ sudo apt-get install gfan
```

```
$ sudo dnf install gfan
```

```
$ sudo pkg install math/gfan
```

```
$ sudo emerge sci-mathematics/gfan
```

```
$ brew install Macaulay2/tap/gfan
```

```
$ nix-env --install gfan
```

```
$ sudo zypper install gfan
```

```
$ sudo xbps-install gfan
```

See https://repology.org/project/gfan/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.115 gfortran: Fortran compiler from the GNU Compiler Collection

Description

This package represents the required Fortran compiler.

Officially we support `gfortran` from GNU Compiler Collection (GCC). It has also been reported that using `flang` (from LLVM) might work.

You can pass the names of compilers to use to `./configure` using the environment variables `CC`, `CXX`, and *`FC`*, for C, C++, and Fortran compilers, respectively.

For example, if your C compiler is `clang`, your C++ compiler is `clang++`, and your Fortran compiler is `flang`, then you would need to run:

```
$ ./configure CC=clang CXX=clang++ FC=flang
```

License

GPL version 2 or version 3

Upstream Contact

http://gcc.gnu.org/

Special Update/Build Instructions

None.

Type

standard

Dependencies

- $(MP_LIBRARY)
- *mpc: Arithmetic of complex numbers with arbitrarily high precision and correct rounding*
- *mpfr: Multiple-precision floating-point computations with correct rounding*

Version Information

package-version.txt:

```
14.2.0
```

Equivalent System Packages

```
$ apk add gfortran
```

```
$ sudo pacman -S gcc-fortran
```

```
$ conda install fortran-compiler
```

```
$ sudo apt-get install gfortran
```

```
$ sudo dnf install gcc-gfortran
```

```
$ sudo pkg install lang/gcc9
```

```
$ brew install gfortran
```

```
$ sudo port install gcc10 +gfortran
```

```
$ sudo zypper install gcc-fortran
```

```
$ sudo slackpkg install gcc-gfortran
```

```
$ sudo xbps-install gcc-fortran
```

See https://repology.org/project/gfortran/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.116 giac: A general purpose computer algebra system

Description

- Giac is a general purpose Computer algebra system by Bernard Parisse. It consists of:
- a C++ library (libgiac).
- a command line interpreter (icas or giac).
- the built of the FLTK-based GUI (xcas) has been disabled in the spkg-install file.
- The english documentation will be installed in:

  $SAGE_LOCAL/share/giac/doc/en/cascmd_en/index.html
- Author's website with debian, ubuntu, macosx, windows package:

  http://www-fourier.ujf-grenoble.fr/~parisse/giac.html
- The Freebsd port is math/giacxcas

Licence

GPLv3+

Note: except the french html documentation which is freely redistributable for non commercial only purposes. This doc has been removed in the Sage package, see spkg-src

Upstream Contact

- Bernard Parisse: http://www-fourier.ujf-grenoble.fr/~parisse/giac.html
- Source file (giac-x.y.z-t.tar.gz) in:
  http://www-fourier.ujf-grenoble.fr/~parisse/debian/dists/stable/main/source/

Dependencies

- gettext, readline
- giac will benefit of ntl, pari, mpfr, gsl, lapack but they should be already installed by sage.
- giac can also benefit of mpfi for arithmetic on intervals.
- The Documentation is pre-built, hevea or latex or … are not needed to install the package.

Special Update/Build Instructions

- Use spkg-src to update this package

Type

optional

Dependencies

- $(MP_LIBRARY)
- *cliquer: Routines for clique searching*
- *curl: Multiprotocol data transfer library and utility*
- *ecm: Elliptic curve method for integer factorization*
- *glpk: GNU Linear Programming Kit*
- *gsl: The GNU Scientific Library*
- *libpng: Bitmap image support*
- *mpfi: Multiple precision interval arithmetic library based on MPFR*
- *mpfr: Multiple-precision floating-point computations with correct rounding*
- *ntl: A library for doing number theory*
- *pari: Computer algebra system for fast computations in number theory*
- *readline: Command line editing library*

Version Information

package-version.txt:

```
1.9.0.15p0
```

Equivalent System Packages

```
$ sudo pacman -S giac
```

```
$ conda install giac
```

```
$ sudo apt-get install libgiac-dev xcas
```

```
$ sudo dnf install giac giac-devel
```

```
$ sudo pkg install math/giacxcas
```

```
$ nix-env --install giac
```

```
$ sudo zypper install giac-devel
```

```
$ sudo xbps-install giac-devel
```

See https://repology.org/project/giac/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.117 git: Version control system

Description

> Git is a fast, scalable, distributed revision control system with an unusually rich command set that provides both high-operations and full access to internals.

- `man git`

Upstream Contact

- Website: https://git-scm.com/

Type

optional

Dependencies

Version Information

Equivalent System Packages

```
$ sudo pacman -S git
```

```
$ conda install git
```

```
$ sudo apt-get install git
```

```
$ sudo dnf install git
```

```
$ sudo pkg install devel/git
```

```
$ brew install git
```

```
$ sudo port install git
```

```
$ sudo zypper install git
```

```
$ sudo slackpkg install git
```

```
$ sudo xbps-install git
```

See https://repology.org/project/git/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.118 github_cli: Command-line interface for GitHub

Description

`gh` is GitHub on the command line. It brings pull requests, issues, and other GitHub concepts to the terminal next to where you are already working with `git` and your code.

License

MIT

Upstream Contact

https://github.com/cli/cli

Type

optional

Dependencies

Version Information

Equivalent System Packages

```
$ apk add github-cli
```

```
$ sudo pacman -S github-cli
```

```
$ conda install gh
```

```
$ sudo apt-get install gh
```

```
$ sudo dnf install gh
```

```
$ sudo pkg install devel/gh
```

```
$ sudo emerge dev-util/github-cli
```

```
$ brew install gh
```

```
$ sudo port install gh
```

```
$ nix-env --install gh
```

```
$ sudo zypper install gh
```

```
$ sudo xbps-install github-cli
```

See https://repology.org/project/github-cli/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.119 gitpython: GitPython is a python library used to interact with Git repositories

Description

GitPython is a python library used to interact with Git repositories

License

BSD

Upstream Contact

https://pypi.org/project/GitPython/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

requirements.txt:

```
GitPython
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.120 givaro: C++ library for arithmetic and algebraic computations

Description

Givaro is a C++ library for arithmetic and algebraic computations. Its main features are implementations of the basic arithmetic of many mathematical entities: Primes fields, Extensions Fields, Finite Fields, Finite Rings, Polynomials, Algebraic numbers, Arbitrary precision integers and rationals (C++ wrappers over gmp) It also provides data-structures and templated classes for the manipulation of basic algebraic objects, such as vectors, matrices (dense, sparse, structured), univariate polynomials (and therefore recursive multivariate).

Website: https://casys.gricad-pages.univ-grenoble-alpes.fr/givaro/

SPKG Repository: https://bitbucket.org/malb/givaro-spkg

License

- GNU GPL

Upstream Contact

- Clement Pernet

Type

standard

Dependencies

- $(MP_LIBRARY)

Version Information

package-version.txt:

```
4.2.1.p1
```

Equivalent System Packages

```
$ conda install givaro
```

```
$ sudo apt-get install libgivaro-dev
```

```
$ sudo dnf install givaro givaro-devel
```

```
$ sudo pkg install math/givaro
```

```
$ sudo emerge sci-libs/givaro
```

```
$ brew install Macaulay2/tap/givaro
```

```
$ nix-env --install givaro
```

```
$ sudo zypper install 'pkgconfig(givaro)'
```

```
$ sudo xbps-install givaro-devel
```

See https://repology.org/project/givaro/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.121 glpk: GNU Linear Programming Kit

Description

The GLPK (GNU Linear Programming Kit) package is intended for solving large-scale linear programming (LP), mixed integer programming (MIP), and other related problems. It is a set of routines written in ANSI C and organized in the form of a callable library.

GLPK supports the GNU MathProg modelling language, which is a subset of the AMPL language.

The GLPK package includes the following main components:

- primal and dual simplex methods
- primal-dual interior-point method
- branch-and-cut method
- translator for GNU MathProg
- application program interface (API)
- stand-alone LP/MIP solver

License

The GLPK package is GPL version 3.

Upstream Contact

GLPK is currently being maintained by:

- Andrew Makhorin (mao@gnu.org, mao@mai2.rcnet.ru)

http://www.gnu.org/software/glpk/#maintainer

Special Update/Build Instructions

- `configure` doesn't support specifying the location of the GMP library to use; only `--with-gmp[=yes]` or `--with-gmp=no` are valid options. (So we *have to* add Sage's include and library directories to `CPPFLAGS` and `LDFLAGS`, respectively.)
- Do we need the `--disable-static`? The stand-alone solver presumably runs faster when built with a static library; also other (stand-alone) programs using it would. (Instead, we should perhaps use `--enable-static --enable-shared` to go safe.)

Type

standard

Dependencies

- $(MP_LIBRARY)

Version Information

package-version.txt:

```
5.0.p1
```

Equivalent System Packages

```
$ apk add glpk-dev
```

```
$ sudo pacman -S glpk
```

```
$ conda install glpk
```

```
$ sudo apt-get install glpk-utils libglpk-dev
```

```
$ sudo dnf install glpk glpk-devel glpk-utils
```

```
$ sudo pkg install math/glpk
```

```
$ sudo emerge sci-mathematics/glpk
```

```
$ brew install glpk
```

```
$ sudo port install glpk
```

```
$ nix-env --install glpk
```

```
$ sudo zypper install glpk glpk-devel
```

install the following packages: glpk

```
$ sudo xbps-install glpk-devel
```

See https://repology.org/project/glpk/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.122 glucose: A SAT solver

Description

Glucose is a SAT solver.

Citing its website:

The name of the solver is a contraction of the concept of "glue clauses", a particular kind of clauses that glucose detects and preserves during search. Glucose is heavily based on Minisat, so please do cite Minisat also if you want to cite Glucose.

License

- nonparallel glucose: MIT
- parallel glucose-syrup: MIT modified with:

  The parallel version of Glucose (all files modified since Glucose 3.0 releases, 2013) cannot be used in any competitive event (sat competitions/evaluations) without the express permission of the authors (Gilles Audemard / Laurent Simon). This is also the case for any competitive event using Glucose Parallel as an embedded SAT engine (single core or not).

Upstream Contact

Website: http://www.labri.fr/perso/lsimon/glucose/

Special Update/Build Instructions

None.

Type

optional

Dependencies

Version Information

package-version.txt:

```
4.1
```

Equivalent System Packages

See https://repology.org/project/glucose/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.123 gmp: Library for arbitrary precision arithmetic

Description

GMP is a free library for arbitrary precision arithmetic, operating on signed integers, rational numbers, and floating-point numbers. There is no practical limit to the precision except the ones implied by the available memory in the machine GMP runs on. GMP has a rich set of functions, and the functions have a regular interface.

License

- LGPL V3

Upstream Contact

- http://gmplib.org

Type

standard

Dependencies

Version Information

package-version.txt:

```
6.3.0
```

Equivalent System Packages

```
$ apk add gmp-dev
```

```
$ conda install gmp
```

```
$ sudo apt-get install libgmp-dev
```

```
$ sudo dnf install gmp gmp-devel
```

```
$ sudo pkg install math/gmp
```

```
$ sudo emerge dev-libs/gmp
```

```
$ brew install gmp
```

```
$ sudo port install gmp
```

```
$ sudo zypper install gmp-devel
```

install the following packages: libgmp

```
$ sudo slackpkg install gmp
```

```
$ sudo xbps-install gmp-devel gmpxx-devel
```

See https://repology.org/project/gmp/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.124 gmpy2: Python interface to GMP/MPIR, MPFR, and MPC

Description

GMP/MPIR, MPFR, and MPC interface to Python

gmpy2 is a C-coded Python extension module that supports multiple-precision arithmetic. In addition to supporting GMP or MPIR for multiple-precision integer and rational arithmetic, gmpy2 adds support for the MPFR (correctly rounded real floating-point arithmetic) and MPC (correctly rounded complex floating-point arithmetic) libraries.

License

LGPL-3.0+

Upstream Contact

https://pypi.org/project/gmpy2/

Type

standard

Dependencies

- $(MP_LIBRARY)
- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *hypothesis: A library for property-based testing*
- *mpc: Arithmetic of complex numbers with arbitrarily high precision and correct rounding*
- *mpfr: Multiple-precision floating-point computations with correct rounding*

Version Information

package-version.txt:

```
2.3.0
```

version_requirements.txt:

```
gmpy2 >=2.1.5
```

Equivalent System Packages

```
$ sudo pacman -S python-gmpy2
```

```
$ conda install gmpy2
```

```
$ sudo apt-get install python3-gmpy2
```

```
$ sudo dnf install python3-gmpy2
```

```
$ sudo pkg install math/py-gmpy2
```

```
$ sudo emerge dev-python/gmpy
```

```
$ sudo port install py-gmpy2
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-gmpy2'
```

```
$ sudo xbps-install python3-gmpy2
```

See https://repology.org/project/python:gmpy2/versions, https://repology.org/project/python:gmpy2-devel/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.125 gnulib: Modules imported from Gnulib

This script package represents the modules imported into the Sage source tree from Gnulib.

Upstream Contact

https://www.gnu.org/software/gnulib/

Type

standard

Dependencies

Version Information

package-version.txt:

```
f9b39c4e337f1dc0dd07c4f3985c476fb875d799
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.126 gnumake_tokenpool: Jobclient and jobserver for the GNU make tokenpool protocol

Description

The project has implementations in multiple languages.

We only install the implementation in Python.

License

MIT

Upstream Contact

- https://github.com/milahu/gnumake-tokenpool (upstream)

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
0.0.9
```

version_requirements.txt:

```
gnumake-tokenpool >= 0.0.4
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.127 gp2c: A compiler for translating GP routines to C

Description

The gp2c compiler is a package for translating GP routines into the C programming language, so that they can be compiled and used with the PARI system or the GP calculator.

License

GPL version 2+

Upstream Contact

- http://pari.math.u-bordeaux.fr/

Dependencies

- PARI
- Perl

Type

optional

Dependencies

- *pari: Computer algebra system for fast computations in number theory*

Version Information

package-version.txt:

```
0.0.14
```

Equivalent System Packages

```
$ sudo apt-get install pari-gp2c
```

```
$ sudo dnf install gp2c
```

```
$ sudo pkg install math/gp2c
```

```
$ sudo emerge sci-mathematics/gp2c
```

```
$ sudo zypper install gp2c
```

```
$ sudo xbps-install gp2c
```

See https://repology.org/project/gp2c/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.128 graph_genus: Graph genus algorithms

Description

graph-genus provides Python bindings for the PAGE and MultiGenus graph embedding algorithms, together with utilities for producing rotation systems and drawings of embeddings.

License

GNU General Public License, version 2

Upstream Contact

https://pypi.org/project/graph-genus/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

requirements.txt:

```
graph-genus
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.129 graphs: A database of combinatorial graphs

Description

A database of graphs. Created by Emily Kirkman based on the work of Jason Grout. Since April 2012 it also contains the ISGCI graph database.

Upstream Contact

- https://jasongrout.org/graph_database
- For ISGCI:

  H.N. de Ridder (hnridder@graphclasses.org)
- For Andries Brouwer's database:

  The data is taken from from Andries E. Brouwer's website (https://www.win.tue.nl/~aeb/). Anything related to the data should be reported to him directly (aeb@cwi.nl)

  The code used to parse the data and create the .json file is available at https://github.com/nathanncohen/strongly_regular_graphs_database.

Type

standard

Dependencies

Version Information

package-version.txt:

```
20210214.p0
```

Equivalent System Packages

```
$ sudo pacman -S sage-data-graphs
```

```
$ conda install sagemath-db-graphs
```

See https://repology.org/project/sagemath-graphs/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.130 graphviz: Graph visualization software

Description

Graphviz is open source graph visualization software. It has several main graph layout programs. They take descriptions of graphs in a simple text language, and make diagrams in several useful formats.

License

Eclipse Public License 1.0

Upstream Contact

https://graphviz.org/about/

Type

optional

Dependencies

Version Information

Equivalent System Packages

```
$ apk add graphviz
```

```
$ sudo pacman -S graphviz
```

```
$ conda install graphviz
```

```
$ sudo apt-get install graphviz
```

```
$ sudo dnf install graphviz
```

```
$ sudo pkg install graphics/graphviz
```

```
$ brew install graphviz
```

```
$ sudo port install graphviz
```

```
$ nix-env --install graphviz
```

```
$ sudo zypper install graphviz
```

```
$ sudo xbps-install graphviz graphviz-devel
```

See https://repology.org/project/graphviz/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.131 gsl: The GNU Scientific Library

Description

The GNU Scientific Library

Website: http://www.gnu.org/software/gsl/

From the website above: The GNU Scientific Library (GSL) is a numerical library for C and C++ programmers. It is free software under the GNU General Public License.

The library provides a wide range of mathematical routines such as random number generators, special functions and least-squares fitting. There are over 1000 functions in total with an extensive test suite. If the variable SAGE_CHECK is exported to the value "yes" when building Sage, GSL's test suite is run.

License

- GPL V3

Upstream Contact

- http://www.gnu.org/software/gsl/

GSL mailing lists:

- Bug-gsl <bug-gsl@gnu.org> mailing list – bug reports for the GNU Scientific Library should be sent to bug-gsl@gnu.org
- Help-gsl <help-gsl@gnu.org> users mailing list – for questions about installation, how GSL works and how it is used, or general questions concerning GSL.
- Info-gsl <info-gsl@gnu.org> mailing list – announcements of new releases are made there.

Special Update/Build Instructions

Type

standard

Dependencies

- $(BLAS)

Version Information

package-version.txt:

```
2.7.1
```

Equivalent System Packages

```
$ apk add gsl-dev
```

```
$ sudo pacman -S gsl
```

```
$ conda install gsl
```

```
$ sudo apt-get install libgsl-dev
```

```
$ sudo dnf install gsl gsl-devel
```

```
$ sudo pkg install math/gsl
```

```
$ sudo emerge sci-libs/gsl
```

```
$ brew install gsl
```

```
$ sudo port install gsl
```

```
$ nix-env --install gsl
```

```
$ sudo zypper install 'pkgconfig(gsl)'
```

install the following packages: libgsl

```
$ sudo slackpkg install gsl
```

```
$ sudo xbps-install gsl-devel
```

See https://repology.org/project/gsl/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.132 h11: A pure-Python, bring-your-own-I/O implementation of HTTP/1.1

Description

A pure-Python, bring-your-own-I/O implementation of HTTP/1.1

License

MIT

Upstream Contact

https://pypi.org/project/h11/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
0.16.0
```

version_requirements.txt:

```
h11
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.133 hatchling: Modern, extensible Python build backend

Description

Provides a plugin for Hatch that uses your VCS (like Git) to determine project versions.

License

MIT

Upstream Contact

https://pypi.org/project/hatch-vcs/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *hatchling: Modern, extensible Python build backend*

Version Information

package-version.txt:

```
25.1.0
```

version_requirements.txt:

```
hatch-fancy-pypi-readme
```

Equivalent System Packages

```
$ apk add py3-hatch-fancy-pypi-readme
```

```
$ sudo pacman -S python-hatch-fancy-pypi-readme
```

```
$ conda install hatchling
```

```
$ sudo apt-get install python-hatch-fancy-pypi-readme
```

```
$ sudo dnf install python-hatch-fancy-pypi-readme
```

```
$ sudo pkg install devel/py-hatch-fancy-pypi-readme
```

```
$ sudo emerge dev-python/hatch-fancy-pypi-readme
```

```
$ sudo pkg_add devel/py-hatch-fancy-pypi-readme
```

```
$ sudo zypper install python-hatch-fancy-pypi-readme
```

```
$ sudo xbps-install hatch-fancy-pypi-readme
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.134 hatchling: Modern, extensible Python build backend

Description

Provides a plugin for Hatch that uses your VCS (like Git) to determine project versions.

License

MIT

Upstream Contact

https://pypi.org/project/hatch-vcs/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *hatchling: Modern, extensible Python build backend*
- *setuptools_scm: Python build system extension to obtain package version from version control*

Version Information

package-version.txt:

```
0.5.0
```

version_requirements.txt:

```
hatch-vcs
```

Equivalent System Packages

```
$ apk add py3-hatch-vcs
```

```
$ sudo pacman -S python-hatch-vcs
```

```
$ conda install hatch-vcs
```

```
$ sudo apt-get install hatch-vcs
```

```
$ sudo dnf install python-hatch-vcs
```

```
$ sudo pkg install devel/py-hatch-vcs
```

```
$ sudo emerge dev-python/hatch-vcs
```

```
$ sudo pkg_add devel/py-hatch-vcs
```

```
$ sudo zypper install python-hatch_vcs
```

```
$ sudo xbps-install hatch-vcs
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.135 hatchling: Modern, extensible Python build backend

Description

Modern, extensible Python build backend

License

MIT

Upstream Contact

https://pypi.org/project/hatchling/

Type

standard

Dependencies

- $(PYTHON)
- *packaging: Core utilities for Python packages*
- *pathspec: Utility library for gitignore style pattern matching of file paths*
- *pip: Tool for installing and managing Python packages*
- *pluggy: Plugin and hook calling mechanisms for python*
- *trove_classifiers: Canonical source for classifiers on PyPI (pypi.org)*

Version Information

package-version.txt:

```
1.29.0
```

version_requirements.txt:

```
hatchling
```

Equivalent System Packages

```
$ apk add py3-hatchling
```

```
$ sudo pacman -S python-hatchling
```

```
$ conda install hatchling
```

```
$ sudo apt-get install hatchling
```

```
$ sudo dnf install python-hatchling
```

```
$ sudo pkg install devel/py-hatchling
```

```
$ sudo emerge dev-python/hatchling
```

```
$ sudo pkg_add devel/py-hatchling
```

```
$ sudo zypper install python-hatchling
```

```
$ sudo xbps-install hatchling
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.136 highs: Linear optimization solver

Description

HiGHS is a high performance serial and parallel solver for large scale sparse linear optimization problems of the form:

$$\min c^T x \quad \text{subject to} \quad L \leq Ax \leq U,\ l \leq x \leq u$$

where c, x, l, u are vectors, L, U are constraint bound vectors, and A is a matrix.

HiGHS has implementations of the dual revised simplex method, primal and dual revised simplex solvers, an interior point solver, and a MIP solver.

This package builds the HiGHS C/C++ library with C API headers for use in Sage's numerical backends.

License

- MIT License

Upstream Contact

- https://github.com/ERGO-Code/HiGHS
- https://www.highs.dev
- Email: hello@highs.dev

Dependencies

- CMake (build dependency)

Special Update/Build Instructions

The library is built using CMake and provides the C API interface (highs_c_api.h) for integration with Sage's backend system.

Type

standard

Dependencies

- *cmake: A cross-platform build system generator*

Version Information

package-version.txt:

```
1.14.0
```

Equivalent System Packages

```
$ sudo pacman -S highs
```

```
$ conda install highs
```

```
$ sudo apt-get install libhighs-dev highs
```

```
$ sudo dnf install coin-or-HiGHS coin-or-HiGHS-devel
```

```
$ sudo pkg install math/highs
```

```
$ sudo emerge sci-mathematics/highs
```

```
$ brew install highs
```

```
$ sudo port install HiGHS
```

```
$ nix-env --install highs
```

```
$ sudo zypper install HiGHS HiGHS-devel
```

See https://repology.org/project/highs/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.137 highspy: Python interface to HiGHS optimizer

Description

HiGHSpy provides a thin set of pybind11 wrappers to the HiGHS linear optimization suite. HiGHS is a high-performance solver for linear programming (LP), mixed-integer programming (MIP), and quadratic programming (QP) models.

License

MIT

Upstream Contact

https://github.com/ERGO-Code/HiGHS

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cmake: A cross-platform build system generator*
- *numpy: Package for scientific computing with Python*
- *pybind11: Create Python bindings to C++ code*
- *scikit_build_core: Build backend for CMake based projects*

Version Information

package-version.txt:

```
1.13.1
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.138 httpcore: Minimal low-level HTTP client

Description

Minimal low-level HTTP client

License

Upstream Contact

https://pypi.org/project/httpcore/

Type

standard

Dependencies

- $(PYTHON)
- *certifi: Python package for providing Mozilla's CA Bundle*
- *h11: A pure-Python, bring-your-own-I/O implementation of HTTP/1.1*
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
1.0.9
```

version_requirements.txt:

```
httpcore
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.139 httpx: The next generation HTTP client.

Description

The next generation HTTP client.

License

Upstream Contact

https://pypi.org/project/httpx/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *anyio: High level compatibility layer for multiple asynchronous event loop implementations*
- *certifi: Python package for providing Mozilla's CA Bundle*
- *httpcore: Minimal low-level HTTP client*
- *idna: Internationalized Domain Names in Applications (IDNA)*
- *sniffio: Sniff out which async library your code is running under*

Version Information

package-version.txt:

```
0.28.1
```

version_requirements.txt:

```
httpx
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.140 hypothesis: A library for property-based testing

Description

A library for property-based testing

License

MPL-2.0

Upstream Contact

https://pypi.org/project/hypothesis/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *attrs: Decorator for Python classes with attributes*
- *sortedcontainers: sorted collections library*

Version Information

package-version.txt:

```
6.151.10
```

version_requirements.txt:

```
hypothesis>=6.123.0
```

Equivalent System Packages

```
$ sudo pacman -S python-hypothesis
```

```
$ sudo apt-get install python-hypothesis
```

```
$ sudo dnf install python-hypothesis
```

```
$ sudo pkg install devel/py-hypothesis
```

```
$ sudo emerge dev-python/hypothesis
```

```
$ sudo zypper install python-hypothesis
```

```
$ sudo xbps-install python3-hypothesis
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.141 iconv: Library for language/country-dependent character encodings

Description

GNU libiconv is a library that is used to enable different languages, with different characters to be handled properly.

License

- GPL 3 and LGPL 3. So we can safely link against the library in Sage.

Upstream Contact

- http://www.gnu.org/software/libiconv/
- Bug reports to bug-gnu-libiconv@gnu.org

Type

standard

Dependencies

Version Information

Equivalent System Packages

```
$ brew install libiconv
```

```
$ sudo port install libiconv
```

install the following packages: libiconv

See https://repology.org/project/libiconv/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.142 idna: Internationalized Domain Names in Applications (IDNA)

Description

Internationalized Domain Names in Applications (IDNA)

License

BSD-3-Clause

Upstream Contact

https://pypi.org/project/idna/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
3.11
```

version_requirements.txt:

```
idna
```

Equivalent System Packages

```
$ sudo pacman -S python-idna
```

```
$ conda install idna
```

```
$ sudo apt-get install python3-idna
```

```
$ sudo dnf install python3-idna
```

```
$ sudo emerge dev-python/idna
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-idna'
```

```
$ sudo xbps-install python3-idna
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.143 igraph: A library for creating and manipulating graphs

Description

igraph is a library for creating and manipulating graphs. It is intended to be as powerful (ie. fast) as possible to enable the analysis of large graphs.

License

GPL version 2

Upstream Contact

http://igraph.org/c/

Dependencies

igraph can optionally use libxml2 for providing a GraphML importer.

Special Update/Build Instructions

Type

optional

Dependencies

- $(BLAS)
- $(MP_LIBRARY)
- *cmake: A cross-platform build system generator*
- *glpk: GNU Linear Programming Kit*

Version Information

package-version.txt:

```
1.0.1
```

Equivalent System Packages

```
$ sudo pacman -S igraph
```

```
$ conda install igraph
```

```
$ sudo apt-get install libigraph-dev
```

```
$ sudo dnf install igraph igraph-devel
```

```
$ sudo pkg install math/igraph
```

```
$ sudo emerge dev-libs/igraph
```

```
$ brew install igraph
```

```
$ sudo port install igraph
```

```
$ sudo xbps-install igraph-devel
```

See https://repology.org/project/igraph/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.144 imagemagick: A collection of tools and libraries for many image file formats

Description

A collection of tools and libraries for many image file formats

License

Copyright [yyyy] [name of copyright owner]

Licensed under the ImageMagick License (the "License"); you may not use this file except in compliance with the License. You may obtain a copy of the License at

https://imagemagick.org/script/license.php

Unless required by applicable law or agreed to in writing, software distributed under the License is distributed on an "AS IS" BASIS, WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied. See the License for the specific language governing permissions and limitations under the License.

Upstream Contact

http://www.imagemagick.org/

Type

optional

Dependencies

Version Information

Equivalent System Packages

```
$ apk add imagemagick
```

```
$ sudo pacman -S imagemagick
```

```
$ conda install imagemagick
```

```
$ sudo apt-get install imagemagick
```

```
$ sudo dnf install ImageMagick
```

```
$ sudo pkg install graphics/ImageMagick7
```

```
$ brew install imagemagick
```

```
$ sudo port install ImageMagick
```

```
$ nix-env --install imagemagick
```

```
$ sudo zypper install ImageMagick
```

```
$ sudo xbps-install ImageMagick
```

See https://repology.org/project/imagemagick/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.145 imagesize: Getting image size from png/jpeg/jpeg2000/gif file

Description

Getting image size from png/jpeg/jpeg2000/gif file

License

MIT

Upstream Contact

https://pypi.org/project/imagesize/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
2.0.0
```

version_requirements.txt:

```
imagesize
```

Equivalent System Packages

```
$ conda install imagesize
```

```
$ sudo dnf install python3-imagesize
```

```
$ sudo emerge dev-python/imagesize
```

```
$ sudo port install py-imagesize
```

```
$ sudo xbps-install python3-imagesize
```

See https://repology.org/project/python:imagesize/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.146 iml: Integer Matrix Library

Description

IML is a free library of C source code which implements algorithms for computing exact solutions to dense systems of linear equations over the integers. IML is designed to be used with the ATLAS/BLAS library and GMP bignum library.

Written in portable C, IML can be used on both 32-bit and 64-bit machines. It can be called from C++.

Website: https://www.cs.uwaterloo.ca/~astorjoh/iml.html

License

- GPLv2+

Upstream Contact

- Zhuliang Chen z4chen@uwaterloo.ca
- Arne Storjohann astorjoh@uwaterloo.ca

Special Update/Build Instructions

- As of version 1.0.4, you need to repackage the upstream tarball using the spkg-src script because there was a bugfix version of 1.0.4 reposted upstream without version number bump.

Patches

- examples.patch: Modified some of the examples.

Type

standard

Dependencies

- $(BLAS)
- $(MP_LIBRARY)

Version Information

package-version.txt:

```
1.0.4p2.p2
```

Equivalent System Packages

```
$ sudo pacman -S iml
```

```
$ conda install iml
```

```
$ sudo apt-get install libiml-dev
```

```
$ sudo dnf install iml iml-devel
```

```
$ sudo pkg install math/iml
```

```
$ sudo emerge sci-libs/iml
```

```
$ brew install dimpase/tap/iml
```

```
$ nix-env --install iml
```

```
$ sudo zypper install iml-devel
```

```
$ sudo xbps-install iml-devel
```

See https://repology.org/project/iml/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.147 info: stand-alone Info documentation reader

Description

GNU Info is the stand-alone "info" reader that is part of the GNU Texinfo suite of tools. Several packages (Maxima, Singular, …) install documentation in "info" format, which can be read either with Emacs, the stand-alone "info" reader, and some other software. In particular, the interactive help system of `singular_console()` uses the `info` program in environments in which a web browser is not available; if `info` is not installed, it falls back to a basic pager with limited capabilities.

Website: https://www.gnu.org/software/texinfo/manual/info-stnd/info-stnd.html

License

GPL-3+ (`info/*.c` comments in the source repository)

Type

standard

Dependencies

- *ncurses: Classic terminal output library*

Version Information

package-version.txt:

```
7.2
```

Equivalent System Packages

```
$ conda install texinfo
```

```
$ sudo apt-get install texinfo
```

```
$ sudo dnf install texinfo info
```

```
$ sudo emerge sys-apps/texinfo
```

```
$ brew install texinfo
```

```
$ sudo port install texinfo
```

```
$ nix-env --install texinfo
```

```
$ sudo zypper install texinfo
```

```
$ sudo xbps-install texinfo
```

See https://repology.org/project/texinfo/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.148 iniconfig: Brain-dead simple config-ini parsing

Description

Brain-dead simple config-ini parsing

License

Upstream Contact

https://pypi.org/project/iniconfig/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
2.3.0
```

version_requirements.txt:

```
iniconfig
```

Equivalent System Packages

```
$ sudo pacman -S python-iniconfig
```

```
$ sudo apt-get install python3-iniconfig
```

```
$ sudo dnf install python3-iniconfig
```

```
$ sudo emerge dev-python/iniconfig
```

```
$ sudo xbps-install python3-iniconfig
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.149 ipykernel: IPython Kernel for Jupyter

Description

This package provides the IPython kernel for Jupyter.

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *appnope: Disable App Nap on macOS >= 10.9*
- *comm: Jupyter Python Comm implementation, for usage in ipykernel, xeus-python etc.*
- *debugpy: Implementation of the Debug Adapter Protocol for Python*
- *hatchling: Modern, extensible Python build backend*
- *ipython: Interactive computing environment with an enhanced interactive Python shell*
- *jupyter_client: Jupyter protocol implementation and client libraries*
- *jupyter_core: Jupyter core package*
- *matplotlib_inline: Inline Matplotlib backend for Jupyter*
- *nest_asyncio: Patch asyncio to allow nested event loops*

- *packaging: Core utilities for Python packages*
- *psutil: Cross-platform lib for process and system monitoring in Python.*
- *pyzmq: Python bindings for the zeromq networking library*
- *tornado: Python web framework and asynchronous networking library*
- *traitlets: Traitlets Python configuration system*

Version Information

package-version.txt:

```
7.2.0
```

version_requirements.txt:

```
ipykernel >=5.2.1
```

Equivalent System Packages

```
$ sudo pacman -S python-ipykernel
```

```
$ conda install ipykernel
```

```
$ sudo apt-get install python3-ipykernel
```

```
$ sudo dnf install python3-ipykernel
```

```
$ sudo pkg install devel/py-ipykernel
```

```
$ sudo emerge dev-python/ipykernel
```

```
$ sudo port install py-ipykernel
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-ipykernel'
```

```
$ sudo xbps-install python3-ipython_ipykernel
```

See https://repology.org/project/python:ipykernel/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.150 ipympl: Matplotlib Jupyter Extension

Description

Matplotlib Jupyter Extension

License

BSD License

Upstream Contact

https://pypi.org/project/ipympl/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *ipython: Interactive computing environment with an enhanced interactive Python shell*
- *ipython_genutils: Vestigial utilities from IPython*
- *ipywidgets: Interactive HTML widgets for Jupyter notebooks and the IPython kernel*
- *matplotlib: Python 2D plotting library*
- *numpy: Package for scientific computing with Python*
- *pillow: Python Imaging Library*
- *traitlets: Traitlets Python configuration system*

Version Information

package-version.txt:

```
0.10.0
```

version_requirements.txt:

```
ipympl
```

Equivalent System Packages

```
$ sudo pacman -S python-ipympl
```

```
$ conda install ipympl
```

```
$ sudo pkg install devel/py-ipympl
```

```
$ sudo port install py-ipympl
```

See https://repology.org/project/python:ipympl/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.151 ipython: Interactive computing environment with an enhanced interactive Python shell

Description

Interactive computing environment with an enhanced interactive Python shell

From the IPython website:

IPython is a multiplatform, Free Software project (BSD licensed) that offers:

- An enhanced Python shell designed for efficient interactive work. It includes many enhancements over the default Python shell, including the ability for controlling interactively all major GUI toolkits in a non-blocking manner.
- A library to build customized interactive environments using Python as the basic language (but with the possibility of having extended or alternate syntaxes).
- A system for interactive distributed and parallel computing (this is part of IPython's new development).

License

BSD

Upstream Contact

http://ipython.org

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *decorator: Python library providing decorators*
- *ipython_pygments_lexers: Pygments lexers for IPython code*
- *jedi: Static analysis tool providing IDE support for Python*
- *matplotlib_inline: Inline Matplotlib backend for Jupyter*
- *pexpect: Python module for controlling and automating other programs*
- *pickleshare: A 'shelve' like datastore with concurrency support*
- *prompt_toolkit: Interactive command lines for Python*
- *pygments: Generic syntax highlighter*
- *pyzmq: Python bindings for the zeromq networking library*
- *stack_data: Extract data from python stack frames and tracebacks for informative displays*
- *tornado: Python web framework and asynchronous networking library*
- *traitlets: Traitlets Python configuration system*
- *wcwidth: Measures the displayed width of unicode strings in a terminal*

Version Information

package-version.txt:

```
9.16.0
```

version_requirements.txt:

```
ipython >=7.13.0
```

Equivalent System Packages

```
$ sudo pacman -S ipython
```

```
$ conda install ipython
```

```
$ sudo apt-get install python3-ipython
```

```
$ sudo dnf install python3-ipython
```

```
$ sudo pkg install devel/ipython
```

```
$ sudo emerge dev-python/ipython
```

```
$ brew install ipython
```

```
$ sudo port install py-ipython
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-ipython'
```

```
$ sudo xbps-install python3-ipython
```

See https://repology.org/project/ipython/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.152 ipython_genutils: Vestigial utilities from IPython

Description

Vestigial utilities from IPython

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
0.2.0
```

version_requirements.txt:

```
ipython_genutils >=0.2.0
```

Equivalent System Packages

```
$ conda install ipython_genutils
```

```
$ sudo dnf install python3-ipython_genutils
```

```
$ sudo emerge dev-python/ipython_genutils
```

```
$ sudo port install py-ipython_genutils
```

```
$ sudo xbps-install python3-ipython_genutils
```

See https://repology.org/project/python:ipython-genutils/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.153 ipython_pygments_lexers: Pygments lexers for IPython code

Description

IPython Pygments Lexers defines a variety of Pygments lexers for highlighting IPython code, including the IPython console session and IPython's special syntax.

License

BSD-3-Clause

Upstream Contact

https://pypi.org/project/ipython-pygments-lexers/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *pygments: Generic syntax highlighter*

Version Information

package-version.txt:

```
1.1.1
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.154 ipywidgets: Interactive HTML widgets for Jupyter notebooks and the IPython kernel

Description

Interactive HTML widgets for Jupyter notebooks and the IPython kernel.

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *comm: Jupyter Python Comm implementation, for usage in ipykernel, xeus-python etc.*
- *ipykernel: IPython Kernel for Jupyter*
- *ipython: Interactive computing environment with an enhanced interactive Python shell*
- *jupyterlab_widgets: Jupyter interactive widgets for JupyterLab*
- *traitlets: Traitlets Python configuration system*
- *widgetsnbextension: Jupyter interactive widgets for Jupyter Notebook*

Version Information

package-version.txt:

```
8.1.8
```

version_requirements.txt:

```
ipywidgets >=7.5.1
```

Equivalent System Packages

```
$ conda install 'ipywidgets>=7.5.1'
```

```
$ sudo dnf install python3-ipywidgets
```

```
$ sudo emerge dev-python/ipywidgets
```

```
$ sudo port install py-ipywidgets
```

```
$ sudo xbps-install python3-jupyter_ipywidgets
```

See https://repology.org/project/python:ipywidgets/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.155 isl: Sets and relations of integer points bounded by affine constraints

Description

isl is a thread-safe C library for manipulating sets and relations of integer points bounded by affine constraints. The descriptions of the sets and relations may involve both parameters and existentially quantified variables. All computations are performed in exact integer arithmetic using GMP.

License

isl is released under the MIT license, but depends on the LGPL GMP library.

Upstream Contact

- http://groups.google.com/group/isl-development

Citation

```
@incollection{Verdoolaege2010isl,
   author = {Verdoolaege, Sven},
   title = {isl: An Integer Set Library for the Polyhedral Model},
   booktitle = {Mathematical Software - ICMS 2010},
   series = {Lecture Notes in Computer Science},
   editor = {Fukuda, Komei and Hoeven, Joris and Joswig, Michael and
   Takayama, Nobuki},
   publisher = {Springer},
   isbn = {978-3-642-15581-9},
   pages = {299-302},
   volume = {6327},
   year = {2010}
}
```

Type

optional

Dependencies

- $(MP_LIBRARY)

Version Information

package-version.txt:

```
0.20
```

Equivalent System Packages

```
$ conda install isl
```

```
$ sudo apt-get install libisl-dev
```

```
$ sudo dnf install isl isl-devel
```

```
$ sudo pkg install devel/isl
```

```
$ sudo emerge dev-libs/isl
```

```
$ brew install isl
```

```
$ sudo port install isl
```

```
$ sudo zypper install 'pkgconfig(isl)'
```

```
$ sudo xbps-install isl-devel
```

See https://repology.org/project/isl/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.156 isoduration: Operations with ISO 8601 durations

Description

Operations with ISO 8601 durations

License

Upstream Contact

https://pypi.org/project/isoduration/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *arrow: Better dates & times for Python*

Version Information

package-version.txt:

```
20.11.0
```

version_requirements.txt:

```
isoduration
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.157 jedi: Static analysis tool providing IDE support for Python

Description

Jedi is a static analysis tool for Python that is typically used in IDEs/editors plugins. Jedi has a focus on autocompletion and goto functionality. Other features include refactoring, code search and finding references.

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *parso: Python Parser*

Version Information

package-version.txt:

```
0.19.2
```

version_requirements.txt:

```
jedi >=0.17.0
```

Equivalent System Packages

```
$ conda install jedi
```

```
$ sudo dnf install python3-jedi
```

```
$ sudo emerge dev-python/jedi
```

```
$ sudo port install py-jedi
```

```
$ sudo xbps-install python3-jedi
```

See https://repology.org/project/jedi/versions, https://repology.org/project/python:jedi/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.158 jinja2: General purpose template engine for Python

Description

Jinja2 is a library for Python 2.4 and onwards that is designed to be flexible, fast and secure.

If you have any exposure to other text-based template languages, such as Smarty or Django, you should feel right at home with Jinja2. It's both designer and developer friendly by sticking to Python's principles and adding functionality useful for templating environments.

License

BSD-3-Clause

Upstream Contact

https://pypi.org/project/Jinja2/

Type

standard

Dependencies

- $(PYTHON)
- *markupsafe: Safely add untrusted strings to HTML/XML markup*
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
3.1.6
```

version_requirements.txt:

```
jinja2 >=3.0
```

Equivalent System Packages

```
$ conda install jinja2
```

```
$ sudo apt-get install python3-jinja2
```

```
$ sudo dnf install python3-jinja2
```

```
$ sudo emerge dev-python/jinja
```

```
$ sudo port install py-jinja2
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-jinja2'
```

```
$ sudo xbps-install python3-Jinja2
```

See https://repology.org/project/python:jinja2/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.159 jmol: Java viewer for chemical structures in 3D

Description

Java viewer for chemical structures in 3D.

This provides files necessary for Jmol (java).

This package does not install JSmol (javascript), which upstream bundles with Jmol.

License

GPLv2+

Upstream Contact

- http://jmol.sourceforge.net
- Bob Hanson
- e-mail: hansonr@stolaf.edu
- Homepage: https://www.stolaf.edu/people/hansonr/
- Development page: https://github.com/BobHanson/Jmol-SwingJS
- Download page: https://sourceforge.net/projects/jmol/files/Jmol/

Dependencies

No build-time dependencies.

The commandline jmol requires java at runtime.

Special Build Instructions

To avoid depending on `unzip` at build time, we have to repack the tarball, see `spkg-src`. We take the opportunity to remove some unnecessary subdirectories, see http://wiki.jmol.org/index.php/Jmol_JavaScript_Object#In_detail

Type

optional

Dependencies

Version Information

package-version.txt:

```
14.29.52
```

Equivalent System Packages

```
$ sudo pacman -S jmol
```

```
$ conda install jmol
```

```
$ sudo dnf install jmol
```

```
$ sudo port install jmol
```

```
$ nix-env --install jmol
```

```
$ sudo zypper install jmol
```

```
$ sudo xbps-install jmol
```

See https://repology.org/project/jmol/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.160 joblib: Lightweight pipelining with Python functions

Description

Joblib is a set of tools to provide lightweight pipelining in Python. It provides utilities for transparent disk-caching of function output values and easy simple parallel computing. It is optimized to be fast and robust for large data, with specific optimizations for numpy arrays.

License

BSD-3-Clause

Upstream Contact

https://pypi.org/project/joblib/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
1.5.3
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.161 json5: Python implementation of the JSON5 data format

Description

Python implementation of the JSON5 data format

License

Apache

Upstream Contact

https://pypi.org/project/json5/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
0.14.0
```

version_requirements.txt:

```
json5
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.162 jsonpointer: Identify specific nodes in a JSON document (RFC 6901)

Description

Identify specific nodes in a JSON document (RFC 6901)

License

Modified BSD License

Upstream Contact

https://pypi.org/project/jsonpointer/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
3.1.1
```

version_requirements.txt:

```
jsonpointer
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.163 jsonschema: Python implementation of JSON Schema

Description

jsonschema is an implementation of JSON Schema for Python

License

MIT License

Upstream Contact

Home page: http://github.com/Julian/jsonschema

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *attrs: Decorator for Python classes with attributes*
- *fqdn: Validates fully-qualified domain names against RFC 1123, so that they are acceptable to modern bowsers*
- *isoduration: Operations with ISO 8601 durations*
- *jsonpointer: Identify specific nodes in a JSON document (RFC 6901)*
- *jsonschema_specifications: The JSON Schema meta-schemas and vocabularies, exposed as a Registry*
- *referencing: JSON Referencing + Python*
- *rpds_py: Python bindings to Rust's persistent data structures*
- *uri_template: RFC 6570 URI Template Processor*
- *webcolors: Library for working with the color formats defined by HTML and CSS*

Version Information

package-version.txt:

```
4.26.0
```

version_requirements.txt:

```
jsonschema >=3.2.0
```

Equivalent System Packages

```
$ sudo pacman -S python-jsonschema
```

```
$ conda install jsonschema
```

```
$ sudo apt-get install python3-jsonschema
```

```
$ sudo dnf install python3-jsonschema
```

```
$ sudo emerge dev-python/jsonschema
```

```
$ sudo port install py-jsonschema
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-jsonschema'
```

```
$ sudo xbps-install python3-jsonschema
```

See https://repology.org/project/python:jsonschema/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.164 jsonschema_specifications: The JSON Schema meta-schemas and vocabularies, exposed as a Registry

Description

The JSON Schema meta-schemas and vocabularies, exposed as a Registry

License

MIT

Upstream Contact

https://pypi.org/project/jsonschema-specifications/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *referencing: JSON Referencing + Python*

Version Information

package-version.txt:

```
2025.9.1
```

version_requirements.txt:

```
jsonschema-specifications
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.165 jupymake: A Python wrapper for the polymake shell

Description

The Python module JuPyMake provides an interface to polymake.

License

- GPL v2

Upstream Contact

https://github.com/polymake/JuPyMake

Special Update/Build Instructions

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *polymake: Computations with polyhedra, fans, simplicial complexes, matroids, graphs, tropical hypersurfaces*

Version Information

package-version.txt:

```
0.9
```

version_requirements.txt:

```
jupymake >=0.9
```

Equivalent System Packages

```
$ sudo dnf install python3-jupymake
```

See https://repology.org/project/jupymake/versions, https://repology.org/project/python:jupymake/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.166 jupyter_client: Jupyter protocol implementation and client libraries

Description

Jupyter protocol implementation and client libraries

License

BSD 3-Clause License

Upstream Contact

https://pypi.org/project/jupyter-client/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *dateutil: Extensions to the standard Python module datetime*
- *jupyter_core: Jupyter core package*
- *pyzmq: Python bindings for the zeromq networking library*
- *tornado: Python web framework and asynchronous networking library*
- *traitlets: Traitlets Python configuration system*

Version Information

package-version.txt:

```
8.8.0
```

version_requirements.txt:

```
jupyter-client
```

Equivalent System Packages

```
$ conda install jupyter_client
```

```
$ sudo dnf install python3-jupyter-client
```

```
$ sudo emerge dev-python/jupyter_client
```

```
$ sudo port install py-jupyter_client
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-jupyter-client'
```

```
$ sudo xbps-install python3-jupyter_client
```

See https://repology.org/project/jupyter-client/versions, https://repology.org/project/python:jupyter-client/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.167 jupyter_core: Jupyter core package

Description

Jupyter core package. A base package on which Jupyter projects rely.

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *hatchling: Modern, extensible Python build backend*
- *platformdirs: Small Python package for determining appropriate platform-specific dirs, e.g. a "user data dir"*
- *traitlets: Traitlets Python configuration system*

Version Information

package-version.txt:

```
5.9.1
```

Equivalent System Packages

```
$ conda install jupyter_core
```

```
$ sudo dnf install python3-jupyter-core
```

```
$ sudo emerge dev-python/jupyter_core
```

```
$ sudo port install py-jupyter_core
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-jupyter-core'
```

```
$ sudo xbps-install python3-jupyter_core
```

See https://repology.org/project/jupyter-core/versions, https://repology.org/project/python:jupyter-core/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.168 jupyter_events: Jupyter Event System library

Description

Jupyter Event System library

License

BSD 3-Clause License

Upstream Contact

https://pypi.org/project/jupyter-events/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *jsonschema: Python implementation of JSON Schema*
- *packaging: Core utilities for Python packages*
- *python_json_logger: Python library adding a json log formatter*
- *pyyaml: YAML parser and emitter for Python*
- *referencing: JSON Referencing + Python*
- *rfc3339_validator: A pure python RFC3339 validator*
- *rfc3986_validator: Pure python rfc3986 validator*
- *traitlets: Traitlets Python configuration system*

Version Information

package-version.txt:

```
0.12.0
```

version_requirements.txt:

```
jupyter-events
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.169 jupyter_jsmol: JSmol viewer widget for Jupyter

Description

JSmol viewer widget for Jupyter

License

MIT

Upstream Contact

https://pypi.org/project/jupyter-jsmol/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *ipywidgets: Interactive HTML widgets for Jupyter notebooks and the IPython kernel*

Version Information

package-version.txt:

```
2022.1.0
```

version_requirements.txt:

```
jupyter-jsmol
```

Equivalent System Packages

```
$ sudo pacman -S jupyter-jsmol
```

```
$ conda install jupyter-jsmol
```

```
$ sudo pkg install science/py-jupyter_jsmol
```

See https://repology.org/project/jupyter-jsmol/versions, https://repology.org/project/python:jupyter-jsmol/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.170 jupyter_lsp: Multi-Language Server WebSocket proxy for Jupyter Notebook/Lab server

Description

Multi-Language Server WebSocket proxy for Jupyter Notebook/Lab server

License

BSD-3-Clause

Upstream Contact

https://pypi.org/project/jupyter-lsp/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *jupyter_server: The backend (core services, APIs, REST endpoints) to Jupyter web applications*

Version Information

package-version.txt:

```
2.3.0
```

version_requirements.txt:

```
jupyter-lsp
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.171 jupyter_server: The backend (core services, APIs, REST endpoints) to Jupyter web applications

Description

The backend, i.e., core services, APIs, and REST endpoints, to Jupyter web applications.

License

BSD 3-Clause License

Upstream Contact

https://pypi.org/project/jupyter-server/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *anyio: High level compatibility layer for multiple asynchronous event loop implementations*
- *argon2_cffi: The secure Argon2 password hashing algorithm*
- *jinja2: General purpose template engine for Python*
- *jupyter_client: Jupyter protocol implementation and client libraries*
- *jupyter_core: Jupyter core package*
- *jupyter_events: Jupyter Event System library*

- *jupyter_server_terminals: A Jupyter Server Extension Providing Terminals*
- *nbconvert: Converting Jupyter Notebooks*
- *nbformat: The Jupyter Notebook format*
- *overrides: Decorator to automatically detect mismatch when overriding a method*
- *packaging: Core utilities for Python packages*
- *platformdirs: Small Python package for determining appropriate platform-specific dirs, e.g. a "user data dir"*
- *prometheus_client: Python client for the systems monitoring and alerting toolkit Prometheus*
- *pyzmq: Python bindings for the zeromq networking library*
- *send2trash: Send file to trash natively under Mac OS X, Windows and Linux*
- *terminado: Tornado websocket backend for the term.js Javascript terminal emulator library*
- *tornado: Python web framework and asynchronous networking library*
- *traitlets: Traitlets Python configuration system*
- *websocket_client: WebSocket client for Python with low level API options*

Version Information

package-version.txt:

```
2.17.0
```

version_requirements.txt:

```
jupyter-server
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.172 jupyter_server_terminals: A Jupyter Server Extension Providing Terminals

Description

A Jupyter Server Extension Providing Terminals.

License

Modified BSD License (also known as New or Revised or 3-Clause BSD)

Upstream Contact

https://pypi.org/project/jupyter-server-terminals/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *terminado: Tornado websocket backend for the term.js Javascript terminal emulator library*

Version Information

package-version.txt:

```
0.5.4
```

version_requirements.txt:

```
jupyter-server-terminals
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.173 jupyter_sphinx: Jupyter Sphinx Extension

Description

Jupyter Sphinx Extension

License

BSD

Upstream Contact

https://pypi.org/project/jupyter-sphinx/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *ipykernel: IPython Kernel for Jupyter*
- *ipython: Interactive computing environment with an enhanced interactive Python shell*
- *ipywidgets: Interactive HTML widgets for Jupyter notebooks and the IPython kernel*
- *nbconvert: Converting Jupyter Notebooks*

- *nbformat: The Jupyter Notebook format*
- *sphinx: Python documentation generator*

Version Information

package-version.txt:

```
0.5.3.p0
```

version_requirements.txt:

```
jupyter-sphinx
```

Equivalent System Packages

```
$ sudo pacman -S python-jupyter-sphinx
```

```
$ conda install jupyter_sphinx
```

```
$ sudo dnf install python3-jupyter-sphinx
```

```
$ sudo pkg install textproc/py-jupyter_sphinx
```

```
$ sudo port install py-jupyter_sphinx
```

```
$ sudo zypper install python-jupyter-sphinx
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.174 jupyterlab: JupyterLab computational environment

Description

JupyterLab computational environment

License

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Semver File License: The semver.py file is from https://github.com/podhmo/python-semver which is licensed under the "MIT" license. See the semver.py file for details.

Upstream Contact

https://pypi.org/project/jupyterlab/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *async_lru: Simple LRU cache for asyncio*
- *httpx: The next generation HTTP client.*
- *ipykernel: IPython Kernel for Jupyter*
- *jinja2: General purpose template engine for Python*
- *jupyter_core: Jupyter core package*
- *jupyter_lsp: Multi-Language Server WebSocket proxy for Jupyter Notebook/Lab server*
- *jupyter_server: The backend (core services, APIs, REST endpoints) to Jupyter web applications*
- *jupyterlab_server: Set of server components for JupyterLab and JupyterLab like applications*
- *notebook_shim: A shim layer for notebook traits and config*
- *packaging: Core utilities for Python packages*
- *tornado: Python web framework and asynchronous networking library*
- *traitlets: Traitlets Python configuration system*

Version Information

package-version.txt:

```
4.5.6
```

version_requirements.txt:

```
jupyterlab
```

Equivalent System Packages

```
$ conda install jupyterlab
```

```
$ sudo dnf install jupyterlab
```

```
$ brew install jupyterlab
```

```
$ sudo port install py-jupyterlab
```

```
$ sudo xbps-install jupyterlab
```

See https://repology.org/project/jupyterlab/versions, https://repology.org/project/python:jupyterlab/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.175 jupyterlab_mathjax2: A MathJax Typesetting provider for JupyterLab 4 and above

Description

A MathJax Typesetting provider for JupyterLab 4 and above

License

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Upstream Contact

https://pypi.org/project/jupyterlab-mathjax2/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
4.0.0
```

version_requirements.txt:

```
jupyterlab-mathjax2
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.176 jupyterlab_pygments: Pygments theme using JupyterLab CSS variables

Description

Pygments theme using JupyterLab CSS variables

License

BSD

Upstream Contact

https://pypi.org/project/jupyterlab-pygments/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *pygments: Generic syntax highlighter*

Version Information

package-version.txt:

```
0.3.0
```

version_requirements.txt:

```
jupyterlab-pygments
```

Equivalent System Packages

```
$ conda install jupyterlab_pygments
```

```
$ sudo dnf install python3-jupyterlab_pygments
```

```
$ sudo emerge dev-python/jupyterlab_pygments
```

```
$ sudo xbps-install python3-jupyterlab_pygments
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.177 jupyterlab_server: Set of server components for JupyterLab and JupyterLab like applications

Description

Set of server components for JupyterLab and JupyterLab like applications

License

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Upstream Contact

https://pypi.org/project/jupyterlab-server/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *babel: Internationalization utilities*
- *jinja2: General purpose template engine for Python*
- *json5: Python implementation of the JSON5 data format*
- *jsonschema: Python implementation of JSON Schema*
- *jupyter_events: Jupyter Event System library*
- *jupyter_server: The backend (core services, APIs, REST endpoints) to Jupyter web applications*
- *packaging: Core utilities for Python packages*
- *requests: An HTTP library for Python*

Version Information

package-version.txt:

```
2.28.0
```

version_requirements.txt:

```
jupyterlab-server
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.178 jupyterlab_widgets: Jupyter interactive widgets for JupyterLab

Description

Jupyter interactive widgets for JupyterLab

License

BSD-3-Clause

Upstream Contact

https://pypi.org/project/jupyterlab-widgets/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
3.0.16
```

version_requirements.txt:

```
jupyterlab-widgets
```

Equivalent System Packages

```
$ sudo pacman -S jupyterlab-widgets
```

```
$ sudo pkg install devel/py-jupyterlab-widgets
```

```
$ sudo port install py-jupyterlab_widgets
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-jupyterlab-widgets'
```

See https://repology.org/project/jupyterlab-widgets/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.179 kenzo: Construct topological spaces and compute homology groups

Description

Kenzo is a package to compute properties (mainly homology groups) of topological spaces. It allows defining spaces created from others by constuctions like loop spaces, classifying spaces and so on.

License

GPL

Upstream Contact

- https://github.com/gheber/kenzo
- https://github.com/miguelmarco/kenzo/

Type

optional

Dependencies

- *ecl: An implementation of the Common Lisp language*

Version Information

package-version.txt:

```
1.1.10
```

Equivalent System Packages

```
$ brew install dimpase/tap/kenzo
```

See https://repology.org/project/kenzo/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.180 khoca: Khoca as pip installable package

Description

Khoca as pip installable package

License

GPLv2+

Upstream Contact

https://pypi.org/project/khoca/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cypari2: Python interface to the number theory library libpari*
- *ipython: Interactive computing environment with an enhanced interactive Python shell*
- sagelib

Version Information

package-version.txt:

```
1.4
```

requirements.txt:

```
khoca
```

version_requirements.txt:

```
khoca
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.181 kissat: SAT solver

Description

From the package README:

```
KISSAT is a "keep it simple and clean bare metal SAT solver" written in C.
It is a port of CaDiCaL back to C with improved data structures, better
scheduling of inprocessing and optimized algorithms and implementation.

Coincidentally 'kissat' also means 'cats' in Finnish.
```

From the website:

```
The Kissat SAT solver is a condensed and improved reimplementation of
CaDiCaL in C.

Kissat won first place in the main track of the SAT Competition 2020 and
first place on unsatisfiable instances.
```

License

MIT license.

Upstream Contact

Website: http://fmv.jku.at/kissat/

Type

optional

Dependencies

Version Information

package-version.txt:

```
3.1.0
```

Equivalent System Packages

```
$ sudo dnf install kissat kissat-devel
```

```
$ sudo emerge sci-mathematics/kissat
```

```
$ nix-env --install kissat
```

See https://repology.org/project/kissat/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.182 kiwisolver: Fast implementation of the Cassowary constraint solver

Description

From https://pypi.org/project/kiwisolver/

A fast implementation of the Cassowary constraint solver

Kiwi is an efficient C++ implementation of the Cassowary constraint solving algorithm. Kiwi is an implementation of the algorithm based on the seminal Cassowary paper. It is not a refactoring of the original C++ solver. Kiwi has been designed from the ground up to be lightweight and fast. Kiwi ranges from 10x to 500x faster than the original Cassowary solver with typical use cases gaining a 40x improvement. Memory savings are consistently > 5x.

In addition to the C++ solver, Kiwi ships with hand-rolled Python bindings.

License

Modified BSD License

Upstream Contact

https://github.com/nucleic/kiwi

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cppy: C++ headers for C extension development*

Version Information

package-version.txt:

```
1.5.0
```

version_requirements.txt:

```
kiwisolver >=1.4.8
```

Equivalent System Packages

```
$ conda install kiwisolver
```

```
$ sudo dnf install python3-kiwisolver
```

```
$ sudo emerge dev-python/kiwisolver
```

```
$ sudo port install py-kiwisolver
```

```
$ sudo xbps-install python3-kiwisolver
```

See https://repology.org/project/python:kiwisolver/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.183 latte_int: Count lattice points, compute volumes, and integrate over convex polytopes

Description

LattE (Lattice point Enumeration) Integrale solves the problems of counting lattice points in and integration over convex polytopes.

License

GPLv2

Upstream Contact

Matthias Köppe, UC Davis, CA, USA

Type

optional

Dependencies

- $(MP_LIBRARY)
- *4ti2: Algebraic, geometric and combinatorial problems on linear spaces*
- *cddlib: Double description method for polyhedral representation conversion*
- *lidia: A library for computational number theory*
- *ntl: A library for doing number theory*

Version Information

package-version.txt:

```
1.7.6
```

Equivalent System Packages

```
$ sudo pacman -S latte-integrale
```

```
$ conda install latte-integrale
```

```
$ sudo dnf install latte-integrale
```

```
$ sudo zypper install latte
```

See https://repology.org/project/latte-integrale/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.184 lcalc: L-function calculator

Description

Michael Rubinstein's L-function calculator.

License

- LGPL V2+

Upstream contact

Michael Rubinstein <mrubinst@uwaterloo.ca>

Sources: http://oto.math.uwaterloo.ca/~mrubinst/L_function_public/L.html

Newer beta version 1.3 (not yet in Sage): http://code.google.com/p/l-calc/

Dependencies

- GMP/MPIR
- MPFR
- PARI
- GNU patch

Special Update/Build Instructions

- There is some garbage in the upstream sources which should be removed:

```
src/include/.Lexplicit_formula.h.swp
src/include/.Lvalue.h.swp
src/include/._.DS_Store
src/include/.DS_Store
src/include/Lexplicit_formula.h.swap.crap
src/include/Lvalue.h.bak
src/src/Makefile.old
src/src/.Makefile.old.swp
src/src/._.DS_Store
src/src/.DS_Store
src/src/.Lcommandline.ggo.swp
src/src/libLfunction.a
```

- We (and apparently also upstream) currently don't build Lcalc's tests (see Makefile), hence there's no spkg-check. This might change in newer upstream versions.
- The original Makefile uses $(CC) to compile C++ (also using $(CCFLAGS)), which it defines to 'g++', and hard-codes 'g++' when linking the shared library. (It should use $(CXX) instead, which might *default* to 'g++'.) We now (lcalc-1.23.p10) patch the Makefile also to use $(CXX) for compiling and linking C++; $(CXX) now *defaults* to 'g++', and $(CC) to 'gcc', but both can be overridden by simply setting their respective environment variables. (Same for $(INSTALL_DIR) btw.)

Patches

- Makefile.patch:

  We change a lot there, since Lcalc doesn't have a 'configure' script, and hence the Makefile is supposed to be edited to customize Lcalc (build options, locations of headers and libraries etc.). Besides that, we

  - put CXXFLAGS into Lcalc's "CCFLAGS" used for compiling C++,
  - remove some stuff involving LDFLAGS1 and LDFLAGS2, setting just LDFLAGS,
  - use $(MAKE) instead of 'make' in the crude build receipts,
  - use CXXFLAG64 when linking the shared library,
  - now use $(CXX) for compiling and linking C++, which *defaults* to 'g++', but can be overridden by setting the environment variable of the same name. ($(CC) now *defaults* to 'gcc', although currently not really used as far as I can see.)
  - $(INSTALL_DIR) can now be overridden by simply setting the environment variable of the same name.

- Lcommon.h.patch:

  Uncomment the definition of lcalc_to_double(const long double& x). (Necessary for GCC >= 4.6.0, cf. #10892.) Comment from there: The reason is the following code horror from src/src/include/Lcommon.h: [...] But somebody who is familiar with the codebase should really rewrite lcalc to not redefine the double() cast, thats just fragile and will sooner or later again fail inside some system headers.

- pari-2.7.patch:

  Various changes to port to newer versions of PARI.

- time.h.patch:

  (Patches src/include/Lcommandline_numbertheory.h) Include also <time.h> in Lcommandline_numbertheory.h (at least required on Cygwin, cf. #9845). This should get reported upstream.

- lcalc-1.23_default_parameters_1.patch: Make Lcalc (1.23) build with GCC 4.9

Type

standard

Dependencies

- *gengetopt: getopt_long parser generator*
- *pari: Computer algebra system for fast computations in number theory*

Version Information

package-version.txt:

```
2.1.0
```

Equivalent System Packages

```
$ sudo pacman -S lcalc
```

```
$ conda install lcalc
```

```
$ sudo apt-get install lcalc liblfunction-dev
```

```
$ sudo dnf install L-function-devel L-function
```

```
$ sudo pkg install math/lcalc
```

```
$ sudo emerge 'sci-mathematics/lcalc[pari]'
```

```
$ nix-env --install lcalc
```

```
$ sudo xbps-install lcalc-devel
```

See https://repology.org/project/lcalc/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.185 libatomic_ops: Access hardware-provided atomic memory update operations

Description

This package provides semi-portable access to hardware-provided atomic memory update operations on a number of architectures.

License

- MIT (core library) + GPL 2.0+ (gpl extension library)

Upstream Contact

https://github.com/bdwgc/libatomic_ops/

Special Update/Build Instructions

None.

Type

standard

Dependencies

Version Information

package-version.txt:

```
7.10.0
```

Equivalent System Packages

```
$ apk add libatomic_ops-dev
```

```
$ sudo pacman -S libatomic_ops
```

```
$ conda install libatomic_ops
```

```
$ sudo apt-get install libatomic-ops-dev
```

```
$ sudo dnf install libatomic_ops libatomic_ops-devel
```

```
$ sudo pkg install devel/libatomic_ops
```

```
$ sudo emerge dev-libs/libatomic_ops
```

```
$ brew install libatomic_ops
```

```
$ sudo port install libatomic_ops
```

```
$ sudo zypper install 'pkgconfig(atomic_ops)'
```

```
$ sudo slackpkg install libatomic_ops
```

```
$ sudo xbps-install libatomic_ops-devel
```

See https://repology.org/project/libatomic-ops/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.186 libbraiding: Computing with braids

Description

libbraiding is a library to compute several properties of braids, including centralizer and conjugacy check.

License

GPLv3+

SPKG Maintainers

- Miguel Marco

Upstream Contact

Miguel Marco (mmarco@unizar.es)

Type

standard

Dependencies

Version Information

package-version.txt:

```
1.3.2
```

Equivalent System Packages

```
$ sudo pacman -S libbraiding
```

```
$ conda install libbraiding
```

```
$ sudo apt-get install libbraiding-dev
```

```
$ sudo dnf install libbraiding-devel
```

```
$ sudo pkg install math/libbraiding
```

```
$ sudo emerge sci-libs/libbraiding
```

```
$ nix-env --install libbraiding
```

```
$ sudo zypper install libbraiding-devel
```

```
$ sudo xbps-install libbraiding-devel
```

See https://repology.org/project/libbraiding/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.187 libffi: A portable foreign-function interface library

Description

Compilers for high level languages generate code that follow certain conventions. These conventions are necessary, in part, for separate compilation to work. One such convention is the "calling convention". The "calling convention" is essentially a set of assumptions made by the compiler about where function arguments will be found on entry to a function. A "calling convention" also specifies where the return value for a function is found.

Some programs may not know at the time of compilation what arguments are to be passed to a function. For instance, an interpreter may be told at run-time about the number and types of arguments used to call a given function. Libffi can be used in such programs to provide a bridge from the interpreter program to compiled code.

The libffi library provides a portable, high level programming interface to various calling conventions. This allows a programmer to call any function specified by a call interface description at run time.

FFI stands for Foreign Function Interface. A foreign function interface is the popular name for the interface that allows code written in one language to call code written in another language. The libffi library really only provides the lowest, machine dependent layer of a fully featured foreign function interface. A layer must exist above libffi that handles type conversions for values passed between the two languages.

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Upstream Contact

- https://sourceware.org/libffi/
- https://github.com/libffi/libffi

Type

standard

Dependencies

Version Information

package-version.txt:

```
3.5.2
```

Equivalent System Packages

```
$ apk add libffi-dev
```

```
$ conda install libffi
```

```
$ sudo apt-get install libffi-dev
```

```
$ sudo dnf install libffi libffi-devel
```

```
$ sudo pkg install devel/libffi
```

```
$ brew install libffi
```

```
$ sudo port install libffi
```

```
$ sudo zypper install 'pkgconfig(libffi)'
```

```
$ sudo slackpkg install libffi
```

```
$ sudo xbps-install libffi-devel
```

See https://repology.org/project/libffi/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.188 libgd: Dynamic graphics generation tool

Description

GD is an open source code library for the dynamic creation of images by programmers. GD is written in C, and "wrappers" are available for Perl, PHP and other languages. GD creates PNG, JPEG, GIF, WebP, XPM, BMP images, among other formats. GD is commonly used to generate charts, graphics, thumbnails, and most anything else, on the fly. While not restricted to use on the web, the most common applications of GD involve website development.

License

- Custom (BSD-ish)

Upstream Contact

- https://libgd.github.io
- Pierre Joye (http://blog.thepimp.net)
- https://github.com/libgd/libgd

Type

standard

Dependencies

- *libpng: Bitmap image support*

Version Information

package-version.txt:

```
2.3.3
```

Equivalent System Packages

```
$ apk add gd-dev
```

```
$ sudo pacman -S gd
```

```
$ conda install libgd expat
```

```
$ sudo apt-get install libgd-dev
```

```
$ sudo dnf install gd gd-devel
```

```
$ sudo pkg install graphics/gd
```

```
$ sudo emerge media-libs/gd
```

```
$ brew install gd
```

No package needed.

```
$ nix-env --install gd
```

```
$ sudo zypper install gd 'pkgconfig(gdlib)'
```

```
$ sudo slackpkg install gd fontconfig libXpm libX11 libxcb libXau  \
      libXdmcp
```

```
$ sudo xbps-install gd-devel
```

See https://repology.org/project/gd/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.189 libgraphviz: Graph visualization software (callable library)

Description

Graphviz is open source graph visualization software. It has several main graph layout programs. They take descriptions of graphs in a simple text language, and make diagrams in several useful formats.

This script package represents the callable library.

License

Eclipse Public License 1.0

Upstream Contact

https://graphviz.org/about/

Type

optional

Dependencies

Version Information

Equivalent System Packages

```
$ apk add graphviz-dev
```

```
$ sudo pacman -S graphviz
```

```
$ conda install graphviz
```

```
$ sudo apt-get install libgraphviz-dev
```

```
$ sudo dnf install graphviz graphviz-devel
```

```
$ sudo pkg install graphics/graphviz
```

```
$ brew install graphviz
```

```
$ sudo port install graphviz
```

```
$ nix-env --install graphviz
```

```
$ sudo zypper install graphviz
```

```
$ sudo xbps-install graphviz
```

See https://repology.org/project/graphviz/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.190 libhomfly: Compute the homfly polynomial of knots and links

Description

libhomfly is a library to compute the homfly polynomial of knots and links.

License

Public domain

SPKG Maintainers

- Miguel Marco

Upstream Contact

Miguel Marco (mmarco@unizar.es)

Type

standard

Dependencies

- *gc: The Boehm-Demers-Weiser conservative garbage collector*

Version Information

package-version.txt:

```
1.04
```

Equivalent System Packages

```
$ sudo pacman -S libhomfly
```

```
$ conda install libhomfly
```

```
$ sudo apt-get install libhomfly-dev
```

```
$ sudo dnf install libhomfly-devel
```

```
$ sudo pkg install math/libhomfly
```

```
$ sudo emerge sci-libs/libhomfly
```

```
$ nix-env --install libhomfly
```

```
$ sudo zypper install libhomfly-devel
```

```
$ sudo xbps-install libhomfly-devel
```

See https://repology.org/project/libhomfly/versions, https://repology.org/project/llibhomfly/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.191 libjpeg: JPEG image support

Description

This dummy package represents the image library `libjpeg`.

We do not have an SPKG for it. The purpose of this dummy package is to associate system package lists with it.

If the system package is installed, the package `pillow` will include support for JPEG images.

Type

optional

Dependencies

Version Information

Equivalent System Packages

```
$ apk add libjpeg-turbo-dev
```

```
$ sudo pacman -S libjpeg-turbo
```

```
$ sudo apt-get install libjpeg-dev
```

```
$ sudo dnf install libjpeg-turbo-devel
```

```
$ sudo emerge media-libs/libjpeg-turbo
```

```
$ brew install jpeg-turbo
```

```
$ nix-env --install libjpeg-turbo
```

```
$ sudo zypper install libjpeg-devel
```

```
$ sudo slackpkg install libjpeg-turbo
```

```
$ sudo xbps-install libjpeg-turbo-devel
```

If the system package is installed, `./configure` will check if it can be used.

6.1.192 libnauty: Find automorphism groups of graphs, generate non-isomorphic graphs (callable library)

Description

Nauty has various tools for finding the automorphism group of a graph, generating non-isomorphic graphs with certain properties, etc.

This script package represents the callable library of nauty.

License

Since version 2.6, nauty license is GPL-compatible, see

http://users.cecs.anu.edu.au/~bdm/nauty/COPYRIGHT.txt

(a copy of this file, called COPYRIGHT, is also present in the tarball)

Special Packaging Instruction

Upstream distribute tarball named nauty${version}.tar.gz. We cannot deal with that so rename it nauty-${version}.tar.gz (notice the "-") without any changes.

Upstream Contact

Brendan D. McKay Computer Science Department Australian National University bdm@cs.anu.edu.au

Adolfo Piperno Dipartimento di Informatica Sapienza - Università di Roma piperno@di.uniroma1.it

See http://cs.anu.edu.au/~bdm/nauty/ or http://pallini.di.uniroma1.it/

Type

optional

Dependencies

- $(SAGE_LOCAL)/$(SPKG_INST_RELDIR)/nauty-$(vers_nauty)

Version Information

Equivalent System Packages

```
$ sudo apt-get install libnauty-dev
```

```
$ sudo dnf install libnauty libnauty-devel
```

```
$ brew install nauty
```

```
$ sudo port install nauty
```

```
$ sudo xbps-install nauty-devel
```

See https://repology.org/project/nauty/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.193 libogg: Library for the Ogg multimedia container format

Description

libogg is the official reference library for the Ogg multimedia container format, and the native file and stream format for the Xiph.org multimedia codecs. As with all Xiph.org technology is it an open format free for anyone to use.

Website: http://www.xiph.org/ogg

License

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Upstream Contact

The Xiph.org mailing lists - see http://lists.xiph.org/mailman/listinfo

Special Update/Build Instructions

- No changes went into src.

Type

optional

Dependencies

Version Information

package-version.txt:

```
1.3.1.p0
```

Equivalent System Packages

```
$ conda install libogg
```

```
$ sudo dnf install libogg libogg-devel
```

```
$ brew install libogg
```

```
$ sudo port install libogg
```

```
$ sudo zypper install 'pkgconfig(ogg)'
```

```
$ sudo xbps-install libogg-devel
```

See https://repology.org/project/libogg/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.194 libpng: Bitmap image support

Description

libpng is the official PNG reference library. It supports almost all PNG features, is extensible, and has been extensively tested for over 13 years. The home site for development versions (i.e., may be buggy or subject to change or include experimental features) is http://libpng.sourceforge.net/, and the place to go for questions about the library is the png-mng-implement mailing list.

Website: http://www.libpng.org/pub/png/libpng.html

License

The libpng license - see http://www.libpng.org/pub/png/src/libpng-LICENSE.txt

Upstream Contact

https://libpng.sourceforge.io

The png mailing lists - see http://www.libpng.org/pub/png/pngmisc.html#lists

Special Update/Build Instructions

- On old versions of Darwin, the symbolic links libpng.* created by libpng16 may interfere with a system-wide libPng.dylib.

  – the following is very likely to be obsolete in 2014 —

  This system-wide library is likely to be a different version and on top of that, the symbols exported there are prefixed with "_cg" (for "Core Graphics"). So even if by chance the functionalities of the two libraries were interchangeable, libraries or applications looking for one and being presented the other won't find the symbols they expect. Note the uppercase "P" which could prevent this conflict; unfortunately, the default filesystem used by Apple is case-insensitive.

  Note there would be no problem if the system-wide library was not looked for when Sage is being built or run, but that's not the case either; it is at least looked for by the "ImageIO" framework:

  - when Python is built with Mac OS extensions, fixed in #4008;
  - when Mercurial is built because it uses $EDITOR, cf. #4678;
  - when R is built and it finds `-lpng`, cf. #4409 and #11696.

  – this is no longer done, as of #27186 —

  As not all of these problems are easily dealt with and new ones may arise, we chose to delete the $SAGE_LOCAL/lib/libpng.* symlinks. Therefore, some packages like Tachyon, which by default look for `-lpng` are patched to look for `-lpng16` instead.

Type

standard

Dependencies

Version Information

package-version.txt:

```
1.6.58
```

Equivalent System Packages

```
$ apk add libpng-dev
```

```
$ conda install libpng
```

```
$ sudo pkg install graphics/png
```

```
$ brew install libpng
```

```
$ sudo port install libpng
```

```
$ sudo zypper install 'pkgconfig(libpng16)'
```

```
$ sudo slackpkg install libpng
```

```
$ sudo xbps-install libpng-devel
```

See https://repology.org/project/libpng/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.195 libsemigroups: Library for semigroups and monoids

Description

C++ library for semigroups and monoids; used in GAP's package Semigroups.

License

GPL-3.0

Upstream Contact

http://james-d-mitchell.github.io/libsemigroups https://github.com/james-d-mitchell/libsemigroups

Type

optional

Dependencies

Version Information

package-version.txt:

```
3.5.1
```

Equivalent System Packages

```
$ conda install libsemigroups
```

```
$ sudo dnf install libsemigroups libsemigroups-devel
```

```
$ sudo pkg install math/libsemigroups
```

```
$ sudo zypper install 'pkgconfig(libsemigroups)'
```

See https://repology.org/project/libsemigroups/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.196 libxml2: XML parser and toolkit

Description

XML C parser and toolkit

License

MIT

Upstream Contact

http://www.xmlsoft.org/index.html

Type

optional

Dependencies

- *iconv: Library for language/country-dependent character encodings*

Version Information

Equivalent System Packages

```
$ apk add libxml2-dev
```

```
$ sudo pacman -S libxml2
```

```
$ sudo apt-get install libxml2-dev
```

```
$ sudo dnf install libxml2-devel
```

```
$ sudo pkg install libxml2
```

```
$ sudo emerge dev-libs/libxml2
```

```
$ brew install libxml2
```

```
$ sudo port install py-libxml2
```

```
$ nix-env --install libxml2
```

```
$ sudo zypper install libxml2
```

install the following packages: libxml

```
$ sudo slackpkg install libxml2
```

```
$ sudo xbps-install libxml2-devel
```

See https://repology.org/project/libxml2/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.197 lidia: A library for computational number theory

Description

A library for computational number theory.

Abandoned upstream and has disappeared from the web at TU Darmstadt.

We use as our new upstream a version minimally maintained for the LattE project.

https://www.math.ucdavis.edu/~latte/software/packages/lidia/current/lidia-2.3.0+latte-patches-2014-10-04.tar.gz

License

lidia is released under the GPL, or so it is claimed. See https://groups.google.com/forum/#!msg/sage-devel/kTxgPSqrbUM/5Txj3_IKhlQJ and https://lists.debian.org/debian-legal/2007/07/msg00120.html

Upstream Contact

Matthias Köppe, UC Davis, CA, USA

Type

optional

Dependencies

- $(MP_LIBRARY)

Version Information

package-version.txt:

```
2.3.0+latte-patches-2019-05-02
```

Equivalent System Packages

See https://repology.org/project/lidia/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.198 lie: Library for the representation theory of complex semisimple Lie groups and algebras

Description

LiE is the name of a software package that enables mathematicians and physicists to perform computations of a Lie group theoretic nature. It focuses on the representation theory of complex semisimple (reductive) Lie groups and algebras, and on the structure of their Weyl groups and root systems.

LiE does not compute directly with elements of the Lie groups and algebras themselves; it rather computes with weights, roots, characters and similar objects. Some specialities of LiE are: tensor product decompositions, branching to subgroups, Weyl group orbits, reduced elements in Weyl groups, distinguished coset representatives and much more. These operations have been compiled into the program which results in fast execution: typically one or two orders of magnitude faster than similar programs written in a general purpose program.

The LiE programming language makes it possible to customise and extend the package with more mathematical functions. A user manual is provided containing many examples.

LiE establishes an interactive environment from which commands can be given that involve basic programming primitives and powerful built-in functions. These commands are read by an interpreter built into the package and passed to the core of the system. This core consists of programs representing some 100 mathematical functions. The interpreter offers on-line facilities which explain operations and functions, and which give background information about Lie group theoretical concepts and about currently valid definitions and values.

(from http://www-math.univ-poitiers.fr/~maavl/LiE/description.html)

License

GNU Lesser General Public License (LGPL), version unspecified

Upstream Contact

- Marc van Leeuwen, http://www-math.univ-poitiers.fr/~maavl/

Dependencies

- readline
- ncurses
- bison (not included in this package or in Sage!)

Type

experimental

Dependencies

- *ncurses: Classic terminal output library*
- *readline: Command line editing library*

Version Information

package-version.txt:

```
2.2.2
```

Equivalent System Packages

```
$ sudo apt-get install lie
```

```
$ sudo emerge sci-mathematics/lie
```

```
$ sudo port install LiE
```

```
$ nix-env --install lie
```

```
$ sudo zypper install LiE
```

See https://repology.org/project/lie/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.199 linbox: Linear algebra with dense, sparse, structured matrices over the integers and finite fields

Description

LinBox is a C++ template library for exact, high-performance linear algebra computation with dense, sparse, and structured matrices over the integers and over finite fields.

License

LGPL V2 or later

Upstream Contact

- https://linalg.org/
- <linbox-devel@googlegroups.com>
- <linbox-use@googlegroups.com>

SPKG Repository

https://bitbucket.org/malb/linbox-spkg

Dependencies

- GNU patch
- GMP/MPIR
- MPFR
- NTL
- fpLLL
- IML
- M4RI
- M4RIE
- Givaro
- FFLAS/FFPACK
- a BLAS implementation such as openblas

Special Update/Build Instructions

TODO:

- spkg-check is disabled for now, should work in the next release after 1.3.2.
- Check whether `make fullcheck` works/builds, is worth running, and doesn't take ages. (Version 1.1.6 doesn't seem to have such a target.)

Type

standard

Dependencies

- $(MP_LIBRARY)
- *fflas_ffpack: Dense linear algebra over word-size finite fields*
- *flint: Fast Library for Number Theory*
- *givaro: C++ library for arithmetic and algebraic computations*
- *iml: Integer Matrix Library*
- *mpfr: Multiple-precision floating-point computations with correct rounding*
- *ntl: A library for doing number theory*

Version Information

package-version.txt:

```
1.7.1
```

Equivalent System Packages

```
$ sudo pacman -S linbox
```

```
$ conda install linbox
```

```
$ sudo apt-get install liblinbox-dev
```

```
$ sudo dnf install linbox linbox-devel
```

```
$ sudo pkg install math/linbox
```

```
$ sudo emerge sci-libs/linbox
```

```
$ brew install Macaulay2/tap/linbox
```

```
$ nix-env --install linbox
```

```
$ sudo zypper install 'pkgconfig(linbox)'
```

```
$ sudo xbps-install linbox-devel
```

See https://repology.org/project/linbox/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.200 llvm: The LLVM Compiler Infrastructure, including the Clang C/C++/Objective-C compiler

Description

The LLVM Project is a collection of modular and reusable compiler and toolchain technologies.

Clang is an "LLVM native" C/C++/Objective-C compiler.

The libc++ and libc++ ABI projects provide a standard conformant and high-performance implementation of the C++ Standard Library, including full support for C++11 and C++14.

License

Apache 2.0 License with LLVM exceptions

Upstream Contact

https://llvm.org/

Type

optional

Dependencies

Version Information

Equivalent System Packages

```
$ apk add clang
```

```
$ sudo pacman -S clang
```

```
$ sudo apt-get install clang
```

```
$ sudo dnf install clang
```

```
$ sudo pkg install devel/llvm
```

```
$ sudo emerge sys-devel/clang
```

```
$ brew install llvm
```

```
$ sudo port install clang
```

```
$ nix-env --install clang
```

```
$ sudo pkg_add devel/llvm
```

```
$ sudo zypper install llvm
```

```
$ sudo slackpkg install llvm
```

```
$ sudo xbps-install clang
```

If the system package is installed, `./configure` will check if it can be used.

6.1.201 lrcalc: Littlewood-Richardson calculator

Description

Littlewood-Richardson Calculator

http://sites.math.rutgers.edu/~asbuch/lrcalc/

License

GNU General Public License V2+

Upstream Contact

Anders S. Buch (asbuch@math.rutgers.edu)

https://bitbucket.org/asbuch/lrcalc

Type

standard

Dependencies

Version Information

package-version.txt:

```
2.1
```

Equivalent System Packages

```
$ sudo pacman -S lrcalc
```

```
$ conda install lrcalc
```

```
$ sudo apt-get install liblrcalc-dev
```

```
$ sudo dnf install lrcalc-devel
```

```
$ sudo pkg install math/lrcalc
```

```
$ sudo emerge sci-mathematics/lrcalc
```

```
$ brew install dimpase/tap/lrcalc
```

```
$ nix-env --install lrcalc
```

```
$ sudo xbps-install lrcalc-devel
```

See https://repology.org/project/lrcalc/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.202 lrcalc_python: Littlewood-Richardson calculator

Description

Python bindings for the Littlewood-Richardson Calculator

http://sites.math.rutgers.edu/~asbuch/lrcalc/

License

GNU General Public License V3

Upstream Contact

Anders S. Buch (asbuch@math.rutgers.edu)

https://bitbucket.org/asbuch/lrcalc

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cython: C-Extensions for Python, an optimizing static compiler*
- *lrcalc: Littlewood-Richardson calculator*

Version Information

package-version.txt:

```
2.1
```

version_requirements.txt:

```
lrcalc ~=2.1
```

Equivalent System Packages

```
$ conda install 'python-lrcalc~=2.1'
```

```
$ sudo dnf install python3-lrcalc
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.203 lrslib: Reverse search algorithm for vertex enumeration and convex hull problems

Description

lrslib implements the linear reverse search algorithm of Avis and Fukuda.

See the homepage (http://cgm.cs.mcgill.ca/~avis/C/lrs.html) for details.

We use an autotoolized version from https://github.com/mkoeppe/lrslib/tree/autoconfiscation

License

lrslib is released under a GPL v2+ license.

Upstream Contact

David Avis, avis at cs dot mcgill dot edu.

Dependencies

If the package finds an MPI C++ compiler script (mpic++), it also builds and installs the "mplrs" binary, a distributed version of lrs using MPI.

(Sage currently does not make use of plrs and mplrs.)

Special Update/Build Instructions

Type

optional

Dependencies

- $(MP_LIBRARY)

Version Information

package-version.txt:

```
071b+autotools-2021-07-13
```

Equivalent System Packages

```
$ sudo pacman -S lrs
```

```
$ conda install lrslib
```

```
$ sudo apt-get install lrslib
```

```
$ sudo dnf install lrslib lrslib-devel
```

```
$ sudo pkg install math/lrslib
```

```
$ sudo emerge sci-libs/lrslib
```

```
$ brew install Macaulay2/tap/lrs
```

```
$ nix-env --install lrs
```

```
$ sudo zypper install lrslib lrslib-devel
```

See https://repology.org/project/lrslib/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.204 m4ri: fast arithmetic with dense matrices over GF(2)

Description

M4RI: Library for matrix multiplication, reduction and inversion over GF(2). (See also m4ri/README for a brief overview.)

License

- GNU General Public License Version 2 or later (see src/COPYING)

Upstream Contact

- Authors: Martin Albrecht et al.
- Email: <m4ri-devel@googlegroups.com>
- Website: https://bitbucket.org/malb/m4ri

Special Update/Build Instructions

- Delete the upstream Mercurial repositories (file m4ri/.hgtags, directory m4ri/.hg).
- Delete the directory m4ri/autom4te.cache (if present).
- Delete m4ri.vcproj (and perhaps other unnecessary baggage).
- Touch m4ri/configure to make sure it is newer than its sources.

Type

standard

Dependencies

- *libpng: Bitmap image support*

Version Information

package-version.txt:

```
20260122
```

Equivalent System Packages

```
$ sudo pacman -S m4ri
```

```
$ conda install m4ri
```

```
$ sudo apt-get install libm4ri-dev
```

```
$ sudo dnf install m4ri-devel
```

```
$ sudo pkg install math/m4ri
```

```
$ sudo emerge 'sci-libs/m4ri[png]'
```

```
$ brew install m4ri
```

```
$ nix-env --install m4ri
```

```
$ sudo zypper install 'pkgconfig(m4ri)'
```

```
$ sudo xbps-install m4ri-devel
```

See https://repology.org/project/libm4ri/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.205 m4rie: Arithmetic with dense matrices over GF(2^e)

Description

M4RIE: Library for matrix multiplication, reduction and inversion over GF(2^k) for 2 <= k <= 10.

License

- GNU General Public License Version 2 or later (see src/COPYING)

Upstream Contact

- Authors: Martin Albrecht
- Email: <m4ri-devel@googlegroups.com>
- Website: http://m4ri.sagemath.org

Dependencies

- M4RI
- Givaro

Type

standard

Dependencies

- *m4ri: fast arithmetic with dense matrices over GF(2)*

Version Information

package-version.txt:

```
20250128
```

Equivalent System Packages

```
$ sudo pacman -S m4rie
```

```
$ conda install m4rie
```

```
$ sudo apt-get install libm4rie-dev
```

```
$ sudo dnf install m4rie-devel
```

```
$ sudo pkg install math/m4rie
```

```
$ sudo emerge sci-libs/m4rie
```

```
$ brew install m4rie
```

```
$ nix-env --install m4rie
```

```
$ sudo zypper install 'pkgconfig(m4rie)'
```

```
$ sudo xbps-install m4rie-devel
```

See https://repology.org/project/libm4rie/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.206 markupsafe: Safely add untrusted strings to HTML/XML markup

Description

Implements a XML/HTML/XHTML Markup safe string for Python

License

BSD-3-Clause

Upstream Contact

https://pypi.org/project/MarkupSafe/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
3.0.3
```

version_requirements.txt:

```
markupsafe >=2.0
```

Equivalent System Packages

```
$ conda install markupsafe
```

```
$ sudo dnf install python3-markupsafe
```

```
$ sudo emerge dev-python/markupsafe
```

```
$ sudo port install py-markupsafe
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-MarkupSafe'
```

```
$ sudo xbps-install python3-MarkupSafe
```

See https://repology.org/project/python:markupsafe/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.207 mathics: General-purpose computer algebra system

Description

General-purpose computer algebra system

License

GPL

Upstream Contact

https://pypi.org/project/Mathics3/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *charset_normalizer: The Real First Universal Charset Detector. Open, modern and actively maintained alternative to Chardet.*

- *dateutil: Extensions to the standard Python module datetime*
- *mpmath: Pure Python library for multiprecision floating-point arithmetic*
- *numpy: Package for scientific computing with Python*
- *pillow: Python Imaging Library*
- *pyyaml: YAML parser and emitter for Python*
- *requests: An HTTP library for Python*
- *sympy: Python library for symbolic mathematics*
- *typing_extensions: Backported and Experimental Type Hints for Python 3.8+*

Version Information

requirements.txt:

```
mathics3
mathics3-scanner
-c ${SAGE_VENV}/var/lib/sage/scripts/numpy/spkg-requirements.txt
```

Equivalent System Packages

```
$ conda install mathics3
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.208 mathjax: A JavaScript library for displaying mathematical formulas

Description

MathJax is a JavaScript library for displaying mathematical formulas.

MathJax is used in the Sage documentation built by Sphinx.

License

Apache License, version 2.0

Upstream Contact

Home page: https://www.mathjax.org/

Type

standard

Dependencies

Version Information

package-version.txt:

```
3.2.0
```

Equivalent System Packages

```
$ conda install mathjax
```

```
$ sudo dnf install mathjax3
```

```
$ sudo emerge '>=dev-libs/mathjax-3'
```

```
$ sudo zypper install mathjax
```

```
$ sudo xbps-install mathjax
```

See https://repology.org/project/mathjax/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.209 matplotlib: Python 2D plotting library

Description

From the Matplotlib website: matplotlib is a python 2D plotting library which produces publication quality figures in a variety of hardcopy formats and interactive environments across platforms. matplotlib can be used in python scripts, the python and ipython shell (ala matlab or mathematica), web application servers, and six graphical user interface toolkits.

License

The Matplotlib license - see https://matplotlib.org/stable/users/project/license.html: Matplotlib only uses BSD compatible code, and its license is based on the PSF license. See the Open Source Initiative licenses page for details on individual licenses. Non-BSD compatible licenses (eg LGPL) are acceptable in matplotlib Toolkits. For a discussion of the motivations behind the licencing choice, see Licenses.

Upstream Contact

https://matplotlib.org

The matplotlib mailing lists: see http://sourceforge.net/projects/matplotlib

Dependencies

- python
- numpy
- setuptools (>= 0.7)
- freetype
- patch (used in spkg-install)
- dateutil
- pyparsing
- tornado
- kiwisolver

Build Instructions/Changes

- NOTE: To drastically cut down on spkg size, we delete the internal testing images. To do this, we repackage the tarball by removing the contents of `lib/matplotlib/tests/baseline_images/*`, this is done by the `spkg-src` script.
- `setup.py.patch`: disable loading of Tests. Otherwise, `setup.py` raises an error because it can't find the deleted files from `src/lib/matplotlib/tests/baseline_images/*`
- NOTE: as of matplotlib-1.0.0 and Sage 4.6, Sage does not use $HOME/.matplotlib by default. Instead, it sets MPLCONFIGDIR to a subdirectory in $DOT_SAGE, see src/bin/sage-env

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *certifi: Python package for providing Mozilla's CA Bundle*
- *contourpy: Python library for calculating contours of 2D quadrilateral grids*
- *cycler: Composable cycles*
- *dateutil: Extensions to the standard Python module datetime*
- *fonttools: Tools to manipulate font files*
- *freetype: A free, high-quality, and portable font engine*
- *kiwisolver: Fast implementation of the Cassowary constraint solver*
- *numpy: Package for scientific computing with Python*
- *packaging: Core utilities for Python packages*
- *pillow: Python Imaging Library*
- *pyparsing: A Python parsing module*
- *qhull: Compute convex hulls, Delaunay triangulations, Voronoi diagrams*
- *setuptools_scm: Python build system extension to obtain package version from version control*
- *tornado: Python web framework and asynchronous networking library*

Version Information

package-version.txt:

```
3.10.8
```

version_requirements.txt:

```
matplotlib >=3.10.1
```

Equivalent System Packages

```
$ sudo pacman -S python-matplotlib
```

```
$ conda install 'matplotlib>=3.5.1'
```

```
$ sudo apt-get install python3-matplotlib
```

```
$ sudo dnf install python3-matplotlib
```

```
$ sudo emerge dev-python/matplotlib
```

```
$ sudo port install py-matplotlib
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-matplotlib'
```

```
$ sudo xbps-install python3-matplotlib
```

See https://repology.org/project/python:matplotlib/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.210 matplotlib_inline: Inline Matplotlib backend for Jupyter

Description

Inline Matplotlib backend for Jupyter

License

BSD 3-Clause

Upstream Contact

https://pypi.org/project/matplotlib-inline/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *traitlets: Traitlets Python configuration system*

Version Information

package-version.txt:

```
0.2.1
```

version_requirements.txt:

```
matplotlib-inline
```

Equivalent System Packages

```
$ conda install matplotlib-inline
```

```
$ sudo dnf install python3-matplotlib-inline
```

```
$ sudo emerge dev-python/matplotlib-inline
```

```
$ sudo xbps-install python3-matplotlib-inline
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.211 matroid_database: Python interface to matroid database

Description

Python interface to matroid database.

This database was retrieved from <https://www-imai.is.s.u-tokyo.ac.jp/~ymatsu/matroid/index.html> (Yoshitake Matsumoto, Database of Matroids, 2012; accessed: 2023.12.02).

License

GPL version 3 or later

Upstream Contact

https://pypi.org/project/matroid-database

https://github.com/gmou3/matroid-database

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
0.3
```

version_requirements.txt:

```
matroid-database
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.212 maxima: System for manipulating symbolic and numerical expressions

Description

Maxima is a system for the manipulation of symbolic and numerical expressions, including differentiation, integration, Taylor series, Laplace transforms, ordinary differential equations, systems of linear equations, polynomials, and sets, lists, vectors, matrices, and tensors. Maxima yields high precision numeric results by using exact fractions, arbitrary precision integers, and variable precision floating point numbers. Maxima can plot functions and data in two and three dimensions.

For more information, see the Maxima web site

http://maxima.sourceforge.net

License

Maxima is distributed under the GNU General Public License, with some export restrictions from the U.S. Department of Energy. See the file COPYING.

Upstream Contact

- The Maxima mailing list - see http://maxima.sourceforge.net/maximalist.html

Special Update/Build Instructions

1. Go to http://sourceforge.net/projects/maxima/files/Maxima-source/ and download the source tarball maxima-x.y.z.tar.gz; place it in the upstream/ directory.
2. Update package-version.txt and run 'sage –package fix-checksum'.
3. Make sure the patches still apply cleanly, and update them if necessary.
4. Test the resulting package.

All patch files in the patches/ directory are applied. Descriptions of these patches are either in the patch files themselves or below.

- infodir.patch: Correct the path to the Info directory. Introduced in Issue #11348 (maxima test fails when install tree is moved).

Type

standard

Dependencies

- *ecl: An implementation of the Common Lisp language*
- *info: stand-alone Info documentation reader*

Version Information

package-version.txt:

```
5.49.0
```

Equivalent System Packages

```
$ sudo pacman -S maxima-fas
```

```
$ conda install maxima
```

```
$ sudo apt-get install maxima-sage maxima
```

```
$ sudo dnf install maxima-runtime-ecl maxima
```

```
$ sudo pkg install math/maxima
```

```
$ sudo emerge  \
     'sci-mathematics/maxima[-sbcl,-clozurecl,-clozurecl64,-gcl,-clisp,ecl]'
```

```
$ brew install dimpase/tap/maxima-ecl
```

```
$ sudo port install maxima
```

```
$ nix-env --install maxima-ecl
```

```
$ sudo zypper install maxima-exec-clisp
```

```
$ sudo xbps-install maxima-ecl maxima-src
```

See https://repology.org/project/maxima/versions, https://repology.org/project/maxima-ecl/versions, https://repology.org/project/maxima-sage/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.213 mcqd: An exact algorithm for finding a maximum clique in an undirected graph

Description

MaxCliqueDyn is a fast exact algorithm for finding a maximum clique in an undirected graph.

License

GPL 3

Upstream Contact

MCQD is currently being maintained by Janez Konc. https://gitlab.com/janezkonc/mcqd

Type

optional

Dependencies

Version Information

package-version.txt:

```
1.0.p0
```

Equivalent System Packages

```
$ sudo pacman -S mcqd
```

```
$ sudo dnf install mcqd mcqd-devel
```

```
$ brew install dimpase/tap/mcqd
```

```
$ sudo zypper install mcqd
```

See https://repology.org/project/mcqd/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.214 meataxe: Library for computing with modular representations

Description

SharedMeatAxe 1.0 is an autotoolized shared library version of C MeatAxe 2.4.24, a set of programs for computing with modular representations. The package comprises a shared library "libmtx", as well as several executables.

See http://users.minet.uni-jena.de/~king/SharedMeatAxe/ for the package documentation.

Licence

The Shared Meat-Axe is free software: you can redistribute it and/or modify it under the terms of the GNU General Public License as published by the Free Software Foundation, either version 2 of the License, or (at your option) any later version. See the file COPYING.

Upstream contact

- Simon King <simon.king@uni-jena.de>

Type

optional

Dependencies

Version Information

package-version.txt:

```
1.0.2
```

Equivalent System Packages

```
$ sudo pacman -S shared_meataxe
```

```
$ sudo dnf install sharedmeataxe
```

See https://repology.org/project/shared-meataxe/versions, https://repology.org/project/sharedmeataxe/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.215 memory_allocator: An extension class to allocate memory easily with Cython

This extension class started as part of the Sage software.

Description

development website: https://github.com/sagemath/memory_allocator

PyPI page: https://pypi.org/project/memory_allocator

License

GPL-3.0

Upstream Contact

https://github.com/sagemath/memory_allocator

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cython: C-Extensions for Python, an optimizing static compiler*

Version Information

package-version.txt:

```
0.2.0
```

version_requirements.txt:

```
memory_allocator
```

Equivalent System Packages

```
$ conda install memory-allocator
```

```
$ sudo emerge dev-python/memory-allocator
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.216 meson: A high performance build system

Description

A high performance build system

License

Apache License, Version 2.0

Upstream Contact

https://pypi.org/project/meson/

Type

standard

Dependencies

- $(PYTHON)
- *flit_core: Distribution-building parts of Flit. See flit package for more information*
- *hatchling: Modern, extensible Python build backend*
- *pip: Tool for installing and managing Python packages*
- *python_build: A simple, correct PEP517 package builder*
- *setuptools: Build system for Python packages*
- *setuptools_scm: Python build system extension to obtain package version from version control*
- *wheel: A built-package format for Python*

Version Information

package-version.txt:

```
1.12.0
```

version_requirements.txt:

```
meson >=1.11.2
meson-python
```

Equivalent System Packages

```
$ apk add meson
```

```
$ sudo pacman -S meson
```

```
$ conda install meson
```

```
$ sudo apt-get install meson
```

```
$ sudo dnf install meson
```

```
$ sudo pkg install devel/meson
```

```
$ sudo emerge dev-build/meson
```

```
$ brew install meson
```

```
$ nix-env --install meson
```

```
$ sudo zypper install meson
```

```
$ sudo slackpkg install meson
```

See https://repology.org/project/meson/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.217 meson_python: Meson Python build backend (PEP 517)

Description

Meson Python build backend (PEP 517)

License

Upstream Contact

https://pypi.org/project/meson-python/

Type

standard

Dependencies

- $(PYTHON)
- *flit_core: Distribution-building parts of Flit. See flit package for more information*
- *meson: A high performance build system*
- *ninja_build: A build system with a focus on speed*
- *packaging: Core utilities for Python packages*
- *patchelf: A small utility to modify the dynamic linker and RPATH of ELF executables*
- *pip: Tool for installing and managing Python packages*
- *pyproject_metadata: PEP 621 metadata parsing*
- *python_build: A simple, correct PEP517 package builder*

- *wheel: A built-package format for Python*

Version Information

package-version.txt:

```
0.19.0
```

version_requirements.txt:

```
meson-python
```

Equivalent System Packages

```
$ apk add py3-meson-python
```

```
$ sudo pacman -S meson-python
```

```
$ conda install meson-python
```

```
$ sudo apt-get install meson-python
```

```
$ sudo dnf install python3-meson-python
```

```
$ sudo pkg install devel/meson-python
```

```
$ sudo emerge dev-python/meson-python
```

```
$ sudo zypper install python-meson-python
```

```
$ sudo xbps-install python3-meson-python
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.218 mistune: Sane and fast Markdown parser with useful plugins and renderers

Description

Sane and fast Markdown parser with useful plugins and renderers

License

BSD-3-Clause

Upstream Contact

https://pypi.org/project/mistune/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cython: C-Extensions for Python, an optimizing static compiler*

Version Information

package-version.txt:

```
3.2.0
```

version_requirements.txt:

```
mistune >=0.8.4
```

Equivalent System Packages

```
$ conda install mistune
```

```
$ sudo dnf install python3-mistune
```

```
$ sudo emerge dev-python/mistune
```

```
$ sudo xbps-install python3-mistune
```

See https://repology.org/project/mistune/versions, https://repology.org/project/python:mistune/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.219 modular_resolution: Modular cohomology rings of finite groups

Description

Modular Cohomology Rings of Finite Groups

The package is located at http://users.fmi.uni-jena.de/cohomology/, that's to say the tarball p_group_cohomology-x.y.tar.xz can be found there and the documentation of the package is provided at http://users.fmi.uni-jena.de/cohomology/documentation/

License

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David J. Green <david.green@uni-jena.de>

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> This code is distributed in the hope that it will be useful, but WITHOUT ANY WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the GNU General Public License for more details.

The full text of the GPL is available at:

> http://www.gnu.org/licenses/

The package includes a data base of cohomology rings of the groups of order 64 and provides access to a data base of cohomology rings of the groups of order 128 and 243, located at

> http://cohomology.uni-jena.de/db/

These data bases are distributed under the Creative Commons Attribution-Share Alike 3.0 License. The full text of this licence is available at

> http://creativecommons.org/licenses/by-sa/3.0/

SPKG Maintainers

Simon A. King <simon.king@uni-jena.de>

Upstream Contact

Simon A. King <simon.king@uni-jena.de> David J. Green <david.green@uni-jena.de>

Acknowledgements

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Since version 1.0.1, the further work on this SPKG was funded by Marie Curie grant MTKD-CT-2006-042685 and was pursued at the National University of Ireland, Galway. Since Novermber 2010, it is moved back to Jena.

We thank William Stein for giving us access to various computers on which we could build test the SPKG and on which some huge computations could be completed, and acknowledge the support by National Science Foundation Grant No. DMS-0821725.

We thank Mathieu Dutour Sikirić for hints on how to use GAP more efficiently.

We owe Peter Symonds the idea of using the Poincaré series in a rather efficient completeness criterion.

We are greatful to John Palmieri for his help on making p_group_cohomology work with python-3.

Dependencies

- The SharedMeatAxe needs to be installed, as a build time dependency.

  This can be met by installing the meataxe spkg

Testing

Our package provides a very short test suite for David Green's routines for the computation of minimal projective resolutions. The majority of this package's tests is formed by doc tests in the Cython code. In fact, any class, method and function is covered by tests.

Note that internet access is required for these tests, as it is attempted to download cohomology rings from a public data base in the web.

The script `spkg-check` calls `sage -t --force_lib` on the files in `pGroupCohomology`.

Documentation

The documentation of this package is automatically built, if the environment variable SAGE_SPKG_INSTALL_DOCS is yes (do "export SAGE_SPKG_INSTALL_DOCS=yes" on the command line before installation). The documents are put into SAGE_ROOT/local/share/doc/p_group_cohomology/.

Type

optional

Dependencies

- *meataxe: Library for computing with modular representations*
- *singular: Computer algebra system for polynomial computations, algebraic geometry, singularity theory*

Version Information

package-version.txt:

```
1.1
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.220 mpc: Arithmetic of complex numbers with arbitrarily high precision and correct rounding

Description

From https://www.multiprecision.org/mpc: GNU MPC is a C library for the arithmetic of complex numbers with arbitrarily high precision and correct rounding of the result. It extends the principles of the IEEE-754 standard for fixed precision real floating point numbers to complex numbers, providing well-defined semantics for every operation. At the same time, speed of operation at high precision is a major design goal.

License

LGPLv3+ for the code and GFDLv1.3+ (with no invariant sections) for the documentation.

Upstream Contact

The MPC website is located at https://www.multiprecision.org/mpc .

The MPC team can be contacted via the MPC mailing list: mpc-discuss@inria.fr

Special Update/Build Instructions

- mpc_mul_faster.patch: Patch from Paul Zimmermann to speed up MPC multiplication (for small precisions) by reducing overhead in MPFR operations.

Type

standard

Dependencies

- $(MP_LIBRARY)
- *mpfr: Multiple-precision floating-point computations with correct rounding*

Version Information

package-version.txt:

```
1.3.1
```

Equivalent System Packages

```
$ apk add mpc1 mpc1-dev
```

```
$ conda install mpc
```

```
$ sudo apt-get install libmpc-dev
```

```
$ sudo dnf install libmpc libmpc-devel
```

```
$ sudo pkg install math/mpc
```

```
$ sudo emerge dev-libs/mpc
```

```
$ brew install libmpc
```

```
$ nix-env --install libmpc
```

```
$ sudo zypper install mpc-devel
```

install the following packages: libmpc

```
$ sudo xbps-install libmpc-devel
```

See https://repology.org/project/gnumpc/versions, https://repology.org/project/libmpc/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.221 mpfi: Multiple precision interval arithmetic library based on MPFR

Description

MPFI is a library for interval arithmetic, which is built upon the MPFR multiple precision floating-point arithmetic.

MPFI is intended to be a portable library written in C for arbitrary precision interval arithmetic with intervals represented using MPFR reliable floating-point numbers. It is based on the GNU MP library and on the MPFR library. The purpose of an arbitrary precision interval arithmetic is on the one hand to get "guaranteed" results, thanks to interval computation, and on the other hand to obtain accurate results, thanks to multiple precision arithmetic. The MPFI library is built

upon MPFR in order to benefit from the correct rounding provided, for each operation or function, by MPFR. Further advantages of using MPFR are its portability and compliance with the IEEE 754 standard for floating-point arithmetic.

License

This version of MPFI is released under the GNU Lesser General Public License. It is permitted to link MPFI to non-free programs, as long as when distributing them the MPFI source code and a means to re-link with a modified MPFI is provided.

Upstream Contact

http://perso.ens-lyon.fr/nathalie.revol/software.html

The MPFI website is located at https://gitlab.inria.fr/mpfi/mpfi

The MPFI team can be contacted via the MPFI mailing list: mpfi-users@inria.fr

Type

standard

Dependencies

- $(MP_LIBRARY)
- *mpfr: Multiple-precision floating-point computations with correct rounding*

Version Information

package-version.txt:

```
1.5.2
```

Equivalent System Packages

```
$ sudo pacman -S mpfi
```

```
$ conda install mpfi
```

```
$ sudo apt-get install libmpfi-dev
```

```
$ sudo dnf install mpfi-devel
```

```
$ sudo pkg install math/mpfi
```

```
$ sudo emerge sci-libs/mpfi
```

```
$ brew install mpfi
```

```
$ nix-env --install mpfi
```

```
$ sudo zypper install mpfi-devel
```

```
$ sudo xbps-install mpfi-devel
```

See https://repology.org/project/mpfi/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.222 mpfr: Multiple-precision floating-point computations with correct rounding

Description

The MPFR library is a C library for multiple-precision floating-point computations with correct rounding. MPFR has continuously been supported by the INRIA and the current main authors come from the Caramba and AriC project-teams at Loria (Nancy, France) and LIP (Lyon, France) respectively; see more on the credit page. MPFR is based on the GMP multiple-precision library.

The main goal of MPFR is to provide a library for multiple-precision floating-point computation which is both efficient and has a well-defined semantics. It copies the good ideas from the ANSI/IEEE-754 standard for double-precision floating-point arithmetic (53-bit significand).

License

MPFR is free. It is distributed under the GNU Lesser General Public License (GNU Lesser GPL), version 3 or later (2.1 or later for MPFR versions until 2.4.x). The library has been registered in France by the Agence de Protection des Programmes under the number IDDN FR 001 120020 00 R P 2000 000 10800, on 15 March 2000. This license guarantees your freedom to share and change MPFR, to make sure MPFR is free for all its users. Unlike the ordinary General Public License, the Lesser GPL enables developers of non-free programs to use MPFR in their programs. If you have written a new function for MPFR or improved an existing one, please share your work!

Upstream Contact

The MPFR website is located at http://mpfr.org/

The MPFR team can be contacted via the MPFR mailing list: mpfr@loria.fr

Special Update/Build Instructions

- Make sure MPFR's settings of `CC` and `CFLAGS` still get properly extracted, currently from its `config.log` in the `src/` directory.
- We should remove the `configure` option `--disable-thread-safe` in case the issues without that have meanwhile been fixed. (Then we should actually pass `--enable-thread-safe`.)

TODO

- `--disable-thread-safe` should be switched to `--enable-thread-safe`, need to check that this works on the buildbot machines

Type

standard

Dependencies

- $(MP_LIBRARY)

Version Information

package-version.txt:

```
4.2.2
```

Equivalent System Packages

```
$ apk add mpfr-dev
```

```
$ conda install mpfr
```

```
$ sudo apt-get install libmpfr-dev
```

```
$ sudo dnf install mpfr-devel
```

```
$ sudo pkg install math/mpfr
```

```
$ sudo emerge dev-libs/mpfr
```

```
$ brew install mpfr
```

```
$ sudo zypper install 'pkgconfig(mpfr)'
```

install the following packages: libmpfr

```
$ sudo slackpkg install mpfr
```

```
$ sudo xbps-install mpfr-devel
```

See https://repology.org/project/mpfr/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.223 mpfrcx: Arithmetic of univariate polynomials over arbitrary precision real or complex numbers

Description

Mpfrcx is a library for the arithmetic of univariate polynomials over arbitrary precision real (Mpfr) or complex (Mpc) numbers, without control on the rounding. For the time being, only the few functions needed to implement the floating point approach to complex multiplication are implemented. On the other hand, these comprise asymptotically fast multiplication routines such as Toom–Cook and the FFT.

License

MPFRCX is distributed under the Gnu Lesser General Public License, either version 2.1 of the licence, or (at your option) any later version (LGPLv2.1+).

Upstream Contact

The MPFRCX website is located at http://www.multiprecision.org/mpfrcx .

Type

optional

Dependencies

- $(MP_LIBRARY)
- *mpc: Arithmetic of complex numbers with arbitrarily high precision and correct rounding*
- *mpfr: Multiple-precision floating-point computations with correct rounding*

Version Information

package-version.txt:

```
0.6.3
```

Equivalent System Packages

```
$ sudo zypper install mpfrcx-devel
```

See https://repology.org/project/mpfrcx/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.224 mpmath: Pure Python library for multiprecision floating-point arithmetic

Description

Mpmath is a pure-Python library for multiprecision floating-point arithmetic. It provides an extensive set of transcendental functions, unlimited exponent sizes, complex numbers, interval arithmetic, numerical integration and differentiation, root-finding, linear algebra, and much more. Almost any calculation can be performed just as well at 10-digit or 1000-digit precision, and in many cases mpmath implements asymptotically fast algorithms that scale well for extremely high precision work. If available, mpmath will (optionally) use gmpy to speed up high precision operations.

Upstream Contact

- Author: Fredrik Johansson
- Email: fredrik.johansson@gmail.com
- https://mpmath.org
- Website: https://github.com/mpmath/mpmath

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
1.4.1
```

version_requirements.txt:

```
mpmath >=1.1.0, <1.4
```

Equivalent System Packages

```
$ sudo pacman -S python-mpmath
```

```
$ conda install mpmath
```

```
$ sudo apt-get install python3-mpmath
```

```
$ sudo dnf install python3-mpmath
```

```
$ sudo emerge dev-python/mpmath
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-mpmath'
```

```
$ sudo xbps-install python3-mpmath
```

See https://repology.org/project/mpmath/versions, https://repology.org/project/python:mpmath/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.225 msolve: Multivariate polynomial system solver

Description

Open source C library implementing computer algebra algorithms for solving polynomial systems (with rational coefficients or coefficients in a prime field).

License

GPL v2+

Upstream Contact

https://github.com/algebraic-solving/msolve

Type

optional

Dependencies

- $(MP_LIBRARY)
- *flint: Fast Library for Number Theory*
- *mpfr: Multiple-precision floating-point computations with correct rounding*

Version Information

package-version.txt:

```
0.9.5
```

Equivalent System Packages

(none known)

If the system package is installed, `./configure` will check if it can be used.

6.1.226 nauty: Find automorphism groups of graphs, generate non-isomorphic graphs

Description

Nauty has various tools for finding the automorphism group of a graph, generating non-isomorphic graphs with certain properties, etc.

License

Since version 2.6, nauty license is GPL-compatible, see

http://users.cecs.anu.edu.au/~bdm/nauty/COPYRIGHT.txt

(a copy of this file, called COPYRIGHT, is also present in the tarball)

Special Packaging Instruction

Upstream distribute tarball named nauty${version}.tar.gz. We cannot deal with that so rename it nauty-${version}.tar.gz (notice the "-") without any changes.

Upstream Contact

Brendan D. McKay, Computer Science Department Australian National University bdm@cs.anu.edu.au

Adolfo Piperno, Dipartimento di Informatica Sapienza - Università di Roma piperno@di.uniroma1.it

See http://cs.anu.edu.au/~bdm/nauty/ Or http://pallini.di.uniroma1.it/

Type

standard

Dependencies

Version Information

package-version.txt:

```
2.9.3
```

Equivalent System Packages

```
$ sudo pacman -S nauty
```

```
$ conda install nauty
```

```
$ sudo apt-get install nauty
```

```
$ sudo dnf install nauty
```

```
$ sudo pkg install math/nauty
```

```
$ brew install nauty
```

```
$ nix-env --install nauty
```

```
$ sudo zypper install nauty nauty-devel
```

```
$ sudo xbps-install nauty
```

See https://repology.org/project/nauty/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.227 nbclient: Client library for executing notebooks. Formerly nbconvert's ExecutePreprocessor

Description

Client library for executing notebooks. Formerly nbconvert's ExecutePreprocessor

License

BSD

Upstream Contact

https://pypi.org/project/nbclient/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *jupyter_client: Jupyter protocol implementation and client libraries*
- *jupyter_core: Jupyter core package*
- *nbformat: The Jupyter Notebook format*
- *traitlets: Traitlets Python configuration system*

Version Information

package-version.txt:

```
0.10.4
```

version_requirements.txt:

```
nbclient
```

Equivalent System Packages

```
$ conda install nbclient
```

```
$ sudo dnf install python3-nbclient
```

```
$ sudo emerge dev-python/nbclient
```

```
$ sudo xbps-install python3-nbclient
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.228 nbconvert: Converting Jupyter Notebooks

Description

jupyter nbconvert converts notebooks to various other formats via Jinja templates.

License

BSD

Upstream Contact

https://pypi.org/project/nbconvert/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *beautifulsoup4: Screen-scraping library*
- *bleach: An HTML-sanitizing tool*
- *defusedxml: Addresses vulnerabilities of XML parsers and XML libraries*
- *jinja2: General purpose template engine for Python*
- *jupyter_client: Jupyter protocol implementation and client libraries*
- *jupyter_core: Jupyter core package*
- *jupyterlab_pygments: Pygments theme using JupyterLab CSS variables*
- *markupsafe: Safely add untrusted strings to HTML/XML markup*
- *mistune: Sane and fast Markdown parser with useful plugins and renderers*
- *nbclient: Client library for executing notebooks. Formerly nbconvert's ExecutePreprocessor*
- *nbformat: The Jupyter Notebook format*
- *packaging: Core utilities for Python packages*
- *pandocfilters: A Python module for writing pandoc filters*
- *pygments: Generic syntax highlighter*
- *tinycss2: A tiny CSS parser*
- *traitlets: Traitlets Python configuration system*

Version Information

package-version.txt:

```
7.17.0
```

version_requirements.txt:

```
nbconvert >=5.6.1
```

Equivalent System Packages

```
$ conda install nbconvert
```

```
$ sudo dnf install python3-nbconvert
```

```
$ sudo emerge dev-python/nbconvert
```

```
$ sudo zypper install jupyter-nbconvert
```

```
$ sudo xbps-install python3-jupyter_nbconvert
```

See https://repology.org/project/nbconvert/versions, https://repology.org/project/python:nbconvert/versions, https://repology.org/project/jupyter-nbconvert/versions, https://repology.org/project/python:jupyter-nbconvert/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.229 nbformat: The Jupyter Notebook format

Description

The Jupyter Notebook format

License

BSD 3-Clause License

Upstream Contact

https://pypi.org/project/nbformat/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *fastjsonschema: Fastest Python implementation of JSON schema*
- *jsonschema: Python implementation of JSON Schema*
- *jupyter_core: Jupyter core package*
- *traitlets: Traitlets Python configuration system*

Version Information

package-version.txt:

```
5.10.4
```

version_requirements.txt:

```
nbformat >=5.0.7
```

Equivalent System Packages

```
$ conda install nbformat
```

```
$ sudo dnf install python3-nbformat
```

```
$ sudo emerge dev-python/nbformat
```

```
$ sudo zypper install jupyter-nbformat
```

```
$ sudo xbps-install python3-jupyter_nbformat
```

See https://repology.org/project/nbformat/versions, https://repology.org/project/python:nbformat/versions, https://repology.org/project/jupyter-nbformat/versions, https://repology.org/project/python:jupyter-nbformat/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.230 ncurses: Classic terminal output library

Description

Ncurses (new curses, pronounced "enn-curses") started as a freely distributable "clone" of System V Release 4.0 (SVr4) curses. It has outgrown the "clone" description, and now contains many features which are not in SVr4 curses. Curses is a pun on the term "cursor optimization". It is a library of functions that manage an application's display on character-cell terminals (e.g., VT100).

The name "ncurses" was first used as the name of the curses library in Pavel Curtis's pcurses, dated 1982. It was apparently developed on a BSD 4.4 system, at Cornell. Parts of pcurses are readily identifiable in ncurses, including the basics for the terminfo compiler (named compile in that package):

- the Caps, used to define the terminfo capabilities
- awk scripts MKcaptab.awk, MKnames.awk
- the library modules used for the terminfo compiler.

Besides ncurses, parts of pcurses still survive in 2010, in recognizable form in Solaris.

Website: http://invisible-island.net/ncurses

License

- MIT-style

Upstream Contact

- bug-ncurses@gnu.org

Special Update/Build Instructions

None

Type

standard

Dependencies

Version Information

package-version.txt:

```
6.6
```

Equivalent System Packages

```
$ apk add ncurses-dev
```

```
$ conda install ncurses
```

```
$ sudo apt-get install libncurses-dev
```

```
$ sudo dnf install ncurses-devel
```

```
$ sudo pkg install devel/ncurses
```

```
$ brew install ncurses
```

```
$ sudo port install ncurses
```

```
$ sudo zypper install 'pkgconfig(ncurses)' 'pkgconfig(ncursesw)'
```

```
$ sudo slackpkg install ncurses
```

```
$ sudo xbps-install ncurses-devel
```

See https://repology.org/project/ncurses/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.231 nest_asyncio: Patch asyncio to allow nested event loops

Description

Patch asyncio to allow nested event loops

License

BSD

Upstream Contact

https://pypi.org/project/nest-asyncio/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
1.6.0
```

version_requirements.txt:

```
nest-asyncio
```

Equivalent System Packages

```
$ conda install nest-asyncio
```

```
$ sudo dnf install python3-nest-asyncio
```

```
$ sudo emerge dev-python/nest_asyncio
```

```
$ sudo xbps-install python3-nest_asyncio
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.232 networkx: Python package for complex networks

Description

NetworkX (NX) is a Python package for the creation, manipulation, and study of the structure, dynamics, and functions of complex networks.

License

BSD

Upstream Contact

https://networkx.github.io/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *scipy: Scientific tools for Python*

Version Information

package-version.txt:

```
3.6.1
```

version_requirements.txt:

```
networkx >=2.4
```

Equivalent System Packages

```
$ sudo pacman -S python-networkx
```

```
$ conda install 'networkx<3.3,>=2.4'
```

```
$ sudo apt-get install python3-networkx
```

```
$ sudo dnf install python3-networkx
```

```
$ sudo emerge dev-python/networkx
```

```
$ sudo port install py-networkx
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-networkx'
```

```
$ sudo xbps-install python3-networkx
```

See https://repology.org/project/python:networkx/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.233 nibabel: Access a multitude of neuroimaging data formats

Description

Access a multitude of neuroimaging data formats

License

MIT License

Upstream Contact

https://pypi.org/project/nibabel/

Type

optional

Dependencies

Version Information

requirements.txt:

```
nibabel
```

Equivalent System Packages

```
$ conda install nibabel
```

```
$ sudo dnf install python3-nibabel
```

```
$ sudo port install py-nibabel
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-nibabel'
```

See https://repology.org/project/nibabel/versions, https://repology.org/project/python:nibabel/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.234 ninja_build: A build system with a focus on speed

Description

Ninja is a small build system with a focus on speed.

License

Apache License 2.0

Upstream Contact

https://ninja-build.org/

Type

standard

Dependencies

- $(PYTHON)

Version Information

package-version.txt:

```
1.13.1
```

Equivalent System Packages

```
$ sudo pacman -S ninja
```

```
$ conda install ninja
```

```
$ sudo apt-get install ninja-build
```

```
$ sudo dnf install ninja-build
```

```
$ sudo pkg install devel/ninja
```

```
$ sudo emerge dev-build/ninja
```

```
$ brew install ninja
```

```
$ sudo port install ninja
```

```
$ sudo zypper install ninja
```

```
$ sudo xbps-install ninja
```

See https://repology.org/project/ninja/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.235 normaliz: Computations in affine monoids, vector configurations, lattice polytopes, and rational cones

Description

Normaliz is a tool for computations in affine monoids, vector configurations, lattice polytopes, and rational cones.

For more details see http://www.mathematik.uni-osnabrueck.de/normaliz/

License

- GPL v3

Upstream Contact

- normaliz@uos.de
- Winfried Bruns <wbruns@uos.de>
- Christof Söger <csoeger@uos.de>
- see also https://www.normaliz.uni-osnabrueck.de/home/contact/

  and https://github.com/Normaliz

Special Update/Build Instructions

- The spkg currently disables features that require packages SCIP and CoCoA, for which we don't have packages (yet).

Type

optional

Dependencies

- $(MP_LIBRARY)
- *e_antic: Real embedded number fields*
- *flint: Fast Library for Number Theory*

- *libnauty: Find automorphism groups of graphs, generate non-isomorphic graphs (callable library)*

Version Information

package-version.txt:

```
3.11.1
```

Equivalent System Packages

```
$ sudo pacman -S normaliz
```

```
$ conda install normaliz
```

```
$ sudo apt-get install libnormaliz-dev
```

```
$ sudo dnf install libnormaliz libnormaliz-devel
```

```
$ sudo emerge sci-mathematics/normaliz
```

```
$ brew install Macaulay2/tap/normaliz
```

```
$ sudo zypper install normaliz-devel
```

See https://repology.org/project/normaliz/versions, https://repology.org/project/libnormaliz/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.236 notebook: Jupyter notebook, a web-based notebook environment for interactive computing

Description

The Jupyter HTML notebook is a web-based notebook environment for interactive computing.

License

BSD 3-Clause License

Upstream Contact

https://pypi.org/project/notebook/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *jupyter_server: The backend (core services, APIs, REST endpoints) to Jupyter web applications*

- *jupyterlab: JupyterLab computational environment*
- *jupyterlab_server: Set of server components for JupyterLab and JupyterLab like applications*
- *notebook_shim: A shim layer for notebook traits and config*
- *tornado: Python web framework and asynchronous networking library*

Version Information

package-version.txt:

```
7.5.5
```

version_requirements.txt:

```
notebook >=6.1.1
```

Equivalent System Packages

```
$ sudo pacman -S jupyter-notebook
```

```
$ conda install notebook
```

```
$ sudo dnf install python3-notebook
```

```
$ sudo emerge dev-python/notebook
```

```
$ sudo port install py-notebook
```

```
$ sudo xbps-install python3-jupyter_notebook
```

See https://repology.org/project/python:notebook/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.237 notebook_shim: A shim layer for notebook traits and config

Description

A shim layer for notebook traits and config

License

BSD 3-Clause License

Upstream Contact

https://pypi.org/project/notebook-shim/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *jupyter_server: The backend (core services, APIs, REST endpoints) to Jupyter web applications*

Version Information

package-version.txt:

```
0.2.4
```

version_requirements.txt:

```
notebook-shim
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.238 notedown: Create IPython notebooks from markdown

Description

Notedown is a simple tool to create IPython notebooks from markdown.

License

BSD 2-Clause License

Upstream Contact

Author: Aaron O'Leary Home page: https://github.com/aaren/notedown

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *nbconvert: Converting Jupyter Notebooks*
- *nbformat: The Jupyter Notebook format*
- *pandoc_attributes: A parser and generator for pandoc block attributes*
- *six: Python 2 and 3 compatibility utilities*

Version Information

package-version.txt:

```
1.5.1
```

version_requirements.txt:

```
notedown >=1.5.1
```

Equivalent System Packages

```
$ conda install notedown
```

See https://repology.org/project/python:notedown/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.239 ntl: A library for doing number theory

Description

NTL is a high-performance, portable C++ library providing data structures and algorithms for manipulating signed, arbitrary length integers, and for vectors, matrices, and polynomials over the integers and over finite fields.

Website: http://www.shoup.net/ntl/

License

- GNU LGPLv2.1+

Upstream Contact

- Victor Shoup - for contact info see http://www.shoup.net/

Special Update/Build Instructions

- None

Type

standard

Dependencies

- $(MP_LIBRARY)
- *gf2x: Fast arithmetic in GF(2)[x] and searching for irreducible/primitive trinomials*

Version Information

package-version.txt:

```
11.6.0
```

Equivalent System Packages

```
$ conda install ntl
```

```
$ sudo apt-get install libntl-dev
```

```
$ sudo dnf install ntl-devel
```

```
$ sudo pkg install math/ntl
```

```
$ sudo emerge dev-libs/ntl
```

```
$ brew install ntl
```

```
$ sudo port install ntl
```

```
$ nix-env --install ntl
```

```
$ sudo zypper install ntl-devel
```

```
$ sudo xbps-install ntl-devel
```

See https://repology.org/project/ntl/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.240 numpy: Package for scientific computing with Python

Description

This package adds numerical linear algebra and other numerical computing capabilities to python.

Upstream Contact

- https://numpy.org/
- Travis Oliphant
- Fernando Perez
- Brian Granger

Special Update/Build Instructions

- Scipy uses numpy's distutils to control its compilation of fortran code.

  Whenever numpy is updated it is necessary to make sure that scipy still builds ok.

Type

standard

Dependencies

- $(BLAS)
- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cython: C-Extensions for Python, an optimizing static compiler*
- *gfortran: Fortran compiler from the GNU Compiler Collection*
- *pkgconfig: Python interface to pkg-config*

Version Information

package-version.txt:

```
2.4.4
```

version_requirements.txt:

```
numpy >=2.2.4
```

Equivalent System Packages

```
$ sudo pacman -S python-numpy
```

```
$ conda install numpy
```

```
$ sudo apt-get install python3-numpy
```

```
$ sudo dnf install python3-numpy
```

```
$ sudo emerge dev-python/numpy
```

```
$ brew install numpy
```

```
$ sudo port install py-numpy
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-numpy'
```

```
$ sudo xbps-install python3-numpy
```

See https://repology.org/project/python:numpy/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.241 onetbb: oneAPI Threading Building Blocks

Description

C++ parallelization library

License

Apache License, Version 2.0

Upstream Contact

https://github.com/oneapi-src/oneTBB

Type

optional

Dependencies

- *cmake: A cross-platform build system generator*

Version Information

package-version.txt:

```
2021.12.0
```

Equivalent System Packages

```
$ apk add libtbb-dev
```

```
$ sudo pacman -S intel-oneapi-tbb
```

```
$ conda install tbb
```

```
$ sudo apt-get install libtbb-dev
```

```
$ sudo dnf install tbb tbb-devel
```

```
$ sudo pkg install devel/onetbb
```

```
$ sudo emerge dev-cpp/tbb
```

```
$ brew install tbb
```

```
$ sudo port install onetbb
```

```
$ nix-env --install tbb
```

```
$ sudo zypper install tbb
```

```
$ sudo xbps-install tbb-devel
```

If the system package is installed, `./configure` will check if it can be used.

6.1.242 openblas: An optimized implementation of BLAS (Basic Linear Algebra Subprograms)

Description

OpenBLAS is an optimized open library implementing the Basic Linear Algebra Subprograms (BLAS) specification.

It is based on GotoBLAS2 1.13 BSD version.

License

3-clause BSD license

SPKG Repository

https://www.openblas.net

GitHub page: https://github.com/xianyi/OpenBLAS

Releases: https://github.com/xianyi/OpenBLAS/releases

Upstream Contact

- OpenBLAS users mailing list:

  https://groups.google.com/forum/#!forum/openblas-users

- OpenBLAS developers mailing list:

  https://groups.google.com/forum/#!forum/openblas-dev

Type

standard

Dependencies

- *gfortran: Fortran compiler from the GNU Compiler Collection*

Version Information

package-version.txt:

```
0.3.28
```

Equivalent System Packages

```
$ apk add openblas-dev
```

```
$ sudo pacman -S openblas lapack cblas
```

```
$ conda install openblas 'blas=2.*=openblas'
```

```
$ sudo apt-get install libopenblas-dev
```

```
$ sudo dnf install openblas-devel
```

```
$ sudo pkg install math/openblas
```

```
$ sudo emerge sci-libs/openblas
```

```
$ brew install openblas
```

```
$ sudo port install OpenBLAS-devel
```

```
$ nix-env --install blas lapack
```

```
$ sudo zypper install openblas-devel
```

install the following packages: openblas

```
$ sudo xbps-install openblas-devel
```

See https://repology.org/project/openblas/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.243 openssl: Implementation of the SSL and TLS protocols

Description

From wikipedia: OpenSSL is an open source implementation of the SSL and TLS protocols. The core library (written in the C programming language) implements the basic cryptographic functions and provides various utility functions. Wrappers allowing the use of the OpenSSL library in a variety of computer languages are available.

License

- Apache License v2 (considered compatible with GPL v3)

Upstream Contact

- http://openssl.org/

Type

standard

Dependencies

Version Information

package-version.txt:

```
3.5.5
```

Equivalent System Packages

```
$ apk add openssl-dev
```

```
$ sudo pacman -S openssl
```

```
$ conda install openssl
```

```
$ sudo apt-get install openssl libssl-dev
```

```
$ sudo dnf install openssl openssl-devel
```

```
$ sudo pkg install security/openssl
```

```
$ brew install openssl
```

```
$ sudo port install openssl
```

```
$ nix-env --install openssl
```

```
$ sudo zypper install libopenssl-3-devel
```

install the following packages: openssl

```
$ sudo slackpkg install openssl openssl-solibs
```

```
$ sudo xbps-install openssl-devel
```

See https://repology.org/project/openssl/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.244 ore_algebra: Ore algebra

Description

A Sage implementation of Ore algebras, Ore polynomials, and differentially finite functions.

Main features for the most common algebras include basic arithmetic and actions; gcrd and lclm; D-finite closure properties; creative telescoping; natural transformations between related algebras; guessing; desingularization; solvers for polynomials, rational functions and (generalized) power series. Univariate differential operators also support the numerical computation of analytic solutions with rigorous error bounds and related features.

License

- GPL-2.0+

Upstream Contact

- Website: https://github.com/mkauers/ore_algebra/
- Sage accounts: mkauers, mmezzarobba

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- $(SAGERUNTIME)

Version Information

requirements.txt:

```
ore_algebra @ git+https://github.com/mkauers/ore_algebra
```

Equivalent System Packages

See https://repology.org/project/ore-algebra/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.245 osqp_python: The Operator Splitting QP Solver (Python wrapper)

Description

This is the Python wrapper for OSQP: The Operator Splitting QP Solver.

It vendors OSQP.

License

Apache 2.0

Upstream Contact

https://pypi.org/project/osqp/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cmake: A cross-platform build system generator*
- *jinja2: General purpose template engine for Python*
- *joblib: Lightweight pipelining with Python functions*
- *numpy: Package for scientific computing with Python*
- *qdldl_python: QDLDL, a free LDL factorization routine (Python wrapper)*
- *scipy: Scientific tools for Python*

Version Information

package-version.txt:

```
1.1.1
```

version_requirements.txt:

```
osqp
```

Equivalent System Packages

```
$ conda install osqp
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.246 overrides: Decorator to automatically detect mismatch when overriding a method

Description

Decorator to automatically detect mismatch when overriding a method

License

Apache License, Version 2.0

Upstream Contact

https://pypi.org/project/overrides/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
7.7.0
```

version_requirements.txt:

```
overrides
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.247 p_group_cohomology: Modular cohomology rings of finite groups

Description

Modular Cohomology Rings of Finite Groups

The package is located at http://users.fmi.uni-jena.de/cohomology/, that's to say the tarball p_group_cohomology-x.y.tar.xz can be found there and the documentation of the package is provided at http://users.fmi.uni-jena.de/cohomology/documentation/

License

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> David J. Green <david.green@uni-jena.de>

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The full text of the GPL is available at:

> http://www.gnu.org/licenses/

The package includes a data base of cohomology rings of the groups of order 64 and provides access to a data base of cohomology rings of the groups of order 128 and 243, located at

> http://cohomology.uni-jena.de/db/

These data bases are distributed under the Creative Commons Attribution-Share Alike 3.0 License. The full text of this licence is available at

> http://creativecommons.org/licenses/by-sa/3.0/

SPKG Maintainers

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Upstream Contact

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We thank Mathieu Dutour Sikirić for hints on how to use GAP more efficiently.

We owe Peter Symonds the idea of using the Poincaré series in a rather efficient completeness criterion.

We are greatful to John Palmieri for his help on making p_group_cohomology work with python-3.

Dependencies

- The SharedMeatAxe needs to be installed, as a build time dependency.

  This can be met by installing the meataxe spkg

Testing

Our package provides a very short test suite for David Green's routines for the computation of minimal projective resolutions. The majority of this package's tests is formed by doc tests in the Cython code. In fact, any class, method and function is covered by tests.

Note that internet access is required for these tests, as it is attempted to download cohomology rings from a public data base in the web.

The script `spkg-check` calls `sage -t --force_lib` on the files in `pGroupCohomology`.

Documentation

The documentation of this package is automatically built, if the environment variable SAGE_SPKG_INSTALL_DOCS is yes (do "export SAGE_SPKG_INSTALL_DOCS=yes" on the command line before installation). The documents are put into SAGE_ROOT/local/share/doc/p_group_cohomology/.

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- $(SAGERUNTIME)
- $(SAGE_SRC)/sage/libs/meataxe.pxd
- $(SAGE_SRC)/sage/matrix/matrix0.pxd
- $(SAGE_SRC)/sage/matrix/matrix_gfpn_dense.pxd
- $(SAGE_SRC)/sage/matrix/matrix_gfpn_dense.pxd
- $(SAGE_SRC)/sage/rings/morphism.pxd
- $(SAGE_SRC)/sage/structure/element.pxd
- *cysignals: Interrupt and signal handling for Cython*
- *cython: C-Extensions for Python, an optimizing static compiler*
- *gap: Groups, Algorithms, Programming - a system for computational discrete algebra*
- *ipywidgets: Interactive HTML widgets for Jupyter notebooks and the IPython kernel*

- *matplotlib: Python 2D plotting library*
- *meataxe: Library for computing with modular representations*
- *modular_resolution: Modular cohomology rings of finite groups*
- *singular: Computer algebra system for polynomial computations, algebraic geometry, singularity theory*

Version Information

package-version.txt:

```
3.3.3.p1
```

version_requirements.txt:

```
p_group_cohomology >=3.3
```

Equivalent System Packages

See https://repology.org/project/python:p-group-cohomology/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.248 packaging: Core utilities for Python packages

Description

Core utilities for Python packages

License

Upstream Contact

https://pypi.org/project/packaging/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
26.0
```

version_requirements.txt:

```
packaging >=24.2
```

Equivalent System Packages

```
$ sudo pacman -S python-packaging
```

```
$ conda install packaging
```

```
$ sudo apt-get install python3-packaging
```

```
$ sudo dnf install python3-packaging
```

```
$ sudo emerge dev-python/packaging
```

```
$ sudo port install py-packaging
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-packaging'
```

```
$ sudo xbps-install python3-packaging
```

See https://repology.org/project/packaging/versions, https://repology.org/project/python:packaging/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.249 palp: A package for Analyzing Lattice Polytopes

Description

A Package for Analyzing Lattice Polytopes (PALP) is a set of C programs for calculations with lattice polytopes and applications to toric geometry.

It contains routines for vertex and facet enumeration, computation of incidences and symmetries, as well as completion of the set of lattice points in the convex hull of a given set of points. In addition, there are procedures specialised to reflexive polytopes such as the enumeration of reflexive subpolytopes, and applications to toric geometry and string theory, like the computation of Hodge data and fibration structures for toric Calabi-Yau varieties. The package is well tested and optimised in speed as it was used for time consuming tasks such as the classification of reflexive polyhedra in 4 dimensions and the creation and manipulation of very large lists of 5-dimensional polyhedra.

While originally intended for low-dimensional applications, the algorithms work in any dimension and our key routine for vertex and facet enumeration compares well with existing packages.

License

- When released, GPL 2 was in force.
- There is a link to a web page, which now points to GPL 3, but would have pointed to GPL 2 at the time the package was released.
- Therefore one can deduce the authors were happy for this to be released under GPL 2 or a later version.

Upstream Contact

- Author: Harald Skarke (skarke@maths.ox.ac.uk)
- Home page: http://hep.itp.tuwien.ac.at/~kreuzer/CY/CYpalp.html

Type

standard

Dependencies

Version Information

package-version.txt:

```
2.21
```

Equivalent System Packages

```
$ sudo pacman -S palp
```

```
$ conda install palp
```

```
$ sudo apt-get install palp
```

```
$ sudo dnf install palp
```

```
$ brew install macaulay2/tap/palp
```

```
$ nix-env --install palp
```

```
$ sudo xbps-install palp
```

See https://repology.org/project/palp/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.250 pandoc: A document converter

Description

This dummy package represents the document converter pandoc.

We do not have an SPKG for it. The purpose of this dummy package is to associate system package lists with it.

Type

optional

Dependencies

Version Information

Equivalent System Packages

```
$ apk add pandoc
```

```
$ sudo pacman -S pandoc
```

```
$ conda install pandoc
```

```
$ sudo apt-get install pandoc
```

```
$ sudo dnf install pandoc
```

```
$ sudo pkg install textproc/hs-pandoc
```

```
$ sudo emerge app-text/pandoc
```

```
$ brew install pandoc
```

```
$ sudo port install pandoc
```

```
$ sudo zypper install pandoc
```

```
$ sudo xbps-install pandoc
```

See https://repology.org/project/pandoc/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.251 pandoc_attributes: A parser and generator for pandoc block attributes

Description

This is a simple parser / emitter for pandoc block attributes, intended for use with pandocfilters.

License

BSD 2-Clause License

Upstream Contact

- Author: Aaron O'Leary
- Home page: https://github.com/aaren/pandoc-attributes

Special Update/Build Instructions

There are no release numbers, hence find the latest commit, download https://github.com/aaren/pandoc-attributes/archive/${COMMIT}.zip and rename it pandoc_attributes-${COMMIT:0:8}.zip

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *pandocfilters: A Python module for writing pandoc filters*

Version Information

package-version.txt:

```
8bc82f6d
```

version_requirements.txt:

```
pandoc_attributes >=8bc82f6d
```

Equivalent System Packages

```
$ conda install pandoc-attributes
```

See https://repology.org/project/pandoc-attributes/versions, https://repology.org/project/python:pandoc-attributes/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.252 pandocfilters: A Python module for writing pandoc filters

Description

A python module for writing pandoc filters.

License

BSD 3-Clause License

Upstream Contact

Author: John MacFarlane Home page: https://github.com/jgm/pandocfilters

Special Update/Build Instructions

Download the last release from https://pypi.python.org/pypi/pandocfilters

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
1.5.1
```

version_requirements.txt:

```
pandocfilters >=1.4.2
```

Equivalent System Packages

```
$ sudo pacman -S python-pandocfilters
```

```
$ conda install pandocfilters
```

```
$ sudo apt-get install python3-pandocfilters
```

```
$ sudo dnf install python3-pandocfilters
```

```
$ sudo emerge dev-python/pandocfilters
```

```
$ sudo port install py-pandocfilters
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-pandocfilters'
```

```
$ sudo xbps-install python3-pandocfilters
```

See https://repology.org/project/python:pandocfilters/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.253 papilo: Parallel presolve for integer and linear optimization

Description

parallel presolve routines for (mixed integer) linear programming problems. The routines are implemented using templates which allows switching to higher precision or rational arithmetic using the boost multiprecision package.

License

LGPL 3.0

Upstream Contact

https://github.com/scipopt/papilo/

Type

optional

Dependencies

- $(BLAS)
- $(MP_LIBRARY)
- *cmake: A cross-platform build system generator*
- *gfortran: Fortran compiler from the GNU Compiler Collection*
- *onetbb: oneAPI Threading Building Blocks*

Version Information

package-version.txt:

```
2.2.1
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.254 pari: Computer algebra system for fast computations in number theory

Description

PARI/GP is a widely used computer algebra system designed for fast computations in number theory (factorizations, algebraic number theory, elliptic curves…), but also contains a large number of other useful functions to compute with mathematical entities such as matrices, polynomials, power series, algebraic numbers etc., and a lot of transcendental functions. PARI is also available as a C library to allow for faster computations.

Originally developed by Henri Cohen and his co-workers (Université Bordeaux I, France), PARI is now under the GPL and maintained by Karim Belabas with the help of many volunteer contributors.

License

GPL version 2+

Upstream Contact

- http://pari.math.u-bordeaux.fr/

Dependencies

- Perl
- MPIR or GMP
- Readline
- GNU patch (shipped with Sage)

Special Update/Build Instructions

See patches/README.txt for a list of patches.

The current upstream tarball was created from the PARI git repository by running “make snapshot”.

Type

standard

Dependencies

- $(MP_LIBRARY)
- *pari_galdata: PARI data package needed to compute Galois groups in degrees 8 through 11*
- *pari_seadata_small: PARI data package needed by ellap for large primes (small version)*

- *readline: Command line editing library*

Version Information

package-version.txt:

```
2.17.1
```

Equivalent System Packages

```
$ apk add pari-dev
```

```
$ sudo pacman -S pari pari-galdata pari-seadata pari-elldata  \
      pari-galpol
```

```
$ conda install 'pari=*=*_pthread' pari-elldata pari-galdata  \
      pari-galpol pari-seadata
```

```
$ sudo apt-get install pari-gp2c libpari-dev pari-doc pari-elldata  \
      pari-galdata pari-galpol pari-seadata
```

```
$ sudo dnf install pari-devel pari-gp pari-galdata pari-galpol  \
      pari-seadata pari-elldata
```

```
$ sudo pkg install math/pari
```

```
$ sudo emerge sci-mathematics/pari sci-mathematics/pari-data
```

```
$ brew install pari pari-elldata pari-galdata pari-galpol pari-seadata
```

No package needed.

```
$ nix-env --install pari
```

```
$ sudo zypper install pari-devel pari-gp
```

```
$ sudo xbps-install pari pari-devel pari-elldata-small pari-galdata  \
      pari-galpol-small pari-seadata
```

See https://repology.org/project/pari/versions, https://repology.org/project/pari-gp/versions, https://repology.org/project/pari-data/versions, https://repology.org/project/pari-elldata/versions, https://repology.org/project/pari-galdata/versions, https://repology.org/project/pari-galpol/versions, https://repology.org/project/pari-nftables/versions, https://repology.org/project/pari-seadata/versions, https://repology.org/project/pari-seadata-big/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.255 pari_elldata: PARI data package for elliptic curves

Description

PARI/GP version of J. E. Cremona Elliptic Curve Data, needed by ellsearch and ellidentify.

License

GNU General Public License (GPL version 2 or any later version).

Upstream Contact

http://pari.math.u-bordeaux.fr/

Dependencies

- Installation: None
- Runtime: PARI/GP

Type

optional

Dependencies

- *pari_seadata_small: PARI data package needed by ellap for large primes (small version)*

Version Information

package-version.txt:

```
20161017
```

Equivalent System Packages

```
$ sudo pacman -S pari-elldata
```

```
$ conda install pari-elldata
```

```
$ sudo dnf install pari-elldata
```

```
$ sudo pkg install math/pari_elldata
```

```
$ sudo zypper install pari-elldata
```

```
$ sudo xbps-install pari-elldata-small
```

See https://repology.org/project/pari-elldata/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.256 pari_galdata: PARI data package needed to compute Galois groups in degrees 8 through 11

Description

PARI package "galdata": Needed by polgalois to compute Galois group in degrees 8 through 11.

License

GPL version 2+

Upstream Contact

http://pari.math.u-bordeaux.fr/

Type

standard

Dependencies

- *pari_seadata_small: PARI data package needed by ellap for large primes (small version)*

Version Information

package-version.txt:

```
20080411.p0
```

Equivalent System Packages

```
$ sudo pacman -S pari-galdata
```

```
$ conda install pari-galdata
```

```
$ sudo dnf install pari-galdata
```

```
$ sudo pkg install pari_galdata
```

```
$ sudo zypper install pari-galdata
```

```
$ sudo xbps-install pari-galdata
```

See https://repology.org/project/pari-galdata/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.257 pari_galpol: PARI data package for polynomials defining Galois extensions of the rationals

Description

PARI package of the GALPOL database of polynomials defining Galois extensions of the rationals, accessed by galoisgetpol, galoisgetgroup, galoisgetname.

License

GNU General Public License (GPL version 2 or any later version).

Upstream Contact

http://pari.math.u-bordeaux.fr/

Dependencies

- Installation: None
- Runtime: PARI/GP

Type

optional

Dependencies

- *pari_seadata_small: PARI data package needed by ellap for large primes (small version)*

Version Information

package-version.txt:

```
20180625
```

Equivalent System Packages

```
$ sudo pacman -S pari-galpol
```

```
$ conda install pari-galpol
```

```
$ sudo dnf install pari-galpol
```

```
$ sudo pkg install math/pari_galpol
```

```
$ sudo zypper install pari-galpol
```

```
$ sudo xbps-install pari-galpol-small
```

See https://repology.org/project/pari-galpol/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.258 pari_jupyter: Jupyter kernel for PARI/GP

Description

Jupyter kernel for PARI/GP

License

GPL version 3 or later

Upstream Contact

- https://github.com/sagemath/pari-jupyter

Dependencies

- Python >= 3.6.1
- Jupyter 4
- PARI version 2.13 or later
- Readline (any version which works with PARI)
- Optional: Cython version 0.25 or later

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cython: C-Extensions for Python, an optimizing static compiler*
- *ipykernel: IPython Kernel for Jupyter*
- *jupyter_core: Jupyter core package*
- *notebook: Jupyter notebook, a web-based notebook environment for interactive computing*
- *pari: Computer algebra system for fast computations in number theory*

Version Information

package-version.txt:

```
1.4.3
```

version_requirements.txt:

```
pari_jupyter >=1.3.2
```

Equivalent System Packages

```
$ conda install pari_jupyter
```

```
$ sudo dnf install python3-pari-jupyter
```

See https://repology.org/project/pari-jupyter/versions, https://repology.org/project/python:pari-jupyter/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.259 pari_nftables: PARI data package for number fields

Description

Repackaging of the historical megrez number field tables (errors fixed, 1/10th the size, easier to use).

License

GNU General Public License (GPL version 2 or any later version).

Upstream Contact

http://pari.math.u-bordeaux.fr/

Dependencies

- Installation: None
- Runtime: PARI/GP

Type

optional

Dependencies

- *pari_seadata_small: PARI data package needed by ellap for large primes (small version)*

Version Information

package-version.txt:

```
20080929
```

Equivalent System Packages

```
$ conda install pari-nftables
```

```
$ sudo dnf install pari-nftables
```

```
$ sudo pkg install math/pari_nftables
```

```
$ sudo zypper install pari-nftables
```

```
$ sudo xbps-install pari-nftables
```

See https://repology.org/project/pari-nftables/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.260 pari_seadata: PARI data package needed by ellap for large primes (full version)

Description

Needed by ellap for large primes. These polynomials were extracted from the ECHIDNA databases and computed by David R. Kohel. This covers finite fields of cardinality q up to 750 bits. PARI/GP 2.9 contains fallback code to go on when all modular polynomials in the database have been exhausted and can handle larger fields (with an important slowdown).

License

GNU General Public License (GPL version 2 or any later version).

Upstream Contact

http://pari.math.u-bordeaux.fr/

Dependencies

- Installation: None
- Runtime: PARI/GP

Type

optional

Dependencies

- *pari_seadata_small: PARI data package needed by ellap for large primes (small version)*

Version Information

package-version.txt:

```
20090618
```

Equivalent System Packages

```
$ sudo pacman -S pari-seadata
```

```
$ conda install pari-seadata
```

```
$ sudo dnf install pari-seadata
```

```
$ sudo pkg install math/pari_seadata
```

```
$ sudo zypper install pari-seadata
```

```
$ sudo xbps-install pari-seadata
```

See https://repology.org/project/pari-seadata/versions, https://repology.org/project/pari-seadata-big/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.261 pari_seadata_small: PARI data package needed by ellap for large primes (small version)

Description

PARI package "seadata_small": Needed by ellap for large primes. This "small" one is a much smaller version that should be suitable for primes up to 350 bits. These polynomials were extracted from the ECHIDNA databases and computed by David R. Kohel.

License

GPL version 2+

Upstream Contact

http://pari.math.u-bordeaux.fr/

Type

standard

Dependencies

Version Information

package-version.txt:

```
20090618.p0
```

Equivalent System Packages

```
$ sudo pacman -S pari-seadata
```

```
$ conda install pari-seadata-small
```

```
$ sudo pkg install math/pari_seadata
```

```
$ sudo xbps-install pari-seadata
```

See https://repology.org/project/pari-seadata-small/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.262 parso: Python Parser

Description

Parso is a Python parser that supports error recovery and round-trip parsing for different Python versions (in multiple Python versions). Parso is also able to list multiple syntax errors in your python file.

License

MIT

Upstream Contact

https://pypi.org/project/parso/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
0.8.6
```

version_requirements.txt:

```
parso >=0.7.0
```

Equivalent System Packages

```
$ conda install parso
```

```
$ sudo dnf install python3-parso
```

```
$ sudo emerge dev-python/parso
```

```
$ sudo port install py-parso
```

```
$ sudo xbps-install python3-parso
```

See https://repology.org/project/python:parso/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.263 patchelf: A small utility to modify the dynamic linker and RPATH of ELF executables

Description

A small utility to modify the dynamic linker and RPATH of ELF executables.

License

GPL-3.0-or-later

Upstream Contact

https://github.com/NixOS/patchelf

Type

standard

Dependencies

Version Information

package-version.txt:

```
0.13.1
```

Equivalent System Packages

```
$ conda install patchelf
```

```
$ sudo apt-get install patchelf
```

```
$ sudo dnf install patchelf
```

```
$ sudo pkg install sysutils/patchelf
```

```
$ sudo emerge dev-util/patchelf
```

```
$ brew install patchelf
```

```
$ sudo port install patchelf
```

```
$ nix-env --install patchelf
```

```
$ sudo zypper install patchelf
```

```
$ sudo xbps-install patchelf
```

See https://repology.org/project/patchelf/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.264 pathspec: Utility library for gitignore style pattern matching of file paths

Description

Utility library for gitignore style pattern matching of file paths

License

MPL 2.0

Upstream Contact

https://pypi.org/project/pathspec/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
1.0.4
```

version_requirements.txt:

```
pathspec
```

Equivalent System Packages

```
$ conda install pathspec
```

```
$ sudo dnf install python3-pathspec
```

```
$ sudo emerge dev-python/pathspec
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.265 perl_cpan_polymake_prereq: Represents all Perl packages that are prerequisites for polymake

Description

This script package represents all Perl packages that are prerequisites for polymake.

License

Various free software licenses

Type

optional

Dependencies

Version Information

Equivalent System Packages

```
$ apk add perl-term-readkey perl-dev perl-file-slurp perl-json  \
      perl-xml-writer perl-xml-libxslt
```

```
$ XML::Writer XML::LibXML XML::LibXSLT File::Slurp JSON SVG  \
      Term::ReadKey
```

```
$ sudo apt-get install libxml-libxslt-perl libxml-writer-perl  \
      libxml2-dev libperl-dev libfile-slurp-perl libjson-perl  \
      libsvg-perl libterm-readkey-perl libterm-readline-gnu-perl
```

```
$ sudo dnf install perl-ExtUtils-Embed perl-File-Slurp perl-JSON  \
      perl-Term-ReadLine-Gnu perl-TermReadKey perl-XML-Writer  \
      perl-XML-LibXML perl-XML-LibXSLT perl-SVG
```

```
$ sudo pkg install textproc/p5-XML-Writer textproc/p5-XML-LibXML  \
      textproc/p5-XML-LibXSLT devel/p5-File-Slurp converters/p5-JSON  \
      textproc/p5-SVG devel/p5-Term-ReadKey
```

```
$ sudo emerge XML-Writer XML-LibXML XML-LibXSLT File-Slurp  \
      dev-perl/Term-ReadLine-Gnu dev-perl/TermReadKey JSON SVG
```

```
$ sudo xbps-install perl-File-Slurp perl-JSON perl-SVG  \
      perl-Term-ReadKey perl-XML-LibXML perl-XML-LibXSLT  \
      perl-XML-Writer
```

If the system package is installed, `./configure` will check if it can be used.

6.1.266 perl_mongodb: A prerequisite for polymake's PolyDB feature

Description

This script package represents the Perl package MongoDB, which is needed for the PolyDB feature of polymake.

License

Various free software licenses

Type

optional

Dependencies

Version Information

Equivalent System Packages

```
$ MongoDB
```

```
$ sudo apt-get install libmongodb-perl
```

```
$ sudo dnf install perl-MongoDB
```

```
$ sudo pkg install databases/p5-MongoDB
```

```
$ sudo emerge dev-perl/MongoDB
```

If the system package is installed, `./configure` will check if it can be used.

6.1.267 perl_term_readline_gnu: Perl extension for the GNU Readline/History libraries

Description

Perl extension for the GNU Readline/History Library

Available on CPAN

License

The Perl 5 License (Artistic 1 & GPL 1)

Upstream Contact

Hiroo HAYASHI

Type

optional

Dependencies

- *readline: Command line editing library*

Version Information

package-version.txt:

```
1.35
```

Equivalent System Packages

```
$ apk add perl-term-readline-gnu
```

```
$ sudo pacman -S perl-term-readline-gnu
```

```
$ Term::ReadLine::Gnu
```

```
$ sudo apt-get install libterm-readline-gnu-perl
```

```
$ sudo dnf install perl-Term-ReadLine-Gnu
```

```
$ sudo pkg install devel/p5-Term-ReadLine-Gnu
```

```
$ sudo emerge dev-perl/Term-ReadLine-Gnu
```

```
$ sudo port install p5-term-readline-gnu
```

```
$ sudo zypper install 'perl(Term::ReadLine::Gnu)'
```

```
$ sudo xbps-install perl-Term-ReadLine-Gnu
```

See https://repology.org/project/perl:term-readline-gnu/versions, https://repology.org/project/perl:termreadline-gnu/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.268 pexpect: Python module for controlling and automating other programs

Description

Pexpect is a pure Python module for spawning child applications; controlling them; and responding to expected patterns in their output.

License

ISC license: http://opensource.org/licenses/isc-license.txt This license is approved by the OSI and FSF as GPL-compatible.

Upstream Contact

- http://pexpect.readthedocs.org/en/stable/
- https://github.com/pexpect/pexpect

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *ptyprocess: Python interaction with subprocesses in a pseudoterminal*

Version Information

package-version.txt:

```
4.9.0
```

version_requirements.txt:

```
pexpect >=4.8.0
```

Equivalent System Packages

```
$ sudo pacman -S python-pexpect
```

```
$ conda install pexpect
```

```
$ sudo dnf install python3-pexpect
```

```
$ sudo pkg install misc/py-pexpect
```

```
$ sudo emerge dev-python/pexpect
```

```
$ sudo port install py-pexpect
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-pexpect'
```

```
$ sudo xbps-install python3-pexpect
```

See https://repology.org/project/pexpect/versions, https://repology.org/project/python:pexpect/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.269 phitigra: Graph editor for SageMath/Jupyter

Description

Graph editor for SageMath/Jupyter

License

Upstream Contact

https://pypi.org/project/phitigra/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *ipywidgets: Interactive HTML widgets for Jupyter notebooks and the IPython kernel*
- *numpy: Package for scientific computing with Python*
- *pillow: Python Imaging Library*

Version Information

requirements.txt:

```
phitigra>=0.2.6
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.270 pickleshare: A 'shelve' like datastore with concurrency support

Description

PickleShare - a small 'shelve' like datastore with concurrency support

Like shelve, a PickleShareDB object acts like a normal dictionary. Unlike shelve, many processes can access the database simultaneously. Changing a value in database is immediately visible to other processes accessing the same database.

Concurrency is possible because the values are stored in separate files. Hence the "database" is a directory where all files are governed by PickleShare.

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
0.7.5
```

version_requirements.txt:

```
pickleshare >=0.7.5
```

Equivalent System Packages

```
$ sudo pacman -S python-pickleshare
```

```
$ conda install pickleshare
```

```
$ sudo apt-get install python3-pickleshare
```

```
$ sudo dnf install python3-pickleshare
```

```
$ sudo emerge dev-python/pickleshare
```

```
$ sudo port install py-pickleshare
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-pickleshare'
```

```
$ sudo xbps-install python3-pickleshare
```

See https://repology.org/project/pickleshare/versions, https://repology.org/project/python:pickleshare/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.271 pillow: Python Imaging Library

Description

Pillow is the "friendly" PIL fork by Alex Clark and Contributors.

The Python Imaging Library (PIL) adds powerful image processing and graphics capabilities to Python. The library supports many file formats.

License

Standard PIL License

Upstream Contact

- Author: Alex Clark <aclark@aclark.net>
- https://python-pillow.org/
- Homepage: http://python-imaging.github.io/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *freetype: A free, high-quality, and portable font engine*
- *pybind11: Create Python bindings to C++ code*
- *trove_classifiers: Canonical source for classifiers on PyPI (pypi.org)*

Version Information

package-version.txt:

```
12.1.1
```

version_requirements.txt:

```
pillow >=7.2.0
```

Equivalent System Packages

```
$ sudo pacman -S python-pillow
```

```
$ conda install pillow
```

```
$ sudo apt-get install python3-pillow
```

```
$ sudo dnf install python3-pillow
```

```
$ sudo emerge dev-python/pillow
```

```
$ brew install pillow
```

```
$ sudo port install py-Pillow
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-Pillow'
```

```
$ sudo xbps-install python3-Pillow
```

See https://repology.org/project/python:pillow/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.272 pip: Tool for installing and managing Python packages

Description

This package installs pip, the tool for installing and managing Python packages, such as those found in the Python Package Index. It's a replacement for easy_install.

License

MIT

Upstream Contact

- Project Page: https://github.com/pypa/pip
- Install howto: https://pip.pypa.io/en/latest/installing.html
- Changelog: https://pip.pypa.io/en/latest/news.html
- Bug Tracking: https://github.com/pypa/pip/issues
- Mailing list: http://groups.google.com/group/python-virtualenv
- Docs: https://pip.pypa.io/

Type

standard

Dependencies

- $(PYTHON)

Version Information

package-version.txt:

```
26.0.1
```

version_requirements.txt:

```
pip >=23.1.0
```

Equivalent System Packages

```
$ sudo pacman -S python-pip
```

```
$ conda install pip
```

```
$ sudo apt-get install python3-pip
```

```
$ sudo dnf install python3-pip
```

```
$ sudo pkg install devel/py-pip
```

```
$ sudo emerge dev-python/pip
```

```
$ sudo port install py-pip
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-pip'
```

```
$ sudo xbps-install python3-pip
```

See https://repology.org/project/pip3/versions, https://repology.org/project/python:pip/versions, https://repology.org/project/python3x-pip/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.273 pkgconfig: Python interface to pkg-config

Description

Pkgconfig is a Python module to interface with the pkg-config command line tool.

License

MIT License

Upstream Contact

https://github.com/matze/pkgconfig

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
1.6.0
```

Equivalent System Packages

```
$ sudo pacman -S python-pkgconfig
```

```
$ conda install pkgconfig
```

```
$ sudo apt-get install python3-pkgconfig
```

```
$ sudo dnf install python3-pkgconfig
```

```
$ sudo pkg install devel/py-pkgconfig
```

```
$ sudo emerge dev-python/pkgconfig
```

```
$ sudo port install py-pkgconfig
```

```
$ sudo zypper install pkg-config
```

```
$ sudo xbps-install python3-pkgconfig
```

See https://repology.org/project/python:pkgconfig/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.274 planarity: Planarity-related graph algorithms

Description

This code project provides a library for implementing graph algorithms as well as implementations of several planarity-related graph algorithms. The origin of this project is the reference implementation for the Edge Addition Planarity Algorithm [1], which is now the fastest and simplest linear-time method for planar graph embedding and planarity obstruction isolation (i.e. Kuratowski subgraph isolation).

[1] http://dx.doi.org/10.7155/jgaa.00091

License

New BSD License

Upstream Contact

- https://github.com/graph-algorithms/edge-addition-planarity-suite/
- John Boyer <John.Boyer.PhD@gmail.com>

Special Update/Build Instructions

The tarballs can be found at, https://github.com/graph-algorithms/edge-addition-planarity-suite/releases sage tarball is repackaged after running autogen.sh

Type

standard

Dependencies

Version Information

package-version.txt:

```
4.0.0.0
```

Equivalent System Packages

```
$ sudo pacman -S planarity
```

```
$ conda install planarity
```

```
$ sudo apt-get install libplanarity-dev planarity
```

```
$ sudo dnf install planarity planarity-devel
```

```
$ sudo pkg install math/planarity
```

```
$ sudo emerge sci-mathematics/planarity
```

```
$ brew install dimpase/tap/planarity
```

```
$ nix-env --install planarity
```

```
$ sudo zypper install edge-addition-planarity-suite  \
     edge-addition-planarity-suite-devel
```

```
$ sudo xbps-install planarity-devel
```

See https://repology.org/project/planarity/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.275 plantri: Generate non-isomorphic sphere-embdedded graphs

Description

Plantri is a program that generates certain types of graphs that are imbedded on the sphere.

Exactly one member of each isomorphism class is output, using an amount of memory almost independent of the number of graphs produced. This, together with the exceptionally fast operation and careful validation, makes the program suitable for processing very large numbers of graphs.

Isomorphisms are defined with respect to the embeddings, so in some cases outputs may be isomorphic as abstract graphs.

License

Plantri is distributed without a license.

Upstream Contact

Gunnar Brinkmann

- University of Ghent
- Gunnar.Brinkmann@ugent.be

Brendan McKay

- Australian National University
- bdm@cs.anu.edu.au

See http://cs.anu.edu.au/~bdm/plantri

Type

optional

Dependencies

Version Information

package-version.txt:

```
5.8
```

Equivalent System Packages

```
$ sudo pacman -S plantri
```

```
$ sudo dnf install plantri
```

See https://repology.org/project/plantri/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.276 platformdirs: Small Python package for determining appropriate platform-specific dirs, e.g. a "user data dir"

Description

Small Python package for determining appropriate platform-specific dirs, e.g. a "user data dir"

License

Upstream Contact

https://pypi.org/project/platformdirs/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
4.9.4
```

version_requirements.txt:

```
platformdirs
```

Equivalent System Packages

```
$ conda install platformdirs
```

```
$ sudo dnf install python3-platformdirs
```

```
$ sudo emerge dev-python/platformdirs
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.277 pluggy: Plugin and hook calling mechanisms for python

Description

Plugin and hook calling mechanisms for python

License

MIT license

Upstream Contact

https://pypi.org/project/pluggy/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
1.6.0
```

version_requirements.txt:

```
pluggy >=1.5
```

Equivalent System Packages

```
$ sudo pacman -S python-pluggy
```

```
$ conda install pluggy
```

```
$ sudo apt-get install python3-pluggy
```

```
$ sudo dnf install python3-pluggy
```

```
$ sudo emerge dev-python/pluggy
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-pluggy'
```

```
$ sudo xbps-install python3-pluggy
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.278 ply: Python Lex & Yacc

Description

Python Lex & Yacc

License

BSD

Upstream Contact

https://pypi.org/project/ply/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
3.11
```

version_requirements.txt:

```
ply
```

Equivalent System Packages

```
$ conda install ply
```

```
$ sudo emerge dev-python/ply
```

```
$ sudo xbps-install python3-ply
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.279 polylib: Operations on unions of polyhedra

Description

The Polyhedral Library (PolyLib for short) operates on objects made up of unions of polyhedra of any dimension. polylib is a C library.

License

GPL v3

Upstream Contact

- https://groups.google.com/forum/#!forum/isl-development

Type

experimental

Dependencies

- $(MP_LIBRARY)
- *mpfr: Multiple-precision floating-point computations with correct rounding*
- *ntl: A library for doing number theory*

Version Information

package-version.txt:

```
5.22.5
```

Equivalent System Packages

```
$ sudo port install polylib
```

```
$ sudo zypper install polylib 'pkgconfig(polylibgmp)'
```

See https://repology.org/project/polylib/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.280 polymake: Computations with polyhedra, fans, simplicial complexes, matroids, graphs, tropical hypersurfaces

Description

polymake is open source software for research in polyhedral geometry. It deals with polytopes, polyhedra and fans as well as simplicial complexes, matroids, graphs, tropical hypersurfaces, and other objects. Supported platforms include various flavors of Linux, Free BSD and Mac OS.

License

- GPL v3

Upstream Contact

- https://polymake.org/

Dependencies

Polymake needs a working installation of Perl, including its shared library and some modules (`XML::Writer XML::LibXML XML::LibXSLT Term::ReadLine::Gnu JSON SVG`). The Polymake interface in Sage additionally needs `File::Slurp`. For full functionality including polymake's polyDB, also the Perl module `MongoDB` is needed.

These are not provided by a Sage package. The dummy package `perl_cpan_polymake_prereq` will signal an error at build time if the required prerequisites are not met.

The `configure` script will inform you about the equivalent system packages that you should install. Otherwise, you can use CPAN (see below).

Sage might install the `Term::ReadLine::Gnu` module, however, when you install polymake, if it is not provided by the system, or if Sage installs its own `readline` library.

A distribution-independent way to install Perl modules (into a user's home directory or `/usr/local`) is using CPAN. This is also the way to install the modules on macOS. For this, if you don't have root access, you will need the `local::lib` Perl module installed:

```
cpan -i XML::Writer XML::LibXML XML::LibXSLT File::Slurp Term::ReadLine::Gnu JSON SVG␣
↪MongoDB
```

Before installing the `polymake` package, refer to the SPKG pages for the following packages to ensure a more featureful Polymake installation:

- [4ti2](https://doc.sagemath.org/html/en/reference/spkg/4ti2.html)
- [latte_int](https://doc.sagemath.org/html/en/reference/spkg/latte_int.html)
- [topcom](https://doc.sagemath.org/html/en/reference/spkg/topcom.html)
- [qhull](https://doc.sagemath.org/html/en/reference/spkg/qhull.html)

For additional software that may enhance your Polymake installation (but for which no Sage package is available), you can manually install the following:

- `azove`
- `porta`
- `vinci`
- `SplitsTree4`

Information on missing Polymake prerequisites after installing polymake:

```
$ sage -sh
(sage-sh) $ polymake
polytope> show_unconfigured;
```

In order to use Polymake from Sage, please refer to the [Jupymake SPKG page](https://doc.sagemath.org/html/en/reference/spkg/jupymake.html) for installation instructions.

Debugging polymake install problems

```
# apt-get install libdevel-trace-perl
$ cd src
$ perl -d:Trace support/configure.pl
```

Type

optional

Dependencies

- $(MP_LIBRARY)
- *bliss: Computing automorphism groups and canonical forms of graphs*
- *cddlib: Double description method for polyhedral representation conversion*
- *libxml2: XML parser and toolkit*
- *lrslib: Reverse search algorithm for vertex enumeration and convex hull problems*
- *ninja_build: A build system with a focus on speed*

- *normaliz: Computations in affine monoids, vector configurations, lattice polytopes, and rational cones*
- *perl_cpan_polymake_prereq: Represents all Perl packages that are prerequisites for polymake*
- *perl_term_readline_gnu: Perl extension for the GNU Readline/History libraries*
- *ppl: Parma Polyhedra Library*

Version Information

package-version.txt:

```
4.14
```

Equivalent System Packages

```
$ sudo pacman -S polymake
```

```
$ sudo apt-get install polymake libpolymake-dev
```

```
$ sudo dnf install polymake
```

```
$ brew install apaffenholz/polymake/polymake
```

```
$ nix-env --install polymake
```

```
$ sudo zypper install polymake
```

See https://repology.org/project/polymake/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.281 polytopes_db: Databases of 2- and 3-dimensional reflexive polytopes

Description

This package includes lists of 2- and 3-dimensional reflexive polytopes.

The list of polygons is quite easy to get and it has been known for a while. The list of 3-polytopes was originally obtained by Maximilian Kreuzer and Harald Skarke using their software PALP, which is included into the standard distribution of Sage. To work with lattice and reflexive polytopes from Sage you can use sage.geometry.lattice_polytope module, which relies on PALP for some of its functionality. To get access to the databases of this package, use ReflexivePolytope and ReflexivePolytopes commands.

License

GPL

Upstream Contact

http://hep.itp.tuwien.ac.at/~kreuzer/CY/CYpalp.html

Type

standard

Dependencies

Version Information

package-version.txt:

```
20170220.p0
```

Equivalent System Packages

```
$ sudo pacman -S sage-data-polytopes_db
```

```
$ conda install sagemath-db-polytopes
```

See https://repology.org/project/sagemath-polytopes-db/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.282 polytopes_db_4d: Database of 4-dimensional reflexive polytopes

Description

This package contains the database of 4-d reflexive polytopes with Hodge numbers as index.

Based on the original list by Maximilian Kreuzer and Harald Skarke using their software PALP.

License

GPL v2+

SPKG Maintainers

Volker Braun <vbraun.name@gmail.com>

Type

optional

Dependencies

Version Information

package-version.txt:

```
1.0
```

Equivalent System Packages

See https://repology.org/project/polytopes-db-4d/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.283 ppl: Parma Polyhedra Library

Description

The Parma Polyhedra Library (PPL) provides numerical abstractions especially targeted at applications in the field of analysis and verification of complex systems. These abstractions include convex polyhedra, defined as the intersection of a finite number of (open or closed) halfspaces, each described by a linear inequality (strict or non-strict) with rational coefficients; some special classes of polyhedra shapes that offer interesting complexity/precision tradeoffs; and grids which represent regularly spaced points that satisfy a set of linear congruence relations. The library also supports finite powersets and products of (any kind of) polyhedra and grids, a mixed integer linear programming problem solver using an exact-arithmetic version of the simplex algorithm, a parametric integer programming solver, and primitives for the termination analysis via the automatic synthesis of linear ranking functions.

It is written in C++, but comes with interfaces to C, Java, OCaml, and Prolog. PPL is one of the fastest implementations of polyhedral computations.

Benchmarks are included in this paper: arXiv cs/1

License

GPL v3+

Upstream Contact

- https://www.bugseng.com/ppl

Core Development Team

- Roberto Bagnara (University of Parma)
- Patricia M. Hill (University of Parma)
- Enea Zaffanella (University of Parma)

Type

standard

Dependencies

- $(MP_LIBRARY)
- *glpk: GNU Linear Programming Kit*

Version Information

package-version.txt:

```
1.2.p1
```

Equivalent System Packages

```
$ sudo pacman -S ppl
```

```
$ conda install ppl
```

```
$ sudo apt-get install libppl-dev ppl-dev
```

```
$ sudo dnf install ppl ppl-devel
```

```
$ sudo pkg install devel/ppl
```

```
$ sudo emerge dev-libs/ppl
```

```
$ brew install ppl
```

```
$ sudo port install ppl
```

```
$ nix-env --install ppl
```

```
$ sudo zypper install ppl-devel
```

```
$ sudo xbps-install ppl-devel
```

See https://repology.org/project/ppl/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.284 pplpy: Python interface to the Parma Polyhedra Library

Description

PPL Python wrapper

This Python package provides a wrapper to the C++ Parma Polyhedra Library (PPL).

The whole package started as a fork of a tiny part of the Sage software.

License

GPL version 3

Upstream Contact

- https://github.com/sagemath/pplpy

Type

standard

Dependencies

- $(MP_LIBRARY)
- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cysignals: Interrupt and signal handling for Cython*
- *gmpy2: Python interface to GMP/MPIR, MPFR, and MPC*
- *mpc: Arithmetic of complex numbers with arbitrarily high precision and correct rounding*
- *mpfr: Multiple-precision floating-point computations with correct rounding*

- *ppl: Parma Polyhedra Library*
- *sphinx: Python documentation generator*

Version Information

package-version.txt:

```
0.9.0
```

Equivalent System Packages

```
$ sudo pacman -S python-pplpy
```

```
$ conda install pplpy
```

```
$ sudo dnf install python3-pplpy
```

```
$ sudo pkg install math/py-pplpy
```

```
$ sudo emerge dev-python/pplpy
```

```
$ sudo xbps-install python3-pplpy
```

See https://repology.org/project/pplpy/versions, https://repology.org/project/python:pplpy/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.285 pplpy_doc: Python interface to the Parma Polyhedra Library (documentation)

Description

PPL Python wrapper (documentation)

License

GPL version 3

Upstream Contact

- https://github.com/sagemath/pplpy

Type

standard

Dependencies

- *cysignals: Interrupt and signal handling for Cython*
- *gmpy2: Python interface to GMP/MPIR, MPFR, and MPC*
- *pplpy: Python interface to the Parma Polyhedra Library*
- *sphinx: Python documentation generator*

Version Information

package-version.txt:

```
0.9.0
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.286 primecount: Algorithms for counting primes

Description

primecount is a C++ implementation of several algorithms for counting primes maintained by Kim Walisch.

Website: https://github.com/kimwalisch/primecount/

License

primecount is licensed BSD 2

Upstream Contact

- https://github.com/kimwalisch/primecount/

Type

standard

Dependencies

- *cmake: A cross-platform build system generator*
- *primesieve: CLI program and C/C++ library for generating primes*

Version Information

package-version.txt:

```
8.6
```

Equivalent System Packages

```
$ sudo pacman -S primecount
```

```
$ conda install primecount
```

```
$ sudo apt-get install libprimecount-dev primecount
```

```
$ sudo dnf install primecount primecount-devel
```

```
$ sudo emerge sci-mathematics/primecount
```

```
$ brew install primecount
```

```
$ sudo zypper install primecount libprimecount-devel
```

```
$ sudo xbps-install primecount-devel
```

See https://repology.org/project/primecount/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.287 primecountpy: Cython interface for C++ primecount library

Description

Cython interface for C++ primecount library

License

GPLv3

Upstream Contact

https://pypi.org/project/primecountpy/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cysignals: Interrupt and signal handling for Cython*
- *cython: C-Extensions for Python, an optimizing static compiler*
- *primecount: Algorithms for counting primes*

Version Information

package-version.txt:

```
0.2.1
```

version_requirements.txt:

```
primecountpy >= 0.2.1
```

Equivalent System Packages

```
$ conda install primecountpy
```

```
$ sudo dnf install python3-primecountpy
```

```
$ sudo emerge dev-python/primecountpy
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.288 primesieve: CLI program and C/C++ library for generating primes

Description

A CLI program and C/C++ library for quickly generating prime numbers. https://github.com/kimwalisch/primesieve

A dependency of the standard spkg primecount.

License

BSD-2-clause

Upstream Contact

- https://github.com/kimwalisch/primesieve

Type

standard

Dependencies

- *cmake: A cross-platform build system generator*

Version Information

package-version.txt:

```
12.15
```

Equivalent System Packages

```
$ apk add primesieve-dev primesieve
```

```
$ sudo pacman -S primesieve
```

```
$ conda install primesieve
```

```
$ sudo apt-get install libprimesieve-dev
```

```
$ sudo dnf install primesieve-devel primesieve
```

```
$ sudo emerge sci-mathematics/primesieve
```

```
$ brew install primesieve
```

```
$ sudo zypper install primesieve
```

```
$ sudo xbps-install primesieve-devel
```

See https://repology.org/project/primesieve/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.289 prometheus_client: Python client for the systems monitoring and alerting toolkit Prometheus

Description

The official Python 2 and 3 client for Prometheus (see https://prometheus.io), an open-source systems monitoring and alerting toolkit.

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
0.24.1
```

version_requirements.txt:

```
prometheus_client >=0.8.0
```

Equivalent System Packages

```
$ conda install prometheus_client
```

```
$ sudo dnf install python3-prometheus_client
```

```
$ sudo emerge dev-python/prometheus_client
```

```
$ sudo port install py-prometheus_client
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-prometheus_client'
```

```
$ sudo xbps-install python3-prometheus_client
```

See https://repology.org/project/python:prometheus-client/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.290 prompt_toolkit: Interactive command lines for Python

Description

Library for building powerful interactive command lines in Python

https://pypi.python.org/pypi/prompt_toolkit

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *wcwidth: Measures the displayed width of unicode strings in a terminal*

Version Information

package-version.txt:

```
3.0.52
```

version_requirements.txt:

```
prompt_toolkit >=3.0.43
```

Equivalent System Packages

```
$ sudo pacman -S python-prompt_toolkit
```

```
$ conda install 'prompt_toolkit>=3.0.38'
```

```
$ sudo apt-get install prompt-toolkit
```

```
$ sudo dnf install python3-prompt-toolkit
```

```
$ sudo pkg install devel/py-prompt_toolkit
```

```
$ sudo emerge dev-python/prompt-toolkit
```

```
$ sudo port install py-prompt_toolkit
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-prompt_toolkit'
```

```
$ sudo xbps-install python3-prompt_toolkit
```

See https://repology.org/project/python:prompt-toolkit/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.291 psutil: Cross-platform lib for process and system monitoring in Python.

Description

Cross-platform lib for process and system monitoring in Python.

License

BSD-3-Clause

Upstream Contact

https://pypi.org/project/psutil/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
7.2.2
```

version_requirements.txt:

```
psutil
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.292 ptyprocess: Python interaction with subprocesses in a pseudoterminal

Description

Launch a subprocess in a pseudo terminal (pty), and interact with both the process and its pty.

Sometimes, piping stdin and stdout is not enough. There might be a password prompt that doesn't read from stdin, output that changes when it's going to a pipe rather than a terminal, or curses-style interfaces that rely on a terminal. If you need to automate these things, running the process in a pseudo terminal (pty) is the answer.

License

Ptyprocess is under the ISC license, as code derived from Pexpect.

> http://opensource.org/licenses/ISC

Upstream Contact

https://github.com/pexpect/ptyprocess

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
0.7.0
```

version_requirements.txt:

```
ptyprocess > 0.5
```

Equivalent System Packages

```
$ sudo pacman -S python-ptyprocess
```

No package needed.

```
$ sudo dnf install python3-ptyprocess
```

```
$ sudo pkg install sysutils/py-ptyprocess
```

```
$ sudo emerge dev-python/ptyprocess
```

```
$ sudo port install py-ptyprocess
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-ptyprocess'
```

```
$ sudo xbps-install python3-ptyprocess
```

See https://repology.org/project/ptyprocess/versions, https://repology.org/project/python:ptyprocess/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.293 pure_eval: Safely evaluate AST nodes without side effects

Description

Safely evaluate AST nodes without side effects

License

MIT

Upstream Contact

https://pypi.org/project/pure-eval/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
0.2.3
```

version_requirements.txt:

```
pure-eval
```

Equivalent System Packages

```
$ conda install pure_eval
```

```
$ sudo dnf install python3-pure-eval
```

```
$ sudo emerge dev-python/pure_eval
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.294 py: Library with cross-python path, ini-parsing, io, code, log facilities

Description

Library with cross-python path, ini-parsing, io, code, log facilities

License

MIT license

Upstream Contact

https://pypi.org/project/py/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
1.11.0
```

version_requirements.txt:

```
py
```

Equivalent System Packages

```
$ sudo pacman -S python-py
```

```
$ conda install py
```

```
$ sudo apt-get install python3-py
```

```
$ sudo dnf install python3-py
```

```
$ sudo emerge dev-python/py
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-py'
```

```
$ sudo xbps-install python3-py
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.295 pybind11: Create Python bindings to C++ code

Description

pybind11 is a lightweight header-only library that exposes C++ types in Python and vice versa, mainly to create Python bindings of existing C++ code. Its goals and syntax are similar to the excellent [Boost.Python](http://www.boost.org/doc/libs/1_58_0/libs/python/doc/) library by David Abrahams: to minimize boilerplate code in traditional extension modules by inferring type information using compile-time introspection.

License

pybind11 is provided under a BSD-style license that can be found in the `LICENSE` file. By using, distributing, or contributing to this project, you agree to the terms and conditions of this license.

Upstream Contact

https://github.com/pybind/pybind11

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
2.13.6
```

version_requirements.txt:

```
pybind11>=2.13.2
```

Equivalent System Packages

```
$ conda install pybind11
```

```
$ sudo dnf install python3-pybind11
```

```
$ sudo emerge dev-python/pybind11
```

```
$ brew install pybind11
```

```
$ sudo port install py-pybind11
```

```
$ sudo xbps-install python3-pybind11
```

See https://repology.org/project/python:pybind11/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.296 pybtex: A BibTeX-compatible bibliography processor in Python

Description

A BibTeX-compatible bibliography processor in Python

License

MIT

Upstream Contact

https://pypi.org/project/pybtex/

Type

optional

Dependencies

Version Information

requirements.txt:

```
pybtex
```

Equivalent System Packages

```
$ conda install pybtex
```

```
$ sudo dnf install python3-pybtex
```

```
$ sudo port install py-pybtex
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-pybtex'
```

See https://repology.org/project/python:pybtex/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.297 pycosat: SAT solver picosat with Python bindings

Description

PicoSAT is a popular SAT solver written by Armin Biere in pure C. This package provides efficient Python bindings to picosat on the C level, i.e. when importing pycosat, the picosat solver becomes part of the Python process itself. For ease of deployment, the picosat source (namely picosat.c and picosat.h) is included in this project. These files have been extracted from the picosat source.

License

MIT

Upstream Contact

- PicoSAT: http://fmv.jku.at/picosat/
- pycosat: https://github.com/ContinuumIO/pycosat

Special Update/Build Instructions

None.

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
0.6.3
```

version_requirements.txt:

```
pycosat >=0.6.3
```

Equivalent System Packages

```
$ apk add py3-pycosat
```

```
$ sudo pacman -S python-pycosat
```

```
$ conda install pycosat
```

```
$ sudo dnf install python3-pycosat
```

```
$ sudo pkg install math/py-pycosat
```

See https://repology.org/project/pycosat/versions, https://repology.org/project/python:pycosat/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.298 pycparser: Parser of the C language in Python

Description

development website: https://github.com/eliben/pycparser

PyPI page: https://pypi.org/project/pycparser/

License

BSD

Upstream Contact

https://github.com/eliben/pycparser

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
3.0
```

version_requirements.txt:

```
pycparser >=2.20
```

Equivalent System Packages

```
$ conda install pycparser
```

```
$ sudo dnf install python3-pycparser
```

```
$ sudo emerge dev-python/pycparser
```

```
$ sudo port install py-pycparser
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-pycparser'
```

```
$ sudo xbps-install python3-pycparser
```

See https://repology.org/project/pycparser/versions, https://repology.org/project/python:pycparser/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.299 pycryptosat: Python module of cryptominisat

Description

Build and install pycryptosat (Python frontend to CryptoMiniSat, which will be built if needed too).

CryptoMiniSat is a SAT solver that aims to become a premiere SAT solver with all the features and speed of successful SAT solvers, such as MiniSat and PrecoSat. The long-term goals of CryptoMiniSat are to be an efficient sequential, parallel and distributed solver. There are solvers that are good at one or the other, e.g. ManySat (parallel) or PSolver (distributed), but we wish to excel at all.

CryptoMiniSat 2.5 won the SAT Race 2010 among 20 solvers submitted by researchers and industry.

License

MIT License

Upstream Contact

- Authors: Mate Soos
- Email: soos.mate@gmail.com
- Website: http://www.msoos.org/
- Releases: https://github.com/msoos/cryptominisat/releases

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cmake: A cross-platform build system generator*
- *libpng: Bitmap image support*
- *m4ri: fast arithmetic with dense matrices over GF(2)*

Version Information

requirements.txt:

```
pycryptosat
```

Equivalent System Packages

```
$ conda install cryptominisat
```

```
$ sudo dnf install python3-pycryptosat
```

```
$ brew install cryptominisat
```

See https://repology.org/project/cryptominisat/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.300 pygments: Generic syntax highlighter

Description

Pygments is a syntax highlighting package written in Python.

It is a generic syntax highlighter suitable for use in code hosting, forums, wikis or other applications that need to prettify source code. Highlights are:

- a wide range of over 300 languages and other text formats is supported

- special attention is paid to details, increasing quality by a fair amount
- support for new languages and formats are added easily
- a number of output formats, presently HTML, LaTeX, RTF, SVG, all image formats that PIL supports and ANSI sequences
- it is usable as a command-line tool and as a library

License

BSD-2-Clause

Upstream Contact

https://pypi.org/project/Pygments/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
2.20.0
```

version_requirements.txt:

```
pygments >=2.12
```

Equivalent System Packages

```
$ sudo pacman -S python-pygments
```

```
$ conda install pygments
```

```
$ sudo apt-get install python3-pygments
```

```
$ sudo dnf install python3-pygments
```

```
$ sudo emerge dev-python/pygments
```

```
$ brew install pygments
```

```
$ sudo port install py-pygments
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-pygments'
```

```
$ sudo xbps-install python3-Pygments
```

See https://repology.org/project/pygments/versions, https://repology.org/project/python:pygments/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.301 pygraphviz: Python interface to Graphviz

Description

Python interface to Graphviz

License

BSD

Upstream Contact

https://pypi.org/project/pygraphviz/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *libgraphviz: Graph visualization software (callable library)*

Version Information

requirements.txt:

```
pygraphviz
```

version_requirements.txt:

```
pygraphviz
```

Equivalent System Packages

```
$ conda install pygraphviz
```

```
$ sudo dnf install python3-pygraphviz
```

```
$ sudo port install py-pygraphviz
```

See https://repology.org/project/python:pygraphviz/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.302 pynormaliz: Python bindings for the normaliz library

Description

The Python module PyNormaliz provides wrappers for normaliz.

License

- GPL v2 or later

Upstream Contact

https://pypi.org/project/PyNormaliz/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *normaliz: Computations in affine monoids, vector configurations, lattice polytopes, and rational cones*

Version Information

package-version.txt:

```
2.24
```

version_requirements.txt:

```
pynormaliz >=2.18
```

Equivalent System Packages

```
$ sudo pacman -S python-pynormaliz
```

```
$ conda install pynormaliz
```

See https://repology.org/project/pynormaliz/versions, https://repology.org/project/python:pynormaliz/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.303 pyparsing: A Python parsing module

Description

A Python Parsing Module

License

MIT License

Upstream Contact

https://pypi.org/project/pyparsing/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
3.3.2
```

version_requirements.txt:

```
pyparsing >=3.2.3
```

Equivalent System Packages

```
$ sudo pacman -S python-pyparsing
```

```
$ conda install pyparsing
```

```
$ sudo dnf install python3-pyparsing
```

```
$ sudo pkg install devel/py-pyparsing
```

```
$ sudo emerge dev-python/pyparsing
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-pyparsing'
```

```
$ sudo xbps-install python3-parsing
```

See https://repology.org/project/pyparsing/versions, https://repology.org/project/python:pyparsing/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.304 pyppeteer: Headless chrome/chromium automation library

Description

Headless chrome/chromium automation library (unofficial port of puppeteer)

License

MIT

Upstream Contact

https://pypi.org/project/pyppeteer/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *appdirs: Small Python module for determining appropriate platform-specific dirs, e.g. a "user data dir"*
- *certifi: Python package for providing Mozilla's CA Bundle*
- *urllib3: HTTP library with thread-safe connection pooling, file post, and more*

Version Information

requirements.txt:

```
pyppeteer
```

Equivalent System Packages

```
$ conda install pyppeteer
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.305 pyproject_api: API to interact with the python pyproject.toml based projects

Description

API to interact with the python pyproject.toml based projects

License

Upstream Contact

https://pypi.org/project/pyproject-api/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *packaging: Core utilities for Python packages*

Version Information

package-version.txt:

```
1.10.0
```

version_requirements.txt:

```
pyproject-api
```

Equivalent System Packages

```
$ sudo pacman -S python-pyproject-api
```

```
$ sudo apt-get install python3-pyproject-api
```

```
$ sudo dnf install python3-pyproject-api
```

```
$ sudo emerge dev-python/pyproject-api
```

```
$ sudo xbps-install python3-pyproject-api
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.306 pyproject_hooks: Wrappers to call pyproject.toml-based build backend hooks.

Description

Wrappers to call pyproject.toml-based build backend hooks.

License

Upstream Contact

https://pypi.org/project/pyproject_hooks/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
1.2.0
```

version_requirements.txt:

```
pyproject_hooks
```

Equivalent System Packages

```
$ sudo pacman -S python-pyproject-hooks
```

```
$ sudo apt-get install python3-pyproject-hooks
```

```
$ sudo dnf install python3-pyproject-hooks
```

```
$ sudo emerge dev-python/pyproject-hooks
```

```
$ sudo xbps-install python3-pyproject-hooks
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.307 pyproject_metadata: PEP 621 metadata parsing

Description

PEP 621 metadata parsing

License

MIT

Upstream Contact

https://pypi.org/project/pyproject-metadata/

Type

standard

Dependencies

- $(PYTHON)
- *packaging: Core utilities for Python packages*
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
0.11.0
```

version_requirements.txt:

```
pyproject-metadata
```

Equivalent System Packages

```
$ sudo pacman -S python-pyproject-metadata
```

```
$ conda install pyproject-metadata
```

```
$ sudo apt-get install python3-pyproject-metadata
```

```
$ sudo dnf install python3-pyproject-metadata
```

```
$ sudo emerge dev-python/pyproject-metadata
```

```
$ sudo xbps-install python3-pyproject-metadata
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.308 pyscipopt: Python interface and modeling environment for SCIP

Description

Python interface and modeling environment for SCIP

License

MIT

Upstream Contact

https://pypi.org/project/PySCIPOpt/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cython: C-Extensions for Python, an optimizing static compiler*
- *numpy: Package for scientific computing with Python*
- *scip: Mixed integer programming solver*

Version Information

package-version.txt:

```
6.1.0
```

version_requirements.txt:

```
PySCIPOpt
```

Equivalent System Packages

```
$ conda install pyscipopt
```

```
$ sudo pkg install math/py-PySCIPOpt
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.309 pysingular: A basic Python interface to Singular

Description

A basic interface to call Singular from python

This python module is meant to be used in Singulars Jupyter interface.

License

GPL version 2 or later

Upstream Contact

- https://github.com/sebasguts/SingularPython

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *singular: Computer algebra system for polynomial computations, algebraic geometry, singularity theory*

Version Information

package-version.txt:

```
0.9.7
```

version_requirements.txt:

```
pysingular >=0.9.5
```

Equivalent System Packages

```
$ conda install pysingular
```

```
$ sudo dnf install python3-pysingular
```

See https://repology.org/project/pysingular/versions, https://repology.org/project/python:pysingular/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.310 pytest: Simple powerful testing with Python

Description

Simple powerful testing with Python

License

MIT

Upstream Contact

https://pypi.org/project/pytest/

Type

standard

Dependencies

- $(PYTHON)
- *colorama: Cross-platform colored terminal text*
- *iniconfig: Brain-dead simple config-ini parsing*
- *packaging: Core utilities for Python packages*
- *pip: Tool for installing and managing Python packages*
- *pluggy: Plugin and hook calling mechanisms for python*
- *pygments: Generic syntax highlighter*

Version Information

package-version.txt:

```
9.0.2
```

version_requirements.txt:

```
pytest
```

Equivalent System Packages

```
$ conda install pytest
```

```
$ sudo dnf install python3-pytest
```

```
$ sudo port install py-pytest
```

```
$ sudo xbps-install python3-pytest
```

See https://repology.org/project/python:pytest/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.311 pytest_mock: Thin-wrapper around the mock package for easier use with pytest

Description

Thin-wrapper around the mock package for easier use with pytest

License

MIT

Upstream Contact

https://pypi.org/project/pytest-mock/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*
- *pytest: Simple powerful testing with Python*

Version Information

package-version.txt:

```
3.15.1
```

version_requirements.txt:

```
pytest-mock
```

Equivalent System Packages

```
$ conda install pytest-mock
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.312 pytest_xdist: Pytest xdist plugin for distributed testing, most importantly across multiple CPUs

Description

Pytest xdist plugin for distributed testing, most importantly across multiple CPUs

License

MIT

Upstream Contact

https://pypi.org/project/pytest-xdist/

Type

standard

Dependencies

- $(PYTHON)
- *execnet: Rapid multi-Python deployment*
- *pip: Tool for installing and managing Python packages*
- *pytest: Simple powerful testing with Python*

Version Information

package-version.txt:

```
3.8.0
```

version_requirements.txt:

```
pytest-xdist
```

Equivalent System Packages

```
$ conda install pytest-xdist
```

```
$ sudo dnf install python3-pytest-xdist
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.313 python_build: A simple, correct PEP517 package builder

Description

`build` is a simple, correct PEP517 package builder

License

MIT

Upstream Contact

https://pypi.org/project/build/

Type

standard

Dependencies

- $(PYTHON)
- *packaging: Core utilities for Python packages*
- *pip: Tool for installing and managing Python packages*
- *pyproject_hooks: Wrappers to call pyproject.toml-based build backend hooks.*

Version Information

package-version.txt:

```
1.4.2
```

version_requirements.txt:

```
build
```

Equivalent System Packages

```
$ conda install python-build
```

```
$ sudo emerge dev-python/build
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.314 python_discovery: Python interpreter discovery

Description

python-discovery provides Python interpreter discovery for tools such as tox and virtualenv. It locates Python installations on the system across different platforms and provides a consistent API for querying interpreter properties.

License

MIT

Upstream Contact

https://pypi.org/project/python-discovery/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *filelock: Platform independent file lock*
- *platformdirs: Small Python package for determining appropriate platform-specific dirs, e.g. a "user data dir"*

Version Information

package-version.txt:

```
1.2.1
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.315 python_flint: Bindings for FLINT and Arb

Description

Bindings for FLINT and Arb

License

MIT

Upstream Contact

https://pypi.org/project/python-flint/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cython: C-Extensions for Python, an optimizing static compiler*
- *flint: Fast Library for Number Theory*

Version Information

package-version.txt:

```
0.8.0
```

version_requirements.txt:

```
python-flint
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.316 python_igraph: Python bindings for igraph

Description

igraph is a library for creating and manipulating graphs. It is intended to be as powerful (ie. fast) as possible to enable the analysis of large graphs.

License

GPL version 2

Upstream Contact

http://igraph.org/python/

Special Update/Build Instructions

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *igraph: A library for creating and manipulating graphs*
- *texttable: Python module for creating simple ASCII tables*

Version Information

package-version.txt:

```
1.0.0
```

version_requirements.txt:

```
igraph
```

Equivalent System Packages

```
$ sudo pacman -S python-igraph
```

```
$ conda install python-igraph
```

```
$ sudo apt-get install python3-igraph
```

```
$ sudo dnf install python-igraph python3-igraph-devel
```

```
$ sudo pkg install math/py-igraph
```

```
$ sudo port install py-igraph
```

See https://repology.org/project/python:igraph/versions, https://repology.org/project/python:python-igraph/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.317 python_json_logger: Python library adding a json log formatter

Description

Python library adding a json log formatter

License

BSD

Upstream Contact

https://pypi.org/project/python-json-logger/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
4.1.0
```

version_requirements.txt:

```
python-json-logger
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.318 pythran: Ahead of Time compiler for numeric kernels

Description

Ahead of Time compiler for numeric kernels

License

BSD 3-Clause

Upstream Contact

https://pypi.org/project/pythran/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *beniget: Extract semantic information about static Python code*
- *gast: Python AST that abstracts the underlying Python version*
- *ply: Python Lex & Yacc*

Version Information

package-version.txt:

```
0.18.1
```

version_requirements.txt:

```
pythran>=0.18.0
```

Equivalent System Packages

```
$ conda install 'pythran>=0.14.0,<0.18'
```

```
$ sudo dnf install pythran
```

```
$ sudo emerge dev-python/pythran
```

```
$ sudo xbps-install pythran
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.319 pytz: Timezone definitions for Python

Description

World Timezone Definitions for Python See https://pypi.org/project/pytz/

Special Update/Build Instructions

The upstream tarball was repackaged after sanitizing the file permissions with

$ chmod go-w

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
2026.1.post1
```

version_requirements.txt:

```
pytz >=2020.1
```

Equivalent System Packages

```
$ sudo pacman -S python-pytz
```

```
$ conda install pytz
```

```
$ sudo apt-get install python3-tz
```

```
$ sudo dnf install python3-pytz
```

```
$ sudo emerge dev-python/pytz
```

```
$ sudo port install py-tz
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-pytz'
```

```
$ sudo xbps-install python3-pytz
```

See https://repology.org/project/python:pytz/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.320 pytz_deprecation_shim: Shims to make deprecation of pytz easier

Description

Shims to make deprecation of pytz easier

License

Apache-2.0

Upstream Contact

https://pypi.org/project/pytz-deprecation-shim/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *dateutil: Extensions to the standard Python module datetime*
- *tzdata: Provider of IANA time zone data*

Version Information

package-version.txt:

```
0.1.0.post0
```

version_requirements.txt:

```
pytz-deprecation-shim
```

Equivalent System Packages

```
$ conda install pytz-deprecation-shim
```

```
$ sudo dnf install python3-pytz-deprecation-shim
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.321 pyx: Generate PostScript, PDF, and SVG files in Python

Description

Python package for the generation of PostScript, PDF, and SVG files

https://pypi.python.org/pypi/PyX

Type

optional

Dependencies

Version Information

requirements.txt:

```
pyx
```

Equivalent System Packages

```
$ sudo port install py-pyx
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-PyX'
```

```
$ sudo xbps-install python3-pyx
```

See https://repology.org/project/python:pyx/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.322 pyyaml: YAML parser and emitter for Python

Description

YAML parser and emitter for Python

License

MIT

Upstream Contact

https://pypi.org/project/PyYAML/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cython: C-Extensions for Python, an optimizing static compiler*

Version Information

package-version.txt:

```
6.0.3
```

version_requirements.txt:

```
PyYAML
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.323 pyzmq: Python bindings for the zeromq networking library

Description

Python bindings for the zeromq networking library.

License

LGPLv3+

Upstream Contact

http://www.zeromq.org

Special Update/Build Instructions

None.

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cffi: Foreign Function Interface for Python calling C code*
- *cython: C-Extensions for Python, an optimizing static compiler*
- *numpy: Package for scientific computing with Python*
- *scikit_build_core: Build backend for CMake based projects*
- *zeromq: A modern networking library*

Version Information

package-version.txt:

```
27.1.0
```

version_requirements.txt:

```
pyzmq >=19.0.2
```

Equivalent System Packages

```
$ sudo pacman -S python-pyzmq
```

```
$ conda install pyzmq
```

```
$ sudo dnf install python3-pyzmq
```

```
$ sudo emerge dev-python/pyzmq
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-pyzmq'
```

```
$ sudo xbps-install python3-pyzmq
```

See https://repology.org/project/pyzmq/versions, https://repology.org/project/python:pyzmq/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.324 qdldl_python: QDLDL, a free LDL factorization routine (Python wrapper)

Description

QDLDL, a free LDL factorization routine.

License

Apache 2.0

Upstream Contact

https://pypi.org/project/qdldl/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cmake: A cross-platform build system generator*
- *numpy: Package for scientific computing with Python*
- *pybind11: Create Python bindings to C++ code*
- *scipy: Scientific tools for Python*

Version Information

package-version.txt:

```
0.1.9.post1
```

version_requirements.txt:

```
qdldl
```

Equivalent System Packages

```
$ conda install qdldl-python
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.325 qepcad: Quantifier elimination by partial cylindrical algebraic decomposition

Description

Qepcad is an implementation of quantifier elimination by partial cylindrical algebraic decomposition

License

ICS Wikipedia article ISC_license

Upstream Contact

- Repository: https://github.com/chriswestbrown/qepcad
- Tarballs: https://www.usna.edu/Users/cs/wcbrown/qepcad/INSTALL/IQ.html
- Website: (outdated) https://www.usna.edu/Users/cs/wcbrown/qepcad/B/QEPCAD.html

Type

optional

Dependencies

- *readline: Command line editing library*
- *saclib: Computations with real algebraic numbers*

Version Information

package-version.txt:

```
1.74
```

Equivalent System Packages

```
$ sudo apt-get install qepcad
```

```
$ sudo dnf install qepcad-B
```

```
$ sudo port install qepcad
```

See https://repology.org/project/qepcad-b/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.326 qhull: Compute convex hulls, Delaunay triangulations, Voronoi diagrams

Description

From the README.txt of Qhull:

Qhull computes convex hulls, Delaunay triangulations, Voronoi diagrams, furthest-site Voronoi diagrams, and halfspace intersections about a point. It runs in 2-d, 3-d, 4-d, or higher. It implements the Quickhull algorithm for computing convex hulls. Qhull handles round-off errors from floating point arithmetic. It can approximate a convex hull.

The program includes options for hull volume, facet area, partial hulls, input transformations, randomization, tracing, multiple output formats, and execution statistics.

Further notes:

The qhull library is already shipped with the Python library scipy (from version 1.4), see

- http://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.ConvexHull.html
- http://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.Delaunay.html
- http://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.Voronoi.html

There is also the Python interface Pyhull available on PyPI https://pypi.python.org/pypi/pyhull (see also documentation at http://pythonhosted.org/pyhull/).

Upstream Contact

http://www.qhull.org/html

C. Bradford Barber bradb@shore.net or qhull@qhull.org

Dependencies

Can be compiled with Qt support, but the Sage version currently doesn't try to do this.

License

Not a standard license, but Sage compatible. See the COPYING.txt file in the source directory for details.

Type

standard

Dependencies

- *cmake: A cross-platform build system generator*

Version Information

package-version.txt:

```
2020-src-8.0.2
```

Equivalent System Packages

```
$ apk add qhull-dev qhull
```

```
$ sudo pacman -S qhull
```

```
$ conda install qhull
```

```
$ sudo apt-get install qhull-bin libqhull-dev
```

```
$ sudo dnf install qhull qhull-devel
```

```
$ sudo pkg install math/qhull
```

```
$ sudo emerge 'media-libs/qhull[tools]'
```

```
$ brew install qhull
```

No package needed.

```
$ nix-env --install qhull
```

```
$ sudo zypper install qhull-devel
```

```
$ sudo xbps-install qhull libqhull-devel
```

See https://repology.org/project/qhull/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.327 r: A free software environment for statistical computing and graphics

Description

R is a language and environment for statistical computing and graphics. It is a GNU project which is similar to the S language and environment which was developed at Bell Laboratories (formerly AT&T, now Lucent Technologies) by John Chambers and colleagues. R can be considered as a different implementation of S. There are some important differences, but much code written for S runs unaltered under R.

(taken from http://www.r-project.org/)

License

- GPL v2 or GPL v3

Upstream Contact

- https://www.r-project.org
- R mailing list, #R in IRC

Special Installation Instructions

In the Sage distribution, `r` is a "dummy" package: It is here to provide information about equivalent system packages. R cannot be installed using the Sage distribution. Please install it manually, either using one of the system package commands shown here or following the upstream instructions at https://www.r-project.org

Type

optional

Dependencies

Version Information

Equivalent System Packages

```
$ apk add R-dev R
```

```
$ sudo pacman -S r
```

```
$ conda install r r-essentials r-lattice
```

```
$ sudo apt-get install r-base r-base-dev libtirpc-dev
```

```
$ sudo dnf install R R-devel
```

```
$ sudo pkg install math/R
```

```
$ sudo emerge dev-lang/R
```

```
$ brew install r
```

No package needed.

```
$ nix-env --install R
```

```
$ sudo zypper install R-base
```

```
$ sudo xbps-install R
```

See https://repology.org/project/r/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.328 r_jupyter: Jupyter kernel for R

Description

This package installs IRkernel, the R Jupyter kernel.

It gets installed via R's package installer on top of Jupyter.

License

MIT

Upstream Contact

- https://github.com/IRkernel/IRkernel
- https://irkernel.github.io/

Dependencies

- R
- notebook

Type

experimental

Dependencies

- *notebook: Jupyter notebook, a web-based notebook environment for interactive computing*
- *rpy2: Python interface to R*

Version Information

Equivalent System Packages

See https://repology.org/project/r:irkernel/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.329 readline: Command line editing library

Description

The GNU Readline library provides a set of functions for use by applications that allow users to edit command lines as they are typed in. Both Emacs and vi editing modes are available. The Readline library includes additional functions to maintain a list of previously-entered command lines, to recall and perhaps reedit those lines, and perform csh-like history expansion on previous commands.

Website: http://tiswww.case.edu/php/chet/readline/rltop.html

License

- GPL V3+

Upstream Contact

- Chet Ramey at http://cnswww.cns.cwru.edu/~chet

Special Update/Build Instructions

We build readline using ncurses. Readline needs to be told to link with libtinfo (part of ncurses), this is what the patch 0002-ltinfo.patch does.

Patches

- 0001-macports.patch: Changes to shobj.conf for OS/X, from macports:

  https://trac.macports.org/browser/trunk/dports/devel/readline/files/patch-shobj-conf.diff

- 0002-ltinfo.patch: We build readline using ncurses, and for that it needs to be told to link with libtinfo (part of ncurses).

Type

standard

Dependencies

- *ncurses: Classic terminal output library*

Version Information

package-version.txt:

```
8.3
```

Equivalent System Packages

```
$ sudo pacman -S readline
```

```
$ conda install readline
```

```
$ sudo apt-get install libreadline-dev
```

```
$ sudo dnf install readline-devel
```

```
$ sudo pkg install devel/readline
```

```
$ brew install readline
```

```
$ sudo port install readline
```

```
$ nix-env --install readline
```

```
$ sudo zypper install readline-devel 'pkgconfig(readline)'
```

```
$ sudo slackpkg install readline
```

```
$ sudo xbps-install readline-devel
```

See https://repology.org/project/readline/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.330 referencing: JSON Referencing + Python

Description

JSON Referencing + Python

License

MIT

Upstream Contact

https://pypi.org/project/referencing/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *attrs: Decorator for Python classes with attributes*
- *rpds_py: Python bindings to Rust's persistent data structures*
- *typing_extensions: Backported and Experimental Type Hints for Python 3.8+*

Version Information

package-version.txt:

```
0.37.0
```

version_requirements.txt:

```
referencing
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.331 regina: Low-dimensional topology: triangulations, knots, links, normal surfaces, angle structures

Description

Regina is a software package for 3-manifold and 4-manifold topologists, with a focus on triangulations, knots and links, normal surfaces, and angle structures.

For 3-manifolds, it includes high-level tasks such as 3-sphere recognition, connected sum decomposition and Hakenness testing, comes with a rich database of census manifolds, and incorporates the SnapPea kernel for working with hyperbolic manifolds. For 4-manifolds, it offers a range of combinatorial and algebraic tools, plus support for normal hypersurfaces. For knots and links, Regina can perform combinatorial manipulation, compute knot polynomials, and work with several import/export formats.

License

GPLv2+

Upstream Contact

Regina: https://regina-normal.github.io/

We use the Python packaging from https://pypi.org/project/regina/ (https://github.com/3-manifolds/regina_wheels)

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *numpy: Package for scientific computing with Python*

Version Information

requirements.txt:

```
regina
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.332 requests: An HTTP library for Python

Description

Python HTTP for Humans.

License

Apache 2.0

Upstream Contact

https://pypi.org/project/requests/

Type

standard

Dependencies

- $(PYTHON)
- *certifi: Python package for providing Mozilla's CA Bundle*
- *charset_normalizer: The Real First Universal Charset Detector. Open, modern and actively maintained alternative to Chardet.*
- *idna: Internationalized Domain Names in Applications (IDNA)*
- *pip: Tool for installing and managing Python packages*
- *urllib3: HTTP library with thread-safe connection pooling, file post, and more*

Version Information

package-version.txt:

```
2.33.0
```

version_requirements.txt:

```
requests >=2.13.0
```

Equivalent System Packages

```
$ sudo pacman -S python-requests
```

```
$ conda install requests
```

```
$ sudo apt-get install python3-requests
```

```
$ sudo dnf install python3-requests
```

```
$ sudo emerge dev-python/requests
```

```
$ sudo port install py-requests
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-requests'
```

```
$ sudo xbps-install python3-requests
```

See https://repology.org/project/requests/versions, https://repology.org/project/python:requests/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.333 rfc3339_validator: A pure python RFC3339 validator

Description

A pure python RFC3339 validator

License

MIT license

Upstream Contact

https://pypi.org/project/rfc3339-validator/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *six: Python 2 and 3 compatibility utilities*

Version Information

package-version.txt:

```
0.1.4
```

version_requirements.txt:

```
rfc3339-validator
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.334 rfc3986_validator: Pure python rfc3986 validator

Description

Pure python rfc3986 validator

License

MIT license

Upstream Contact

https://pypi.org/project/rfc3986-validator/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
0.1.1
```

version_requirements.txt:

```
rfc3986-validator
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.335 roman-numerals: Manipulate well-formed Roman numerals

Description

A library for manipulating well-formed Roman numerals.

Integers between 1 and 3,999 (inclusive) are supported. Numbers beyond this range will return an OutOfRangeError.

The classical system of roman numerals requires that the same character may not appear more than thrice consecutively, meaning that 'MMMCMXCIX' (3,999) is the largest well-formed Roman numeral. The smallest is 'I' (1), as there is no symbol for zero in Roman numerals.

License

This project is licenced under the terms of either the Zero-Clause BSD licence or the CC0 1.0 Universal licence

Upstream Contact

https://pypi.org/project/roman-numerals/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
4.1.0
```

version_requirements.txt:

```
roman-numerals
```

Equivalent System Packages

```
$ sudo pacman -S python-roman-numerals
```

```
$ conda install roman-numerals
```

```
$ sudo dnf install python-roman-numerals
```

```
$ sudo pkg install textproc/py-roman-numerals
```

```
$ sudo emerge dev-python/roman-numerals
```

```
$ sudo port install py-roman_numerals
```

```
$ sudo zypper install python-roman-numerals
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.336 rpds_py: Python bindings to Rust's persistent data structures

Description

Python bindings to the Rust rpds crate for persistent data structures.

rpds-py provides efficient, immutable data structures including:

- `HashTrieMap` - Persistent hash map
- `HashTrieSet` - Persistent hash set
- `List` - Persistent list with efficient operations

These data structures are backed by Rust implementations for high performance while maintaining a Pythonic API. They are particularly useful for functional programming patterns and situations requiring immutable, persistent collections.

The library is used by projects like the referencing library (part of the Python JSON Schema ecosystem) as a faster alternative to pyrsistent.

License

MIT License

Upstream Contact

- Author: Julian Berman <Julian+rpds@GrayVines.com>
- Home page: https://github.com/crate-py/rpds
- PyPI: https://pypi.org/project/rpds-py/
- Documentation: https://rpds.readthedocs.io/
- Upstream Rust crate: https://github.com/orium/rpds

Dependencies

Python (>= 3.12)

Build dependencies: pip, packaging

Special Notes

This package provides platform-specific binary wheels for multiple Python versions and platforms:

- Python 3.12, 3.13, 3.14 (including free-threaded 3.14t)
- Linux (x86_64, aarch64, armv7l, ppc64le, s390x, i686, musllinux variants)
- macOS (x86_64, arm64)
- Windows (win32, win_amd64, win_arm64)

The Sage build system automatically selects and downloads the appropriate wheel for your platform and Python version using the packaging library's compatibility tags. If no compatible binary wheel is available for your platform, the building system will raise an error indicating that no matching wheels were found.

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *packaging: Core utilities for Python packages*
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
0.30.0
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.337 rpy2: Python interface to R

Description

rpy2 is a redesign and rewrite of rpy. It is providing a low-level interface to R, a proposed high-level interface, including wrappers to graphical libraries, as well as R-like structures and functions.

License

- GPL 2+

Upstream Contact

- https://github.com/rpy2/rpy2

Special Installation Instructions

In the Sage distribution, `rpy2` is a "semi-standard" package: It will be automatically installed by the Sage distribution if a suitable system installation of R is detected by `configure`. (Note that Sage no longer ships and installs its own copy of R.)

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cffi: Foreign Function Interface for Python calling C code*
- *jinja2: General purpose template engine for Python*
- *packaging: Core utilities for Python packages*
- *pycparser: Parser of the C language in Python*
- *pytz: Timezone definitions for Python*
- *rpy2_rinterface: Low-level interface from Python to R*
- *rpy2_robjects: High-level interface from Python to R*
- *tzlocal: Python timezone information for the local timezone*

Version Information

package-version.txt:

```
3.6.7
```

version_requirements.txt:

```
rpy2 >=3.3
```

Equivalent System Packages

```
$ sudo pacman -S python-rpy2
```

```
$ conda install rpy2 r-lattice
```

```
$ sudo apt-get install python3-rpy2
```

```
$ sudo pkg install math/py-rpy2
```

```
$ sudo port install py-rpy2
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-rpy2'
```

See https://repology.org/project/rpy2/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.338 rpy2_rinterface: Low-level interface from Python to R

Description

rpy2-rinterface provides the low-level CFFI-based interface between Python and the R statistical computing environment. It handles the embedding of R within a Python process and the translation of data structures between the two languages.

This package was split out from rpy2 starting with version 3.6.

License

GPL-2.0-or-later

Upstream Contact

https://pypi.org/project/rpy2-rinterface/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

- *cffi: Foreign Function Interface for Python calling C code*
- *pycparser: Parser of the C language in Python*

Version Information

package-version.txt:

```
3.6.6
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.339 rpy2_robjects: High-level interface from Python to R

Description

rpy2-robjects provides the high-level object-oriented interface between Python and R:r-project:. It offers R objects mapped as Python objects, conversion rules, and integration with common Python data science packages such as pandas and numpy.

This package was split out from rpy2 starting with version 3.6.

License

GPL-2.0-or-later

Upstream Contact

https://pypi.org/project/rpy2-robjects/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *jinja2: General purpose template engine for Python*
- *rpy2_rinterface: Low-level interface from Python to R*
- *tzlocal: Python timezone information for the local timezone*

Version Information

package-version.txt:

```
3.6.5
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.340 rst2ipynb: Convert reStructuredText files to Jupyter notebooks

Description

The rst2pynb program converts a standalone reStructuredText file to a Jupyter notebook file.

This is currently achieved by converting to markdown with pandoc and then to Jupyter notebook using notedown, plus some configuration and tweaks.

License

BSD 3-Clause License

Upstream Contact

Authors: Scott Sievert and Nicolas M. Thiéry Home page: https://github.com/nthiery/rst-to-ipynb

Special Update/Build Instructions

Fetch tarball from https://pypi.python.org/pypi/rst2ipynb/

As it is written in Haskell, pandoc must be installed from the distro.

The main rationale for having a notedown package in Sage (rather than just let pip fetch it) is that the version on pipy (1.5.0, 2015-10-07) is outdated and lacks important features / fixes for us.

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *notedown: Create IPython notebooks from markdown*
- *pandoc: A document converter*
- *pandocfilters: A Python module for writing pandoc filters*

Version Information

package-version.txt:

```
0.2.3
```

version_requirements.txt:

```
rst2ipynb >=0.2.2
```

Equivalent System Packages

See https://repology.org/project/python:rst2ipynb/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.341 rubiks: Programs for Rubik's cube

Description

There are several programs for working with Rubik's cubes, by three different people. Look inside the directories under /src to see specific info and licensing. In summary the three contributers are:

Michael Reid (GPL) http://www.cflmath.com/~reid/Rubik/optimal_solver.html

- optimal - uses many pre-computed tables to find an optimal solution to the 3x3x3 Rubik's cube

Dik T. Winter (MIT License)

- cube - uses Kociemba's algorithm to iteratively find a short solution to the 3x3x3 Rubik's cube
- size222 - solves a 2x2x2 Rubik's cube

Eric Dietz (GPL) https://web.archive.org/web/20121212175710/http://www.wrongway.org/?rubiksource

- cu2 - A fast, non-optimal 2x2x2 solver
- cubex - A fast, non-optimal 3x3x3 solver
- mcube - A fast, non-optimal 4x4x4 solver

Type

optional

Dependencies

Version Information

package-version.txt:

```
20070912.p21
```

Equivalent System Packages

```
$ sudo pacman -S rubiks
```

```
$ conda install rubiks
```

```
$ sudo apt-get install rubiks
```

```
$ sudo dnf install rubiks
```

```
$ sudo pkg install math/rubiks
```

```
$ nix-env --install rubiks
```

See https://repology.org/project/rubiks/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.342 rw: Compute rank-width and rank-decompositions

Description

rw is a program that calculates rank-width and rank-decompositions.

https://sourceforge.net/projects/rankwidth/

License

GPL version 2 or later

Upstream Contact

Philipp Klaus Krause (philipp@informatik.uni-frankfurt.de)

Type

standard

Dependencies

Version Information

package-version.txt:

```
0.10
```

Equivalent System Packages

```
$ sudo pacman -S rankwidth
```

```
$ conda install rw
```

```
$ sudo apt-get install librw-dev
```

```
$ sudo dnf install rw-devel
```

```
$ sudo pkg install math/rankwidth
```

```
$ nix-env --install rankwidth
```

```
$ sudo xbps-install rankwidth-devel
```

See https://repology.org/project/rankwidth/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.343 saclib: Computations with real algebraic numbers

Description

Saclib is a library of C programs for computer algebra derived from the SAC2 system. It is mainly used as a dependency of qepcad.

License

ICS Wikipedia article ISC_license

Upstream Contact

- Repository: https://github.com/chriswestbrown/saclib
- Tarballs: https://www.usna.edu/Users/cs/wcbrown/qepcad/INSTALL/IQ.html
- Website: (outdated) https://www.usna.edu/Users/cs/wcbrown/qepcad/B/QEPCAD.html

Type

optional

Dependencies

Version Information

package-version.txt:

```
2.2.8
```

Equivalent System Packages

```
$ sudo apt-get install libsaclib-dev
```

```
$ sudo dnf install saclib saclib-devel
```

See https://repology.org/project/saclib/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.344 sage_docbuild: Build system of the Sage documentation

Note

If you are a developer and want to build the SageMath documentation from source, refer to developer's guide.

About SageMath

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> Copyright (C) 2005-2020 The Sage Development Team
>
> https://www.sagemath.org

SageMath fully supports all major Linux distributions, recent versions of macOS, and Windows (using Windows Subsystem for Linux).

The traditional and recommended way to install SageMath is from source via Sage-the-distribution (https://www.sagemath.org/download-source.html). Sage-the-distribution first builds a large number of open source packages from source (unless it finds suitable versions installed in the system) and then installs the Sage Library (sagelib, implemented in Python and Cython).

About this pip-installable source distribution

This is the build system of the Sage documentation, based on Sphinx.

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- sagelib
- *sphinx: Python documentation generator*

Version Information

package-version.txt:

```
10.10
```

version_requirements.txt:

```
sage-docbuild ~= 10.10rc2
```

Equivalent System Packages

See https://repology.org/project/sage-docbuild/versions, https://repology.org/project/python:sage-docbuild/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.345 sage_flatsurf: Flat surfaces in SageMath

Description

SageMath package for studying the geometry of flat surfaces and the dynamics of their foliations.

License

GNU General Public License, version 2

Upstream Contact

https://pypi.org/project/sage-flatsurf/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- $(SAGERUNTIME)
- *surface_dynamics: dynamics on surfaces (measured foliations, interval exchange transformation, Teichmüller flow, etc)*

Version Information

requirements.txt:

```
sage-flatsurf
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.346 sage_numerical_backends_coin: COIN-OR backend for Sage MixedIntegerLinearProgram

Description

COIN-OR backend for Sage MixedIntegerLinearProgram

License

GPLv2+

Upstream Contact

https://pypi.org/project/sage-numerical-backends-coin/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- $(SAGERUNTIME)
- $(SAGE_SRC)/sage/cpython/string.pxd
- $(SAGE_SRC)/sage/cpython/string_impl.h
- $(SAGE_SRC)/sage/numerical/backends/generic_backend.pxd

- *cbc: COIN-OR branch and cut solver for mixed-integer programs*
- *cysignals: Interrupt and signal handling for Cython*
- *cython: C-Extensions for Python, an optimizing static compiler*
- *ipywidgets: Interactive HTML widgets for Jupyter notebooks and the IPython kernel*

Version Information

package-version.txt:

```
10.4
```

version_requirements.txt:

```
sage_numerical_backends_coin >=9.0b12
```

Equivalent System Packages

See https://repology.org/project/sage-numerical-backends-coin/versions, https://repology.org/project/python:sage-numerical-backends-coin/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.347 sage_numerical_backends_cplex: Cplex backend for Sage MixedIntegerLinearProgram

Description

Cplex backend for Sage MixedIntegerLinearProgram

License

GPLv2+

Upstream Contact

https://pypi.org/project/sage-numerical-backends-cplex/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- $(SAGERUNTIME)
- $(SAGE_SRC)/sage/cpython/string.pxd
- $(SAGE_SRC)/sage/cpython/string_impl.h
- $(SAGE_SRC)/sage/numerical/backends/generic_backend.pxd
- *cysignals: Interrupt and signal handling for Cython*

- *cython: C-Extensions for Python, an optimizing static compiler*
- *ipywidgets: Interactive HTML widgets for Jupyter notebooks and the IPython kernel*

Version Information

package-version.txt:

```
10.4
```

version_requirements.txt:

```
sage_numerical_backends_cplex >=9.0b12
```

Equivalent System Packages

See https://repology.org/project/python:sage-numerical-backends-cplex/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.348 sage_numerical_backends_gurobi: Gurobi backend for Sage MixedIntegerLinearProgram

Description

Gurobi backend for Sage MixedIntegerLinearProgram

License

GPLv2+

Upstream Contact

https://pypi.org/project/sage-numerical-backends-gurobi/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- $(SAGERUNTIME)
- $(SAGE_SRC)/sage/cpython/string.pxd
- $(SAGE_SRC)/sage/cpython/string_impl.h
- $(SAGE_SRC)/sage/numerical/backends/generic_backend.pxd
- *cysignals: Interrupt and signal handling for Cython*
- *cython: C-Extensions for Python, an optimizing static compiler*
- *ipywidgets: Interactive HTML widgets for Jupyter notebooks and the IPython kernel*

Version Information

package-version.txt:

```
10.4
```

version_requirements.txt:

```
sage_numerical_backends_gurobi >=9.0.0
```

Equivalent System Packages

See https://repology.org/project/sage-numerical-backends-gurobi/versions, https://repology.org/project/python:sage-numerical-backends-gurobi/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.349 sage-setup: Build system of the SageMath library

This is the build system of the Sage library, based on setuptools.

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cython: C-Extensions for Python, an optimizing static compiler*
- *jinja2: General purpose template engine for Python*
- *pkgconfig: Python interface to pkg-config*

Version Information

package-version.txt:

```
10.10
```

version_requirements.txt:

```
sage-setup ~= 10.10rc2
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.350 sagemath_categories: Sage categories and basic rings

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> https://www.sagemath.org

SageMath fully supports all major Linux distributions, recent versions of macOS, and Windows (Windows Subsystem for Linux).

See https://doc.sagemath.org/html/en/installation/index.html for general installation instructions.

About this pip-installable distribution package

The pip-installable distribution package $sagemath-categories$ is a distribution of a small part of the Sage Library.

It provides a small subset of the modules of the Sage library ("sagelib", $sagemath-standard$) that is a superset of $sagemath-objects$ (providing Sage objects, the element/parent framework, categories, the coercion system and the related metaclasses), making various additional categories available without introducing dependencies on additional mathematical libraries.

Dependencies

When building from source, development packages of gmp, $mpfr$, and mpc are needed.

Documentation

- Categories
- Structure
- Coercion
- Classes, Metaclasses

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cython: C-Extensions for Python, an optimizing static compiler*
- *pkgconfig: Python interface to pkg-config*
- *python_build: A simple, correct PEP517 package builder*
- sage_setup
- sagemath_environment
- sagemath_objects

Version Information

package-version.txt:

```
10.10
```

version_requirements.txt:

```
sagemath-categories ~= 10.10rc2
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.351 sagemath_doc_html: SageMath documentation in HTML format

Upon installation, this package builds the SageMath documentation in HTML format.

It is a standard package. It is built on every invocation of `make` or `make all`, but not on `make build`. The documentation build can also be run separately using `make doc-html`.

Type

optional

Dependencies

- $(SAGERUNTIME)
- *conway_polynomials: Python interface to Frank Lübeck's Conway polynomial database*
- *elliptic_curves: Databases of elliptic curves*
- *fpylll: Python interface for FPLLL*
- *furo: A clean customizable Sphinx documentation theme*
- *graphs: A database of combinatorial graphs*
- *ipykernel: IPython Kernel for Jupyter*
- *ipywidgets: Interactive HTML widgets for Jupyter notebooks and the IPython kernel*
- *jupyter_client: Jupyter protocol implementation and client libraries*
- *mathjax: A JavaScript library for displaying mathematical formulas*
- *matplotlib: Python 2D plotting library*
- *maxima: System for manipulating symbolic and numerical expressions*
- *mpmath: Pure Python library for multiprecision floating-point arithmetic*
- *networkx: Python package for complex networks*
- *pillow: Python Imaging Library*
- *pplpy_doc: Python interface to the Parma Polyhedra Library (documentation)*
- sage_docbuild
- sagelib

- *scipy: Scientific tools for Python*
- *sphinx: Python documentation generator*
- *sphinx_copybutton: Add a copy button to each of your code cells*
- *sphinx_inline_tabs: Add inline tabbed content to your Sphinx documentation*
- *sympy: Python library for symbolic mathematics*
- *tachyon: A ray tracing system*
- *typing_extensions: Backported and Experimental Type Hints for Python 3.8+*

Version Information

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.352 sagemath_doc_pdf: SageMath documentation in PDF format

Upon installation, this package builds the SageMath documentation in PDF format.

It is an optional package. It can be enabled at configuration time using `./configure --enable-sagemath_doc_pdf`. Alternatively, it can be installed by using `make doc-pdf`.

Type

optional

Dependencies

- *free_fonts: a free family of scalable outline fonts*
- sagemath_doc_html
- *texlive: A comprehensive TeX system*
- *texlive_luatex: LuaTeX packages*
- *xindy: a general-purpose index processor*

Version Information

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.353 sagemath_environment: System and software environment

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SageMath fully supports all major Linux distributions, recent versions of macOS, and Windows (Windows Subsystem for Linux).

See https://doc.sagemath.org/html/en/installation/index.html for general installation instructions.

About this pip-installable distribution package

The pip-installable distribution package $sagemath-environment$ is a distribution of a small part of the Sage Library.

It provides a small, fundamental subset of the modules of the Sage library ("sagelib", $sagemath-standard$), providing the connection to the system and software environment. It also includes the $sage$ script for launching the Sage REPL and accessing various developer tools (see $sage--help$).

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *python_build: A simple, correct PEP517 package builder*

Version Information

package-version.txt:

```
10.10
```

version_requirements.txt:

```
sagemath-environment ~= 10.10rc2
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.354 sagemath_giac: Giac integration

This pip-installable source distribution `sagemath-giac` is a small optional distribution for use with `sagemath-standard`.

It provides a Cython interface to the `libgiac` library for the purpose of symbolic integration and certain Groebner basis calculations.

To install this as part of the SageMath distribution, you can either pass `--enable-sagemath_giac` to `./configure`, or run `make sagemath_giac` later (after building Sage) at your convenience.

For other types of installations (meson, pip, linux distro, etc.) you can download the latest release from the Github repository,

- https://github.com/sagemath/sagemath-giac/

and follow the instructions in its `README.rst`.

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cysignals: Interrupt and signal handling for Cython*
- *cython: C-Extensions for Python, an optimizing static compiler*
- *giac: A general purpose computer algebra system*
- *gmpy2: Python interface to GMP/MPIR, MPFR, and MPC*
- sagelib

Version Information

package-version.txt:

```
0.1.3
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.355 sagemath_objects: Sage objects, elements, parents, categories, coercion, metaclasses

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SageMath fully supports all major Linux distributions, recent versions of macOS, and Windows (Windows Subsystem for Linux).

See https://doc.sagemath.org/html/en/installation/index.html for general installation instructions.

About this pip-installable distribution package

The pip-installable distribution package $sagemath-objects$ is a distribution of a small part of the Sage Library.

It provides a small, fundamental subset of the modules of the Sage library ("sagelib", $sagemath-standard$), making Sage objects, the element/parent framework, categories, the coercion system and the related metaclasses available.

Dependencies

When building from source, development packages of gmp, $mpfr$, and mpc are needed.

Documentation

- Categories
- Structure
- Coercion
- Classes, Metaclasses

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cysignals: Interrupt and signal handling for Cython*
- *cython: C-Extensions for Python, an optimizing static compiler*
- *gmpy2: Python interface to GMP/MPIR, MPFR, and MPC*
- *pkgconfig: Python interface to pkg-config*
- *python_build: A simple, correct PEP517 package builder*
- sage_setup
- sagemath_environment

Version Information

package-version.txt:

```
10.10
```

version_requirements.txt:

```
sagemath-objects ~= 10.10rc2
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.356 sagemath_repl: IPython kernel, Sage preparser, doctester

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https://www.sagemath.org

SageMath fully supports all major Linux distributions, recent versions of macOS, and Windows (Windows Subsystem for Linux).

See https://doc.sagemath.org/html/en/installation/index.html for general installation instructions.

About this pip-installable distribution package

The pip-installable source distribution $sagemath-repl$ is a distribution of a small part of the Sage Library.

It provides a small, fundamental subset of the modules of the Sage library ("sagelib", $sagemath-standard$), providing the IPython kernel, Sage preparser, and doctester.

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *ipython: Interactive computing environment with an enhanced interactive Python shell*
- *ipywidgets: Interactive HTML widgets for Jupyter notebooks and the IPython kernel*
- *python_build: A simple, correct PEP517 package builder*
- sagemath_environment
- sagemath_objects

Version Information

package-version.txt:

```
10.10
```

version_requirements.txt:

```
sagemath-repl ~= 10.10rc2
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.357 sagenb_export: Convert legacy SageNB notebooks to Jupyter notebooks and other formats

Description

This is a tool to convert SageNB notebooks to other formats, in particular IPython/Jupyter notebooks.

It includes a Jupyter notebook extension to provide a UI for the import of SageNB notebooks.

Upstream Contact

https://github.com/vbraun/ExportSageNB

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *ipython: Interactive computing environment with an enhanced interactive Python shell*
- *nbconvert: Converting Jupyter Notebooks*
- *notebook: Jupyter notebook, a web-based notebook environment for interactive computing*
- *six: Python 2 and 3 compatibility utilities*

Version Information

package-version.txt:

```
3.3
```

version_requirements.txt:

```
git+https://github.com/vbraun/ExportSageNB.git#egg=sagenb_export
```

Equivalent System Packages

See https://repology.org/project/sagenb-export/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.358 sagetex: Embed code, results of computations, and plots from Sage into LaTeX documents

Description

The SageTeX package allows you to embed code, results of computations, and plots from Sage into LaTeX documents.

License

The *source code* of the SageTeX package may be redistributed and/or modified under the terms of the GNU General Public License as published by the Free Software Foundation, either version 2 of the License, or (at your option) any later version. To view a copy of this license, see http://www.gnu.org/licenses/ or send a letter to the Free Software Foundation, Inc., 51 Franklin Street, Fifth Floor, Boston, MA 02110-1301, USA.

The *documentation* of the SageTeX package is licensed under the Creative Commons Attribution-Share Alike 3.0 License. To view a copy of this license, visit http://creativecommons.org/licenses/by-sa/3.0/ or send a letter to Creative Commons, 171 Second Street, Suite 300, San Francisco, California, 94105, USA.

SPKG Maintainers

Dan Drake (dr.dan.drake at gmail) and SageMath developers (sage-devel@googlegroups.com)

Upstream Contact

Author: Dan Drake.

Web: https://github.com/sagemath/sagetex

Dependencies

To install, nothing more than a standard Sage install. The `spkg-check` script will exit without actually testing anything if it cannot find "latex" in your path.

Notes

To use SageTeX, both Sage and LaTeX need to know about it. SageTeX comes standard with Sage, so you only need to make sure LaTeX can find what it needs. Full details are in the Sage installation guide at http://doc.sagemath.org/html/en/installation/ and http://doc.sagemath.org/html/en/tutorial/sagetex.html .

The directory `$SAGE_ROOT/venv/share/doc/sagetex` contains documentation and an example file. See `$SAGE_ROOT/venv/share/texmf/tex/latex/sagetex` for the source code and some possibly useful scripts. If you have problems or suggestions see the sage-support group.

If you want to help develop SageTeX, please clone the github repository (see the "Upstream Contact" above) and send me patches based on that.

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *matplotlib: Python 2D plotting library*
- *maxima: System for manipulating symbolic and numerical expressions*
- *pillow: Python Imaging Library*
- *pyparsing: A Python parsing module*
- *scipy: Scientific tools for Python*
- *tachyon: A ray tracing system*

Version Information

package-version.txt:

```
3.6.1
```

version_requirements.txt:

```
sagetex >=3.5
```

Equivalent System Packages

```
$ conda install sagetex
```

See https://repology.org/project/sagetex/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.359 sbcl: a lisp compiler and runtime system

Description

Steel Bank Common Lisp (SBCL) is a high performance Common Lisp compiler. It is open source / free software, with a permissive license (see https://www.sbcl.org/history.html). In addition to the compiler and runtime system for ANSI Common Lisp, it provides an interactive environment including a debugger, a statistical profiler, a code coverage tool, and many other extensions.

(taken from https://www.sbcl.org)

License

- a mix of BSD-style and public domain

Upstream Contact

- https://www.sbcl.org

Type

optional

Dependencies

Version Information

Equivalent System Packages

```
$ apk add sbcl
```

```
$ sudo pacman -S sbcl
```

```
$ conda install sbcl
```

```
$ sudo apt-get install sbcl
```

```
$ sudo dnf install sbcl
```

```
$ sudo pkg install lang/sbcl
```

```
$ sudo emerge dev-lisp/sbcl
```

```
$ brew install sbcl
```

```
$ sudo port install sbcl
```

```
$ nix-env --install sbcl
```

```
$ sudo pkg_add lang/sbcl
```

```
$ sudo zypper install sbcl
```

```
$ sudo xbps-install sbcl
```

See https://repology.org/project/sbcl/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.360 scikit_build_core: Build backend for CMake based projects

Description

scikit-build-core is a build backend for Python packages that use CMake as their build system. It provides a modern, PEP 517 compliant replacement for the original scikit-build, with improved support for cross-compilation, editable installs, and modern CMake features.

License

Apache-2.0

Upstream Contact

https://github.com/scikit-build/scikit-build-core

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cmake: A cross-platform build system generator*
- *hatchling: Modern, extensible Python build backend*
- *packaging: Core utilities for Python packages*
- *pathspec: Utility library for gitignore style pattern matching of file paths*

Version Information

package-version.txt:

```
0.12.2
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.361 scip: Mixed integer programming solver

Description

SCIP is currently one of the fastest open source mixed integer programming (MIP) solvers. It is also a framework for constraint integer programming and branch-cut-and-price. It allows total control of the solution process and the access of detailed information down to the guts of the solver.

License

Apache 2.0

Upstream Contact

https://scipopt.org/#scipoptsuite

Dependencies

scip brings its own patched version of the bliss library. This will conflict with the optional package bliss.

Type

optional

Dependencies

- $(MP_LIBRARY)
- *bliss: Computing automorphism groups and canonical forms of graphs*
- *cmake: A cross-platform build system generator*
- *papilo: Parallel presolve for integer and linear optimization*
- *readline: Command line editing library*
- *soplex: Linear optimization solver using the revised simplex method*

Version Information

package-version.txt:

```
9.0.1
```

Equivalent System Packages

```
$ conda install scip
```

```
$ sudo dnf install scip
```

See https://repology.org/project/scipoptsuite/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.362 scip_sdp: Mixed integer semidefinite programming plugin for SCIP

Description

SCIP-SDP allows to solve MISDPs using a nonlinear branch-and-bound approach or a linear programming cutting-plane approach.

- In the first case (the default), the semidefinite programming (SDP) relaxations are solve using interior-point SDP-solvers.
- In the second case, cutting planes based on eigenvector are generated.

SCIP-SDP is based on the branch-and-cut framework SCIP. In addition to providing a constraint handler for SDP-constraints and a relaxator to solve continuous SDP-relaxations using interior-point solvers, SCIP-SDP adds several heuristics and propagators to SCIP.

License

Apache 2.0

Upstream Contact

http://www.opt.tu-darmstadt.de/scipsdp/

https://github.com/scipopt/SCIP-SDP

Type

optional

Dependencies

- *cmake: A cross-platform build system generator*
- *dsdp: Semidefinite programming solver*
- *scip: Mixed integer programming solver*

Version Information

package-version.txt:

```
4.3.0
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.363 scipy: Scientific tools for Python

Description

SciPy (pronounced "Sigh Pie") is open-source software for mathematics, science, and engineering. The SciPy library depends on NumPy, which provides convenient and fast N-dimensional array manipulation. The SciPy library is built to work with NumPy arrays, and provides many user-friendly and efficient numerical routines such as routines for numerical integration and optimization. Together, they run on all popular operating systems, are quick to install, and are free of charge. NumPy and SciPy are easy to use, but powerful enough to be depended upon by some of the world's leading scientists and engineers.

License

SciPy's license is free for both commercial and non-commercial use, under the BSD terms. See http://www.scipy.org/License_Compatibility

Upstream Contact

https://www.scipy.org/

Dependencies

- Python, which in Sage has numerous dependencies
- Numpy
- Fortran
- GNU patch

Special Update/Build Instructions

- None.

Type

standard

Dependencies

- $(BLAS)
- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cython: C-Extensions for Python, an optimizing static compiler*
- *gfortran: Fortran compiler from the GNU Compiler Collection*
- *numpy: Package for scientific computing with Python*
- *pybind11: Create Python bindings to C++ code*
- *pythran: Ahead of Time compiler for numeric kernels*

Version Information

package-version.txt:

```
1.17.1
```

version_requirements.txt:

```
scipy >=1.12
```

Equivalent System Packages

```
$ sudo pacman -S python-scipy
```

```
$ conda install 'scipy>=1.12'
```

```
$ sudo apt-get install python3-scipy
```

```
$ sudo dnf install python3-scipy
```

```
$ sudo emerge dev-python/scipy
```

```
$ brew install scipy
```

```
$ sudo port install py-scipy
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-scipy'
```

```
$ sudo xbps-install python3-scipy
```

See https://repology.org/project/python:scipy/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.364 scs: Splitting conic solver

Description

scs: splitting conic solver

License

MIT

Upstream Contact

https://pypi.org/project/scs/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cmake: A cross-platform build system generator*
- *numpy: Package for scientific computing with Python*
- *scipy: Scientific tools for Python*

Version Information

package-version.txt:

```
3.2.11
```

version_requirements.txt:

```
scs
```

Equivalent System Packages

```
$ conda install scs
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.365 semigroups: An optional GAP package

Description

Installing this SPKG will install the corresponding GAP package, but before you can use them in Sage, they still have to be loaded into either the GAP interface or libgap:

```
sage: gap.eval('LoadPackage("semigroups")')  # optional - semigroups
'true'
sage: libgap.LoadPackage("semigroups")       # optional - semigroups
true
```

```
>>> from sage.all import *
>>> gap.eval('LoadPackage("semigroups")')  # optional - semigroups
'true'
>>> libgap.LoadPackage("semigroups")       # optional - semigroups
true
```

Those correspond to:

```
gap> LoadPackage("semigroups");
```

within the GAP interface and libgap, respectively.

Upstream Contact

See https://semigroups.github.io/Semigroups/

Dependencies

- GAP (a standard spkg), gap_packages and libsemigroups (optional packages)

Notes

This is a GAP package for semigroups, and monoids. There are particularly efficient methods for finitely presented semigroups and monoids, and for semigroups and monoids consisting of transformations, partial permutations, bipartitions, partitioned binary relations, subsemigroups of regular Rees 0-matrix semigroups, and matrices of various semirings including boolean matrices, matrices over finite fields, and certain tropical matrices. Semigroups contains efficient methods for creating semigroups, monoids, and inverse semigroups and monoids, calculating their Green's structure, ideals, size, elements, group of units, small generating sets, testing membership, finding the inverses of a regular element, factorizing elements over the generators, and so on. It is possible to test if a semigroup satisfies a particular property, such as if it is regular, simple, inverse, completely regular, and a large number of further properties. There are methods for finding presentations for a semigroup, the congruences of a semigroup, the maximal subsemigroups of a finite semigroup, smaller degree partial permutation representations, and the character tables of inverse semigroups. There are functions for producing pictures of the Green's structure of a semigroup, and for drawing graphical representations of certain types of elements. (Authors: James Mitchell, Marina Anagnostopoulou-Merkouri, Thomas Breuer, Stuart Burrell, Reinis Cirpons, Tom Conti-Leslie, Joseph Edwards, Attila Egri-Nagy, Luke Elliott, Fernando Flores Brito, Tillman Froehlich, Nick Ham, Robert Hancock, Max Horn, Christopher Jefferson, Julius Jonusas, Chinmaya Nagpal, Olexandr Konovalov, Artemis Konstantinidi, Hyeokjun Kwon, Dima V. Pasechnik, Markus Pfeiffer, Christopher Russell, Jack Schmidt, Sergio Siccha, Finn Smith, Ben Spiers, Nicolas Thiéry, Maria Tsalakou, Chris Wensley, Murray Whyte, Wilf A. Wilson, Tianrun Yang, Michael Young and Fabian Zickgraf)

Type

optional

Dependencies

- $(SAGERUNTIME)
- *gap: Groups, Algorithms, Programming - a system for computational discrete algebra*
- *gap_packages: A collection of GAP packages*
- *libsemigroups: Library for semigroups and monoids*

Version Information

package-version.txt:

```
5.6.0
```

Equivalent System Packages

```
$ conda install gap
```

```
$ sudo dnf install gap-pkg-cohomolo gap-pkg-corelg gap-pkg-crime  \
      gap-pkg-cryst gap-pkg-ctbllib gap-pkg-design gap-pkg-factint  \
      GAPDoc gap-pkg-gbnp gap-pkg-grape gap-pkg-guava gap-pkg-hap  \
      gap-pkg-hapcryst gap-pkg-hecke gap-pkg-laguna gap-pkg-liealgdb  \
```

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```
    gap-pkg-liepring gap-pkg-liering gap-pkg-loops gap-pkg-mapclass  \
    gap-pkg-polymaking gap-pkg-qpa gap-pkg-quagroup gap-pkg-repsn  \
    gap-pkg-sla gap-pkg-sonata gap-pkg-toric
```

```
$ sudo emerge dev-gap/semigroups
```

See https://repology.org/project/gap/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.366 send2trash: Send file to trash natively under Mac OS X, Windows and Linux

Description

Send file to trash natively under Mac OS X, Windows and Linux.

License

BSD License

Upstream Contact

https://pypi.org/project/Send2Trash/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
2.1.0
```

version_requirements.txt:

```
send2trash >=1.5.0
```

Equivalent System Packages

```
$ conda install send2trash
```

```
$ sudo dnf install python3-send2trash
```

```
$ sudo emerge dev-python/send2trash
```

```
$ sudo port install py-send2trash
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-Send2Trash'
```

```
$ sudo xbps-install python3-send2trash
```

See https://repology.org/project/send2trash/versions, https://repology.org/project/python:send2trash/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.367 setuptools: Build system for Python packages

Description

setuptools is the classical build system for Python packages, a collection of enhancements to the Python distutils.

License

MIT License

Upstream Contact

http://pypi.python.org/pypi/setuptools/

https://github.com/pypa/setuptools

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
82.0.1
```

Equivalent System Packages

```
$ sudo pacman -S python-setuptools
```

```
$ conda install setuptools
```

```
$ sudo apt-get install python3-setuptools
```

```
$ sudo dnf install python3-setuptools
```

```
$ sudo emerge dev-python/setuptools
```

```
$ sudo port install py-setuptools
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-setuptools'
```

```
$ sudo xbps-install python3-setuptools
```

See https://repology.org/project/python:setuptools/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.368 setuptools_scm: Python build system extension to obtain package version from version control

Description

the blessed package to manage your versions by scm tags

License

MIT

Upstream Contact

https://pypi.org/project/setuptools-scm/

Type

standard

Dependencies

- $(PYTHON)
- *packaging: Core utilities for Python packages*
- *pip: Tool for installing and managing Python packages*
- *setuptools: Build system for Python packages*
- *typing_extensions: Backported and Experimental Type Hints for Python 3.8+*
- *vcs_versioning: Version management from VCS metadata*

Version Information

package-version.txt:

```
10.0.5
```

version_requirements.txt:

```
setuptools_scm >=7
```

Equivalent System Packages

```
$ sudo pacman -S python-setuptools-scm
```

```
$ conda install setuptools_scm
```

```
$ sudo apt-get install python3-setuptools-scm
```

```
$ sudo dnf install python3-setuptools_scm
```

```
$ sudo pkg install devel/py-setuptools_scm
```

```
$ sudo emerge dev-python/setuptools-scm
```

```
$ sudo port install py-setuptools_scm
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-setuptools_scm'
```

```
$ sudo xbps-install python3-setuptools_scm
```

See https://repology.org/project/python:setuptools-scm/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.369 singular: Computer algebra system for polynomial computations, algebraic geometry, singularity theory

Description

Singular is a computer algebra system for polynomial computations, with special emphasis on commutative and non-commutative algebra, algebraic geometry, and singularity theory.

License

GPLv2 or GPLv3

Upstream Contact

libsingular-devel@mathematik.uni-kl.de

https://www.singular.uni-kl.de/

Special Update/Build Instructions

Other notes:

- If the environment variable SAGE_DEBUG is set to "yes", then omalloc will be replaced by xalloc. The resulting Singular executable and libsingular library will be slower than with omalloc, but allow for easier debugging of memory corruptions.

Type

standard

Dependencies

- $(MP_LIBRARY)
- *cddlib: Double description method for polyhedral representation conversion*
- *flint: Fast Library for Number Theory*
- *mpfr: Multiple-precision floating-point computations with correct rounding*
- *ntl: A library for doing number theory*
- *readline: Command line editing library*

Version Information

package-version.txt:

```
4.4.1
```

Equivalent System Packages

```
$ sudo pacman -S singular
```

```
$ conda install singular
```

```
$ sudo apt-get install singular singular-doc libsingular4-dev
```

```
$ sudo dnf install Singular Singular-devel Singular-doc
```

```
$ sudo pkg install math/singular
```

```
$ sudo emerge 'sci-mathematics/singular[readline]'
```

```
$ brew install singular
```

```
$ sudo port install singular
```

```
$ nix-env --install singular
```

```
$ sudo xbps-install singular
```

See https://repology.org/project/singular/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.370 singular_jupyter: Jupyter kernel for Singular

Description

This is a beta version of a jupyter kernel for Singular.

License

GPL version 2 or later

Upstream Contact

- https://github.com/sebasguts/jupyter_kernel_singular

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *ipython: Interactive computing environment with an enhanced interactive Python shell*
- *ipywidgets: Interactive HTML widgets for Jupyter notebooks and the IPython kernel*
- *jupyter_client: Jupyter protocol implementation and client libraries*
- *pysingular: A basic Python interface to Singular*

Version Information

package-version.txt:

```
0.9.7
```

version_requirements.txt:

```
singular_jupyter >=0.9.7
```

Equivalent System Packages

```
$ conda install jupyter-kernel-singular
```

```
$ sudo dnf install python3-jupyter-kernel-singular
```

See https://repology.org/project/jupyter-singular/versions, https://repology.org/project/python:jupyter-kernel-singular/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.371 sirocco: Compute topologically certified root continuation of bivariate polynomials

Description

sirocco is a library to compute topologically certified root continuation of bivariate polynomials.

License

GPLv3+

SPKG Maintainers

- Miguel Marco

Upstream Contact

Miguel Marco (mmarco@unizar.es)

Dependencies

- gcc

Type

optional

Dependencies

- *mpfr: Multiple-precision floating-point computations with correct rounding*

Version Information

package-version.txt:

```
2.1.1
```

Equivalent System Packages

```
$ sudo pacman -S sirocco
```

```
$ conda install sirocco
```

```
$ sudo dnf install sirocco sirocco-devel
```

```
$ sudo zypper install sirocco-devel
```

See https://repology.org/project/sirocco/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.372 six: Python 2 and 3 compatibility utilities

Description

Python 2 and 3 compatibility utilities

License

MIT License

Upstream Contact

- Author: Benjamin Peterson
- Home page: http://pypi.python.org/pypi/six/

Dependencies

Python

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
1.17.0
```

version_requirements.txt:

```
six >=1.16.0
```

Equivalent System Packages

```
$ sudo pacman -S python-six
```

```
$ conda install six
```

```
$ sudo apt-get install python3-six
```

```
$ sudo dnf install python3-six
```

```
$ sudo emerge dev-python/six
```

```
$ sudo port install py-six
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-six'
```

```
$ sudo xbps-install python3-six
```

See https://repology.org/project/python:six/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.373 slabbe: Sébastien Labbé's Research code

Description

This SageMath package contains various modules for experimentation with

- discrete dynamical systems
- combinatorics
- digital geometry
- visualization
- miscellaneous development tools

License

GPLv2+

Upstream Contact

https://pypi.org/project/slabbe/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- $(SAGERUNTIME)

Version Information

requirements.txt:

```
slabbe
```

Equivalent System Packages

See https://repology.org/project/python:slabbe/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.374 snappy: Topology and geometry of 3-manifolds, with a focus on hyperbolic structures

Description

Studying the topology and geometry of 3-manifolds, with a focus on hyperbolic structures.

License

GPLv2+

Upstream Contact

https://pypi.org/project/snappy/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cypari2: Python interface to the number theory library libpari*
- *decorator: Python library providing decorators*
- *ipython: Interactive computing environment with an enhanced interactive Python shell*
- sagelib

Version Information

requirements.txt:

```
snappy
cypari !=2.4.0
snappy_15_knots
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.375 sniffio: Sniff out which async library your code is running under

Description

Sniff out which async library your code is running under

License

MIT OR Apache-2.0

Upstream Contact

https://pypi.org/project/sniffio/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
1.3.1
```

version_requirements.txt:

```
sniffio
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.376 snowballstemmer: Stemmer algorithms for natural language processing in Python

Description

This package provides 29 stemmers for 28 languages generated from Snowball algorithms.

License

BSD-3-Clause

Upstream Contact

https://pypi.org/project/snowballstemmer/

This is a pure Python stemming library. If PyStemmer is available, this module uses it to accelerate.

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
3.0.1
```

version_requirements.txt:

```
snowballstemmer >=1.2.1
```

Equivalent System Packages

```
$ conda install snowballstemmer
```

```
$ sudo emerge dev-python/snowballstemmer
```

```
$ sudo port install py-snowballstemmer
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-snowballstemmer'
```

```
$ sudo xbps-install python3-snowballstemmer
```

See https://repology.org/project/python:snowballstemmer/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.377 soplex: Linear optimization solver using the revised simplex method

Description

SoPlex is an optimization package for solving linear programming problems (LPs) based on an advanced implementation of the primal and dual revised simplex algorithm. It provides special support for the exact solution of LPs with rational input data.

License

Apache License, Version 2.0

Upstream Contact

https://github.com/scipopt/soplex

Type

optional

Dependencies

- $(MP_LIBRARY)
- *cmake: A cross-platform build system generator*
- *mpfr: Multiple-precision floating-point computations with correct rounding*
- *papilo: Parallel presolve for integer and linear optimization*

Version Information

package-version.txt:

```
7.0.1
```

Equivalent System Packages

```
$ conda install soplex
```

```
$ sudo dnf install soplex
```

```
$ sudo pkg install math/SoPlex
```

See https://repology.org/project/soplex/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.378 sortedcontainers: sorted collections library

Description

Sorted collections library, written in pure-Python

License

Apache2

Upstream Contact

https://pypi.org/project/sortedcontainers

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
2.4.0
```

version_requirements.txt:

```
sortedcontainers
```

Equivalent System Packages

```
$ apk add py3-sortedcontainers
```

```
$ sudo pacman -S python-sortedcontainers
```

```
$ sudo apt-get install sortedcontainers
```

```
$ sudo dnf install python-sortedcontainers
```

```
$ sudo pkg install devel/py-sortedcontainers
```

```
$ sudo emerge dev-python/sortedcontainers
```

```
$ sudo pkg_add devel/py-sortedcontainers
```

```
$ sudo zypper install python-sortedcontainers
```

```
$ sudo xbps-install python3-sortedcontainers
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.379 soupsieve: Modern CSS selector implementation for Beautiful Soup

Description

Modern CSS selector implementation for Beautiful Soup

License

Upstream Contact

https://pypi.org/project/soupsieve/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
2.8.3
```

version_requirements.txt:

```
soupsieve
```

Equivalent System Packages

```
$ conda install soupsieve
```

```
$ sudo dnf install python3-soupsieve
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.380 sphinx: Python documentation generator

Description

Sphinx is a tool that makes it easy to create intelligent and beautiful documentation for Python projects (or other documents consisting of multiple reStructuredText sources), written by Georg Brandl. It was originally created to translate the new Python documentation, but has now been cleaned up in the hope that it will be useful to many other projects.

License

Modified BSD; see e.g. its egg-info file for other options

Upstream Contact

https://pypi.org/project/Sphinx/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *alabaster: Default theme for the Sphinx documentation system*
- *babel: Internationalization utilities*
- *colorama: Cross-platform colored terminal text*
- *docutils: Processing plaintext documentation into useful formats, such as HTML or LaTeX*
- *imagesize: Getting image size from png/jpeg/jpeg2000/gif file*
- *jinja2: General purpose template engine for Python*
- *packaging: Core utilities for Python packages*
- *pygments: Generic syntax highlighter*
- *requests: An HTTP library for Python*
- *roman-numerals: Manipulate well-formed Roman numerals*
- *snowballstemmer: Stemmer algorithms for natural language processing in Python*
- *sphinxcontrib_applehelp: Sphinx extension which outputs Apple help book*

- *sphinxcontrib_devhelp: Sphinx extension which outputs Devhelp documents*
- *sphinxcontrib_htmlhelp: Sphinx extension which outputs HTML help book*
- *sphinxcontrib_jsmath: Sphinx extension which renders display math in HTML via JavaScript*
- *sphinxcontrib_qthelp: Sphinx extension which outputs QtHelp documents*
- *sphinxcontrib_serializinghtml: Sphinx extension which outputs serialized HTML files*
- *sphinxcontrib_websupport: Sphinx API for Web apps*

Version Information

package-version.txt:

```
9.2.0
```

version_requirements.txt:

```
sphinx >=9.1, <10
```

Equivalent System Packages

```
$ sudo pacman -S python-sphinx
```

```
$ conda install 'sphinx>=9.1,<10'
```

```
$ sudo apt-get install sphinx
```

```
$ sudo dnf install python3-sphinx
```

```
$ sudo pkg install textproc/py-sphinx
```

```
$ sudo emerge dev-python/sphinx
```

```
$ brew install sphinx-doc
```

```
$ sudo port install py-sphinx
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-Sphinx'
```

```
$ sudo xbps-install python3-Sphinx
```

See https://repology.org/project/python:sphinx/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.381 sphinx_basic_ng: Modern skeleton for Sphinx themes

Description

Modern skeleton for Sphinx themes

License

Upstream Contact

https://pypi.org/project/sphinx-basic-ng/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *sphinx: Python documentation generator*

Version Information

package-version.txt:

```
1.0.0b2
```

version_requirements.txt:

```
sphinx-basic-ng
```

Equivalent System Packages

```
$ sudo pacman -S python-sphinx-basic-ng
```

```
$ conda install sphinx-basic-ng
```

```
$ sudo apt-get install sphinx-basic-ng
```

```
$ sudo dnf install python3-sphinx-basic-ng
```

```
$ sudo pkg install textproc/py-sphinx-basic-ng
```

```
$ sudo emerge dev-python/sphinx-basic-ng
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.382 sphinx_copybutton: Add a copy button to each of your code cells

Description

Add a copy button to each of your code cells

License

MIT License

Upstream Contact

https://pypi.org/project/sphinx-copybutton/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *sphinx: Python documentation generator*

Version Information

package-version.txt:

```
0.5.2
```

version_requirements.txt:

```
sphinx-copybutton
```

Equivalent System Packages

```
$ apk add py3-sphinx-copybutton
```

```
$ sudo pacman -S python-sphinx-copybutton
```

```
$ conda install sphinx-copybutton
```

```
$ sudo dnf install python3-sphinx-copybutton
```

```
$ sudo pkg install textproc/py-sphinx-copybutton
```

```
$ sudo emerge dev-python/sphinx-copybutton
```

```
$ sudo xbps-install python3-sphinx-copybutton
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.383 sphinx_inline_tabs: Add inline tabbed content to your Sphinx documentation

Description

Add inline tabbed content to your Sphinx documentation

License

Upstream Contact

https://pypi.org/project/sphinx-inline-tabs/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *sphinx: Python documentation generator*

Version Information

package-version.txt:

```
2025.12.21.14
```

version_requirements.txt:

```
sphinx-inline-tabs
```

Equivalent System Packages

```
$ conda install sphinx-inline-tabs
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.384 sphinxcontrib_applehelp: Sphinx extension which outputs Apple help book

Description

Sphinx extension which outputs Apple help book

License

BSD

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
2.0.0
```

version_requirements.txt:

```
sphinxcontrib_applehelp >=1.0.2
```

Equivalent System Packages

```
$ sudo pacman -S python-sphinxcontrib-applehelp
```

```
$ conda install sphinxcontrib-applehelp
```

```
$ sudo pkg install textproc/py-sphinxcontrib-applehelp
```

```
$ sudo emerge dev-python/sphinxcontrib-applehelp
```

```
$ sudo port install py-sphinxcontrib-applehelp
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-sphinxcontrib-applehelp'
```

```
$ sudo xbps-install python3-sphinxcontrib-applehelp
```

See https://repology.org/project/python:sphinxcontrib-applehelp/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.385 sphinxcontrib_devhelp: Sphinx extension which outputs Devhelp documents

Description

Sphinx extension which outputs Devhelp documents

License

BSD

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
2.0.0
```

version_requirements.txt:

```
sphinxcontrib_devhelp >=1.0.2
```

Equivalent System Packages

```
$ sudo pacman -S python-sphinxcontrib-devhelp
```

```
$ conda install sphinxcontrib-devhelp
```

```
$ sudo dnf install python3-sphinxcontrib-devhelp
```

```
$ sudo pkg install textproc/py-sphinxcontrib-devhelp
```

```
$ sudo emerge dev-python/sphinxcontrib-devhelp
```

```
$ sudo port install py-sphinxcontrib-devhelp
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-sphinxcontrib-devhelp'
```

```
$ sudo xbps-install python3-sphinxcontrib-devhelp
```

See https://repology.org/project/python:sphinxcontrib-devhelp/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.386 sphinxcontrib_htmlhelp: Sphinx extension which outputs HTML help book

Description

Sphinx extension which outputs HTML help book

License

BSD

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
2.1.0
```

version_requirements.txt:

```
sphinxcontrib_htmlhelp >=2.0.6
```

Equivalent System Packages

```
$ sudo pacman -S python-sphinxcontrib-htmlhelp
```

```
$ conda install sphinxcontrib-htmlhelp
```

```
$ sudo dnf install python3-sphinxcontrib-htmlhelp
```

```
$ sudo pkg install textproc/py-sphinxcontrib-htmlhelp
```

```
$ sudo emerge dev-python/sphinxcontrib-htmlhelp
```

```
$ sudo port install py-sphinxcontrib-htmlhelp
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-sphinxcontrib-htmlhelp'
```

```
$ sudo xbps-install python3-sphinxcontrib-htmlhelp
```

See https://repology.org/project/python:sphinxcontrib-htmlhelp/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.387 sphinxcontrib_jsmath: Sphinx extension which renders display math in HTML via JavaScript

Description

Sphinx extension which renders display math in HTML via JavaScript

License

BSD

Upstream Contact

https://pypi.org/project/sphinxcontrib-jsmath/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
1.0.1
```

version_requirements.txt:

```
sphinxcontrib_jsmath >=1.0.1
```

Equivalent System Packages

```
$ sudo pacman -S python-sphinxcontrib-jsmath
```

```
$ conda install sphinxcontrib-jsmath
```

```
$ sudo pkg install textproc/py-sphinxcontrib-jsmath
```

```
$ sudo emerge dev-python/sphinxcontrib-jsmath
```

```
$ sudo port install py37-sphinxcontrib-jsmath
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-sphinxcontrib-jsmath'
```

```
$ sudo xbps-install python3-sphinxcontrib-jsmath
```

See https://repology.org/project/python:sphinxcontrib-jsmath/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.388 sphinxcontrib_qthelp: Sphinx extension which outputs QtHelp documents

Description

Sphinx extension which outputs QtHelp documents

License

BSD

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
2.0.0
```

version_requirements.txt:

```
sphinxcontrib_qthelp >=1.0.3
```

Equivalent System Packages

```
$ sudo pacman -S python-sphinxcontrib-qthelp
```

```
$ conda install sphinxcontrib-qthelp
```

```
$ sudo dnf install python3-sphinxcontrib-qthelp
```

```
$ sudo pkg install textproc/py-sphinxcontrib-qthelp
```

```
$ sudo emerge dev-python/sphinxcontrib-qthelp
```

```
$ sudo port install py-sphinxcontrib-qthelp
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-sphinxcontrib-qthelp'
```

```
$ sudo xbps-install python3-sphinxcontrib-qthelp
```

See https://repology.org/project/python:sphinxcontrib-qthelp/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.389 sphinxcontrib_serializinghtml: Sphinx extension which outputs serialized HTML files

Description

Sphinx extension which outputs serialized HTML files

License

BSD

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
2.0.0
```

version_requirements.txt:

```
sphinxcontrib_serializinghtml >=1.1.5
```

Equivalent System Packages

```
$ sudo pacman -S python-sphinxcontrib-serializinghtml
```

```
$ conda install sphinxcontrib-serializinghtml
```

```
$ sudo dnf install python3-sphinxcontrib-serializinghtml
```

```
$ sudo pkg install textproc/py-sphinxcontrib-serializinghtml
```

```
$ sudo emerge dev-python/sphinxcontrib-serializinghtml
```

```
$ sudo port install py-sphinxcontrib-serializinghtml
```

```
$ sudo zypper install  \
      'python3${PYTHON_MINOR}-sphinxcontrib-serializinghtml'
```

```
$ sudo xbps-install python3-sphinxcontrib-serializinghtml
```

See https://repology.org/project/python:sphinxcontrib-serializinghtml/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.390 sphinxcontrib_websupport: Sphinx API for Web apps

Description

sphinxcontrib-websupport provides a Python API to easily integrate Sphinx documentation into your Web application.

License

BSD

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *jinja2: General purpose template engine for Python*
- *sphinxcontrib_serializinghtml: Sphinx extension which outputs serialized HTML files*

Version Information

package-version.txt:

```
2.0.0
```

version_requirements.txt:

```
sphinxcontrib_websupport >=1.2.1
```

Equivalent System Packages

```
$ apk add py3-sphinxcontrib-websupport
```

```
$ sudo pacman -S python-sphinxcontrib-websupport
```

```
$ conda install sphinxcontrib-websupport
```

```
$ sudo apt-get install python3-sphinxcontrib.websupport
```

```
$ sudo dnf install python3-sphinxcontrib-websupport
```

```
$ sudo pkg install textproc/py-sphinxcontrib-websupport
```

```
$ sudo emerge dev-python/sphinxcontrib-websupport
```

```
$ sudo port install py-sphinxcontrib-websupport
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-sphinxcontrib-websupport'
```

See https://repology.org/project/python:sphinxcontrib-websupport/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.391 sqlalchemy: A database abstraction library

Description

Database Abstraction Library

License

MIT

Upstream Contact

https://pypi.org/project/SQLAlchemy/

Type

optional

Dependencies

Version Information

requirements.txt:

```
sqlalchemy
```

Equivalent System Packages

```
$ conda install sqlalchemy
```

```
$ sudo dnf install python3-sqlalchemy
```

```
$ sudo port install py-sqlalchemy
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-SQLAlchemy'
```

```
$ sudo xbps-install python3-SQLAlchemy
```

See https://repology.org/project/python:sqlalchemy/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.392 stack_data: Extract data from python stack frames and tracebacks for informative displays

Description

Extract data from python stack frames and tracebacks for informative displays

License

MIT

Upstream Contact

https://pypi.org/project/stack-data/

Type

standard

Dependencies

- $(PYTHON)
- *asttokens: Annotate AST trees with source code positions*
- *executing: Get the currently executing AST node of a frame, and other information*
- *pip: Tool for installing and managing Python packages*
- *pure_eval: Safely evaluate AST nodes without side effects*

Version Information

package-version.txt:

```
0.6.3
```

version_requirements.txt:

```
stack-data
```

Equivalent System Packages

```
$ conda install stack_data
```

```
$ sudo dnf install python3-stack-data
```

```
$ sudo emerge dev-python/stack-data
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.393 suitesparse: A suite of sparse matrix software

SuiteSparse is a collection of software to deal with sparse matrix. It is hosted at https://people.engr.tamu.edu/davis/suitesparse.html with source code now on github at https://github.com/DrTimothyAldenDavis/SuiteSparse

This spkg does a minimal install of suitesparse, only installing the following

- AMD
- CAMD
- COLAMD
- CCOLAMD
- CHOLMOD
- UMFPACK

Those are all the packages needed for cvxopt.

Other configurations are self explanatory.

License: because SuiteSparse is a collection, it comes with a variety of licenses. Find below a copy of the "LICENSES.txt" shipped with SuiteSparse.

AMD/Doc/License.txt

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Availability:

http://www.suitesparse.com

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Availability:

http://www.suitesparse.com

CCOLAMD/Doc/License.txt

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Type

standard

Dependencies

- $(BLAS)
- $(MP_LIBRARY)
- *cmake: A cross-platform build system generator*
- *gfortran: Fortran compiler from the GNU Compiler Collection*
- *mpfr: Multiple-precision floating-point computations with correct rounding*

Version Information

package-version.txt:

```
7.8.0
```

Equivalent System Packages

```
$ apk add suitesparse-dev
```

```
$ sudo pacman -S suitesparse
```

```
$ conda install suitesparse
```

```
$ sudo apt-get install libsuitesparse-dev
```

```
$ sudo dnf install suitesparse suitesparse-devel
```

```
$ sudo pkg install math/suitesparse
```

```
$ sudo emerge sci-libs/amd sci-libs/cholmod sci-libs/suitesparseconfig  \
     sci-libs/umfpack
```

```
$ brew install suite-sparse
```

No package needed.

```
$ sudo zypper install suitesparse-devel
```

install the following packages: suitesparse

```
$ sudo xbps-install SuiteSparse-devel
```

See https://repology.org/project/suitesparse/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.394 surface_dynamics: dynamics on surfaces (measured foliations, interval exchange transformation, Teichmüller flow, etc)

Description

Dynamics on surfaces.

License

GPLv2+

Upstream Contact

https://gitlab.com/videlec/surface_dynamics https://pypi.org/project/surface-dynamics/

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- $(SAGERUNTIME)
- $(SAGE_SRC)/sage/ext/stdsage.pxd
- $(SAGE_SRC)/sage/rings/integer.pxd
- *cysignals: Interrupt and signal handling for Cython*
- *pplpy: Python interface to the Parma Polyhedra Library*

Version Information

requirements.txt:

```
surface_dynamics
```

Equivalent System Packages

See https://repology.org/project/python:surface-dynamics/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.395 symengine: A C++ symbolic manipulation library

Description

SymEngine is a standalone fast C++ symbolic manipulation library.

License

BSD 3-clause

Upstream Contact

https://github.com/symengine/symengine

Type

optional

Dependencies

- $(MP_LIBRARY)
- *cmake: A cross-platform build system generator*
- *ecm: Elliptic curve method for integer factorization*
- *flint: Fast Library for Number Theory*
- *mpc: Arithmetic of complex numbers with arbitrarily high precision and correct rounding*
- *mpfr: Multiple-precision floating-point computations with correct rounding*

Version Information

package-version.txt:

```
0.14.0
```

Equivalent System Packages

```
$ sudo pacman -S symengine
```

```
$ conda install symengine
```

```
$ sudo pkg install math/symengine
```

```
$ sudo emerge sci-libs/symengine
```

```
$ brew install symengine
```

```
$ sudo port install symengine
```

```
$ nix-env --install symengine
```

```
$ sudo zypper install symengine
```

See https://repology.org/project/symengine/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.396 symengine_py: Python wrappers for SymEngine

Description

Python wrappers for SymEngine

License

symengine.py is MIT licensed and uses several LGPL, BSD-3 and MIT licensed libraries

Upstream Contact

https://github.com/symengine/symengine.py

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cmake: A cross-platform build system generator*
- *cython: C-Extensions for Python, an optimizing static compiler*
- *symengine: A C++ symbolic manipulation library*

Version Information

package-version.txt:

```
0.14.1
```

version_requirements.txt:

```
symengine.py >= 0.6.1
```

Equivalent System Packages

```
$ sudo pacman -S python-symengine
```

```
$ conda install python-symengine
```

```
$ sudo pkg install math/py-symengine
```

```
$ sudo emerge dev-python/symengine
```

See https://repology.org/project/python:symengine/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.397 symmetrica: Library for representation theory

Description

Symmetrica is a Collection of C routines for representation theory.

It is a program developed by Lehrstuhl Mathematik II of the University of Bayreuth. It has routines to handle the following topics:

- ordinary representation theory of the symmetric group and related groups (2/11/04)
- ordinary representation theory of the classical groups
- modular representation theory of the symmetric group
- projective representation theory of the symmetric group
- combinatorics of tableaux
- symmetric functions and polynomials (7/22/04)
- commutative and non commutative Schubert polynomials
- operations of finite groups.
- ordinary representation theory of Hecke algebras of type A_n

For more details check http://www.algorithm.uni-bayreuth.de/en/research/SYMMETRICA

Updated package on https://gitlab.com/sagemath/symmetrica/-/releases with changes to modernize the source and the build system.

License

Public Domain (see the above web site)

Upstream Contact

- (passed away in 2013) Axel Kohnert - see http://www.mathe2.uni-bayreuth.de/axel/

Type

standard

Dependencies

Version Information

package-version.txt:

```
3.1.0
```

Equivalent System Packages

```
$ sudo pacman -S symmetrica
```

```
$ conda install symmetrica
```

```
$ sudo apt-get install libsymmetrica-dev
```

```
$ sudo dnf install symmetrica-devel
```

```
$ sudo pkg install math/symmetrica
```

```
$ sudo emerge sci-libs/symmetrica
```

```
$ brew install dimpase/tap/symmetrica
```

```
$ nix-env --install symmetrica
```

```
$ sudo xbps-install symmetrica-devel
```

See https://repology.org/project/symmetrica/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.398 sympow: Computes special values of symmetric power elliptic curve L-functions

Description

SYMPOW is a package to compute special values of symmetric power elliptic curve L-functions. It can compute up to about 64 digits of precision.

License

- See the file src/COPYING

Upstream Contact

SYMPOW does not appear to be maintained any longer. Mark Watkins, the package author, now works at Magma. Previous (possibly still usable) email is watkins@maths.usyd.edu.au

New upstream: https://gitlab.com/rezozer/forks/sympow

Dependencies

- GNU patch

Special Update/Build Instructions

- Some of the code is very dubious, and it is anyones guess really what the compiler does with it. For example, the following line exists in src/eulerfactors.c:

  if ((HECKE) && (d==1)) return hecke_good(p,ap,m,v);

  But since hecke_good is defined as returning void, it's hard to know exactly how this code behaves. I would not be surprised by any bugs that might show up. I (David Kirkby) would personally not trust this code much at all.

- This is a difficult package to maintain. A github issue (#9758) has been opened to implement Watkins-Delaunay's algorithm for computing modular degrees in Sage. Once implemented, it should be possible to remove this package.
- The package is configured such that the data files are in a directory below where 'sympow' is installed. If Sage is installed globally, then it will be impossible to create the data files without being root. This has been fixed in the Gentoo Linux distribution. Some information from Christopher can be seen on Issue #1 This package will generate binary versions of all shipped datafiles, so these will work. However, creating totally new datafiles from scratch will not work.

Type

standard

Dependencies

- *pari: Computer algebra system for fast computations in number theory*

Version Information

package-version.txt:

```
2.023.6
```

Equivalent System Packages

```
$ sudo pacman -S sympow
```

```
$ conda install sympow
```

```
$ sudo apt-get install sympow
```

```
$ sudo dnf install sympow
```

```
$ sudo emerge sci-mathematics/sympow
```

```
$ brew install dimpase/tap/sympow
```

```
$ nix-env --install sympow
```

```
$ sudo zypper install sympow
```

```
$ sudo xbps-install sympow
```

See https://repology.org/project/sympow/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.399 sympy: Python library for symbolic mathematics

Description

SymPy is a Python library for symbolic mathematics. It aims to become a full-featured computer algebra system (CAS) while keeping the code as simple as possible in order to be comprehensible and easily extensible. SymPy is written entirely in Python and does not require any external libraries, except optionally for plotting support.

Website

https://sympy.org/

License

New BSD: http://www.opensource.org/licenses/bsd-license.php

Upstream Contact

https://pypi.org/project/sympy/

Type

standard

Dependencies

- $(PYTHON)
- *mpmath: Pure Python library for multiprecision floating-point arithmetic*
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
1.14.0
```

version_requirements.txt:

```
sympy >=1.6, <2.0
```

Equivalent System Packages

```
$ sudo pacman -S python-sympy
```

```
$ conda install sympy
```

```
$ sudo apt-get install python3-sympy
```

```
$ sudo dnf install python3-sympy
```

```
$ sudo emerge dev-python/sympy
```

```
$ sudo port install py-sympy
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-sympy'
```

```
$ sudo xbps-install python3-sympy
```

See https://repology.org/project/python:sympy/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.400 tachyon: A ray tracing system

Description

Tachyon is a raytracer developed by John E. Stone. Tachyon supports the typical ray tracer features, most of the common geometric primitives, shading and texturing modes, etc. It also supports less common features such as HDR image output,

ambient occlusion lighting, and support for various triangle mesh and volumetric texture formats beneficial for molecular visualization (e.g. rendering VMD scenes).

Currently not all of Tachyon's functionality is exported by the Sage interface.

License

Copyright (c) 1994-2010 John E. Stone All rights reserved.

Redistribution and use in source and binary forms, with or without modification, are permitted provided that the following conditions are met:

1. Redistributions of source code must retain the above copyright notice, this list of conditions and the following disclaimer.
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3. The name of the author may not be used to endorse or promote products derived from this software without specific prior written permission.

Upstream Contact

- http://jedi.ks.uiuc.edu/~johns/raytracer/
- http://www.photonlimited.com/~johns/raytracer/
- John Stone <johns@ks.uiuc.edu>

Dependencies

This spkg depends on:

- libpng

Special Update/Build Instructions

- Delete the scenes directory, which has lots of cool examples.
- Delete the msvc directory, which is also large and not used within Sage.
- The CVS subdirectories are currently (almost) empty, but should otherwise be deleted.
- The upstream files had strange permissions, i.e. some source files were executable, while almost all files weren't world-readable.
- There's seems to be some crap like `tachyon.html.tar.gz` and a few `.#*` files I haven't [yet] deleted, since they're not that large.
- TODO: Check whether building multi-threaded versions on MacOS X meanwhile works. (This was said to fail with an old beta.)
- TODO: Use `patch` instead of copying over pre-patched files.
- TODO: [Optionally] also install some of the documentation.
- TODO: I doubt the CFLAGS set for AIX and HP-UX won't get overridden by the created Makefile, but that's a minor issue. -leif

Type

standard

Dependencies

- *libpng: Bitmap image support*

Version Information

package-version.txt:

```
0.99.5.p0
```

Equivalent System Packages

```
$ sudo pacman -S tachyon
```

```
$ conda install tachyon
```

```
$ sudo apt-get install tachyon
```

```
$ sudo dnf install tachyon tachyon-devel
```

```
$ sudo pkg install graphics/tachyon
```

```
$ sudo emerge media-gfx/tachyon
```

```
$ brew install dimpase/tap/tachyon
```

```
$ nix-env --install tachyon
```

```
$ sudo zypper install tachyon
```

```
$ sudo xbps-install tachyon
```

See https://repology.org/project/tachyon/versions, https://repology.org/project/tachyon-opengl/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.401 tdlib: Algorithms for computing tree decompositions of graphs

Description

This library, now known as treedec, provides algorithms concerning tree decompositions.

License

- GNU General Public License v2
- GNU General Public License v3

Upstream Contact

- https://gitlab.com/freetdi/treedec
- https://github.com/freetdi/tdlib
- https://github.com/felix-salfelder

Type

optional

Dependencies

Version Information

package-version.txt:

```
0.9.3.p0
```

Equivalent System Packages

```
$ sudo pacman -S tdlib
```

```
$ sudo dnf install python3-tdlib python3-tdlib-devel
```

See https://repology.org/project/python:tdlib/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.402 terminado: Tornado websocket backend for the term.js Javascript terminal emulator library

Description

This is a Tornado websocket backend for the term.js Javascript terminal emulator library.

It evolved out of pyxterm, which was part of GraphTerm (as lineterm.py), v0.57.0 (2014-07-18), and ultimately derived from the public-domain Ajaxterm code, v0.11 (2008-11-13) (also on Github as part of QWeb).

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *hatchling: Modern, extensible Python build backend*
- *ptyprocess: Python interaction with subprocesses in a pseudoterminal*
- *tornado: Python web framework and asynchronous networking library*

Version Information

package-version.txt:

```
0.18.1
```

version_requirements.txt:

```
terminado >=0.8.3
```

Equivalent System Packages

```
$ conda install terminado
```

```
$ sudo emerge dev-python/terminado
```

```
$ sudo port install py-terminado
```

```
$ sudo xbps-install python3-terminado
```

See https://repology.org/project/terminado/versions, https://repology.org/project/python:terminado/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.403 texlive: A comprehensive TeX system

Description

TeX Live is an easy way to get up and running with the TeX document production system. It provides a comprehensive TeX system with binaries for most flavors of Unix, including GNU/Linux, and also Windows. It includes all the major TeX-related programs, macro packages, and fonts that are free software, including support for many languages around the world.

This package installs all texlive packages required to build Sage. If necessary, texlive itself is installed.

License

Various FSF-approved free software licenses. See https://www.tug.org/texlive/copying.html for details.

Upstream Contact

Home page: https://www.tug.org/texlive

Dependencies

- python

Special Update/Build Instructions

This package requires internet access to download texlive packages for the TeX mirrors.

Type

optional

Dependencies

Version Information

Equivalent System Packages

```
$ apk add texlive
```

```
$ sudo pacman -S texlive-core texlive-latexextra texlive-langjapanese  \
      texlive-langcyrillic texlive-langchinese
```

```
$ sudo apt-get install texlive-latex-extra texlive-xetex latexmk dvipng  \
      tex-gyre texlive-fonts-recommended texlive-lang-cyrillic  \
      texlive-lang-english texlive-lang-european texlive-lang-french  \
      texlive-lang-german texlive-lang-italian texlive-lang-japanese  \
      texlive-lang-polish texlive-lang-portuguese texlive-lang-spanish  \
      texlive-lang-chinese
```

```
$ sudo dnf install latexmk texlive texlive-collection-latexextra  \
      texlive-collection-langcyrillic texlive-collection-langeuropean  \
      texlive-collection-langfrench texlive-collection-langgerman  \
      texlive-collection-langitalian texlive-collection-langjapanese  \
      texlive-collection-langpolish texlive-collection-langportuguese  \
      texlive-collection-langspanish texlive-collection-langcjk
```

```
$ sudo emerge dev-tex/latexmk app-text/texlive app-text/dvipng  \
      dev-texlive/texlive-langcjk dev-texlive/texlive-langcyrillic  \
      dev-texlive/texlive-langenglish dev-texlive/texlive-langeuropean  \
      dev-texlive/texlive-langfrench dev-texlive/texlive-langgerman  \
      dev-texlive/texlive-langitalian dev-texlive/texlive-langjapanese  \
      dev-texlive/texlive-langportuguese  \
      dev-texlive/texlive-langspanish dev-texlive/texlive-latexextra  \
      dev-texlive/texlive-latexrecommended  \
      dev-texlive/texlive-mathscience dev-texlive/texlive-langchinese
```

```
$ sudo port install texlive
```

```
$ sudo zypper install texlive
```

```
$ sudo slackpkg install texlive
```

```
$ sudo xbps-install texlive
```

See https://repology.org/project/texlive/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.404 texlive_luatex: LuaTeX packages

Description

Packages for LuaTeX, a TeX engine using Lua as an embedded scripting and extension language, with native support for Unicode, OpenType/TrueType fonts, and both PDF and DVI output.

The purpose of this dummy package is to associate system package lists with it.

License

GNU General Public License version 2.0 (GPLv2)

Upstream Contact

https://www.luatex.org/

Type

optional

Dependencies

- *texlive: A comprehensive TeX system*

Version Information

Equivalent System Packages

```
$ apk add texlive-luatex
```

```
$ sudo pacman -S texlive-luatex
```

```
$ sudo apt-get install texlive-luatex
```

```
$ sudo dnf install texlive-luatex
```

```
$ sudo emerge dev-texlive/texlive-luatex
```

```
$ sudo port install texlive-luatex
```

```
$ sudo zypper install texlive-luatex
```

See https://repology.org/project/texlive-luatex/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.405 texttable: Python module for creating simple ASCII tables

Description

Python module for creating simple ASCII tables

License

MIT License (MIT)

Upstream Contact

https://github.com/foutaise/texttable/

Dependencies

- python

Special Update/Build Instructions

Type

optional

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
1.7.0
```

version_requirements.txt:

```
texttable >=1.6.3
```

Equivalent System Packages

```
$ sudo pacman -S python-texttable
```

```
$ conda install texttable
```

```
$ sudo apt-get install python3-texttable
```

```
$ sudo dnf install python3-texttable
```

```
$ sudo emerge dev-python/texttable
```

```
$ sudo port install py-texttable
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-texttable'
```

```
$ sudo xbps-install python3-texttable
```

See https://repology.org/project/texttable/versions, https://repology.org/project/python:texttable/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.406 threejs: JavaScript library to display 3D graphics in the browser

Description

Three.js is a JavaScript library to display 3D graphics in the browser.

License

MIT License

Upstream Contact

Home page: http://threejs.org

Dependencies

None.

Special Update/Build Instructions

None.

Type

standard

Dependencies

Version Information

package-version.txt:

```
r122.p0
```

Equivalent System Packages

```
$ conda install 'threejs-sage=122.*'
```

See https://repology.org/project/threejs/versions, https://repology.org/project/threejs-sage/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.407 tides: Integration of ODEs

Description

TIDES is a library for integration of ODEs with high precision.

License

GPLv3+

Upstream Contact

- Marcos Rodriguez (marcos@unizar.es)

Dependencies

- gcc
- mpfr
- gmp

Special Update/Build Instructions

minc_tides.patch changes the size of the name of the temporal files, so there is no problem in systems that use long names. Also solves a bug in the inverse function.

Type

optional

Dependencies

- $(MP_LIBRARY)
- *mpfr: Multiple-precision floating-point computations with correct rounding*

Version Information

package-version.txt:

```
2.0.p0
```

Equivalent System Packages

See https://repology.org/project/tides/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.408 tinycss2: A tiny CSS parser

Description

A tiny CSS parser

License

Upstream Contact

https://pypi.org/project/tinycss2/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *webencodings: Character encoding aliases for legacy web content*

Version Information

package-version.txt:

```
1.5.1
```

version_requirements.txt:

```
tinycss2
```

Equivalent System Packages

```
$ apk add py3-tinycss2
```

```
$ sudo pacman -S python-tinycss2
```

```
$ conda install tinycss2
```

```
$ sudo apt-get install python3-tinycss2
```

```
$ sudo dnf install python3-tinycss2
```

```
$ sudo pkg install textproc/py-tinycss2
```

```
$ sudo emerge dev-python/tinycss2
```

```
$ sudo port install py-tinycss2
```

```
$ sudo zypper install python-tinycss2
```

```
$ sudo xbps-install python3-tinycss2
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.409 tomli_w: A TOML writer for Python

Description

tomli-w is a Python library for writing TOML files. It is the write counterpart to tomllib (included in the standard library since Python 3.11) and produces standards-compliant TOML output.

License

MIT

Upstream Contact

https://pypi.org/project/tomli-w/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
1.2.0
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.410 topcom: Compute triangulations of point configurations and oriented matroids

Description

TOPCOM is a collection of clients to compute Triangulations Of Point Configurations and Oriented Matroids, resp.

The algorithms use only combinatorial data of the point configuration as is given by its oriented matroid. Some basic commands for computing and manipulating oriented matroids can also be accessed by the user.

It was very much inspired by the maple program PUNTOS, which was written by Jesus de Loera. TOPCOM is entirely written in C++, so there is a significant speed up compared to PUNTOS.

License

GPL v2

Upstream Contact

```
Prof. Dr. Jörg Rambau <Joerg.Rambau@uni-bayreuth.de>
Lehrstuhl für Wirtschaftsmathematik
Raum FAN-D.1.29 (Sekretariat: FAN-D.1.30)
Universität Bayreuth
D-95440 Bayreuth
Germany
Tel: +49-921-55-7350, Fax: +49-921-55-7352
http://www.rambau.wm.uni-bayreuth.de
```

Dependencies

- gmp, libcdd

Special Update/Build Instructions

See spkg-src

Type

optional

Dependencies

- *cddlib: Double description method for polyhedral representation conversion*

Version Information

package-version.txt:

```
1.1.2
```

Equivalent System Packages

```
$ brew install Macaulay2/tap/topcom
```

See https://repology.org/project/topcom/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.411 tornado: Python web framework and asynchronous networking library

Description

Python web framework and asynchronous networking library

License

Apache License

Upstream Contact

Home page: http://www.tornadoweb.org

Dependencies

Python

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *certifi: Python package for providing Mozilla's CA Bundle*

Version Information

package-version.txt:

```
6.5.5
```

version_requirements.txt:

```
tornado >=6.0.4
```

Equivalent System Packages

```
$ conda install tornado
```

```
$ sudo dnf install python3-tornado
```

```
$ sudo emerge www-servers/tornado
```

```
$ sudo port install py-tornado
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-tornado'
```

```
$ sudo xbps-install python3-tornado
```

See https://repology.org/project/python:tornado/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.412 tox: tox is a generic virtualenv management and test command line tool

Description

tox is a generic virtualenv management and test command line tool

License

MIT

Upstream Contact

https://pypi.org/project/tox/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *cachetools: Extensible memoizing collections and decorators*
- *chardet: Universal encoding detector for Python 3*
- *colorama: Cross-platform colored terminal text*
- *filelock: Platform independent file lock*
- *packaging: Core utilities for Python packages*
- *platformdirs: Small Python package for determining appropriate platform-specific dirs, e.g. a "user data dir"*
- *pluggy: Plugin and hook calling mechanisms for python*
- *pyproject_api: API to interact with the python pyproject.toml based projects*
- *python_discovery: Python interpreter discovery*
- *tomli_w: A TOML writer for Python*
- *virtualenv: Virtual Python Environment builder*

Version Information

package-version.txt:

```
4.51.0
```

version_requirements.txt:

```
tox >= 4.2.7
```

Equivalent System Packages

```
$ sudo pacman -S python-tox
```

```
$ conda install tox
```

```
$ sudo apt-get install tox
```

```
$ sudo dnf install tox
```

```
$ sudo pkg install tox
```

```
$ sudo emerge dev-python/tox
```

```
$ brew install tox
```

```
$ sudo port install py-tox
```

```
$ sudo slackpkg install tox
```

```
$ sudo xbps-install tox
```

See https://repology.org/project/python:tox/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.413 traitlets: Traitlets Python configuration system

Description

Traitlets Python configuration system

License

BSD

Upstream Contact

https://pypi.org/project/traitlets/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
5.14.3
```

version_requirements.txt:

```
traitlets >=4.3.3
```

Equivalent System Packages

```
$ conda install traitlets
```

```
$ sudo emerge dev-python/traitlets
```

```
$ sudo port install py-traitlets
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-traitlets'
```

```
$ sudo xbps-install python3-traitlets
```

See https://repology.org/project/traitlets/versions, https://repology.org/project/python:traitlets/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.414 trove_classifiers: Canonical source for classifiers on PyPI (pypi.org)

Description

Canonical source for classifiers on PyPI (pypi.org)

License

Upstream Contact

https://pypi.org/project/trove-classifiers/

Type

standard

Dependencies

- $(PYTHON)
- *calver: Setuptools extension for CalVer package versions*
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
2026.1.14.14
```

version_requirements.txt:

```
trove-classifiers >= 2025.2
```

Equivalent System Packages

```
$ apk add py3-trove-classifiers
```

```
$ sudo pacman -S python-trove-classifiers
```

```
$ sudo dnf install python3-trove-classifiers
```

```
$ sudo pkg install devel/py-trove-classifiers
```

```
$ sudo emerge dev-python/trove-classifiers
```

```
$ sudo port install py-trove-classifiers
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.415 types_python_dateutil: Typing stubs for python-dateutil

Description

Typing stubs for python-dateutil

License

Apache-2.0 license

Upstream Contact

https://pypi.org/project/types-python-dateutil/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
2.9.0.20260323
```

version_requirements.txt:

```
types-python-dateutil
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.416 typing_extensions: Backported and Experimental Type Hints for Python 3.8+

Description

Backported and Experimental Type Hints for Python 3.8+

License

PSF

Upstream Contact

https://pypi.org/project/typing-extensions/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
4.15.0
```

version_requirements.txt:

```
typing_extensions >= 4.4.0; python_version<'3.11'
```

Equivalent System Packages

```
$ sudo pacman -S python-typing_extensions
```

```
$ conda install typing_extensions
```

```
$ sudo apt-get install python3-typing-extensions
```

```
$ sudo dnf install python3-typing-extensions
```

```
$ sudo pkg install devel/py-typing-extensions
```

```
$ sudo emerge dev-python/typing-extensions
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-typing_extensions'
```

```
$ sudo xbps-install python3-typing_extensions
```

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.417 tzdata: Provider of IANA time zone data

Description

Provider of IANA time zone data

License

Apache-2.0

Upstream Contact

https://pypi.org/project/tzdata/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
2025.3
```

version_requirements.txt:

```
tzdata
```

Equivalent System Packages

```
$ conda install python-tzdata
```

```
$ sudo dnf install tzdata
```

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.418 tzlocal: Python timezone information for the local timezone

Description

tzinfo object for the local timezone

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *pytz_deprecation_shim: Shims to make deprecation of pytz easier*
- *tzdata: Provider of IANA time zone data*

Version Information

package-version.txt:

```
5.3.1
```

version_requirements.txt:

```
tzlocal >=2.1
```

Equivalent System Packages

```
$ sudo pacman -S python-tzlocal
```

```
$ conda install tzlocal
```

```
$ sudo apt-get install python3-tzlocal
```

```
$ sudo dnf install python3-tzlocal
```

```
$ sudo pkg install devel/py-tzlocal
```

```
$ sudo emerge dev-python/tzlocal
```

```
$ sudo port install py-tzlocal
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-tzlocal'
```

```
$ sudo xbps-install python3-tzlocal
```

See https://repology.org/project/tzlocal/versions, https://repology.org/project/python:tzlocal/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.419 uri_template: RFC 6570 URI Template Processor

Description

RFC 6570 URI Template Processor

License

MIT License

Upstream Contact

https://pypi.org/project/uri-template/

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
1.3.0
```

version_requirements.txt:

```
uri-template
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.420 urllib3: HTTP library with thread-safe connection pooling, file post, and more

Description

HTTP library with thread-safe connection pooling, file post, and more

License

MIT

Upstream Contact

https://pypi.org/project/urllib3/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
2.6.3
```

version_requirements.txt:

```
urllib3
```

Equivalent System Packages

```
$ sudo pacman -S python-urllib3
```

```
$ conda install urllib3
```

```
$ sudo apt-get install python3-urllib3
```

```
$ sudo dnf install python3-urllib3
```

```
$ sudo pkg install net/py-urllib3
```

```
$ sudo emerge dev-python/urllib3
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-urllib3'
```

```
$ sudo xbps-install python3-urllib3
```

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.421 valgrind: Memory error detector, call graph generator, runtime profiler

Description

Valgrind is an instrumentation framework for building dynamic analysis tools. There are Valgrind tools that can automatically detect many memory management and threading bugs, and profile your programs in detail. You can also use Valgrind to build new tools.

The Valgrind distribution currently includes six production-quality tools: a memory error detector, two thread error detectors, a cache and branch-prediction profiler, a call-graph generating cache and branch-prediction profiler, and a heap profiler. It also includes three experimental tools: a heap/stack/global array overrun detector, a second heap profiler that examines how heap blocks are used, and a SimPoint basic block vector generator.

License

Valgrind is Open Source / Free Software, and is freely available under the GNU General Public License, version 2.

Upstream Contact

- http://www.valgrind.org/
- valgrind-user, valgrind-devel mailing lists

Type

optional

Dependencies

Version Information

Equivalent System Packages

```
$ apk add valgrind
```

```
$ sudo dnf install valgrind vagrind-devel
```

```
$ brew install valgrind
```

```
$ sudo port install valgrind
```

```
$ sudo zypper install valgrind
```

```
$ sudo xbps-install valgrind
```

See https://repology.org/project/valgrind/versions

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.422 vcs_versioning: Version management from VCS metadata

Description

vcs-versioning extracts and manages project version numbers from version control system (VCS) metadata such as Git tags. It is used by setuptools-scm and other packaging tools to automatically derive the project version without hardcoding it.

License

MIT

Upstream Contact

https://pypi.org/project/vcs-versioning/

Type

standard

Dependencies

- $(PYTHON)
- *packaging: Core utilities for Python packages*
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
1.1.1
```

Equivalent System Packages

(none known)

However, these system packages will not be used for building Sage because `spkg-configure.m4` has not been written for this package; see Issue #27330 for more information.

6.1.423 virtualenv: Virtual Python Environment builder

Description

Virtual Python Environment builder

License

MIT

Upstream Contact

https://pypi.org/project/virtualenv/

Type

standard

Dependencies

- $(PYTHON)
- *distlib: Distribution utilities*
- *filelock: Platform independent file lock*
- *pip: Tool for installing and managing Python packages*
- *platformdirs: Small Python package for determining appropriate platform-specific dirs, e.g. a "user data dir"*
- *python_discovery: Python interpreter discovery*

Version Information

package-version.txt:

```
21.2.0
```

version_requirements.txt:

```
virtualenv
```

Equivalent System Packages

```
$ conda install virtualenv
```

```
$ sudo dnf install python3-virtualenv
```

```
$ sudo xbps-install python3-virtualenv
```

If the system package is installed, `./configure` will check if it can be used.

6.1.424 wcwidth: Measures the displayed width of unicode strings in a terminal

Description

Measures the displayed width of unicode strings in a terminal

License

MIT

Upstream Contact

https://pypi.org/project/wcwidth/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
0.6.0
```

version_requirements.txt:

```
wcwidth >=0.1.7
```

Equivalent System Packages

```
$ conda install wcwidth
```

```
$ sudo dnf install python3-wcwidth
```

```
$ sudo emerge dev-python/wcwidth
```

```
$ sudo port install py-wcwidth
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-wcwidth'
```

```
$ sudo xbps-install python3-wcwidth
```

See https://repology.org/project/wcwidth/versions, https://repology.org/project/python:wcwidth/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.425 webcolors: Library for working with the color formats defined by HTML and CSS

Description

Library for working with the color formats defined by HTML and CSS

License

BSD-3-Clause

Upstream Contact

https://pypi.org/project/webcolors/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
25.10.0
```

version_requirements.txt:

```
webcolors
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.426 webencodings: Character encoding aliases for legacy web content

Description

Character encoding aliases for legacy web content.

License

BSD License

Upstream Contact

Home Page: https://github.com/gsnedders/python-webencodings

Dependencies

Python

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
0.5.1
```

version_requirements.txt:

```
webencodings >=0.5.1
```

Equivalent System Packages

```
$ sudo pacman -S python-webencodings
```

```
$ conda install webencodings
```

```
$ sudo apt-get install python3-webencodings
```

```
$ sudo dnf install python3-webencodings
```

```
$ sudo emerge dev-python/webencodings
```

```
$ sudo port install py-webencodings
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-webencodings'
```

```
$ sudo xbps-install python3-webencodings
```

See https://repology.org/project/python:webencodings/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.427 websocket_client: WebSocket client for Python with low level API options

Description

WebSocket client for Python with low level API options

License

Apache-2.0

Upstream Contact

https://pypi.org/project/websocket-client/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)

Version Information

package-version.txt:

```
1.9.0
```

version_requirements.txt:

```
websocket-client
```

Equivalent System Packages

(none known)

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.428 wheel: A built-package format for Python

Description

A built-package format for Python

License

MIT

Upstream Contact

https://pypi.org/project/wheel/

Type

standard

Dependencies

- $(PYTHON)
- *packaging: Core utilities for Python packages*
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
0.46.3
```

Equivalent System Packages

```
$ sudo pacman -S python-wheel
```

```
$ conda install wheel
```

```
$ sudo apt-get install python3-wheel
```

```
$ sudo dnf install python3-wheel
```

```
$ sudo emerge dev-python/wheel
```

```
$ sudo port install py-wheel
```

```
$ sudo zypper install 'python3${PYTHON_MINOR}-wheel'
```

```
$ sudo xbps-install python3-wheel
```

See https://repology.org/project/wheel/versions, https://repology.org/project/python:wheel/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.429 widgetsnbextension: Jupyter interactive widgets for Jupyter Notebook

Description

Jupyter interactive widgets for Jupyter Notebook

License

BSD 3-Clause License

Upstream Contact

https://pypi.org/project/widgetsnbextension/

Type

standard

Dependencies

- $(PYTHON)
- $(PYTHON_TOOLCHAIN)
- *jupyter_core: Jupyter core package*

Version Information

package-version.txt:

```
4.0.15
```

version_requirements.txt:

```
widgetsnbextension
```

Equivalent System Packages

```
$ sudo pacman -S jupyter-widgetsnbextension
```

```
$ conda install widgetsnbextension
```

```
$ sudo dnf install python3-widgetsnbextension
```

```
$ sudo pkg install devel/py-widgetsnbextension
```

```
$ sudo emerge dev-python/widgetsnbextension
```

```
$ sudo port install py-widgetsnbextension
```

```
$ sudo zypper install jupyter-widgetsnbextension
```

```
$ sudo xbps-install python3-jupyter_widgetsnbextension
```

See https://repology.org/project/python:widgetsnbextension/versions, https://repology.org/project/jupyter-widgetsnbextension/versions, https://repology.org/project/python:jupyter-widgetsnbextension/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

6.1.430 xindy: a general-purpose index processor

Description

This dummy package represents xindy: xindy was developed after an impasse had been encountered in the attempt to complete internationalisation of makeindex. Xindy can be used to process indexes for documents marked up using (LA)TEX, Nroff family and SGML-based languages. Xindy is highly configurable, both in markup terms and in terms of the collating order of the text being processed.

We do not have an SPKG for it. The purpose of this dummy package is to associate system package lists with it.

License

GNU General Public License version 2.0 (GPLv2)

Upstream Contact

http://www.xindy.org/

Type

optional

Dependencies

Version Information

Equivalent System Packages

```
$ sudo apt-get install xindy
```

```
$ sudo dnf install texlive-xindy
```

```
$ sudo port install xindy
```

```
$ sudo zypper install xindy
```

See https://repology.org/project/xindy/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.431 zeromq: A modern networking library

Description

A modern networking library. Also known as 0mq or zmq. The same API is provided by http://www.crossroads.io, though we currently use the http://www.zeromq.org implementation.

License

LGPLv3+

Upstream Contact

http://www.zeromq.org

Dependencies

A working compiler.

Special Update/Build Instructions

N/A

Type

standard

Dependencies

Version Information

package-version.txt:

```
4.3.5
```

Equivalent System Packages

```
$ apk add zeromq-dev
```

```
$ sudo pacman -S zeromq
```

```
$ conda install zeromq
```

```
$ sudo apt-get install libzmq3-dev
```

```
$ sudo dnf install zeromq zeromq-devel
```

```
$ sudo pkg install net/libzmq4
```

```
$ sudo emerge net-libs/zeromq
```

```
$ brew install zeromq
```

No package needed.

```
$ sudo zypper install zeromq zeromq-tools 'pkgconfig(libzmq)'
```

```
$ sudo xbps-install zeromq-devel
```

See https://repology.org/project/zeromq/versions

If the system package is installed, `./configure` will check if it can be used.

6.1.432 zipp: A pathlib-compatible zipfile object wrapper

Description

A pathlib-compatible Zipfile object wrapper. A backport of the Path object.

License

MIT License

Upstream Contact

Home page: https://github.com/jaraco/zipp

Dependencies

Python, Setuptools

Type

standard

Dependencies

- $(PYTHON)
- *pip: Tool for installing and managing Python packages*

Version Information

package-version.txt:

```
3.23.0
```

version_requirements.txt:

```
zipp >=0.5.2
```

Equivalent System Packages

```
$ conda install zipp
```

```
$ sudo dnf install python3-zipp
```

```
$ sudo emerge dev-python/zipp
```

```
$ sudo port install py-zipp
```

```
$ sudo xbps-install python3-zipp
```

See https://repology.org/project/python:zipp/versions

If the system package is installed and if the (experimental) option `--enable-system-site-packages` is passed to `./configure`, then `./configure` will check if the system package can be used.

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